

Mediatek Multi-AP Application Note

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Version History

Version	Date	Author	Description
2.3.0	2022-02-11	Sonal Arora	• Application note revamped according to new template and requirement. • Multiple review comments fixed.
2.3.0	2022-02-22	Yogender Pathak	• Example diagram added for user skip channel list in section 7.3.1.4 • Updated Diagram for user configurable restricted BSS in section 6.3.1
2.3.1	2022-02-23	Faping	• Service prioritization feature added in section 8.2
2.3.1	2022-03-16	Jignesh Gosai	• WPS Unconfig feature added in section 3.11
3.2	2022-03-24	Eric PC/Dipanshu	• Mesh SR feature added in section 3.12
3.2	2022-03-27	Faping	• Example diagram added for auto role in section 3.5.3 • Example diagram added for switch setting in section 3.7.2, 3.7.3 • Example diagram added for DHCP in section 3.9.1
3.3	2022-05-25	Deepak Kumar	• Added Third Party connection limitations new points 6 in section 3.3.4.
3.3	2022-05-27	Dipanshu Mittal	• Added Note 7.2.3.1 “Limitation with channel monitoring”
3.3	2022-05-30	Sonal Arora	• Updated section 6.2.5 point 2 , to clearly indicate that Pre-assoc Band Steering is not supported in MAP Turnkey
3.3	2022-05-30	Mayank Arora	• Added section 3.13 for Single band operations.
3.3	2022-08-17	Jie Shen	• Update 3.7 for GSW and DSA setting
3.3	2022-9-13	Jing Chang	• Customer Feature to Enable/Disable SSID on Specific Interface

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4.0	2022-10-07	Saidul Choudhury Yogender Pathak Sonal Arora Dipanshu Mittal	• Updated 3.2.2.2, Special notes for WPS • Updated 7.2.6, Radar handling • Updated Figure 10-9 for wts_bss_info_config for 6G • Updated 4.3.2.2.2 for BH priority on 6G and added NOTE to make connection on 6G • Added new point 12 under section 4.3.2.1 for connection procedure. • Added new section for ax7800 10.3.1.5 • Updated Section 4.4.3 • Updated Section 7.3.1.1 • Update section 1.3, MAP variants add with MAP TK support • Added in section 3.2.1 Important Note about 6G Radio • Updated section 7.2.9 update commands and flags for channel planning feature • Updated 7.2.9.3. screen shots • Appendix table 10.3 updated. • Updated 5.3.2.1 screen shot of client database • Update Multi-AP steering. Sections: 6.2.3.1, 6.2.3.2, 6.2.3.4 • Updated parameters in 6.2.4.4 • Update example in section 6.2.4.5
4.0	2022-11-28	Nakul Dhingra	• Added new section for DPP Reconfiguration. (3.2.3.9).
4.0	2022-12-13	Wailiu Huang	• 3rd party controller feature added in section 10.
4.0	2023-1-5	Sonal Arora	• Added note in section 3.2.1 “Important Note about Max BSS number in the mesh network”
4.1	2023-4-6	Sonal	• Added important note about WEP, in section 3.2.1 • Added one Known IOT issue with Redmi note pro5 phone in section 6.2.6 • Added mapcli command for assoc sta list in section 6.2.4.5 • In section 6.2.2 Added sub points(6 and 7) in 6.2.3.2 and BS is not supported for MLD client • Updated note in section 7.2.8 , Divergent channel planning disable • Section 1.4 added MLO MAP device • Section 1.5 , sub point added for MLD group creation. • Section 11.13 , add acl_ctrl commands

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4.1	2023-4-7	Saidul	• Added new section 3.2.2 Configuring BSS for MLO and new WTS file format • Added NOTE for WIFI7 SKU in the section 4.3.2 and 4.3.2.1 • Added NOTE for WIFI7 SKU for sample WTS file format.
4.1	2023-4-10	Deepak	• Added changes related to UCI support in latest SKU (Eagle) in multiple features. • Added 3.2.2 , 11.19 and 11.20
4.1	2023-4-10	Manoj	• Added dump topology, client capability and DE new screen shots for MLD specific new configuration added.
4.1	2023-5-18	Deepak	• Static Puncturing handling in channel planning
4.1	2023-5-24	Yogender	• Network optimization and BH priority coexistence for MLD agents Section 4.4.4, Point:6
4.1	2023-6-19	Sonal	• Add MLD regroup feature details section 3.14

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Table of Contents

Version History **2**
Table of Contents **5**
List of Figures **11**
1 MAP Introduction **19**
1.1 Introduction 19
1.2 Feature Overview 19
1.3 MAP Variants 20
1.4 MLO MAP device 22
1.5 EasyMesh (MAP) WebUI Introduction 24
2 MAP Architecture **30**
2.1 Introduction 30
2.2 Driver 31
2.3 WAPP 31
2.3.1 Wappctrl 31
2.3.2 Libwpactrl 32
2.4 1905 daemon 32
2.4.1 1905_ctrl 32
2.5 Mapfilter 32
2.6 MAP Daemon 32
2.6.1 MAPD_ctrl 32
2.6.2 Libmapd 33
2.7 Topologies supported 33
3 MAP Network formation and support **36**
3.1 Introduction 36
3.2 MAP Onboarding and Auto-Configuration 36
3.2.1 Configuration Settings 36
3.2.1.1 Configuring a SSID having space or special character **40**
3.2.1.2 Configuring a BSS in WEP security **40**
3.2.1.3 Configuring a Fronthaul BSS in Mixed Mode security **41**
3.2.1.4 Configuring BSS for MLO **41**
3.2.1.1 Configuring BSS using UCI **43**
3.2.2 WPS Onboarding for MAP 43
3.2.2.1 WPS Onboarding Daemon interactions **44**
3.2.2.2 WPS Onboarding daemon interactions for hostapd supported SKU **46**
3.2.2.3 Onboarding and Autoconfiguration procedure **47**
3.2.2.4 WPS trigger on a remote agent via Controller **51**
3.2.3 DPP Onboarding for MAP 51
3.2.3.1 MAP Turnkey Model **52**
3.2.3.2 DPP Onboarding Daemon interactions **52**
3.2.3.3 Onboarding Method via a Multi-AP Logical Ethernet Interface with out-of-band DPP bootstrapping **54**
3.2.3.4 Onboarding method via Wi-Fi with out-of-band DPP bootstrapping **56**
3.2.3.5 Onboarding method via Wi-Fi with inband DPP bootstrapping using Push Button **59**
3.2.3.6 DPP timer **60**
3.2.3.7 DPP User failure Events **61**
SEC_CONNECTION_FAILURE **61**
USER_FAIL_EVENTS **61**
3.2.3.8 DPP State Machine **62**
3.2.3.9 DPP Reconfiguration **65**

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3.2.3.10 DPP Onboarding Important Points 65
3.2.3.11 DPP Onboarding Procedure 65
3.2.3.12 DPP onboarding after Profile-1 or Profile-2 onboarding 76
3.2.4 Behavior at boot-up 78
3.2.5 Backhaul Configuration Save 78
3.2.6 Re-onboard an already on-boarded device 78
3.2.7 BSS interface-SSID matching 78
3.2.8 Hidden SSID 79
3.2.9 Error handling for Auto-Configuration 80
3.3 Mediatek Multi-AP Controller Connectivity with Non-MAP devices 81
3.3.1 Ethernet BH link for controller with non-MAP uplink device 81
3.3.2 Wireless BH for controller with non-MAP uplink device: 81
3.3.2.1 Procedure 82
For this feature to work some settings need to be enabled in the Configuration file interf 82
3.3.3 Third Party Connection Example 82
3.3.3.1 Connection by Ethernet 82
3.3.3.2 Connection by WiFi 83
3.3.4 Limitations : 83
3.4 Trigger WPS at front-haul BSS of an Agent 84
3.4.1 Main flow 84
3.4.2 Procedure 84
Command Line Interface 85
3.5 Auto Role Selection 85
3.5.1 Main flow 86
3.5.2 Procedure 86
3.5.3 Configure & Test 87
3.6 Multi-AP Renew 88
3.6.1 Main flow of first time set-up 88
3.6.2 Main flow of wireless setting update 89
3.6.3 Procedure 89
3.6.4 Limitations 90
3.6.5 Customer Feature to Enable/Disable SSID on Specific Interface 90
3.7 Variations of switch/GMAC combination support 90
3.7.1 Switch OPS 92
3.7.1.1 Switch OPS arch 92
3.7.1.2 Switch OPS description 94
3.7.1.3 get_port_type 96
3.7.1.4 get_table_entry 97
3.7.1.5 search_table 99
3.7.1.6 get_port_status 102
3.7.1.7 set_switch_ts_vlan 104
3.7.1.8 set_switch_transparent_vlan 104
3.7.1.9 get_port_speed 105
3.7.1.10 set_port_block 106
3.7.1.11 clear_table 106
3.7.1.12 clear_port_table 107
3.7.1.13 deinit 107
3.7.2 Default setting 107
3.7.2.1 customer setting(wan port 0) 108
3.7.3 ext_phy_port setting 109
3.7.4 Per-Port-Per-Vlan setting for Non-DSA Architecture 111
3.7.5 Per-Port-Per-Vlan setting for DSA Architecture 111
3.7.5.1 Customer setting(wan port 0) 112
3.7.6 Related features 114

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3.7.7 Support status 114
3.8 Automatic BH Switching between Ethernet and Wi-Fi 114
3.8.1.1 Manual BH Switching (without Onboarding) 115
3.9 DHCP Control 115
3.9.1 Usage 115
3.10 Tx/Rx higher layer protocol data 117
3.10.1 Data flow 117
3.10.2 Usage 118
3.11 WPS Unconfig feature Support 118
3.11.1 Procedure 118
3.12 Mesh Spatial Reuse 121
3.12.1 Introduction 121
3.12.2 Operating Mode 122
3.12.3 Mesh SRG Protect Mode 122
3.12.4 Mesh Spatial Reuse 124
3.12.5 Profile Configuration Related to Mesh SR 126
3.12.6 Commands for Debugging 126
3.13 Single Band Operation 128
3.13.1 Limitation 128
3.14 MLD Dynamic Regroup 128
3.14.1 Configuring BSS for MLO regroup 130
4 MAP Network Maintenance 133
4.1 Introduction 133
4.2 Network Healing 133
4.2.1 Procedure 133
4.2.2 Feature Management Interfaces 134
4.2.2.1 WEB UI Interface 134
4.3 Backhaul Link Steering and Prioritization 135
4.3.1 Backhaul Link Steering 135
4.3.1.1 Feature Management Interfaces 135
4.3.2 Backhaul Link Prioritization 137
4.3.2.1 Main Flow 138
4.3.2.2 Main flow for Hostapd and MLO supported SKU 139
4.3.2.3 Feature Management Interfaces 139
4.4 Network Optimization 141
4.4.1 Network Optimization for MLDs in mesh network 141
4.4.2 Trigger conditions 141
4.4.3 Fail Safe Mechanism for network optimization 142
4.4.4 Important points 142
4.4.5 Procedure 143
4.4.6 Command Line Interface 144
4.5 ETH Backhaul between Agents (Loop detection and avoidance) 144
4.5.1 Main Flow 145
Configuration Interface 146
4.5.2 Limitations 146
5 Database and Diagnostics 147
5.1 Introduction 147
5.2 Topology Database 147
5.2.1 Feature Details 147
5.2.2 Feature Management Interfaces 148
5.2.2.1 Command line Interface 148
5.2.2.2 WEBUI Interface 173
5.3 Client Database 175
5.3.1 Feature details 175

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5.3.2 Feature Management Interfaces 175
5.3.2.1 Command line Interface 175
5.3.2.2 WEBUI Interface 177
5.4 Data Elements 178
5.4.1 MAP R2 DE Support 178
5.4.2 MAP R3 DE Support 179
5.4.2.1 R3 DE Spec Details 179
5.4.2.2 R3 DE design 180
5.4.2.3 R3 DE SW Stack 180
5.4.3 Feature Management Interfaces 181
5.4.3.1 MAP R3 DE value changes are per SKU 181
5.4.3.2 Command Line Interface 181
5.4.3.3 WEB UI Interfaces 181
5.5 Wi-Fi6 Support 184
5.5.2 Wi-Fi 6 Details 184
5.5.3 Wi-Fi 6 SW Stack 185
5.6 DPP Configuration file 187
5.7 1905 DPP parameters stored 189
5.7.1 csign key 189
5.7.2 Csign key 189
5.7.3 private net-access-key(pk) 189
5.7.4 Connector 190
5.7.5 fh_connector 190
5.7.6 bh_connector 190
6 Steering 191
6.1 Introduction 191
6.2 Centralized Steering feature details 191
6.2.1 Decision for Multi-AP Steering or Standalone Band Steering 192
6.2.2 Band Steering 193
6.2.2.1 Band steering for MLD clients 193
6.2.2.2 Idle post-association steering 193
6.2.2.3 Active post-association steering. 194
6.2.2.4 Band Steering Examples 195
6.2.3 Multi-AP Steering 198
6.2.3.1 Selection of Measurement Channels 199
6.2.3.2 Selection of Best BSS 200
6.2.3.3 Selection of Best BSS for MLD Clients 200
6.2.3.4 Steer coordination in Multi-AP environment 200
6.2.3.5 Tri-band detection 201
6.2.3.6 Multi AP Steering example 201
6.2.4 Feature Management Interfaces 202
6.2.4.1 Compile Time Options 202
6.2.4.2 Configuration file interface 202
6.2.4.3 Configuration Parameters for Band Steering 202
6.2.4.4 Configuration Parameters for Multi AP steering 204
6.2.4.5 Command Line Interface 206
6.2.4.6 WEBUI Interface 210
6.2.5 Limitations 211
6.2.6 Known IOT Issues 211
6.3 User Configurable Target BSS of Client Steering 213
6.3.1 Feature Details 213
6.3.2 Feature Management Interfaces 213
6.3.2.1 Configuration file interface 213
6.3.2.2 Command Line Interface 214

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6.4 User configured ACL 214
6.4.1 Feature Management Interfaces 214
6.4.1.1 Command Line Interface 214
7 Multi-AP Channel planning 217
7.1 Introduction 217
7.2 Channel Planning feature details 217
7.2.1 Message flow 217
7.2.2 When is channel planning run 218
7.2.3 Channel planning stages 219
7.2.3.1 Channel Monitoring 219
7.2.3.2 Channel Scanning 219
7.2.3.3 Channel Switching 220
7.2.4 Channel Planning with BW consideration 221
7.2.5 User Preferred Channel Configuration 221
7.2.6 RADAR Handling in channel planning 221
7.2.7 Channel Planning examples 223
7.2.7.1 Channel Planning run at Bootup 223
7.2.7.2 Channel Planning run due to User Prefer Channel command 223
7.2.7.3 Channel Planning handling when Radar on Agent (Ethernet BH) 224
7.2.7.4 Channel Planning handling when Radar on Agent (WiFi BH) 224
7.2.7.5 Channel Planning handling when Radar on Controller 225
7.2.8 Divergent channel planning 226
7.2.8.1 Divergent Channel planning example 226
7.2.8.2 RADAR handling for Divergent channel planning 227
7.2.9 Static Puncturing handling in channel planning 227
7.2.10 Feature Management Interfaces 229
7.2.10.1 Configuration file interface 229
7.2.10.2 Command Line Interface 231
7.2.10.3 WEB UI Interfaces 233
7.2.10.4 UCI Interfaces 238
7.3 User Configurable Skip Channel List 239
7.3.1 Feature Management Interfaces 240
7.3.1.1 Configuration file interface 240
7.3.1.2 Command Line Interface 240
7.3.1.3 WEBUI Interface 240
7.3.1.4 User configurable skip channel list Example 240
7.4 DFS CAC 241
7.4.1 Switching Channel to DFS 242
7.4.1.1 Which device will perform CAC algorithm Examples 243
7.4.2 CAC Enhancements 244
7.4.2.1 CAC by BW 244
7.4.2.2 CAC at boot time 244
7.4.2.3 CAC reset after HWRDD moved 245
7.4.3 RADAR Detection 245
7.4.3.1 Agent Radar Detection 245
7.4.3.2 Controller Radar Detection 245
7.4.4 DFS CAC in case of Bandwidth 160MHz 246
7.4.5 Feature Management Interfaces 247
7.4.5.1 Configuration File Interface 247
8 Traffic Separation and Service Prioritization 248
8.1 Traffic Separation 248
8.1.1 Introduction 248
8.1.2 Operating Mode 248
8.1.3 IPTV Service 249

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8.1.4 Topology and command combination 250
8.1.5 Limitations 251
8.1.6 Feature Management Interfaces 253
8.1.6.1 Configuration Parameters 253
8.1.6.2 WEB UI Interfaces 253
8.2 Service prioritization 255
8.2.1 Introduction 255
8.2.2 Rule category 255
8.2.2.1 Static rule 255
8.2.2.2 Dynamic rule 257
8.2.2.3 Dscp to pcp mapping table 258
8.2.3 SP configuration file 258
8.2.4 Feature Management Interfaces 259
8.2.4.1 WEB UI Interfaces 259
8.2.4.2 8.10.4.2 Command Line Interfaces 259
9 Stand-Alone Band Steering (Easy Mesh Disabled) 260
9.1 Introduction 260
9.2 Components 260
9.3 Feature details 260
9.3.1 Pre-association steering 260
9.3.2 Idle post-association steering 261
9.3.3 Active post-association steering. 262
9.3.3.1 Active Non-Overloaded Steering 262
9.3.3.2 Active Overloaded Steering [Offloading] 262
9.4 Limitations 263
9.5 Feature Management Interfaces 263
9.5.1 Enable Bandsteering 2.0 263
9.5.2 Configuration Parameters 263
9.5.3 Command Line Interface 265
9.6 Examples 268
9.7 Known IOT Issues 271
10 3rd party controller co-exists with MTK agent 273
10.1 Introduction 273
10.2 Setup Environment 273
10.2.1 Add virtual interface for controller 273
10.2.2 Add veth0 to the interface list of agent1 273
10.3 Onboarding and Auto Configuration 274
10.3.1 Agent1 onboarding and auto configuration 274
10.3.2 Agent2 and other enrollee onboarding and auto configuration 274
10.3.2.1 Wireless onboarding 274
10.3.2.2 Ethernet onboarding 274
10.4 Get Topology 274
11 Appendix 275
11.1 Configuration difference between LSDK and OpenWRT 275
11.2 How to enable MAP 275
11.3 How to enable MAP R2 276
11.3.1.1 Compile time flag 276
11.3.1.2 AX1800 276
11.3.1.3 AX3600 276
11.3.1.4 AX3200 276
11.3.1.5 AX7800 277
11.3.2 Run Time option 278
11.3.3 Dat file settings on each board 278
11.4 How to enable MAP R3 278

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11.4.1.1 Compile time flag 278
11.4.2 Run Time option 280
11.5 How to enable MAP R4 280
11.5.1.1 Compile time flag 280
11.5.2 Run Time option 281
11.6 How to use WAPP daemon 282
11.7 How to use 1905 daemon 282
11.8 How to use MAP Daemon 282
11.9 MAP device Bring-up 283
11.9.1 EasyMesh (LSDK) 283
11.9.1.1 Steps to bring up MAP device (LSDK) 283
11.9.2 EasyMesh (OpenWRT) 284
11.9.2.1 Steps to bring up MAP device (OpenWrt) 285
11.10 Steps to bring up Stand Alone Band Steering 285
11.11 Sample data of wts_bss_info_config file 286
11.12 DPP APIs 288
11.13 List of Configuration file parameters 295
11.14 List of MAPD_CLI Commands 298
11.15 Things to remember 305
11.16 Sample Application for receiving events from MAP Daemon 305
11.16.1 List of Currently Supported User Events 306
11.17 Sample mapd_default.txt file 308
11.18 Sample mapd_strng.txt file 309
11.19 Sample 1905d.cfg file 309
11.20 MAP integration with UCI system 310
11.20.1 Sample mapd UCI file 310
11.20.2 Sample 1905d_cfg UCI file 313
Exhibit 1 Terms and Conditions 315

List of Figures

Figure 1- 1. Easymesh (MAP) WFA History 19
Figure 1- 2. Single Chip Dual Band repeater 21
Figure 1- 3. Dual Chip Dual Band repeater 21
Figure 1- 4. Triband Repeater 21
Figure 1- 5. AP and Non-AP MLD vs. SLD 23
Figure 1- 6. Easymesh network with 2 MLD devices 23
Figure 1- 7. Easymesh network with 3 MLD devices, 1 controller and 2 agents connected in daisy chain 23
Figure 1- 8. EasyMesh Web UI drop-down 24
Figure 1- 9. Web UI default page 25
Figure 1- 10. Web UI page after adding MBSS for each radio 26
Figure 1- 11. Web UI page Need to reload 2.4G band 26
Figure 1- 12. Web UI page need to reload 5G band 27
Figure 1- 13. Web UI page to add MLD groups 27
Figure 1- 14. Web UI page enable Easymesh feature 28
Figure 1- 15. Web UI page for controller 28
Figure 1- 16. Web UI page for agent 29
Figure 2- 1. MAP Turnkey Model 30
Figure 2- 2. MAP Turnkey Model with hostapd/supplicant and cfg802.11 support 31
Figure 2- 3. Topology with Agent having wireless BH with Controller 33

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Figure 2- 4. Topology with Agent having wired BH with Controller 34
Figure 2- 5. Topology with some Agents having wireless BH and some having wired BH with Controller 34
Figure 2- 6. Mixed Topology with Controller having connection with Non-MAP AP 35

Figure 3- 1. AP MLD creation 41
Figure 3- 2. APCLI MLD creation 42
Figure 3- 3. WTS file for MLO connection 42
Figure 3- 4. Snapshot of etc/config/map file for configuring the BSS via UCI 43
Figure 3- 5. MAP WPS Onboarding and Autoconfiguration Procedure 44
Figure 3- 6. WPS Onboarding Interactions between daemons at the enrollee agent. 45
Figure 3- 7. WPS Onboarding Interactions between daemons at the controller. 45
Figure 3- 8. WPS Onboarding Interactions between daemons at the enrollee agent for Hostapd/Supplicant supported SKU. 46
Figure 3- 9. WPS Onboarding Interactions between daemons at the controller for Hostapd supported SKU. 47
Figure 3- 10. Agent WEB UI page shows Back-Haul Connection status as Ethernet Backhaul 48
Figure 3- 11. Agent and Controller connected via Ethernet Backhaul and 48
Figure 3- 12. Agent WEB UI page shows Back-Haul Connection status as Disconnected initially 49
Figure 3- 13. Agent WEB UI page shows Back-Haul Connection status as 5GL 49
Figure 3- 14. Agent and Controller connected via WIFI Backhaul 50
Figure 3- 15. Agent's WEB UI Advanced tab ALMAC of agent display 50
Figure 3- 16. Agent's and controller's WEB UI Display topology page show new topo 50
Figure 3- 17. WPS trigger on a remote agent via Controller flow diagram 51
Figure 3- 18. MTK Turnkey Model for DPP 52
Figure 3- 19. DPP Onboarding Controller Daemon Interaction 53
Figure 3- 20. DPP Onboarding Proxy Agent Daemon Interaction 53
Figure 3- 21. DPP Onboarding Enrollee agent Daemon Interaction 54
Figure 3- 22. DPP Onboarding Method via a Multi-AP Logical Ethernet Interface with out-of-band DPP bootstrapping 55
Figure 3- 23. DPP Onboarding Method via WiFi with out-of-band DPP bootstrapping 57
Figure 3- 24. Presence band priority parameter 58
Figure 3- 25. DPP CCE Indication message 59
Figure 3- 26 . DPP CCE Procedure 59
Figure 3- 27 . DPP initiator state machine 62
Figure 3- 28. DPP responder state machine 63
Figure 3- 29 . DPP configurator state machine 64
Figure 3- 30. DPP Enrollee state machine 64
Figure 3- 31. Controller's dpp_cfg.txt file example snapshot 66
Figure 3- 32. Agent's dpp_cfg.txt file example snapshot 66
Figure 3- 33. Generate DPP URI for Agent WEBUI snapshot 66
Figure 3- 34 . Retrieve DPP URI of Agent WEBUI snapshot 68
Figure 3- 35. Add and Submit Agent's URI on Controller's WEB UI page snapshot 69
Figure 3- 36. Controller's dpp_cfg.txt file example after addition of agent's URI snapshot 69
Figure 3- 37. Easymesh topology display after ether onboarding 70
Figure 3- 38. Post ethernet onboarding dpp_cfg.txt snapshot at Controller 70
Figure 3- 39. Post ethernet onboarding dpp_cfg.txt snapshot at Agent 70
Figure 3- 40. Trigger DPP Onboarding button at Agent's WEBUI page snapshot 71
Figure 3- 41. Topology after Agent onboards to controller via WiFi 71

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Figure 3- 42. Post WiFi onboarding dpp_cfg.txt snapshot at Agent 72
Figure 3- 43. Post ethernet onboarding dpp_cfg.txt snapshot at Controller 72
Figure 3- 44. Controller WEB UI page snapshot to trigger WPS onboarding 73
Figure 3- 45. Agent WEB UI page snapshot to trigger WPS onboarding 74
Figure 3- 46. Topology display after WPS onboarding success 74
Figure 3- 47. Post WPS onboarding dpp_cfg.txt snapshot at Agent 75
Figure 3- 48. Post WPS onboarding dpp_cfg.txt snapshot at Controller 75
Figure 3- 49. DPP Profile saved in mapd_user.cfg file after successful onboarding snapshot 78
Figure 3- 50. “wts_bss_info_config” file present at the controller 79
Figure 3- 51. MAP network topology tree 80
Figure 3- 52. MAP network topology tree A3 disconnects 80
Figure 3- 53. MAP network topology tree A3 reconnects 81
Figure 3- 54. Third Party Connection with Ethernet Example 83
Figure 3- 55. Third Party Connection with WiFi Example 83
Figure 3- 56. WEB UI button to trigger WPS at fronthaul of an agent 85
Figure 3- 57. Chain Topology with direct internet connection to MAP device 85
Figure 3- 58. Chain Topology with PON 85
Figure 3- 59. STAR Topology with PON 86
Figure 3- 60. WEB UI button to set MAP device role 87
Figure 3- 61. BSS configuration renew Web UI Interface 89
Figure 3- 63. Higher Layer Data flow daemon interaction 117
Figure 3- 64. Pin Generation 119
Figure 3- 65. Select AP to configure 119
Figure 3- 66. Enter the PIN generated on AP 120
Figure 3- 67. Name the SSID to configure on AP 120
Figure 3- 68 . Configured the Key & SSID 121
Figure 3- 69. Controller's BSS & WTS_BSS_Config_Info file updated 121
Figure 3- 70. Mesh Mode Switch Criteria 122
Figure 3- 71. Mesh SRG SR (not reuse MAP internal traffic, able to reuse OBSS Traffic) 122
Figure 3- 72. SRG/non-SRG PD threshold 123
Figure 3- 73. Mesh Spatial Reuse – reuse mesh internal fronthaul and external traffic 124
Figure 3- 74. Limited scenario of “Mesh internal SR on only near-end STAs” 125
Figure 3- 75. RSSI limitation of “Mesh internal SR on only near-end STAs” 125
Figure 3- 76. RSSI limitation of “Mesh internal SR not allowed on far-end STAs” 126
Figure 3- 77. Types of MLD groups 129
Figure 3- 77. Daemon event interaction diagram for MLD Reconfig/Regroup 129
Figure 3- 78. Sample Default wts_bss_info_config file 130
Figure 3- 79. Example of the MLO groups 131
Figure 3- 80. Example how to modify the group 131
Figure 3- 81. Modified wts_bss_info_config_file for BSS1 of 6G band 132
Figure 3- 82. Single Group made for BSS 1 of 6G band 132
Figure 4- 1. Link Fail over example Image 1 and Image 2 134
Figure 4- 2. Link failover example Image 3 134
Figure 4- 3. WEB UI button to trigger Best AP selection at the controller 134
Figure 4- 4. BH Steer example Image 1 135
Figure 4- 5. BH Steer example Image 2 136
Figure 4- 6. BH Steer example Image 3 136

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Figure 4- 7. BH Steer example Image 4 136
Figure 4- 8. BH Steer example Image 5 136
Figure 4- 9. BH steer command Controller WEB UI button snapshot 137
Figure 4- 10. WEB UI Button for changing BH priority/preference 141
Figure 4- 11. Network optimization example Image 1 143
Figure 4- 12. Network optimization example Image 2 143
Figure 4- 13. Network optimization example Image 3 144
Figure 4- 14. Loop Detection 145
Figure 4- 15. Un-supported Topology 146

Figure 5- 1. Topology Message exchange example 148
Figure 5- 2. Dump topology command console log snapshot 149
Figure 5- 3 . Dump topology command console log with MLD parameters 154
Figure 5- 4. Dump_topology_v1 command JSON file snapshot 157
Figure 5- 5. Dump_topology_v1 command JSON file snapshot with MLD parameters 166
Figure 5- 6. runtime topology with NON-MLD STA, MLD STA and MLD Agent 167
Figure 5- 7. show_bh_link_metric command console log 171
Figure 5- 8. show_sta_metric command console log 172
Figure 5- 9. Topology display WEB UI page snapshot 1 173
Figure 5- 10. Topology display WEB UI page snapshot 2 174
Figure 5- 11. MIB sta_seen_list command console logs snapshot 175
Figure 5- 12. MIB sta <id> command console logs snapshot 176
Figure 5- 13. MIB sta <id> command console logs snapshot with MLD parameters 176
Figure 5- 14. WEB UI Client capability display button 177
Figure 5- 15. WEB UI Client capability page snapshot 177
Figure 5- 16. WEB UI Client capability page snapshot with MLD parameters 177
Figure 5- 17. R3 DE SW Stack Msg flow diagram 180
Figure 5- 18. Click on topology display bubble for DE display page 182
Figure 5- 19. Data Elements (DE) Display Page WEB UI snapshot 183
Figure 5- 20. WiFi 6 SW Stack 185
Figure 5- 21. WiFi 6 Msg Flow Image 1 186
Figure 5- 22. WiFi 6 Msg Flow Image 2 186
Figure 5- 23. dpp config file after device is configured as Controller 187
Figure 5- 24. dpp config file after device is configured as Agent 187
Figure 5- 25. dpp config file of Controller device when Agent onboard to Controller 187
Figure 5- 26. dpp config file of Agent device when Agent onboard to Controller 188
Figure 5- 27. 1905_dpp_keys.txt file snapshot 189

Figure 6- 1. Pictorial description of scope of STA steering(red highlight) in MultiAP steering vs Standalone Band Steering 193
Figure 6- 2. Band Steering example Image 1 196
Figure 6- 3. Band Steering example Image 2 196
Figure 6- 4. Band Steering example Image 3 197
Figure 6- 5. Band Steering example Image 4 198
Figure 6- 6. Band Steering example Image 5 198
Figure 6- 7. Flow Chart for AP Steering Procedure 199
Figure 6- 8. Steering coordination message exchange between different MAP devices 201

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Figure 6- 9. Multi AP centralized steering procedure deployment example 202
Figure 6- 10. Controller side console logs for Mandate Steering 207
Figure 6- 11. Agent side console logs for Mandate Steering 207
Figure 6- 12. Mapd_cli mib command console logs 208
Figure 6- 13. . Mapd_cli mib sta_seen_list command console logs 209
Figure 6- 14. Mapd_cli mib sta <id> command console logs 209
Figure 6- 15. Mapd_cli mib bl_list command console logs 209
Figure 6- 16. Mapd_cli mib assoc_sta_list concole logs 210
Figure 6- 17. WEB UI method to change the AP Steering RSSI Threshold 210
Figure 6- 18. WEB UI display parameter to show Steering status 211
Figure 6- 19. WEB UI button to give mandate steering command 211
Figure 6- 20. Flow chart for restricted BSSID feature 213

Figure 7- 1. Channel scan initialization message flow 220
Figure 7- 2. Channel scan request/report message flow 220
Figure 7- 3. Radar handling by Channel planning flow when Radar is detected at the agent Image 1 222
Figure 7- 4. Radar handling by Channel planning flow when Radar is detected at the agent Image 2 222
Figure 7- 5. Channel Planning run at bootup example 223
Figure 7- 6. User Preferred Channel command example 224
Figure 7- 7. Radar found on ethernet agent example 224
Figure 7- 8. Radar found on Agent with WiFi BH example 225
Figure 7- 9. Radar found on Controller example 226
Figure 7- 10. Divergent Channel planning example 227
Figure 7- 11. Static puncturing of 20MHz channel when iner BSS traffix is detected 227
Figure 7- 12. EHT Operation Information field format 228
Figure 7- 13. Agent punctured Channel due to Interference 228
Figure 7- 14. Inform Punctured Channel Information to Controller 229
Figure 7- 15. Controller move to new channel based on static punctured when channel planning run 229
Figure 7- 16. Channel statistics WEB UI button snapshot 233
Figure 7- 17. Channel statistics example image 1 233
Figure 7- 18. Channel statistics example image 2 234
Figure 7- 19. Channel statistics example image 3 235
Figure 7- 20. Display Run time topology button on WEB UI 235
Figure 7- 21. Current operating channel value on WEB UI topology display example 236
Figure 7- 22. User triggered channel planning WEB UI button image 1 236
Figure 7- 23. User triggered channel planning image 2 236
Figure 7- 24. User triggered channel planning image 3 237
Figure 7- 25. User triggered channel planning image 4 237
Figure 7- 26. User triggered channel planning image 5 238
Figure 7- 27. Channel planning parameters in UCI file system 238
Figure 7- 28. Channel Skip List flow chart 239
Figure 7- 29. WEB UI button to configure User Preferred channel using drop down 240
Figure 7- 30. User configurable skip channel list Example 241
Figure 7- 31. User Configurable Skip channel list, with user preferred channel 241
Figure 7- 32. Selecting CAC device Flow Chart 243
Figure 7- 33. Device selection for CAC example case 1 243
Figure 7- 34. Device selection for CAC example case 2 243

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Figure 7- 35. Device selection for CAC example case 3 243
Figure 7- 36. Device selection for CAC example case 4 244
Figure 7- 37. Device selection for CAC example case 5 244
Figure 7- 38. Device selection for CAC example case 6 244
Figure 7- 39. Device selection for CAC example case 7 244
Figure 7- 40. Channel groups based on BW 246

Figure 8- 1. Traffic Separation Image 1 248
Figure 8- 2. Traffic Separation Image 2 249
Figure 8- 3. Traffic Separation Image 3 249
Figure 8- 4. Traffic Separation Image 4 250
Figure 8- 5. Traffic Separation Image 5 251
Figure 8- 6. Traffic Separation Image 6 251
Figure 8- 7. Traffic Separation Limitation Image 1 252
Figure 8- 8. Traffic Separation Limitation Image 2 252
Figure 8- 9. Traffic Separation Limitation Image 3 252
Figure 8- 10. Traffic Separation feature configuration via wts_bss_file example 253
Figure 8- 11. Traffic Separation parameter in 1905d.cfg snapshot 253
Figure 8- 12. Traffic Separation WEB UI Image 1 254
Figure 8- 13. Traffic Separation WEB UI Image 2 254
Figure 8- 14. Traffic Separation WEB UI Image 3 254
Figure 8- 15. Traffic Separation WEB UI Image 4 255
Figure 8- 16. Service Prioritization Data Flow with Static Rule 256
Figure 8- 17. Service Prioritization Data Flow with Dynamic Rule 257
Figure 8- 18. Service Prioritization UI Setting 259

Figure 9- 1. Mapd_strng.conf file example 263
Figure 9- 2. Mapd_cli mib command console logs 266
Figure 9- 3. Mapd_cli mib sta_seen_list command console logs 267
Figure 9- 4. Mapd_cli mib sta <id> command console logs 267
Figure 9- 5. Mapd_cli mib bl_list command console logs 267
Figure 9- 6. Band Steering example Image 1 268
Figure 9- 7. Band Steering example Image 2 269
Figure 9- 8. Band Steering example Image 3 269
Figure 9- 9. Band Steering example Image 4 270
Figure 9- 10. Band Steering example Image 5 271

Figure 10- 1. 3rd party controller architecture overview 273

Figure 11- 1. How to enable MAP Image 1 275
Figure 11- 2. How to enable MAP Image 2 275
Figure 11- 3. How to enable MAP R2 Image 1 277
Figure 11- 4. How to enable MAP R2 Image 2 277
Figure 11- 5. How to enable Turnkey via MapMode in ax7800 277
Figure 11- 6. Don't increase BSSID for ra0 radio in ax7800. 277
Figure 11- 7. How to enable MAP R3 Image 1 279
Figure 11- 8. How to enable MAP R3 Image 2 279

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Figure 11- 9. How to enable MAP R4 Image 1 281
Figure 11- 10. How to enable MAP R4 Image 2 281
Figure 11- 11. 1905daemon uses 1905d.cfg file 282
Figure 11- 12. wts_bss_info_config file example 1 287
Figure 11- 13. wts_bss_info_config file example 2 287
Figure 11- 14. wts_bss_info_config file example 3 288
Figure 11- 15. wts_bss_info_config file example 3 288
Figure 11- 16. Console log output example of command dpp dpp_bootstrap_gen 289
Figure 11- 17. Console log output example of command dpp dpp_bootstrap_get_uri 290
Figure 11- 18. Console log output example of command dpp_bootstrap_remove 290
Figure 11- 19. Console log output example of command dpp_qr_code 291
Figure 11- 20. Console log output example of command dpp_cce_indication 292
Figure 11- 21. Console log output example of command dpp_onboard_type 292
Figure 11- 22. Console log output example of command dpp dpp_start 293
Figure 11- 23. Console log output example of command dpp_chirp_stop 293
Figure 11- 24. Console log output example of command dpp_chirp_ch_en en 294
Figure 11- 25. Console log output example of command dpp dpp_chirp_notif 294
Figure 11- 26. Console log output example of command dpp_reset_dpp_config 295
Figure 11- 27. mapd_default.txt file sample 308
Figure 11- 28. Mapd_strng.conf file sample 309
Figure 11- 29. 1905d.cfg file sample 309
Figure 11- 30. UCI system for Easymesh 310
Figure 11- 31. etc/config/mapd file example 313
Figure 11- 32. etc/config/1905d_cfg example 313

List of Tables

Table 1- 1. EasyMesh (MAP) R1/R2/R3/R4 wise feature list 20
Table 1- 2. Different SKU with MAP support and corresponding Linux Kernel version 21

Table 3- 1. “wts_bss_info_config” file format 37
Table 3- 2. AuthMode and EncrypType configuration 38
Table 3- 3. AuthMode and EncrypType configuration 38
Table 3- 5. DPP Timer Settings 60
Table 3- 4. Mesh SR Mode Lists 122
Table 3-5. Mesh SR Profile Configuration 126
Table 3-6. Mesh SR Debug Commands 126

Table 5- 1. Dump Topology Parameter Description 167
Table 5- 2. Backhaul Link Metric description 168
Table 5- 3. Per BH metric information description 168
Table 5- 4. Radio Information description 168
Table 5- 5. Backhaul Information description 169
Table 5- 6. MAP R2 Data elements 178

Table 6- 1. Band Steering Steering decision based on client action 194
Table 6- 2. Band Steering Configurable Parameters 203

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Table 6- 3. Multi AP Steering Configurable Parameters 204
Table 6- 4. Steering known IOT issues 211
Table 6- 5. ACL command allowed values 214

Table 7- 1. MAP R1 vs. MAP R2 Channel planning differences 217

Table 9- 1. Pre-Assoc Steering action 261
Table 9- 2. Post-Assoc Steering action 261
Table 9- 3. Band Steering Configuration parameters 264
Table 9- 4. Known IOT issue for steering 271

Table 11- 1. SKU that support MAP R2 version ..... 錯誤! 尚未定義書籤。
Table 11- 2. SKUs that support MAP R3 version ..... 錯誤! 尚未定義書籤。
Table 11- 3. Configuration file parameters description 295
Table 11- 4. MAPD_CLI command list 298
Table 11- 5. Sample Application user events list 306

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1 MAP Introduction

1.1 Introduction

MAP (Multi-AP) is a WFA specification whose purpose is to enable inter-operability across Wi-Fi access points (APs) from different vendors in a Wi-Fi network deployment comprising of multiple APs. It defines the control protocols between Wi-Fi access points (APs) as well as the data objects necessary to enable onboarding, provisioning, control and management of APs in a Multiple-AP Wi-Fi network. This specification also defines the mechanism to route traffic between Wi-Fi access points within the Multi-AP network.

A Multi-AP network consists of two types of logical entities:

- One Multi-AP Controller and
- One or more Multi-AP Agents

A Multi-AP Controller is a logical entity in a Multi-AP network that implements logic for controlling the fronthaul APs and backhaul links in the Multi-AP network

A Multi-AP Agent is a logical entity that executes the Multi-AP Controller functions, and reports measurements and capabilities data for fronthaul APs, clients and backhaul links to a Multi-AP Controller and/or to other Multi-AP Agents

Mediatek MAP turnkey solution provides implementation of MAP specification features as well as proprietary features which help in smooth functioning of Multi-AP networks.

Wi-Fi Alliance has published multiple revisions for the MAP specification. Each revision is named as R1, R2, R3 and so on. Compared with R1, R2 has further improved and standardized Wi-Fi functions, such as Wi-Fi diagnostics, channel selection and traffic separation. Similarly, with R3 some new features were added like DPP onboarding for providing more flexibility and security. Then MediaTek Turnkey solution has also kept enhancing along with each MAP revision to provide enhanced functionality of existing features and spec compliance directed by new revisions. Below is the WFA timeline history for MAP revisions:



Figure 1- 1. Easymesh (MAP) WFA History

1.2 Feature Overview

MediaTek MAP Turnkey solution is an umbrella of multiple components/features. All these features are explained in detail in following sections of the document. Names of these have been listed in below table in accordance with the MAP Spec revision in which they were introduced in MediaTek MAP solution.

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Table 1- 1. EasyMesh (MAP) R1/R2/R3/R4 wise feature list

Category	MAP R1 Certifications	MAP R1 Turnkey	MAP R2 Certification	MAP R2 Turnkey	MAP R3 Spec	MAP R3 TK	MAP R4 TK
Network Setup (*Single Backhaul)	MAP Onboarding (WPS Based & Ethernet)	Dual-band Concurrent WPS BH preference configuration			*DPP onboarding (QR Code) via Eth & WiFi BH *Push button based secure onboarding	*Simultaneous Onboarding eth/wifi, DPP/WPS *Mixed topology	DPP Reconfig support
	MAP Discovery	Topology Database					
	MAP Configuration	FH/BH Configuration renew					
	Mesh Spatial Reuse Configuration						Update bss color of mesh network and use spatial reuse in better way. (support given only on SKU Panther AX4200)
Security			WPA3 Support	WPA3 Support	Control Message Integrity		
Channel Planning	Channel Selection	Convergent/Divergent Channel Planning	Channel Scan Request & Report Coordinated CAC for DFS	Environment based Channel Selection Channel Scan Request & Report Coordinated CAC for DFS		AutoChannelSkipList User configurable channel list	
Device Capability Collection	Capability Information Collection (AP/Client)	AP Database Client Database					
Statistics	Link Metric Collection (Backhaul, per-AP, per-STA)		Channel status Data Elements R1 (22)	Channel status Data Elements R1 (22)	Date Element (1)		
Client Steering	Map Client Steering *11kvu-assisted *MAP Controller/MAP Agent coordinated Steering	Controller coordinated Steering Standalone Band Steering Agile Multiband (MBO)	MBO Support in MAP Agents	MBO Support in MAP Agents • Dynamic network monitoring • Intelligent steering • Fast transitions		User configurable target BSS	
Backhaul Optimization				Centralized automatic Backhaul optimization			
Traffic Management			Traffic Separation – Guest SSID Support (Wi-Fi Only)	Traffic Separation – Guest SSID Support (Wi-Fi & Eth)	Traffic Separation – Guest SSID Support (Eth)		
Service Prioritization					Service prioritization basic, medium		
Wi-Fi6					Wi-Fi6 capability TLV		

1.3 MAP Variants

A Mediatek Turnkey MAP device supports both Ethernet and Wi-Fi backhaul connections. It has an optional APCLI interface on each radio to create a wireless backhaul connection (BH) with controller or other MAP devices. It also has at least one fronthaul (FH) BSS to accept connections from wireless stations. Below is the image of all the possible MAP variants.

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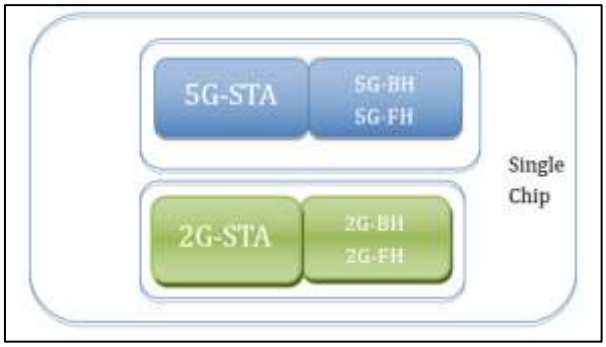


Figure 1- 2. Single Chip Dual Band repeater

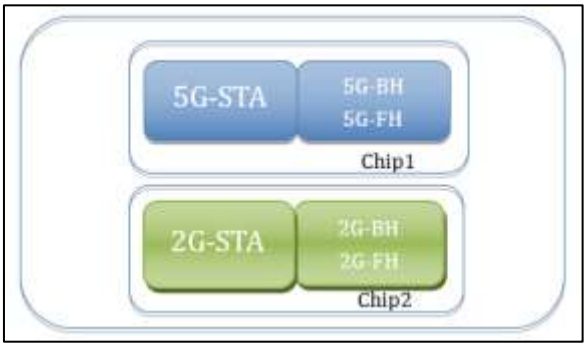


Figure 1- 3. Dual Chip Dual Band repeater



Figure 1- 4. Triband Repeater

Till now MAP support is available on all the below MediaTek SKUs. The corresponding Linux Kernel version and SDK information is listed in below table for reference. For configuration difference between LSDK and OpenWRT, you may refer to Appendix section 10.1

Table 1- 2. Different SKU with MAP support and corresponding Linux Kernel version

SKU	SDK	Linux Kernel
MT7621 + MT7615D	LSDK	Linux 3.10.14
MT7621 +MT7615 + MT7615	LSDK	Linux 3.10.14
MT7621 + MT7615D	mtk-openwrt-cc-v3.6	Linux 3.10.108
MT7621 +MT7615 + MT7615	mtk-openwrt-cc-v3.6	Linux 3.10.108
MT7621 +MT7603 + MT7615	mtk-openwrt-cc-v3.6	Linux 3.10.108
MT7621 +MT7603 + MT7613	mtk-openwrt-cc-v3.6	Linux 3.10.108
MT7629	mtk-openwrt-lede-4.2.0.0	Linux 4.4.198

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MT7622 + MT7615	mtk-openwrt-lede-v4.1.1.0	Linux 4.4.177
MT7622 + MT7615 + MT7615	mtk-openwrt-lede-v4.1.1.0	Linux 4.4.177
MT7621+MT7915D	mtk-openwrt-lede-4.2.1.0	Linux 4.4.198
MT7621+MT7915+MT7915	mtk-openwrt-lede-4.2.1.0	Linux 4.4.198
MT7622+MT7915	mtk-openwrt-lede-4.2.1.0	Linux 4.4.198
MT7986 (Panther), variants AX6000 and AX4200	mtk-openwrt-lede-openwrt-21.02	Linux 5.4.158
MT7622+MT7916(Merlin) , variant AX3000	mtk-openwrt-lede-4.2.1.0	Linux 4.4.198
MT7981(Cheetah) , variant AX3000	mtk-openwrt-lede-openwrt-21.02	Linux 5.4.158
MT7986(Panther 5G) + MT7916 (Merlin 2G+6G) , variant AX7800	mtk-openwrt-lede-openwrt-21.02	Linux 5.4.219
[Jaguar+Eagle] BE19000 MT7988-MT7990	OpenWrt 21.02-SNAPSHOT r0-8d995b3b **With hostapd and supplicant support** **This SKU is Wifi 7 Capable and also supports Multi Link Operation (MLO)** **With this SKU it is mandatory to configure MAP config files via UCI.**	Linux 5.4.231

1.4 MLO MAP device

802.11be introduces a concept of a Multi-Link Device (MLD), they support Multi Link Operation (MLO). Legacy devices which support Single Link Operation (SLO) is referred as Single-Link Devices (SLD) in specificiation. Some key points about a MLD are as follows:

- MLD consists of several so-called affiliated Wi-Fi devices (each has a PHY interface to the wireless media), but with a single interface to the LLC layer. The upper-layer protocols consider the MLD as a single device.
- Despite having multiple physical radio interfaces, MLD has a single MAC address, and the sequence numbers are generated uniquely from the same sequence number space. They allow packet retransmission on any link regardless of the link of the initial transmission of the packet.
- Each link of an MLD is regarded as a conventional single-link station (STA), and an MLD is a device affiliated with more than one STA. In contrast, an SLD (legacy device) is affiliated with only a single STA.
- MLDs can be further classified into two types based on the type of the affiliated STAs:
    - AP multi-link device (AP MLD): each of the affiliated STAs is an AP.
    - Non-AP multi-link device (non-AP MLD): each of the affiliated STAs is a non-AP (client) STA.

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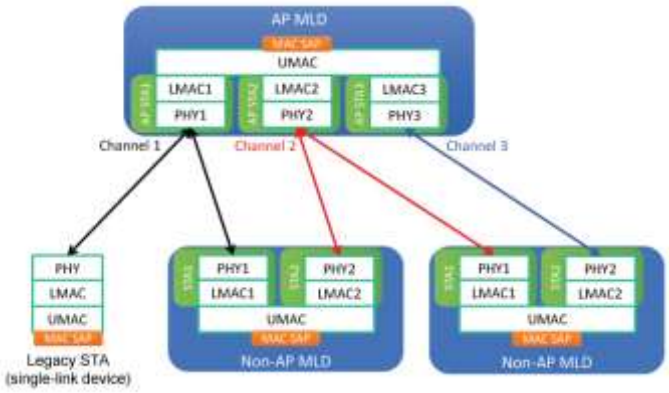


Figure 1- 5. AP and Non-AP MLD vs. SLD

For a mesh network each device will have both AP MLD and non-AP MLD (APCLI MLD). All the existing features will now work with MLO enabled devices as well In an easymesh network made with triband 802.11be devices which are all MLDs, and all bands part of MLO groups, we will have basic topologies in below figures

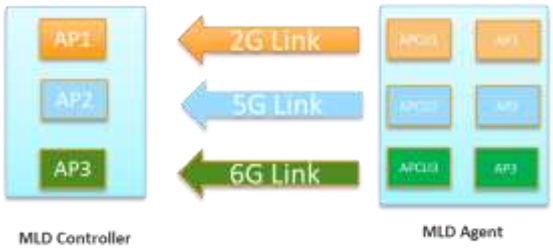


Figure 1- 6. Easymesh network with 2 MLD devices



Figure 1- 7. Easymesh network with 3 MLD devices, 1 controller and 2 agents connected in daisy chain

For existing MAP feature to coexist for an MLD, enhancement has been done in the existing features. The reference to the sub section can be found below:

1. Configuration settings
    a. Configuring BSS for MLO (Section 3.2.1.4)
2. Main flow for Hostapd and MLO supported SKU (Section 4.3.2 and 4.3.2.2)
3. Network Optimization for MLDs in a Mesh network (Section 4.4.1)
4. Steering
    a. Band steering for MLD clients (Section 6.2.2.1)
    b. Selection of Best BSS for MLD clients (Section 6.2.3.3)
5. Data base and Diagnostics (Non-Setup Metrics info display when device is MLD enabled Agent Section 5.2.2)
6. Client Database for MLD client (Section 5.3)

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1.5 EasyMesh (MAP) WebUI Introduction

The Web Page of this device can be accessed by the browser by typing the IP address of the br0 interface.

- User name : admin
- Password : admin

Easy Mesh WEB UI can be accessed from the button in below image:

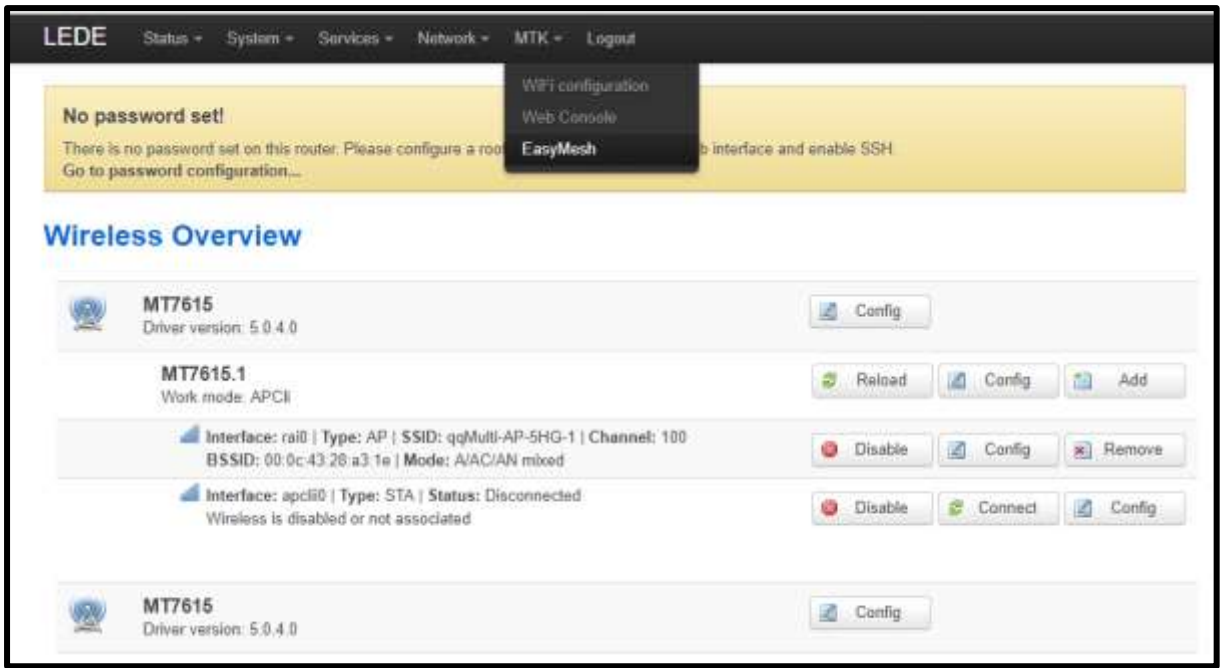


Figure 1- 8. EasyMesh Web UI drop-down

- Please use latest Chrome or Firefox browser, otherwise some features will not work.
- On this page you can see three different tabs:
  - Basic
  - Advanced
  - Status
- If you have downloaded the image on a fresh board or have just done a factory reset, then the NVRAM will not contain the MAP configuration settings, so you need to:
  - Enable Easy Mesh
  - Select Device Role
  - Save and Apply
- You can select the Device role by the Drop-down menu: Controller / Agent / Auto
- Device role can be changed later also.
- The controller has more functions than an Agent, hence their pages UI page differs.
- Details about all the WEB UI features will be introduced in respective feature sections in this document.

If not mentioned then step is common for all devices irrespective of device to be made as Controller or Agent.

a. Default configuration after flashing image:

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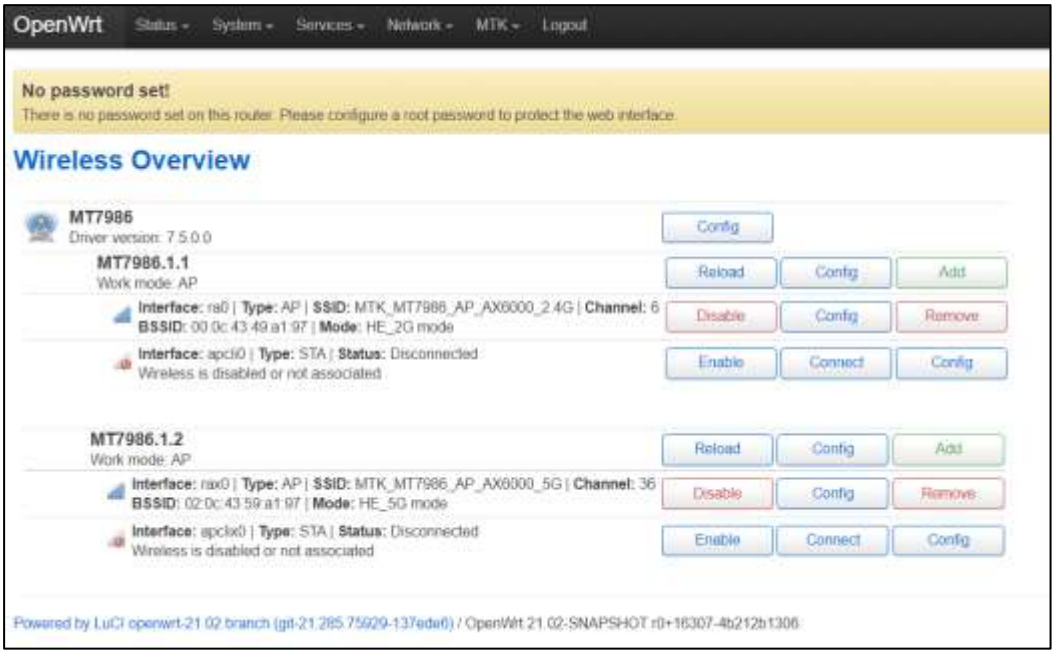


Figure 1- 9. Web UI default page

b. Add multiple BSS (minimum 2) using WEB UI on Boards for both controller and agents.

- Example added four BSS per radio. On creating BSS press save button, as in Step c reload will help to add all bss in driver.

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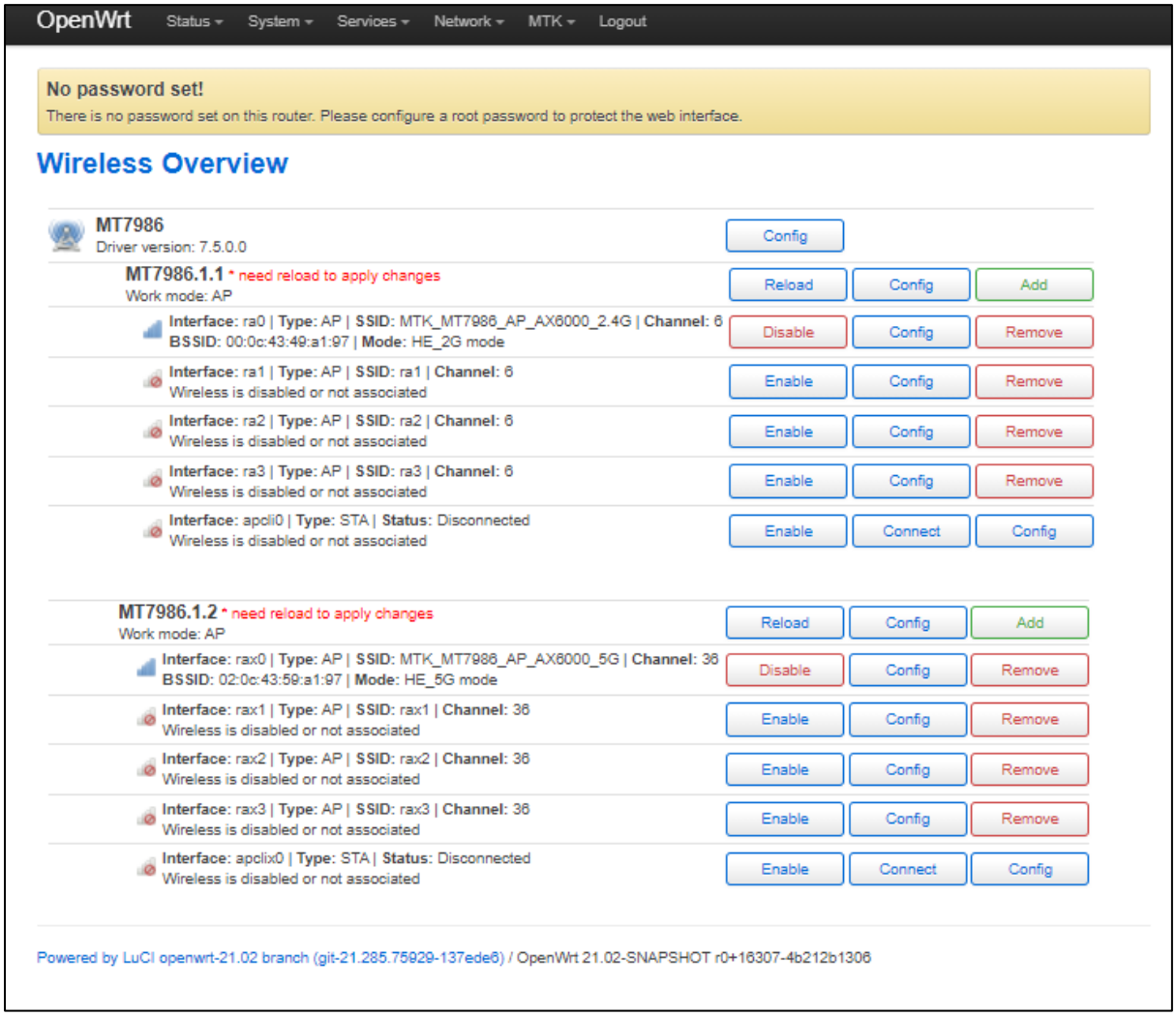


Figure 1- 10. Web UI page after adding MBSS for each radio

c. Need to reload the device from UI (for 2.4 Ghz , 5Ghz and 6GHz (if present) separately) for both agents and controller.

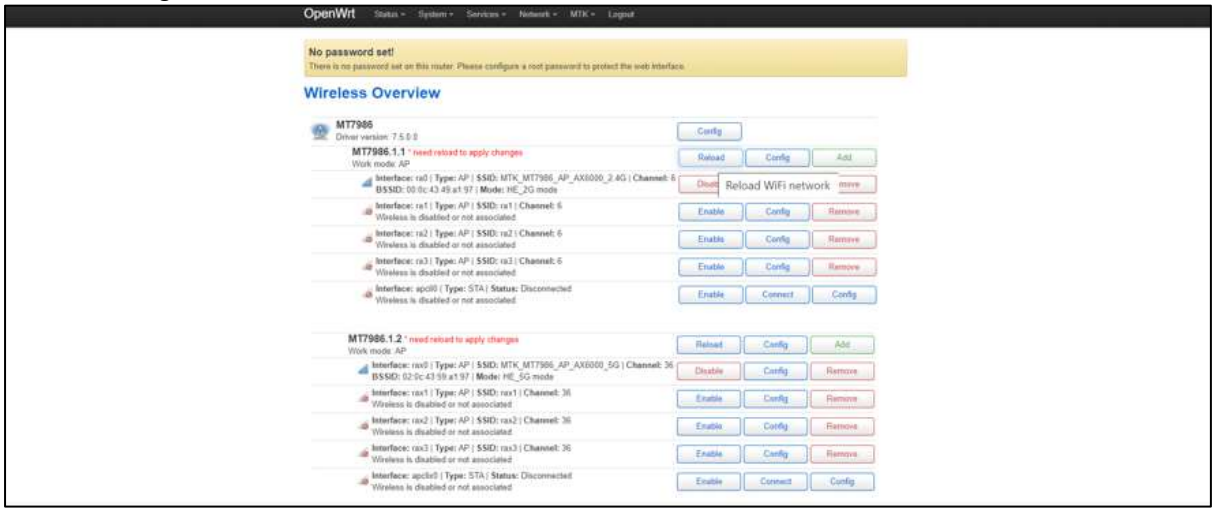


Figure 1- 11. Web UI page Need to reload 2.4G band

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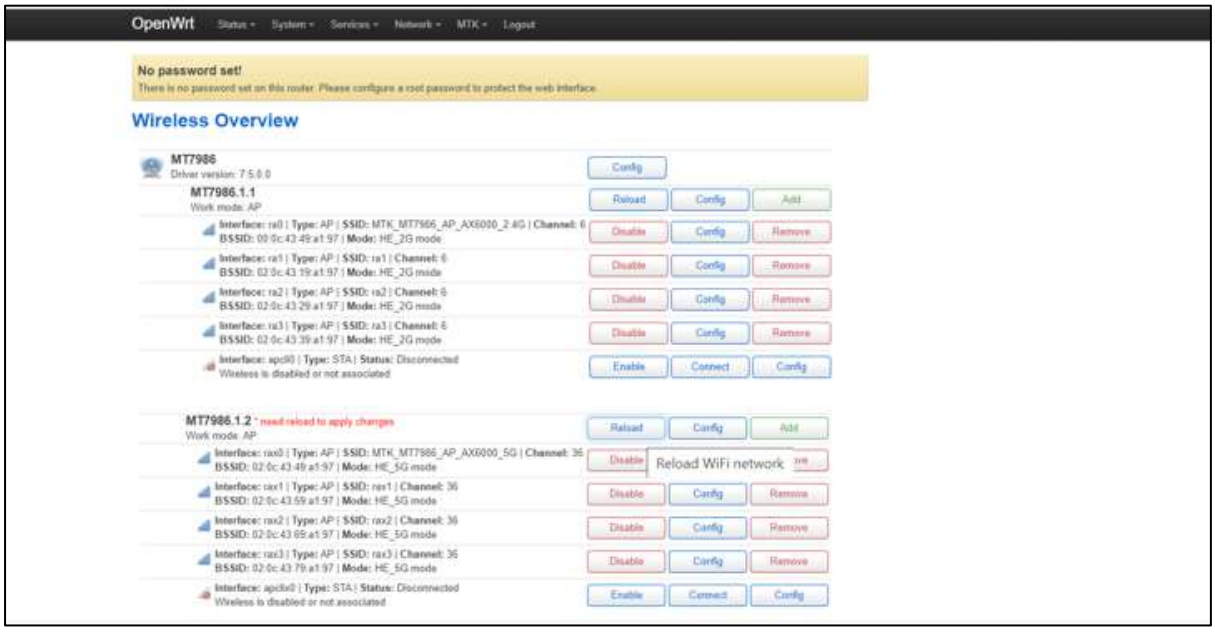


Figure 1- 12. Web UI page need to reload 5G band

d. Prepare MLD groups if needed from this page

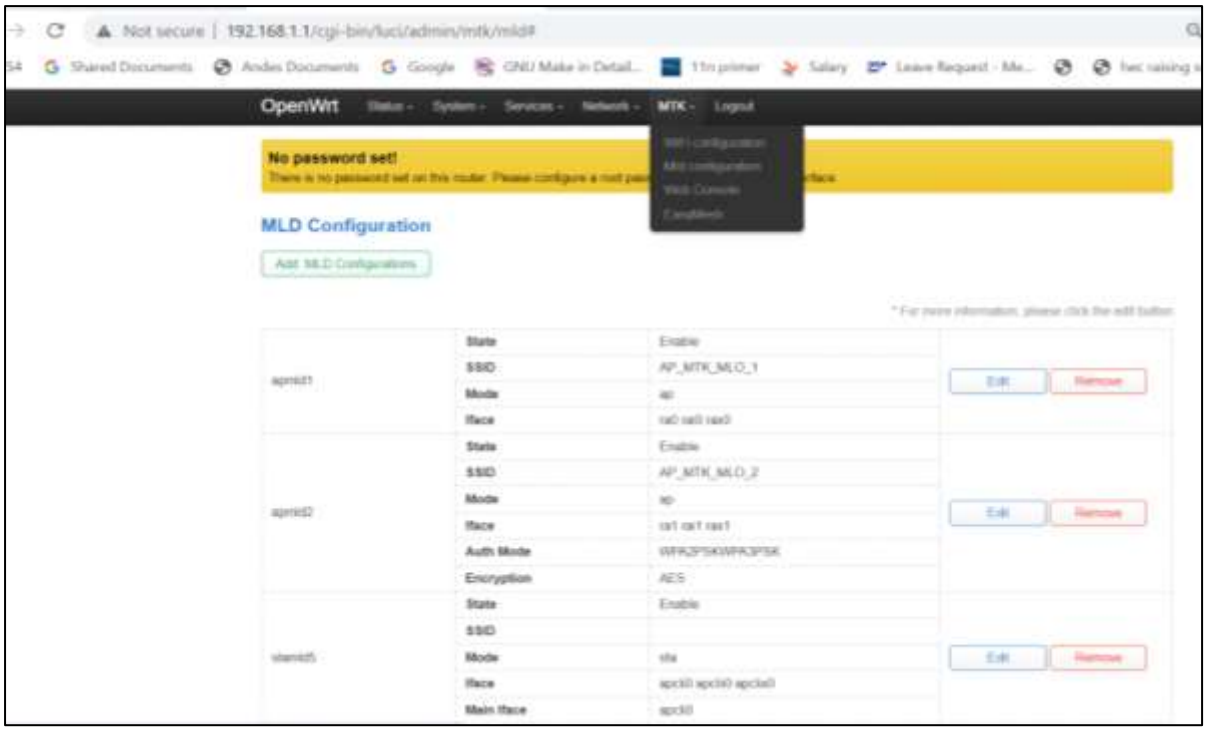


Figure 1- 13. Web UI page to add MLD groups

e. Enable MAP Turnkey on the boards from UI by selecting Map Turnkey Option and then clicking on Save and Apply button.

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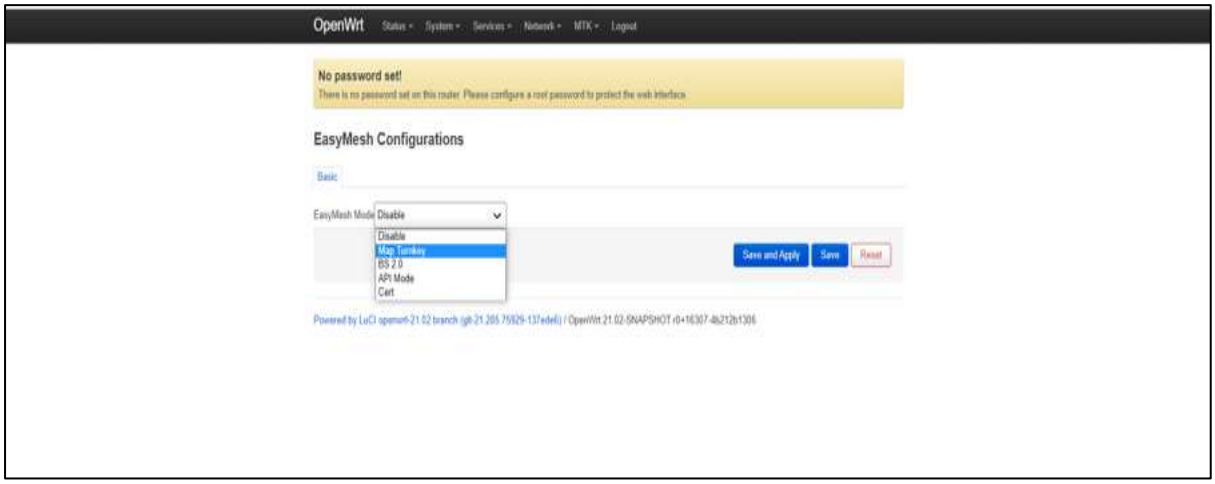


Figure 1- 14. Web UI page enable Easymesh feature

f. **Controller:** Configure the Device mode as Controller using UI. Set Device Role as Controller and click on Save and Apply button.

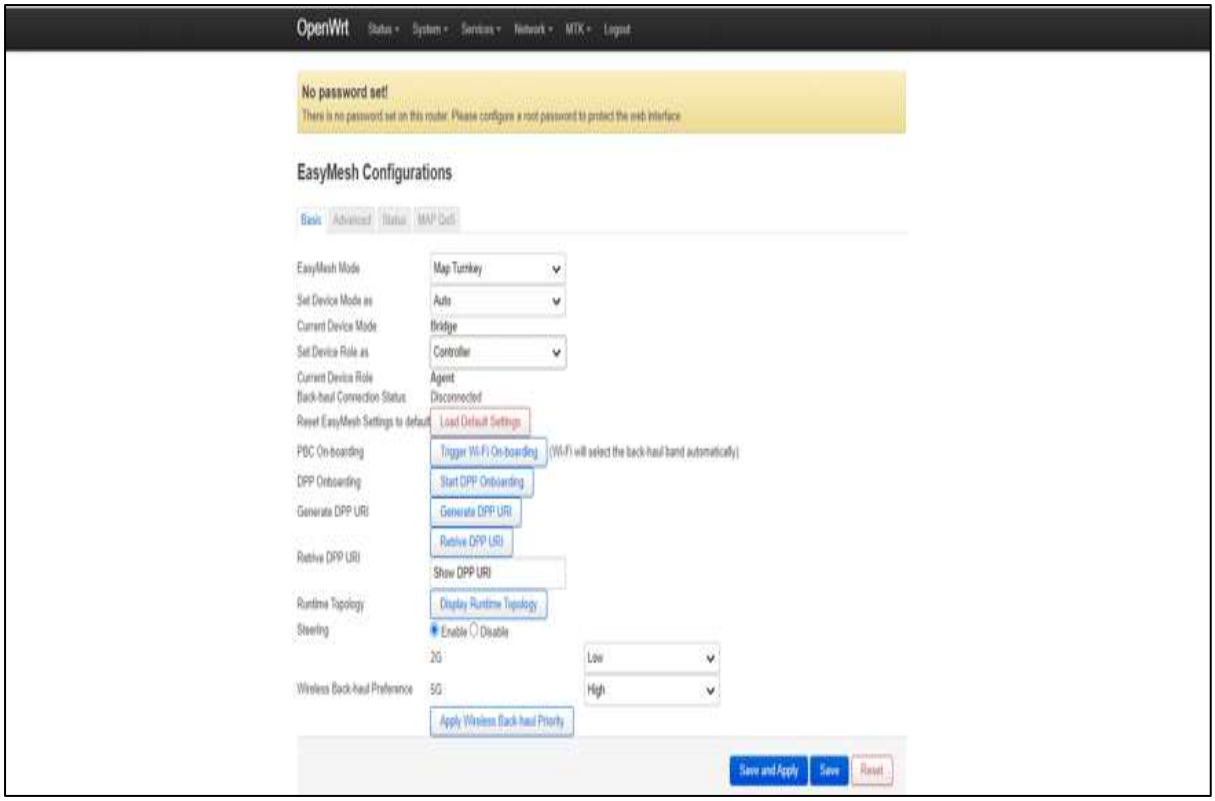


Figure 1- 15. Web UI page for controller

g. **Agent:** Configure the Device mode as Agent using UI. Set Device Role as Agent and click on Save and Apply button.

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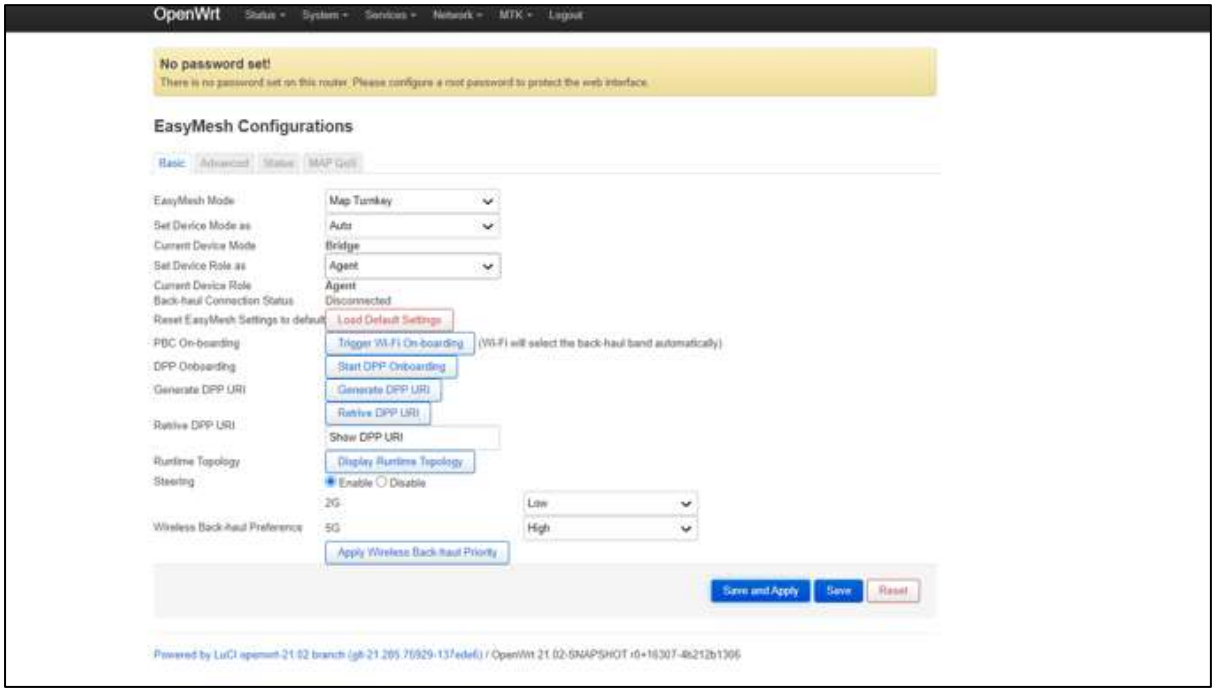


Figure 1- 16. Web UI page for agent

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2 MAP Architecture

2.1 Introduction

Mediatek MAP turnkey software consists of various components. There are 3 major applications running namely mapd, wapp and 1905 Daemon. These are responsible for handling MAP functionality in user space. There is a new kernel module mapfilter.ko which helps in 1905 related frame forwarding. Then there is MAP feature addition in the driver to take care of 11kv and WPS based onboarding. Details of all the modules involved are given in below sub sections. The MediaTek MAP turnkey architecture can be represented in the below model.

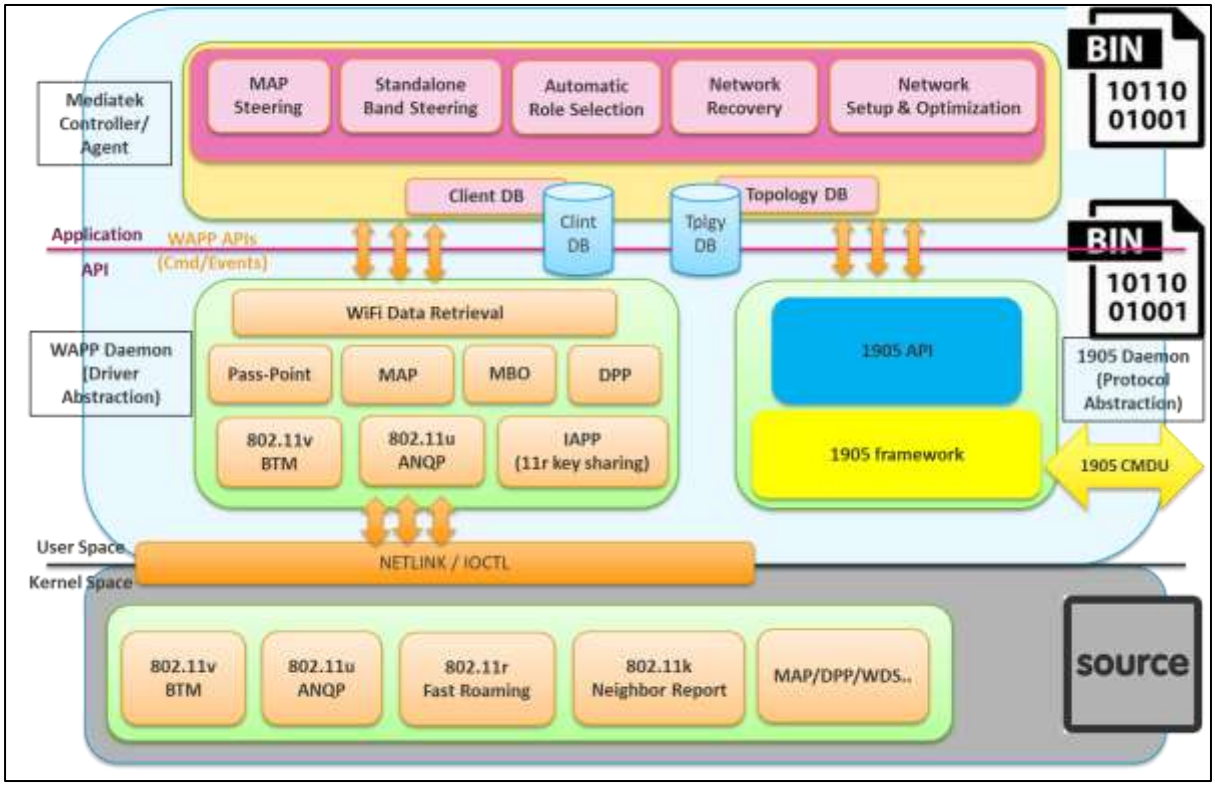


Figure 2- 1. MAP Turnkey Model

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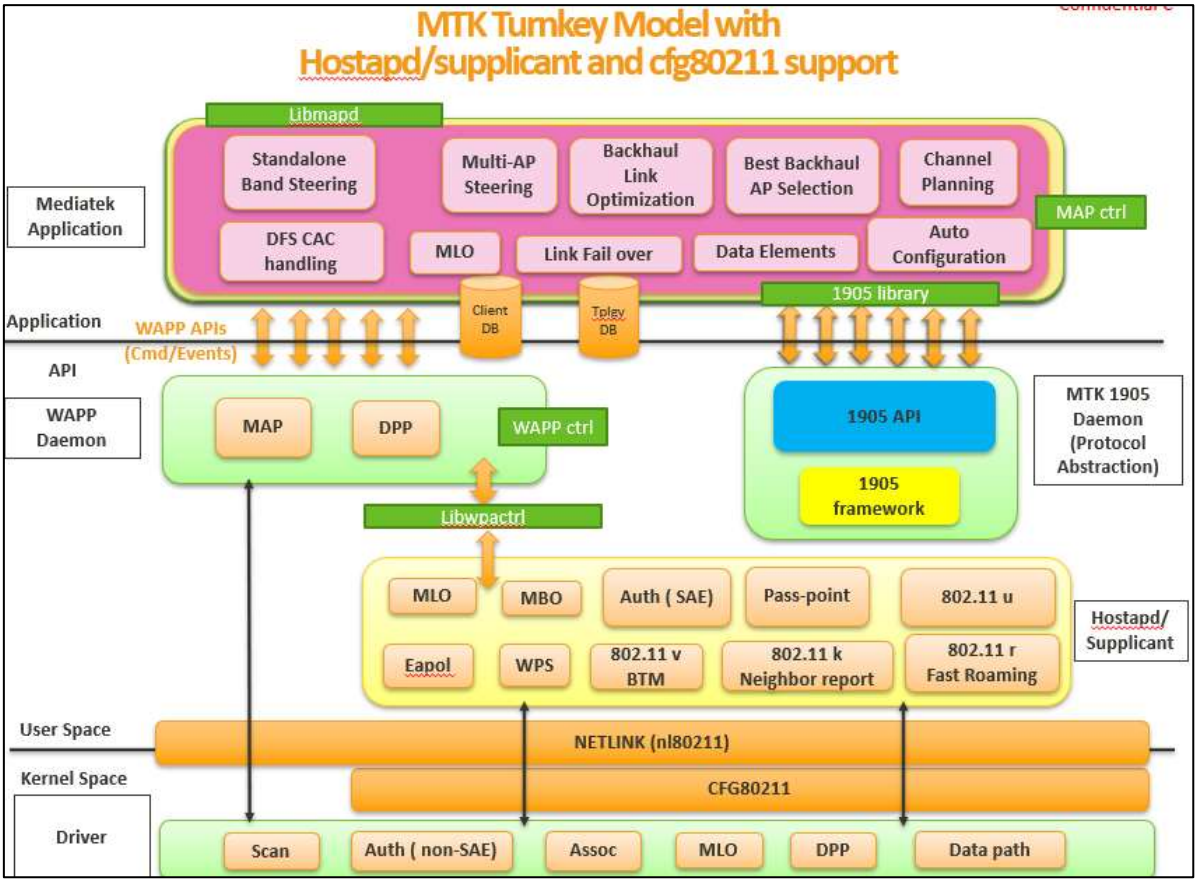


Figure 2- 2. MAP Turnkey Model with hostapd/supplicant and cfg802.11 support

2.2 Driver

Driver exposes interfaces for WAPP to get/set parameters, send/receive data. It uses ioctl and netlink to interact with WAPP. Driver also provides the capability of 802.11V BTM, 802.11 fast roaming, 802.11 neighbor report, WPS procedure, WDS 4 address format, MBO.

2.3 WAPP

WAPP (Wi-Fi App) encapsulates different ioctl and netlink event which are access to Wi-Fi driver part, so it may supply services for other application. Application accessing Wi-Fi resources through WAPP does not need to know the difference of Wi-Fi drivers, this makes it easier for upper layer applications to port from one generation of Wi-Fi chipset to another.
WAPP collects common Wi-Fi information of all Wi-Fi interfaces on the same device. Information such as BSS capability, RSSI, TX power, operating channel, number of clients, are managed in WAPP. For Wi-Fi functions that requires to get information from driver before making decision, they can take the common information stored in WAPP instead of send queries to drivers. WAPP also handles IAPP Daemon which is mandatory for 802.11R.

2.3.1 Wappctrl

WAPP control interface is used to control WAPP by using commands of another application called “wappctrl”. These commands, which are implemented in WAPP with a pre-defined format, can be used to do manual test, feature development, or debug.

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For the details and usage of all the wappctrl commands you can refer to the "Multi-AP_APIs_and_Events_v3.1.docx"

2.3.2 Libwpactrl

This library is present only for the SKUs that support HostApd and supplicant. It acts as an interfacing layer between the WAPP daemon and Hostapd. WAPP daemon communicates with Hostapd by the help of libwpactrl APIs.

2.4 1905 daemon

1905 daemon is used to receive, send and process protocol related messages. It has two kinds of role, controller and agent. That is to say different role has different logic in message processing. User needs explicitly assign a role to the daemon when starting up it. Every 1905 device needs to maintain topology database. The 1905 daemon will announce its interface list and attribute by topology response to help other device learn the fronthaul bss of the device. And step by step, every device in the same topology knows the information of adjacent and non-adjacent device. There is no need for user to care about wireless interface. The information of wireless interface will be created by WAPP. User only need explicitly assign Ethernet interface in 1905d.cfg. In this file, there are two more parameters that need to be configured by the user.

• Every 1905 device needs to assign one unique abstract mac which is called ALMAC. 1905 daemon needs to receive packets from driver.

• All net devices are bridged to bridge interface. 1905 needs to get the bridge interface to create raw socket to monitor packets.

For more details and layout of 1905d.cfg file, please refer to Appendix section 10.18.

2.4.1 1905_ctrl

1905 ctrl daemon is used to send cmd to 1905 daemon. This daemon will only be used for passing WFA MAP certification. In MAP certification, UCC will send bss configuration information and 1905 raw data to 1905 daemon. This information will be written to a file in specific format. 1905 ctrl daemon will send cmd with file path to 1905 daemon. Then 1905 daemon will parse the event and do related works.

2.5 Mapfilter

Mapfilter is a kernel module that needs to be inserted into kernel before 1905 daemon is started. 1905 daemon needs to get the net device mac of received packets. It will use this information to maintain neighbor list and send message back to peer. Mapfilter will maintain a table based on source address of the packets and the net device mac address which received the packets. 1905 daemon will get net device mac based on source mac by sockopt.

2.6 MAP Daemon

MAP daemon is an application which implements turnkey features of Mediatek Turnkey MAP solution. It interacts with both WAPP and 1905 daemon to implement both driver specific and MAP specification specific features respectively.

2.6.1 MAPD_ctrl

MAPD ctrl daemon is used to send commands to MAP daemon. This daemon is used for testing/configuring various turnkey features by means of mapd_cli commands. A cumulative list of all mapd_cli commands can be in the Appendix section 10.13.

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2.6.2 Libmapd

Libmapd is a static library which is used by MAPD ctrl to Map daemon. More detail about each API provided in this library can be found in the mapd_library document.

2.7 Topologies supported

Mediatek solution supports the following topologies:

1. **Wireless backhaul connections**

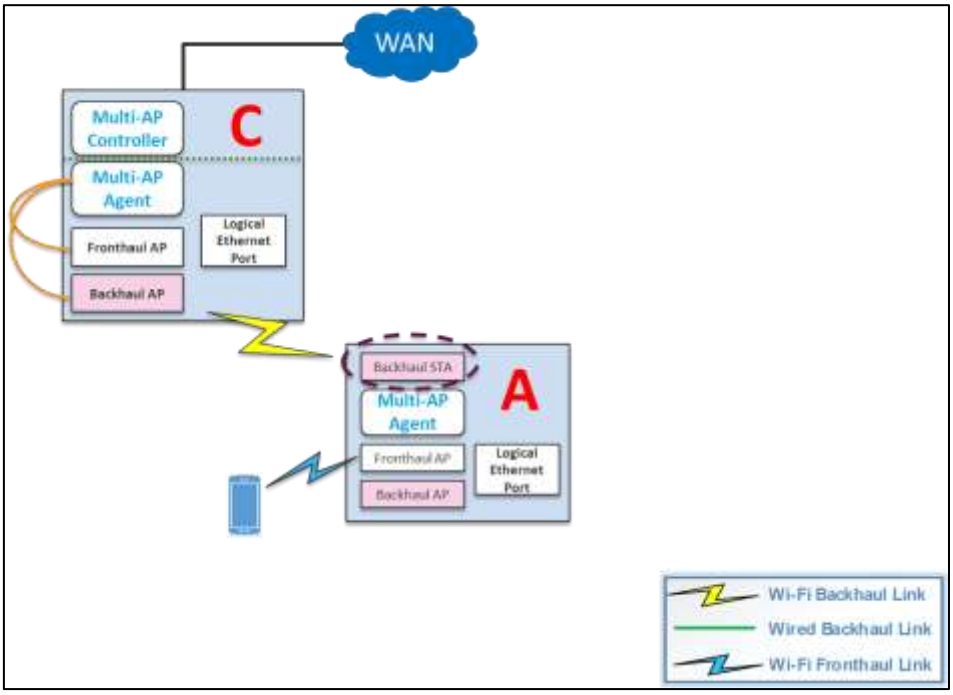


Figure 2- 3. Topology with Agent having wireless BH with Controller

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2. **Wired backhaul connections**

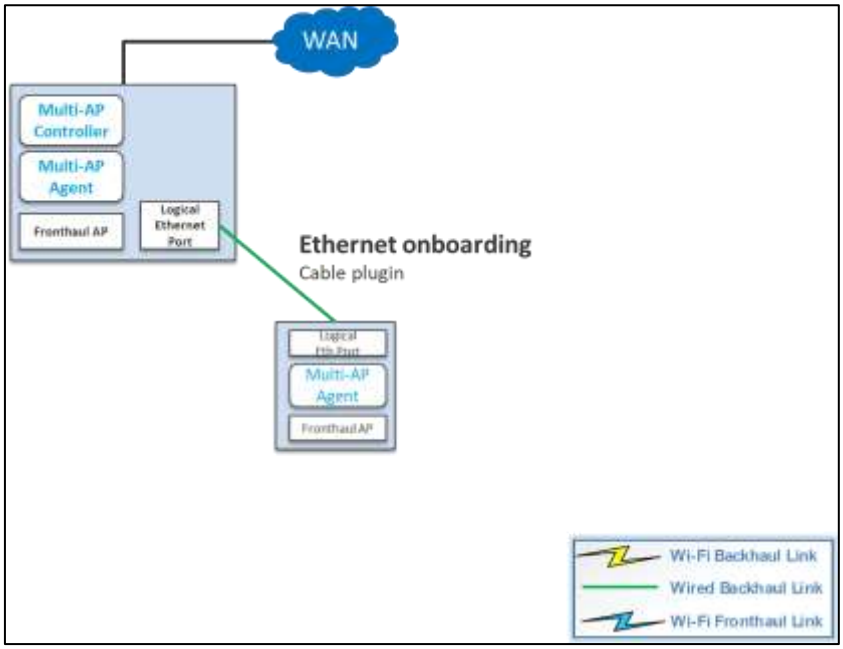


Figure 2- 4. Topology with Agent having wired BH with Controller

3. **Mixed network with ethernet and wifi onboarded devices**

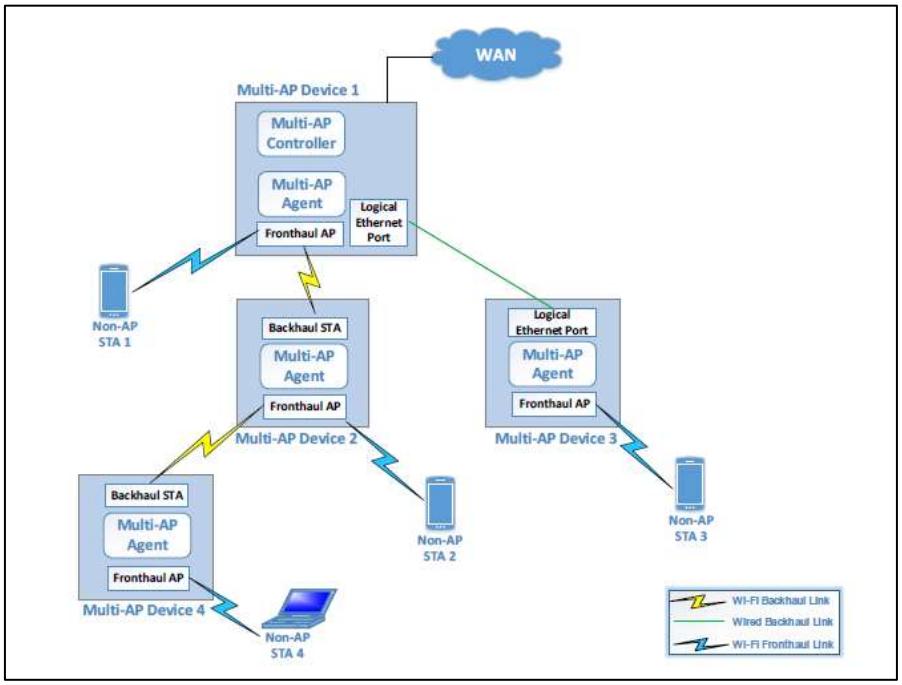


Figure 2- 5. Topology with some Agents having wireless BH and some having wired BH with Controller

4. **Controller connection to Non MAP root AP/gateway**

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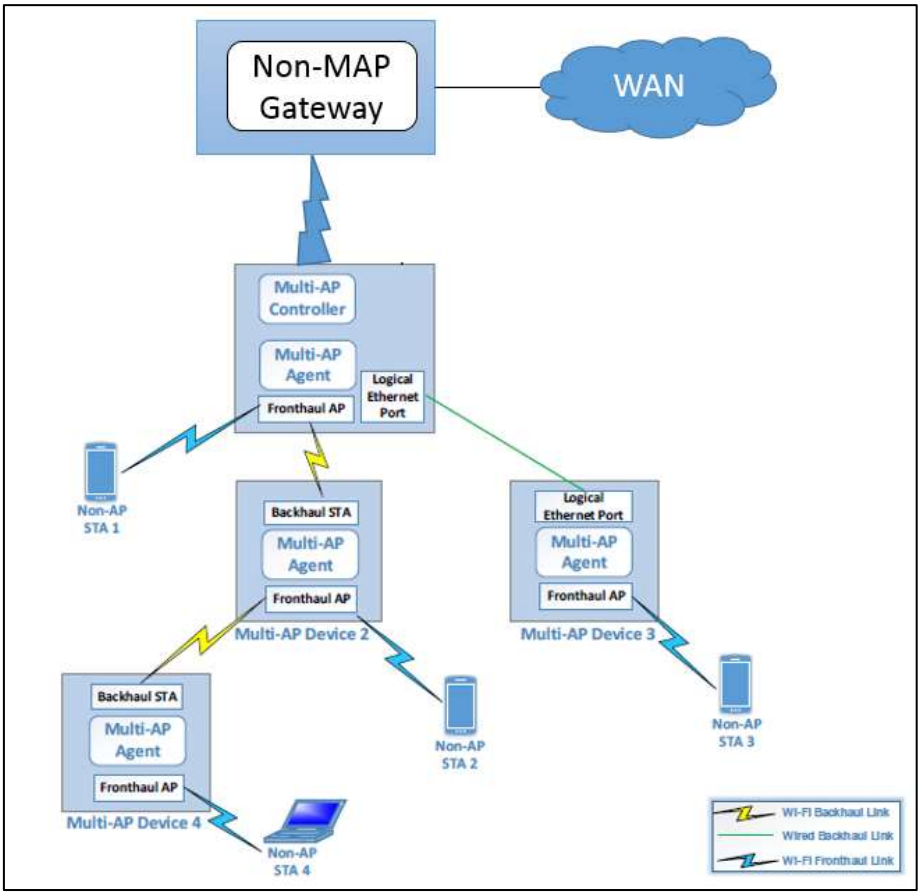


Figure 2- 6. Mixed Topology with Controller having connection with Non-MAP AP

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3 MAP Network formation and support

3.1 Introduction

MAP network formation refers to the process of agents getting onboarded to the MAP network via a controller or existing agents. An onboarded Multi-AP Agent gains layer-2 connectivity onto a Multi-AP network through Wi-Fi or wired connectivity. Till MAP R1/R2 only one onboarding technique was used i.e. WPS Onboarding. With Map R3 specification a new onboarding technique has also been added i.e. DPP based onboarding. This chapter describes all aspects of the various onboarding techniques as provided by MediaTek's turnkey solution. Later sections of this chapter describe the other support feature which help in network formation.

3.2 MAP Onboarding and Auto-Configuration

Onboarding is the process by which a Multi-AP Agent gains layer-2 connectivity onto a Multi-AP network through Wi-Fi or wired connectivity. The specification defines methods for onboarding by which a Multi-AP device can onboard to a Multi-AP network and they have been made part of the MediaTek turnkey solution. Till MAP R1/R2 only one onboarding technique was used i.e. WPS Onboarding. With Map R3 specification a new onboarding technique has also been added i.e. DPP based onboarding. Each of these techniques have been introduced in below sections.

Once layer-2 connectivity is established, the multi-AP agent commences the discovery of Multi-AP controller. The specification enables Multi-AP controller to configure 802.11 interfaces (BSS) on each radio(s) of a Multi-AP agent. For this process, Agent will use the AP-Autoconfiguration process defined in the 1905 specification.

3.2.1 Configuration Settings

Onboarding and Autoconfiguration require some configuration settings to be done at the controller. When the controller performs auto configuration of an enrollee agent, then it consults the wts_bss_info_config file which stores the configuration that controller will apply to new agents.
User needs to set this configuration in the file. There are 2 ways of setting the configuration:

1. Agent specific configuration:
    - User can set the configuration for a particular Agent by specifying the AL MAC address of the agent in the configuration
    - User need make sure the backhaul bss setting on agent need be the same with the whole MAP network. The fronthaul bss setting can be different with the other MAP network. Agent need use its backhaul bss setting to reconnect to MAP network in wifi case.
2. Generic configuration:
    - For rest of the agents, generic configuration is used which is specified by mentioning wildcard AL MAC in the configuration (FF:FF:FF:FF:FF:FF)

For sample data of the "wts_bss_info_config" file , please refer to the Appendix section 10.10.
If there is a space in SSID, the user needs to enter \ as ESC in front of it in wts file. This is because space is regarded as a separator in wts file. If ssid has \, then user needs to add \ as ESC in front of it.
For WPS to be successful, there must be at least one NON-WEP, NON-TKIP and NON-HIDDEN Front haul BSS per radio.

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The format of the file is as follows:

Table 3- 1. "wts_bss_info_config" file format

Index	ALMAC	Radio	SSID	AuthMode	EncryptType	PSK	BH BSS	FH BSS	Hidden Y/N	VLAN	PVID	PCP	MLD Group ID
Values of each column and example can be seen in table below													

Field	Value description
Index	Line number
ALMAC	6 octets of unique abstract layer mac address of agent which need be configured
Radio	band which agent operating on. 8x means 24G; 11x means 5GL; 12xmeans 5GH; 13x means 6G.
SSID	SSID to be configured on this BSS
AuthMode	0x0001 - Open 0x0002 - WPA - Personal 0x0020 - WPA2 – Personal 0x0022 - WPA – Personal/ WPA2 – Personal 0x0040 – WPA3 SAE only # 0x0060 – WPA3 transition# 0x0080 – DPP Only* 0x00C0 – DPP+SAE* 0x00E0 – DPP+SAE+PSK*
EncryptType	0x0001 - NONE 0x0002 - WEP 0x0004 - TKIP 0x0008 - AES 0x000c – TKIP/AES
PSK	PASSPHRASE for this BSS (MAX 64 byte)
BH BSS	enable this bit if the bss support backhaul bss
FH BSS	enable this bit if the bss support fronthaul bss
Hidden Y/N	hidden-Y , if BSS is hidden otherwise hidden-N
VLAN	FH VLAN ID, valid values: 3~4094, disabled values: 4095
PVID	default value: N/A, if pvid means VLAN means primary vlan id
PCP	default value: N/A, valid values: 0~7,only valid for primary vlan id
MLD Group ID	This field is optional and can be skipped if the MLD groups are preconfigured. It contains the group index of each BSS , same group index vaue for same BSS.

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Table 3- 2. AuthMode and EncrypType configuration

AuthMode	EncrypType	Fronthaul support for (R1/R2)	Backhaul Support for (R1/R2)	Fronthaul support for (R3)	Backhaul Support for (R3)	MAP version in which Support was added
0x0001 - Open	0x0001 - NONE	Y	Y	N	N	R1
	0x0002 - WEP	Y	N	N	N	R1
0x0002 - WPA - Personal	0x0004 - TKIP	Y	N	N	N	R1
0x0020 - WPA2 – Personal	0x0008 - AES	Y	Y	Y	Y	R1
	0x000c – TKIP/AES	Y	Y	N	N	R1
0x0022 - WPA – Personal/ WPA2 – Personal (Mixed)	0x0008 - AES	Y	Y	N	N	R1
	0x000c – TKIP/AES	Y	Y	N	N	R1
0x0040 – WPA3 SAE only	0x0008 - AES	Y	Y	Y	Y	R2
0x0060 – WPA3 transition	0x0008 - AES	Y	Y	Y	Y	R2
0x0080 – DPP Only*	0x0008 - AES	N	N	N	Y	R3
0x00A0 – DPP+WPA2PSK*	0x0008 - AES	N	N	N	Y	R3
0x00C0 – DPP+SAE*	0x0008 - AES	N	N	N	Y	R3
0x00E0 – DPP+SAE+WPA2PSK*	0x0008 - AES	N	N	N	Y	R3

Important Note about WPA3

This feature can be used in MAP R2 and above MAP versions and we can configure the device to WPA3 security either in two modes one is WPA3 Transition mode (PSK+SAE) or in SAE only mode. According to MAP R2 specification, if a Multi-AP Controller sends an M2 indicating Fronthaul BSS, it shall not set the Authentication Type attribute in that M2 to 0x0040 (SAE only). The Multi-AP Controller must not configure SAE-only mode on fronthaul BSSs. However, if the Agent indicates support for SAE in the M1, the Controller might configure a fronthaul BSS to 0x0060 (PSK+SAE), aka WPA3 Transition Mode. However, user can configure device in any WPA3 security modes based on its requirement, but it may have interoperability issues with other vendor solutions.

Important Note about DPP

As DPP Spec allows config for DPP/SAE/WPA2-PSK. So make sure authmode is set accordingly. DPP+SAE+PSK is suggested combination to be set to cover all cases for Backhaul and for Fronthaul use SAE Transition mode as many Mobile STA don't support DPP and not connect if DPP is set by AP.

Make sure atleast 1 fronthaul exist as DPP Onboarding uses fronthaul. So minimum of 2 BSS is needed for DPP onboarding, one as Backhaul BSS and one as front haul BSS.

Important Note about 6G Radio

As 6G band operation as per spec allows minimum WPA3 SAE Auth Mode support, so we can allows the below Auth modes encryptype combinations as per table below for radio 0x13 (6G). DPP only is also supported as spec does not mention any restriction about its support on Backhaul.

Table 3- 3. AuthMode and EncrypType configuration

AuthMode	EncrypType	Supported with 6G band

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0x0001 - Open	0x0001 - NONE	No
	0x0002 - WEP	No
0x0002 - WPA - Personal	0x0004 - TKIP	No
0x0020 - WPA2 – Personal	0x0008 - AES	No
	0x000c – TKIP/AES	No
0x0022 - WPA – Personal/ WPA2 – Personal (Mixed)	0x0008 - AES	No
	0x000c – TKIP/AES	No
0x0040 – WPA3 SAE only	0x0008 - AES	Yes
0x0060 – WPA3 transition	0x0008 - AES	No
0x0080 – DPP Only*	0x0008 - AES	Yes (only BH)
0x00A0 – DPP+WPA2PSK*	0x0008 - AES	No
0x00C0 – DPP+SAE*	0x0008 - AES	Yes (only BH)
0x00E0 – DPP+SAE+WPA2PSK*	0x0008 - AES	No

Important Note about Max BSS number in the mesh network

Maximum number of BSS supported in topology: 76
This restriction is due to the size limit of combined infra metrics sent by the controller.

For example for a 3 device MAP network with each device supporting 3 radios each, BSS distribution can be as follows:

Controller (16 BSS on Radio1 + 8 BSS on Radio2 + 8 BSS on Radio3)
Agent 1 (16 BSS on Radio1 + 8 BSS on Radio2 + 8 BSS on Radio3)
Agent 2 (4 BSS on Radio1 + 4 BSS on Radio2 + 4 BSS on Radio3)

So , total BSS number in network is 76.
If BSS are more, number of MAP devices supported in the network will be less. But max number of BSS in topology should not exceed 76. Similarly vice versa, if we need to support more devices in mesh network then bss on some devices will have to be reduced.
Till now testing of 76 BSS has been done only with max device in mesh network supported as 3.

In MAP (inside mapd_i.h) , 1905daemon(inside p1905_database.h) and WAPP (inside mapd.h) daemons we have a macro with the name :
MAX_SET_BSS_INFO_NUM

This macro's value can be tuned based on the requirement
Based on this value various buffers (especially related to the combine infra metrics response, onboarding related BSS configuration variables) are allocated in the daemon code.

Important Note about *WEP*

WEP support has been removed from default build since hostapd v2.10. So all SKUs which have hostapd enabled SDK will have this feature off.

Important Note about Fronthaul BSS
For WPS to be successful, there must be at least one NON-WEP, NON-TKIP and NON-HIDDEN Front haul BSS per radio.

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Important Notes about BSS configurations

- For WPS onboarding to be successful, you need to have at least one fronthaul BSS present on the device.
- For recent changes done for MAP R3, now wts_bss_config file has been updated and the first BSS is only Backhaul BSS (it does not have FH enable). So it is advisable to do onboarding testing in MBSS case if you want to use the default wts_bss_config file settings for your test , otherwise WPS will fail.
- For devices using 6G band, WPS must happen on either 24G or 5G. Since WPS uses WPA2 security and WPA3 is mandatory in 6E so WPS is not allowed in 6E band. So we need to keep minimum one fronthaul BSS on 2G or 5G radio with WPS can happen through it even when 6G BH is needed.
- Addition or deletion of MBSS is not allowed in MAP at run time with MAP Enable. Need to disable MAP and then add or delete the BSS.If you need to add or delete an MBSS in MAP, then you need follow below process:
    - Disable MAP from WEB GUI
    - Add/Delete MBSS from WEB UI.
    - Then enable MAP.
- MTK solution provides a number of configurable parameters and APIs to control MAP Turnkey behavior. The parameters need to be configure either through configuration file or through command line interface. If a parameter is configured through configuration file, the change will be reflected only after restarting the MAP Turnkey applications.
- Certain configuration parameters of Agents is controlled by the controller. These parameters should not be changed from the Agent's GUI or iwpriv/mwctl commands
    - SSID
    - Channel
    - Security
    - Hidden SSID

  If these parameters need to be changed, do it through the GUI of controller.

3.2.1.1 Configuring a SSID having space or special character

If there is a space in SSID, the user needs to enter \ as ESC in front of it in wts file. This is because space is regarded as a separator in wts file. If ssid has \, then user needs to add \ as ESC in front of it.
So User needs to follow below rules:

1. add \ before space
2. add \ before \

For example:

If the ssid is: hello world \!
In wts file, it should be hello\ world\ \\!

3.2.1.2 Configuring a BSS in WEP security

A fronthaul BSS can be configured to support WEP security, by modifying the wts_bss_info_config file in correct format. But the following points must be kept in mind:

1. At least one Front haul BSS must be NON- WEP, so that WPS can be functional.
2. If a BSS is backhaul , then it should not support WEP.
3. There is support for only one Default key ID, DefaultKeyID=1.

In wts_bss_info_config file , if **EncrypType is WEP(*0x0002* corresponds to WEP)**

Eg : *6,ff:ff:ff:ff:ff:ff 8x Multi-AP-24G-2 0x0001* ***0x0002*** *mapro 0 1 hidden-N*

In this case the key can be a string of length 5 or 13 (in ascii)

Eg : *6,ff:ff:ff:ff:ff:ff 8x Multi-AP-24G-2 0x0001 0x0002* ***mapro*** *0 1 hidden-N*

Or

The key can be a string of length 10 or 26 (in hexadecimal)

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Eg : *6,ff:ff:ff:ff:ff:ff 8x Multi-AP-24G-2 0x0001 0x0002* ***1111111111*** *0 1 hidden-N*

3.2.1.3 Configuring a Fronthaul BSS in Mixed Mode security

A fronthaul BSS can be configured to support Mixed mode (both WPA-Personal and WPA2-Personal enabled) security, by modifying the wts_bss_info_config file in correct format. But the following points must be kept in mind:

1. At least one Front haul BSS must be AES, so that WPS can be functional.
2. If a BSS is backhaul , then it should not support WEP/TKIP.
3. In wts_bss_info_config file, AuthMode and **EncrypType** will be **Mixed Mode** as below:

6,ff:ff:ff:ff:ff:ff 8x Multi-AP-24G-2 **0x0022 0x000c** mapro 0 1 hidden-N

Authmode = 0x0022 *and* ***EncrypType= 0x000c*** *corresponds to Mixed Mode security*

3.2.1.4 Configuring BSS for MLO

For WIFI7 (802.11be) device which supports Multi link operation (MLO) connection for which after creating normal BSS we need to create AP and STA (Multi link device) MLD groups
Please refer this path in WebUI: MTK->Mld Configuration-> Add MLD Configuration



Figure 3- 1. AP MLD creation

In the above diagram one MLD group has been formed using ra1/rai1/rax1. This MLD group have same SSID and security configurations. Similarly, we can create 4 MLD groups for 4MBSS.

We also need to create STA MLD at Agent side for MLO connection, below diagram shows how to create STA MLD
Please refer this path in WebUI: MTK->Mld Configuration-> Add MLD Configuration

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Figure 3- 2. APCLI MLD creation

After creating STA Mld, Normal Auth, Assoc and EAPOL handshakes will happen only on primary link, but actual data transfer can happen on all the Bands.

For Multi link operation (MLO) supported SKU's the "wts_bss_configuration" file should follow the below example. The security configurations like SSID, Authmode, passphrase, encryption type should be same for the interfaces associated with one MLD groups. You can refer section 3.2.2.1 for WTS modification using UCI.

Example:
If the MLD groups are formed as below setting:

ra0, rai0, rax1 have config as "Multi-AP-1", "0x00E0 / 0x00C0 ", "maprocks1"
ra1, rai1, rax1 have config as "Multi-AP-2", "0x0060 / 0x0040 ", "maprocks2"
ra2, rai2, rax2 have config as "Multi-AP-3", "0x0060 / 0x0040 ", "maprocks3"
ra3, rai3, rax3 have config as "Multi-AP-4", "0x0060 / 0x0040 ", "maprocks4"

- ➔ Here since on 6G radio the minimum Authmode support is SAE only so we need to keep the 2G and 5G band I WPA2+WPS3 transition mixed mode support.
- ➔ SSID is same for each BSS of the MLD group
- ➔ Password is same for each BSS of the MLD group
- ➔ Similarly other configurations must also match.

```
root@OpenWrt:/# cat /etc/map/wts_bss_info_config
#ucc_bss_info
1,ff:ff:ff:ff:ff:ff 11x Multi-AP-1 0x00E0 0x0008 maprocks1 1 0 hidden-N 4095 pvid 5 mld_groupID 1
2,ff:ff:ff:ff:ff:ff 11x Multi-AP-2 0x0060 0x0008 maprocks2 0 1 hidden-N 4095 N/A N/A mld_groupID 2
3,ff:ff:ff:ff:ff:ff 11x Multi-AP-3 0x0060 0x0008 maprocks3 0 1 hidden-N 4095 N/A N/A mld_groupID 3
4,ff:ff:ff:ff:ff:ff 11x Multi-AP-4 0x0060 0x0008 maprocks4 0 1 hidden-N 4095 N/A N/A mld_groupID 4
5,ff:ff:ff:ff:ff:ff 8x Multi-AP-1 0x00E0 0x0008 maprocks1 1 0 hidden-N 4095 N/A N/A mld_groupID 1
6,ff:ff:ff:ff:ff:ff 8x Multi-AP-2 0x0060 0x0008 maprocks2 0 1 hidden-N 4095 N/A N/A mld_groupID 2
7,ff:ff:ff:ff:ff:ff 8x Multi-AP-3 0x0060 0x0008 maprocks3 0 1 hidden-N 4095 N/A N/A mld_groupID 3
8,ff:ff:ff:ff:ff:ff 8x Multi-AP-4 0x0060 0x0008 maprocks4 0 1 hidden-N 4095 N/A N/A mld_groupID 4
9,ff:ff:ff:ff:ff:ff 12x Multi-AP-1 0x00E0 0x0008 maprocks1 1 0 hidden-N 4095 N/A N/A mld_groupID 1
10,ff:ff:ff:ff:ff:ff 12x Multi-AP-2 0x0060 0x0008 maprocks2 0 1 hidden-N 4095 N/A N/A mld_groupID 2
11,ff:ff:ff:ff:ff:ff 12x Multi-AP-3 0x0060 0x0008 maprocks3 0 1 hidden-N 4095 N/A N/A mld_groupID 3
12,ff:ff:ff:ff:ff:ff 12x Multi-AP-4 0x0060 0x0008 maprocks4 0 1 hidden-N 4095 N/A N/A mld_groupID 4
13,ff:ff:ff:ff:ff:ff 13x Multi-AP-1 0x00C0 0x0008 maprocks1 1 0 hidden-N 4095 N/A N/A mld_groupID 1
14,ff:ff:ff:ff:ff:ff 13x Multi-AP-2 0x0040 0x0008 maprocks2 0 1 hidden-N 4095 N/A N/A mld_groupID 2
15,ff:ff:ff:ff:ff:ff 13x Multi-AP-3 0x0040 0x0008 maprocks3 0 1 hidden-N 4095 N/A N/A mld_groupID 3
16,ff:ff:ff:ff:ff:ff 13x Multi-AP-4 0x0040 0x0008 maprocks4 0 1 hidden-N 4095 N/A N/A mld_groupID 4
```

Figure 3- 3. WTS file for MLO connection

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3.2.1.1 Configuring BSS using UCI

If device has Universal Configuration Interface (UCI) support then the BSS can be configured by modifying the /etc/config/mapd in correct format as shown below:

```
config iface '1'
        option mac 'ff:ff:ff:ff:ff:ff'
        option radio '11x'
        option ssid 'Multi-AP-1'
        option authmode '0x00E0'
        option EncryptType '0x0008'
        option PSK 'maprocks1'
        option bhbss '1'
        option fhbss '0'
        option hidden 'N'
        option vlan '4095'
        option pvid 'pvid'
        option pcp '5'
        option mld_groupID '1'
```

Figure 3- 4. Snapshot of etc/config/map file for configuring the BSS via UCI

For more details please refer to the Appendix section : 11.19

Eg: change security of bss:
uci set mapd.7.authmode='0x00c0'
uci commit mapd

Eg: change ssid name
uci set mapd.7.ssid='Multi-AP'
uci commit mapd

3.2.2 WPS Onboarding for MAP

MAP specification defines WPS based onboarding as a mandatory method for onboarding.
Following are the message exchange that occur during the MAP onboarding and auto configuration process on WLAN link using WPS.

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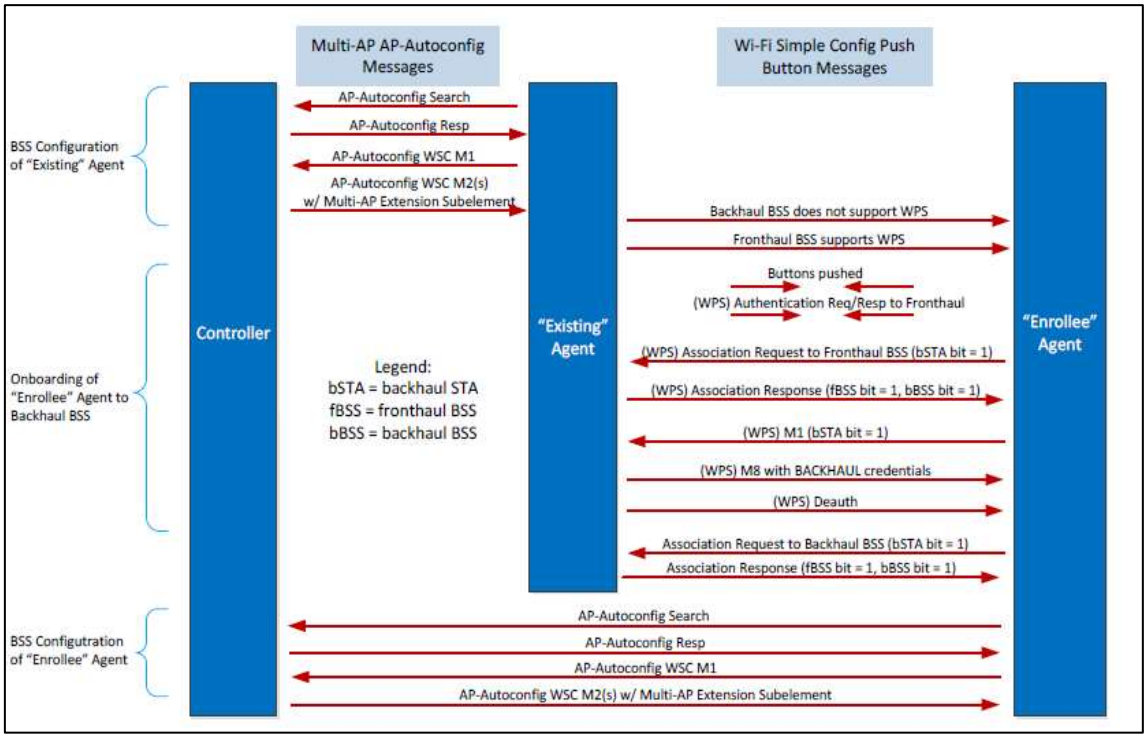


Figure 3- 5. MAP WPS Onboarding and Autoconfiguration Procedure

3.2.2.1 WPS Onboarding Daemon interactions

This section demonstrates the interactions that happen between software modules during the onboarding and auto configuration process both at Enrollee Agent and controller.

Interactions at Enrollee Agent

Following are the interactions between modules during WPS based MAP onboarding of an enrollee agent:

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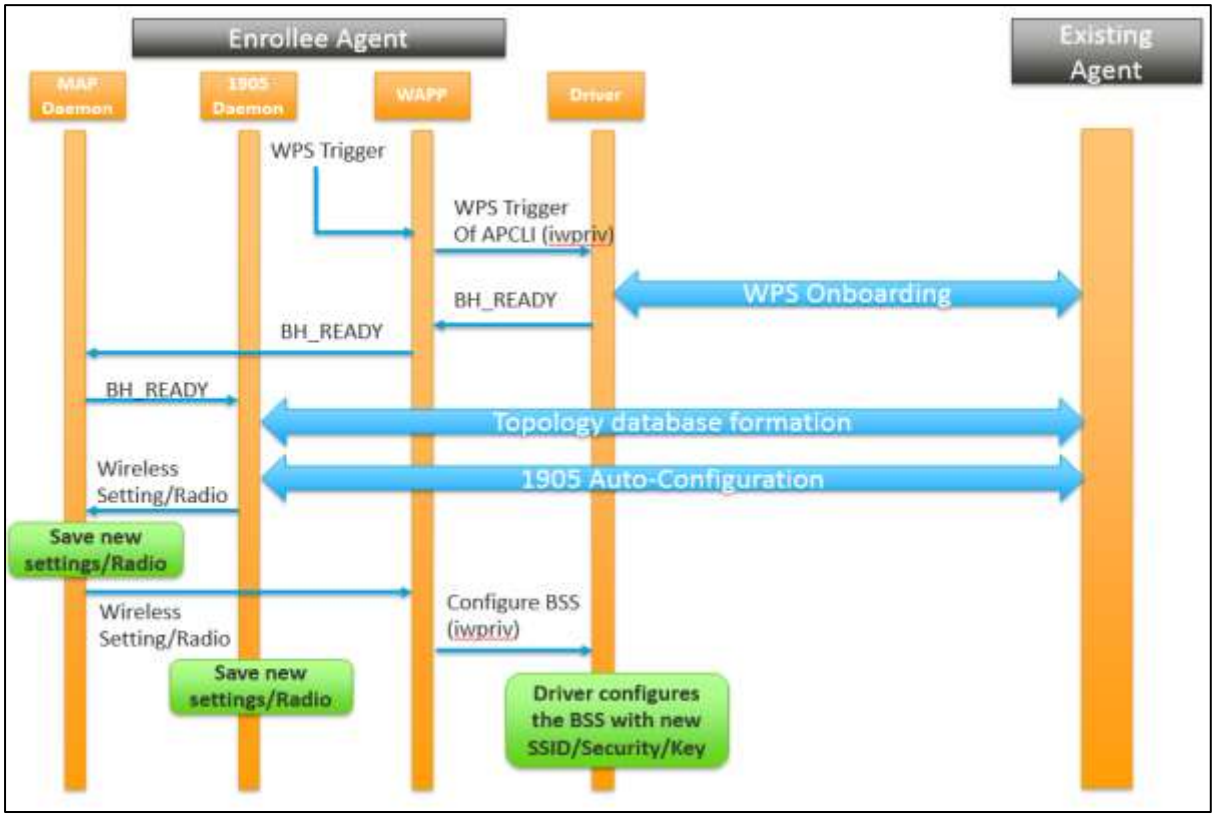


Figure 3- 6. WPS Onboarding Interactions between daemons at the enrollee agent.

Interactions at controller

Following are the interactions between modules at the Controller when an enrollee agent tries to get configuration

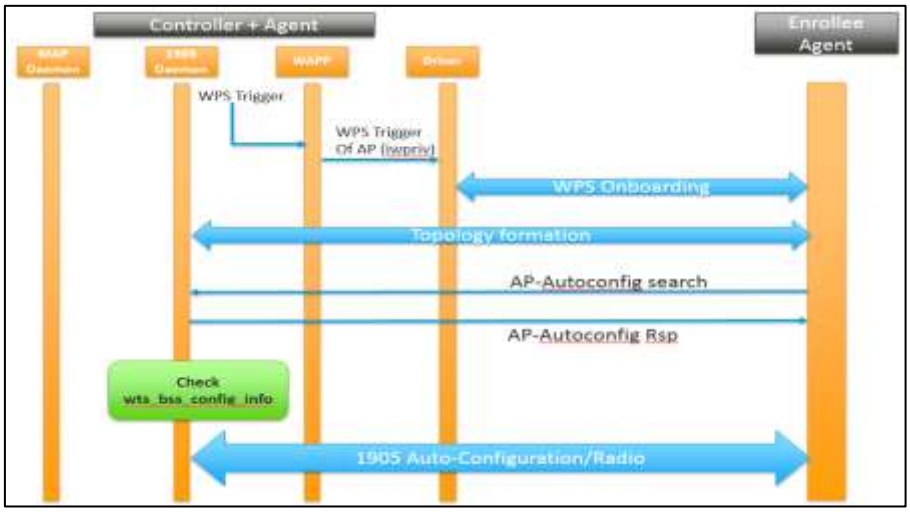


Figure 3- 7. WPS Onboarding Interactions between daemons at the controller.

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3.2.2.2 WPS Onboarding daemon interactions for hostapd supported SKU

This section demonstrates the interactions that happen between software modules during the onboarding and auto configuration process both at Enrollee Agent and controller. If the architure supports HostAPD/Supplicant then the flow for WPS Ponboarding will be as follows.

Interactions at Enrollee Agent

Following are the interaction between modules for SKU supporting hostapd/Supplicant during WPS based MAP onboarding

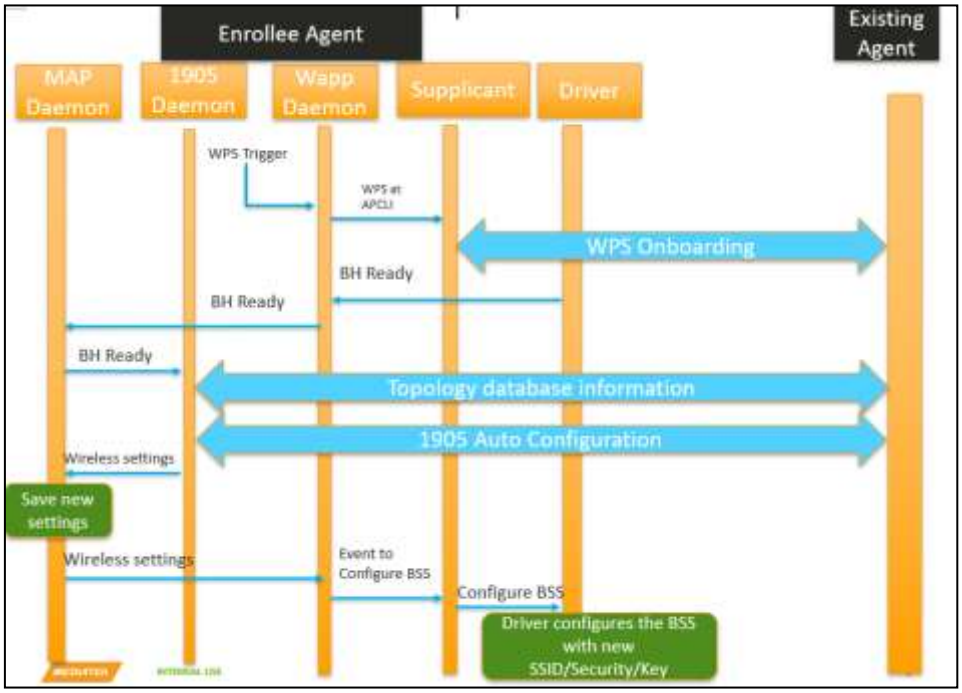


Figure 3- 8. WPS Onboarding Interactions between daemons at the enrollee agent for Hostapd/Supplicant supported SKU.

Interactions at controller

Following are the interaction between modules when SKU supports Hostapd/Supplicant when an enrollee does WPS based MAP onboarding

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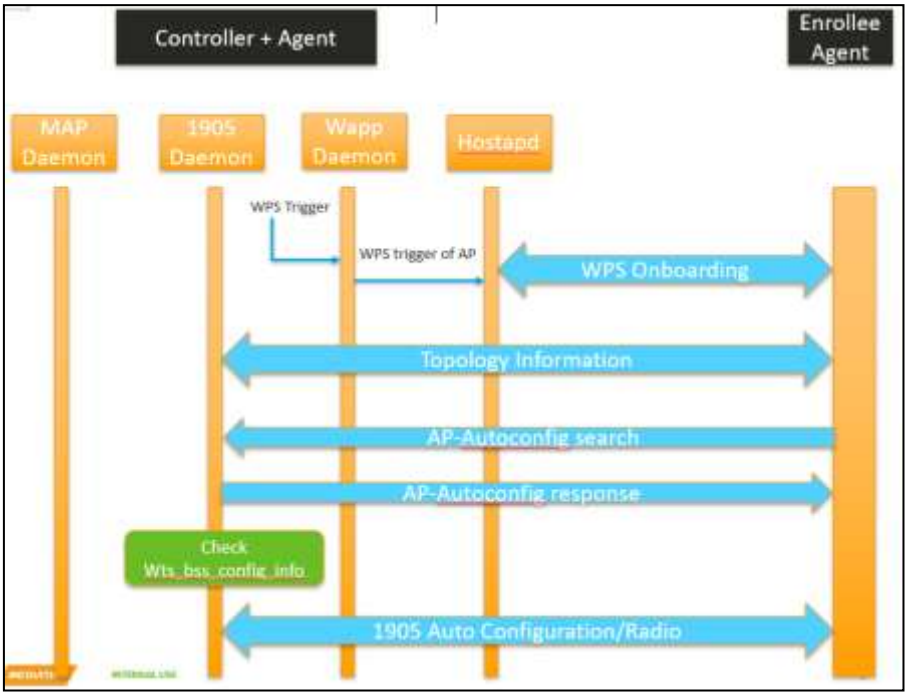


Figure 3- 9. WPS Onboarding Interactions between daemons at the controller for Hostapd supported SKU.

3.2.2.3 Onboarding and Autoconfiguration procedure

In this section steps to trigger onboarding is explained with the help of two scenarios:

1. **Ethernet based onboarding and auto configuration**
2. **WiFi based onboarding and auto configuration**

1. <u>Ethernet based onboarding and auto configuration</u>

a. Bring up 2 MAP enabled devices, 1 as existing agent (can be controller) and 1 as enrollee agent by following the steps mentioned in section 3.1
b. Connect the Ethernet cable between the two devices.
c. After plugging in Ethernet cable, MAP onboarding and auto configuration will start.
    i. Agent will update the fronthaul BSS configurations based on the configuration given by the controller.

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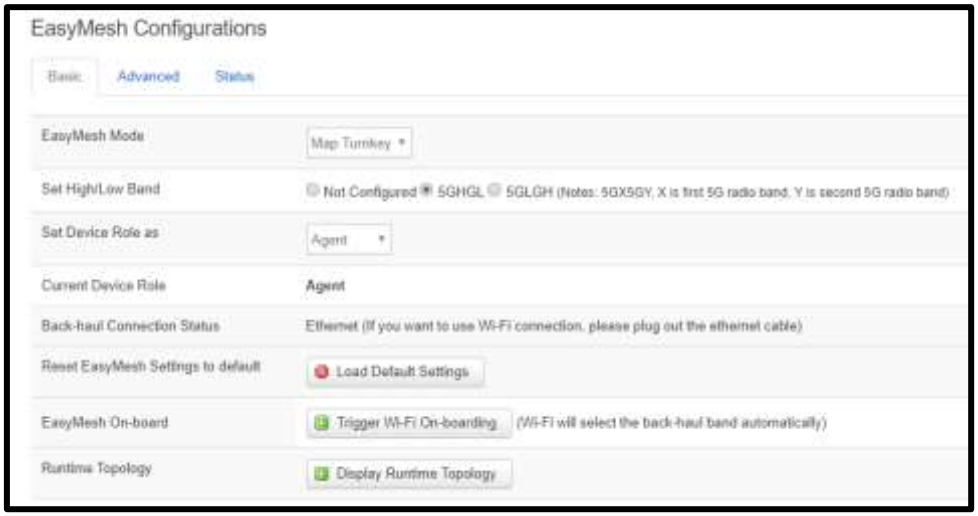


Figure 3- 10. Agent WEB UI page shows Back-Haul Connection status as Ethernet Backhaul

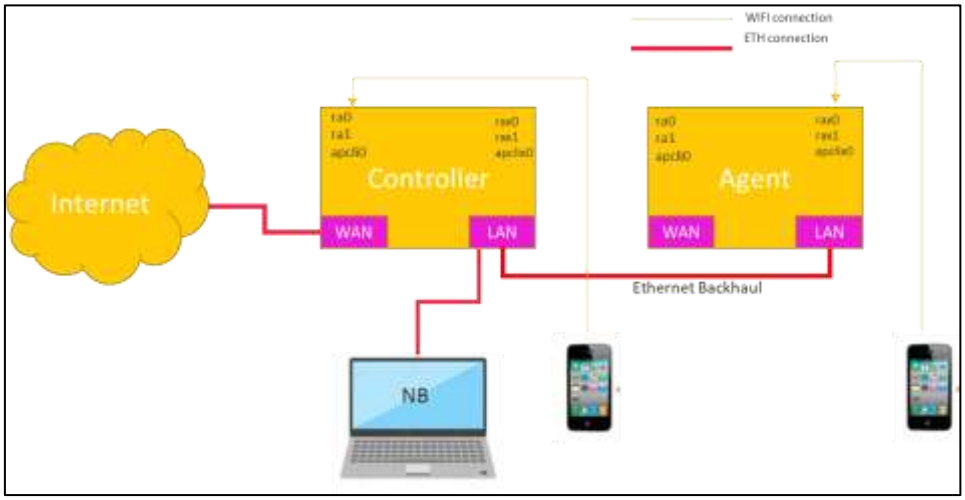


Figure 3- 11. Agent and Controller connected via Ethernet Backhaul and Third party NB and Phone connect via LAN and Fronthaul

2. **<u>WiFi based onboarding and auto configuration</u>**

   a. Bring up 2 MAP enabled devices, 1 as existing agent (can be controller) and 1 as enrollee agent by following the steps mentioned in section 3.1.
   b. Trigger WPS based onboarding between the two devices by issuing the following command on both the devices:
      i. wappctrl ra0 wps_pbc
         or
      ii. Click on the Agent's web page on the following button choose "Trigger Wi-Fi On-boarding"

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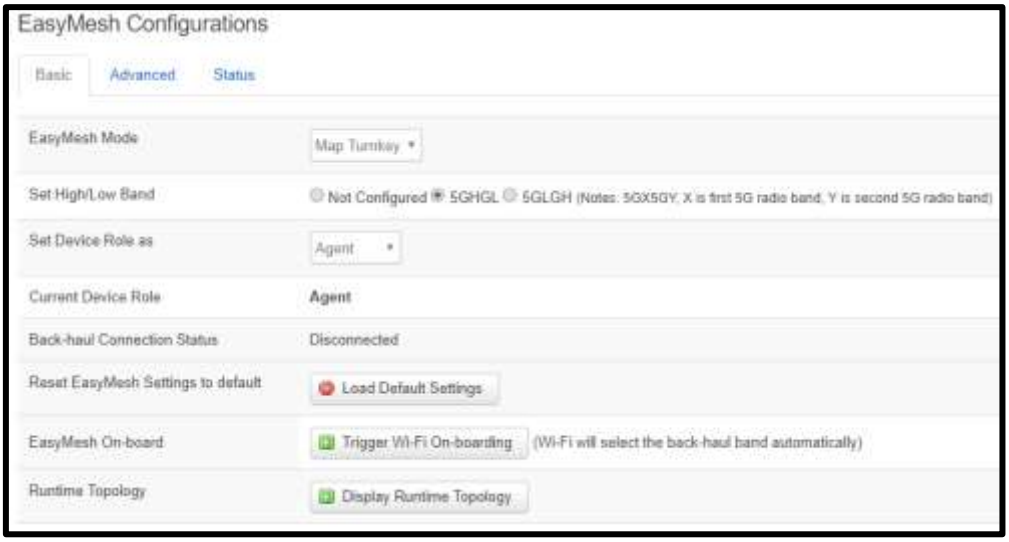


Figure 3- 12. Agent WEB UI page shows Back-Haul Connection status as Disconnected initially

c. After executing this command, MAP onboarding and auto configuration will start.
   i. Existing Agent (or controller) will start WPS registrar on AP interface
   ii. Enrollee Agent will start WPS enrollee on APCLI interface
   iii. Agent will update the fronthaul BSS configurations based on the configuration given by the controller.
   iv. Agent will save the configuration in the nvram for LSDK and mapd_cfg for Open WRT and use it to automatically connect to the same configuration after reboot.
   v. To remove the previously stored configuration, give the following command :
      1. EashMesh.sh default
         OR
      2. You can click on “Load Default Settings” Button from Web UI:

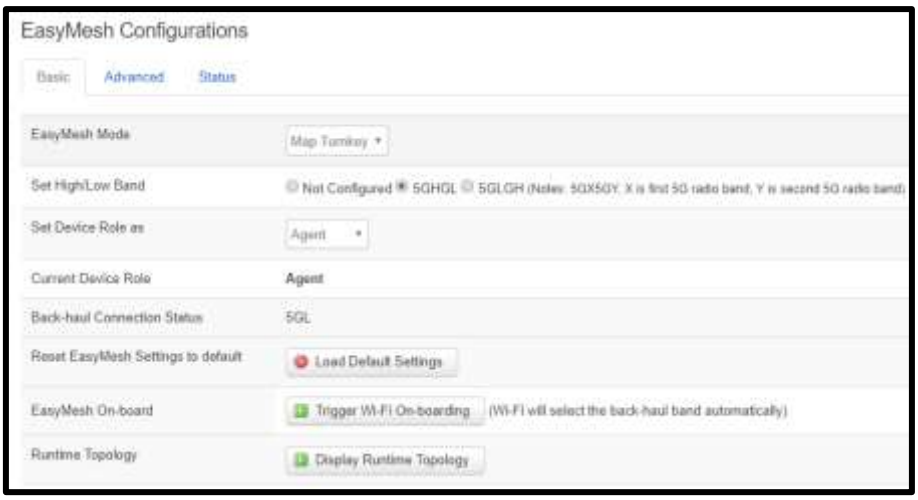


Figure 3- 13. Agent WEB UI page shows Back-Haul Connection status as 5GL

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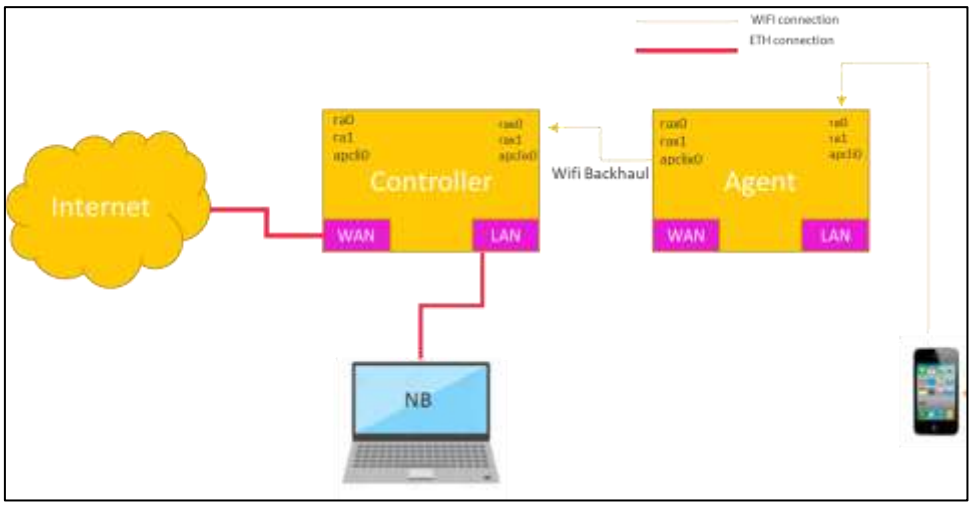


Figure 3- 14. Agent and Controller connected via WIFI Backhaul

Now we can see Agent's AL_MAC in Advanced tab as below :



Figure 3- 15. Agent's WEB UI Advanced tab ALMAC of agent display

After this the Run-Time Topology Display will show as follows:

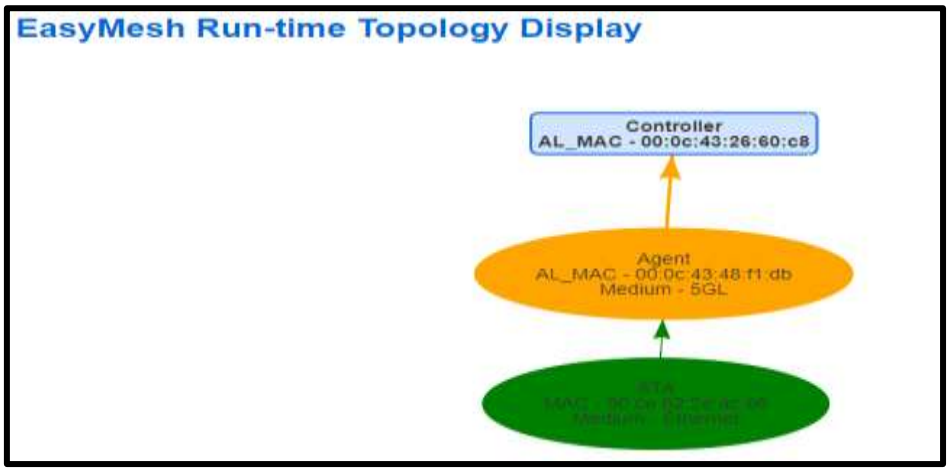


Figure 3- 16. Agent's and controller's WEB UI Display topology page show new topo

Some points to note for WPS Onboarding on Agent's APCLI :

a. WPS onboarding on Agent's APCLI will only be triggered when Agent is not in configured state. Once it reaches configured state i.e. it has received the configurations from the controller, then wps will always be triggered on the AP interface.
b. To trigger wps on APCLI interface after the Agent is configured, user must give the following command before triggering WPS:
    i. wappctrl ra0 map reset_default

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3.2.2.4 WPS trigger on a remote agent via Controller

We provide one API of MAPD to launch remoted agent's WPS from Controller. Customers could integrate 3rd-party APPs with Smartphone and MTK MAP framework to have full functionality. The flow is as below:

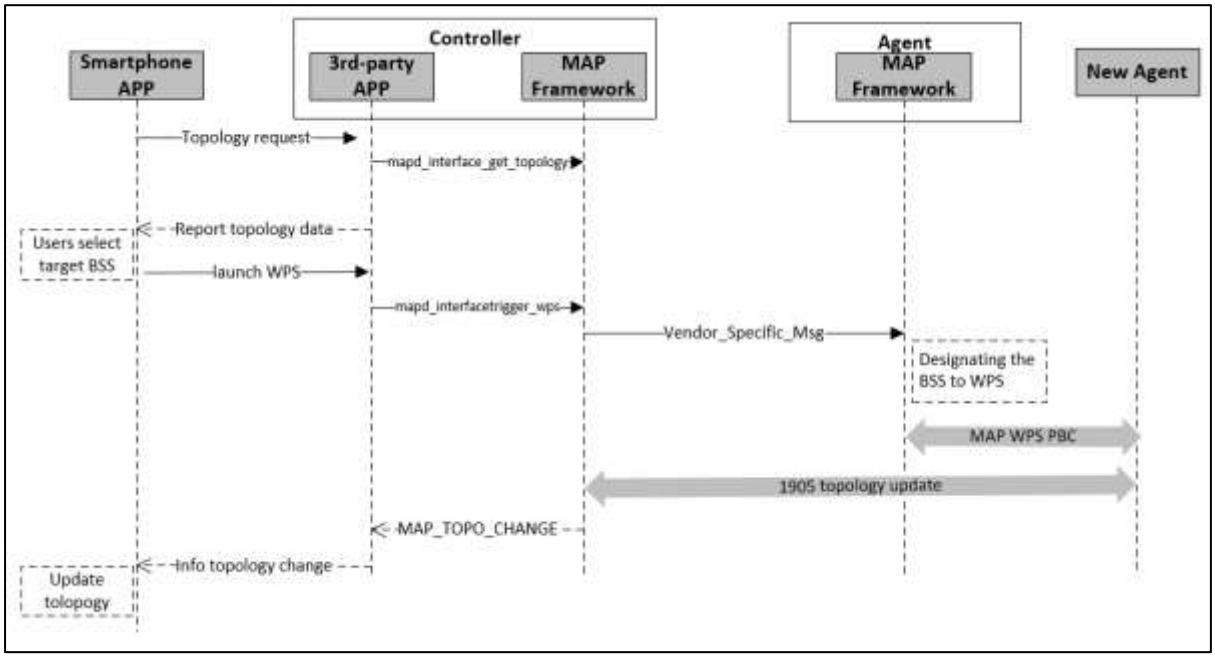


Figure 3- 17. WPS trigger on a remote agent via Controller flow diagram

The steps in above figure are explained as below:

1. The mobile APP connects to the MAP network and queries the Controller for the
2. Network topology.
3. The Controller feedbacks the obtained topology information (JSON format) to the mobile
4. APP.
5. User selects Fronthaul BSS that needs to start WPS
6. Mobile APP notifies Controller to start WPS for specified BSS.
7. Controller sends the request by vendor_specific_Msg to the target Agent to lunch WPS.
8. Vendor_specific_Msg carries WPS vendor-specific TLV which is used to specify the BSS.
9. Agent receives vendor-specific message and starts WPS for the specified BSS.
10. The new networking device starts WPS and establishes a connection.
11. MAP network updates topology information.
12. Controller notifies mobile APP of topology changes.
13. Mobile APP refresh topology.

3.2.3 DPP Onboarding for MAP

The MAP R3 specification defines new DPP method for onboarding by which a Multi-AP device can onboard to a Multi-AP network. This DPP onboarding is using out-of-band bootstrapping (wifi and ethernet based).

The Multi-AP DPP onboarding method enables a Multi-AP Agent that implements Profile-3 to use the DPP Protocol to onboard to a Multi-AP network using a Multi-AP Controller that implements Profile-3. The onboarding can occur over Ethernet or over Wi-Fi via a Multi-AP Agent that implements Profile-3 acting as a Proxy Agent.

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Since the DPP Protocol supports configuration of DPP capable devices using a variety of Wi-Fi technologies, onboarding of Multi-AP devices has the DPP "wi-fi_tech" field set to "map". Multi-AP-specific extensions to the DPP Configuration Request object and DPP Configuration object parameters are defined in the MAP-R3 specification.

3.2.3.1 MAP Turnkey Model

Below figure shows which part is handled by MAP Turnkey for DPP i.e. complete DPP connection and provide a WEB UI/API mechanism to pass URI to the system for out of band DPP bootstrapping. Mobile APP or any other way of passing QR code information to system is out of scope. MTK provide's API to user to pass collected QR information to the system.

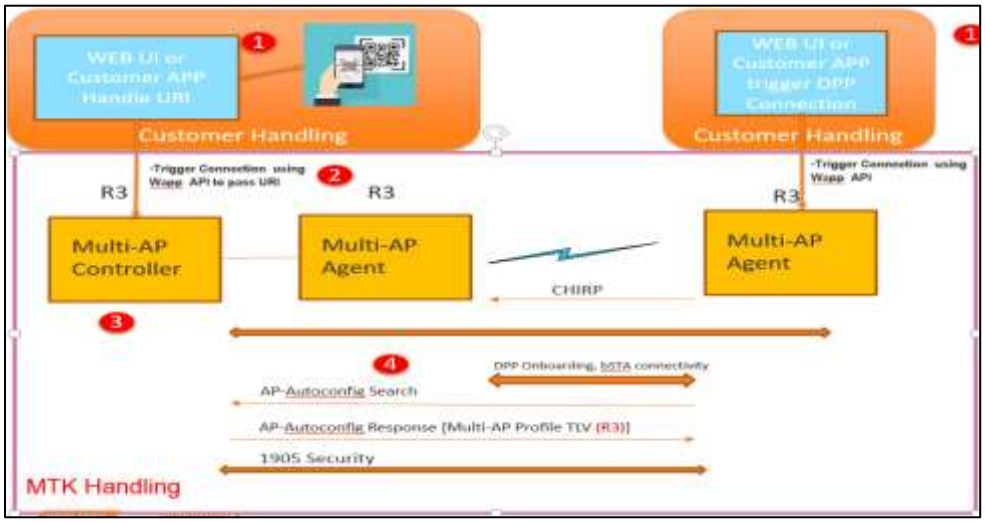


Figure 3- 18. MTK Turnkey Model for DPP

3.2.3.2 DPP Onboarding Daemon interactions

MAP R3 specification defines DPP based onboarding. Following sections describe SW architecture and the message exchange that occur during the MAP onboarding via Wi-Fi and via Ethernet.

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Controller

A Multi-AP compliant logical entity that implements logic for controlling the operation of the Multi-AP network.

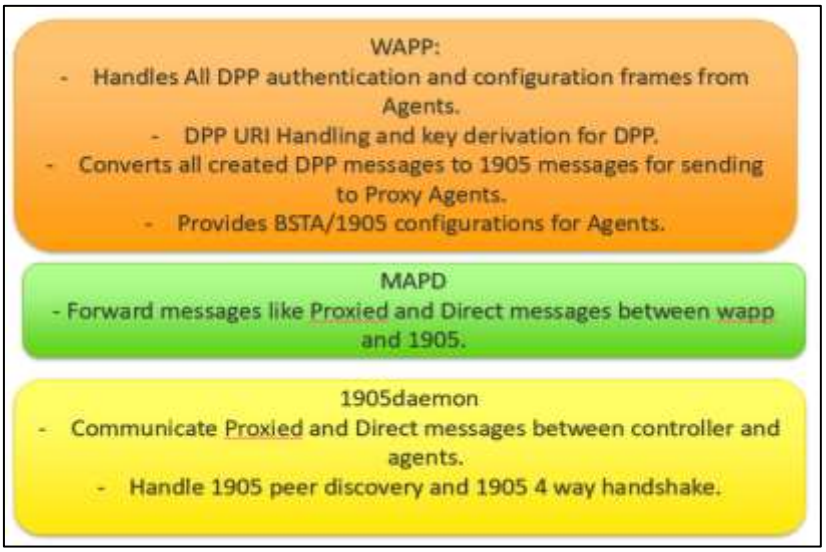


Figure 3- 19. DPP Onboarding Controller Daemon Interaction

Proxy Agent

A Multi-AP Agent that translates between 1905 CMDUs and 802.11 frames to and from an Enrollee Multi-AP Agent.



Figure 3- 20. DPP Onboarding Proxy Agent Daemon Interaction

Enrollee Agent

A Multi-AP compliant logical entity that executes AP control functions and provides Multi-AP specific control information.

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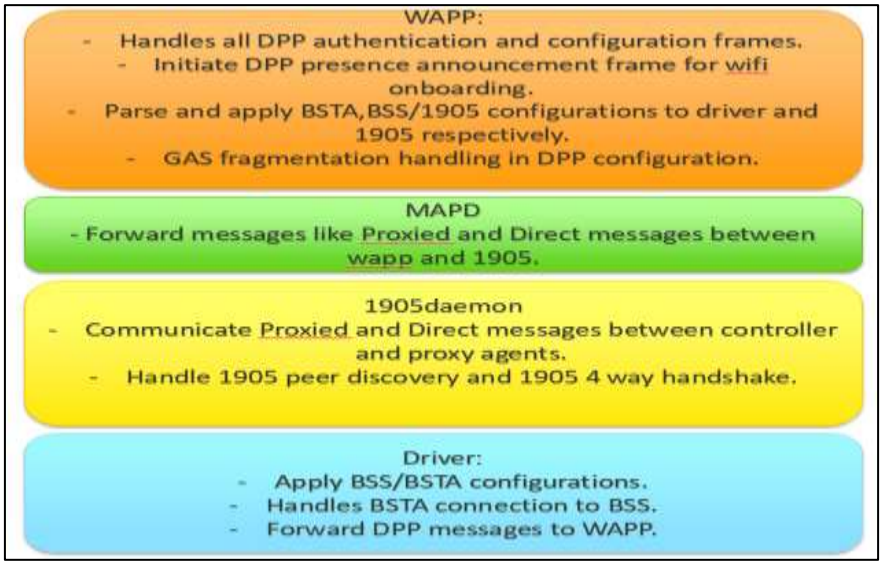


Figure 3- 21. DPP Onboarding Enrollee agent Daemon Interaction

3.2.3.3 Onboarding Method via a Multi-AP Logical Ethernet Interface with out-of-band DPP bootstrapping

Following are the brief steps following during Ethernet DPP QR based onboarding. It is expected that controller to have QR code of Enrollee before triggering onboarding:

- Once the Agent is connected to the controller through Ethernet, the Enrollee Multi-AP Agent shall search for the Multi-AP Controller by sending 1905 AP-Autoconfiguration Search messages and will include chirp hash in search message.
- Controller checks for the chirp hash with the QR code key it already has. Once match Controller send 1905 AP-Autoconfiguration Search response (Multi-AP Profile TLV is set to “Multi-AP Profile-3") and also initiate the DPP Authentication with Agent.
- After DPP authentication, The Agent sends DPP configuration request. Upon receiving it the controller sends and DPP config response which contains the config response object consisting of 1905 connector. The 1905 connector will be used by 1905 to establish secure 1905 connection between devices.
- All DPP Authentication and Configuration frames are converted to direct encap messages between the controller and agent.
- After DPP authentication and configuration, the controller and agent establish secure 1905 connection using 1905 peer discovery and 1905 4 way handshake.

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- After secure connection establishment, Agent sends BSS configuration request to which Controller sends BSS config response consist of FBSS and BBSS credentials which are to be configured on Agent.

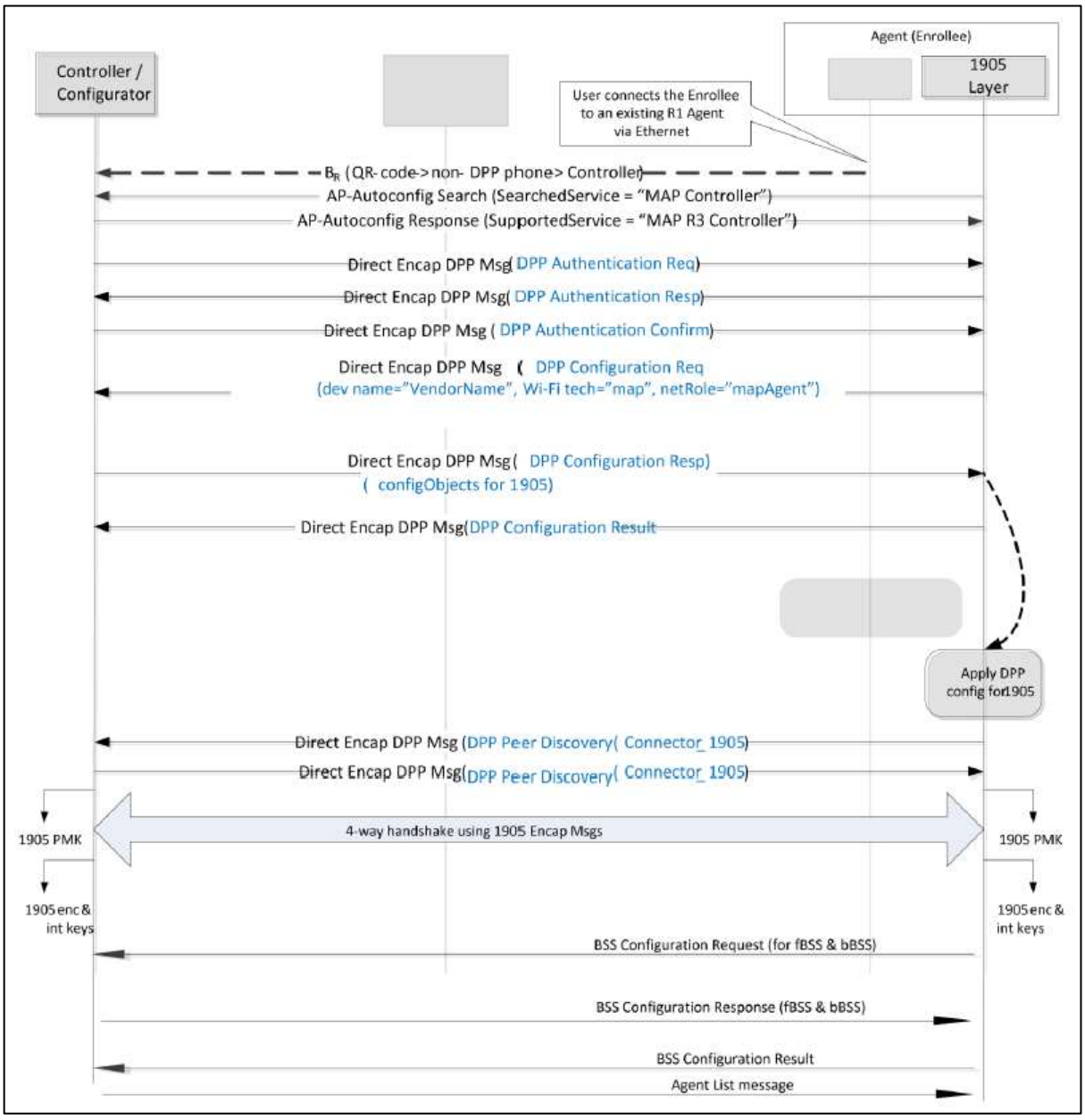


Figure 3- 22. DPP Onboarding Method via a Multi-AP Logical Ethernet Interface with out-of-band DPP bootstrapping

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3.2.3.4 Onboarding method via Wi-Fi with out-of-band DPP bootstrapping

Following are the brief steps following during Wi-Fi DPP QR based onboarding. It is expected that controller to have QR code of Enrollee before triggering onboarding:

- Here previously on boarded agent acts as a Proxy Agent which forwards the packets from Controller to new Enrollee onboarding agent.
- In Wi-Fi onboarding the Enrollee Agent initiate sending presence announcement frame on its channel list(which can be updated periodically based on CCE field from Proxy agent beacon/probe responses).
- Proxy Agent encapsulate the DPP frames coming from Enrollee Agent into the Proxied DPP encap messages for controller and decapsulate the Proxied DPP messages in to the DPP frames for Enrollee agent.
- After DPP authentication, the enrollee agent sends DPP configuration request with object. Upon receiving it the controller sends DPP config response.
- DPP configuration response consist of 1905 connector and bsta Connector. Enrollee agent uses the BSTA connector to extract Backhaul BSS information for BSTA connection using AKM specified in config response. The 1905 connector will be used by 1905 to establish secure 1905 connection between devices.
- All DPP Authentication and Configuration frames between controller and proxy agent are converted to 1905 proxied DPP encap messages.
- After DPP authentication and configuration and BSTA connection, the controller and Agent establish secure 1905 connection using 1905 peer discovery and 1905 4 way handshake.
- After secure connection establishment, Agent sends BSS configuration request to which Controller sends BSS config response consist of FBSS and BBSS credentials which are to be configured on Agent.

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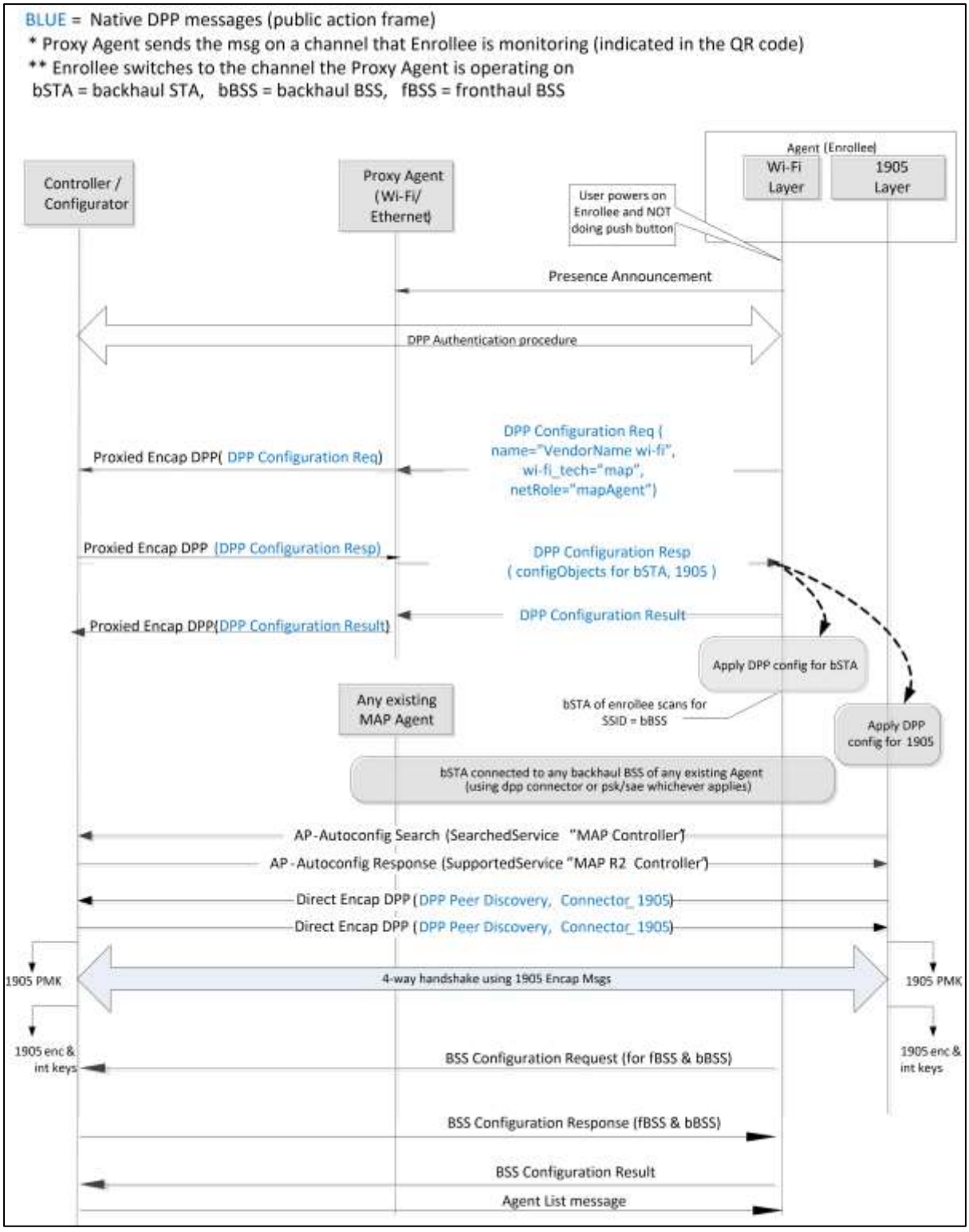


Figure 3- 23. DPP Onboarding Method via WiFi with out-of-band DPP bootstrapping

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3.2.3.4.1 DPP Presence Announcement and channel list selection procedure

A channel list on which to send DPP Presence Announcement frames by the enrollee shall be derived From the following procedure:

1. One from each of the following channels, as supported and configurable:
    a. 2.4 GHz: Channel 6 (2.437 GHz)
    b. 5 GHz: Channel 44 (5.220 GHz) if local regulations permit operation only in the 5.150 – 5.250 GHz band and Channel 149 (5.745 GHz) otherwise
2. Scan all supported bands and add each channel on which an AP is advertising the Configurator Connectivity element to the channel list
3. DPP chirping will be started based on priority flag present in /etc/dpp_cfg.txt or in /etc/config/mapd file if SKU supports UCI i.e presence_band_priority. If presence_band_priority=5g, then presence announcement will be started on 5G channels first then on 2G channels. EasyMesh Daemons needs to be restarted post this change or need to reboot the device in order to apply this change.
   In default /etc/dpp_cfg.txt, presence_band_priority will not be present so default value will be used as 5g.



Figure 3- 24. **Presence band priority parameter**

4. The Enrollee shall follow the procedure below for sending the Presence Announcement frames:
    a. For each channel in the channel list generated, the Enrollee, shall send a DPP Presence Announcement frame and listen for 2 seconds to receive a DPP Authentication Request frame. If a valid DPP Authentication Request frame is not received, it shall repeat the presence announcement for the next channel in the channel list.
    b. When all channels in the channel list have been exhausted, the Enrollee shall pause for at least 30 seconds before repeating the procedure in step a above.
    c. If the device has cycled through the procedure (Steps a & b) four times without receipt of a valid DPP Authentication Request frame, it shall generate a new channel list using the steps 1,2 specified in this section, prior to resuming the presence announcement procedure.

For DPP channel used in the MAP R3 controller, device Channels will be selected based on Channel planning not by bootstrap information.

3.2.3.4.2 CCE Indication

A Multi-AP Controller instruct the existing Multi-AP Agents to start advertising the CCE in Beacon and Probe Response frames to facilitate seamless (re)configuration of backhaul STA:

1. upon receipt of an Enrollee's DPP URI
2. a user specific request to WAPP

On boot up - CCE will be enabled on Controller/connected Agents. Also enable CCE whenever any new Agent connection is formed.

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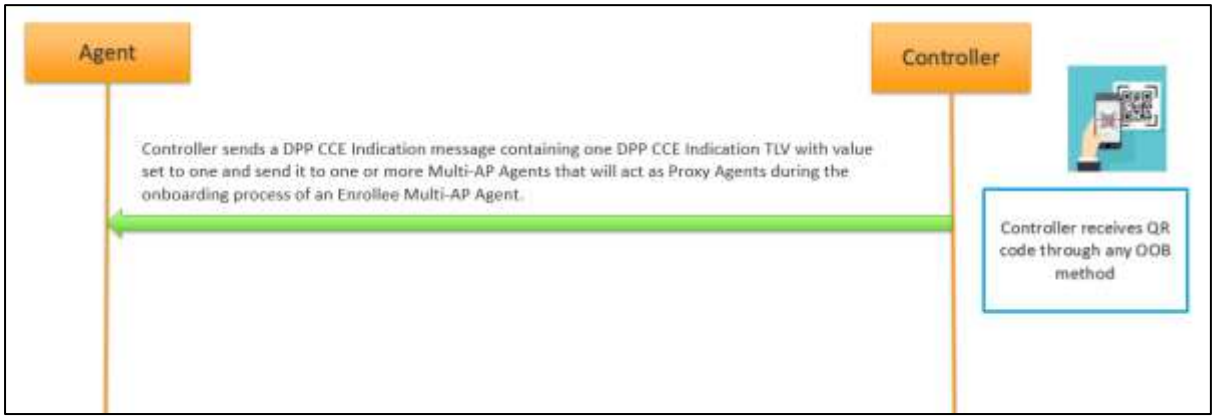


Figure 3- 25. DPP CCE Indication message

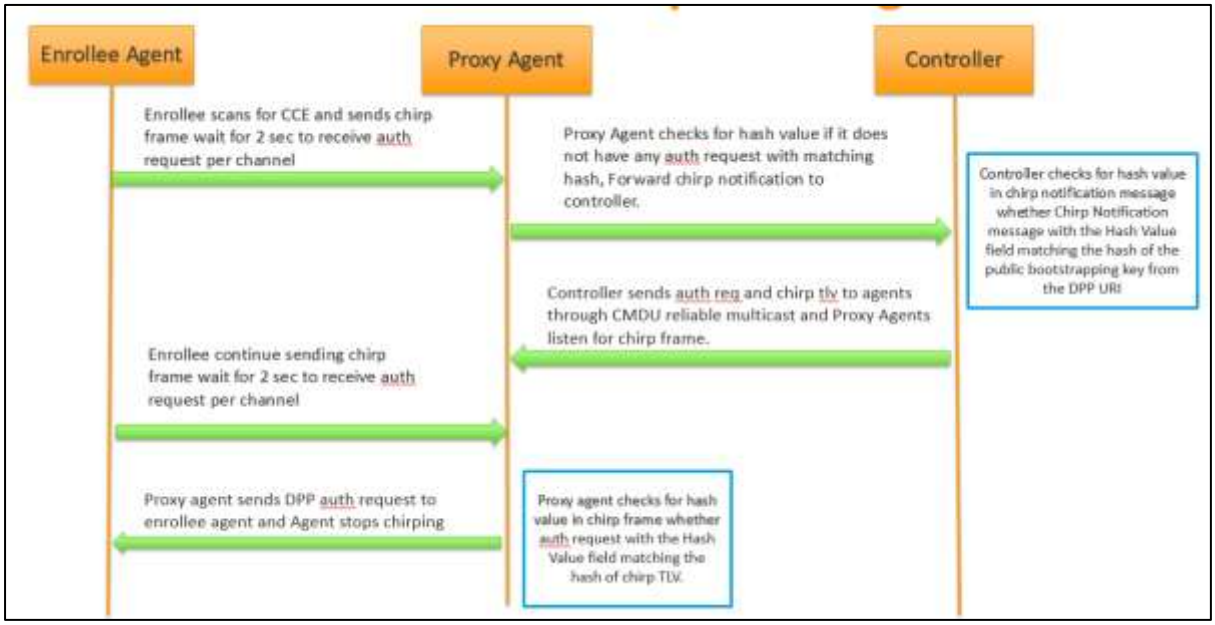


Figure 3- 26 . DPP CCE Procedure

3.2.3.5 Onboarding method via Wi-Fi with inband DPP bootstrapping using Push Button

If a Multi-AP Agent that implements Profile-3 performs the Multi-AP PBC Backhaul STA Onboarding procedure with another Multi-AP Agent, the Enrollee Multi-AP Agent shall include its DPP URI in the Encrypted Settings of M7. By default MAP-R3 device will send the DPP URI in M7 message.

When a Multi-AP R3 controller receives a DPP URI from an Enrollee Multi-AP Agent during Multi-AP PBC Onboarding procedure, it shall perform the following:

- Send an M8 message to the Enrollee Multi-AP Agent without any backhaul BSS credentials.

If an Enrollee Multi-AP Agent which implements Profile-3 receives an M8 without any backhaul credentials, it should trigger the R3 DPP onboarding.

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Legacy WPS Onboarding:
If a user want the Multi-AP agent that implements Profile-3 with generated URI to start the Legacy WPS onboarding (independent of the Proxy agent setting) then user need to set the onboarding type to 1 before triggering the WPS onboarding. API details can be referred in Appendix section 10.11 .
"**wappctrl ra0 dpp dpp_onboard_type type=1**"

Note:-
If the user don't generate any DPP URI on bootup of the R3 agent, it will follow the legacy WPS onboarding method.

If a controller or Proxy Agent does not receive the DPP URI in the M7 message then, it shall perform the following:

- Send an M8 message to the Enrollee Multi-AP Agent with the backhaul BSS credentials.

NOTE: under this condition, the Enrollee Multi-AP Agent falls back to PBC onboarding procedure without securing the 1905-layer.

3.2.3.5.1 Onboarding using Proxy Agent:

When the new agent is to be onboarded to network using Proxy agent, and the proxy agent is previously R3 onboarded (which means it is 1905 secured with the controller), then upon receiving the DPP URI in M7, Proxy Agent will send the DPP URI Notification message to the controller and do not send the BSS credentials in the M8 message to the R3 enrollee agent.

And then Enrollee agent triggers DPP onboarding upon receiving M8 without BH credentials.

If a proxy agent is R1/R2 onboarded then it will send the BH credentials in the WPS M8 message and legacy WPS onboarding will start.

3.2.3.6 DPP timer

In MAP-R3 DPP onboarding there are some timers used to maintain the state machine on controller for each agents and on agent for onboarding state.

Table 3- 4. DPP Timer Settings

Sr. No.	DPP Role	Type	Value	Description
1	Enrollee	Time	2 seconds	Time to wait for each channel on the channel list for auth req.
2	Enrollee	Time	30 seconds	Time to start presence announcement again after each channels in channel list has been exhausted.
3	Enrollee	Count	4	Number of cycles to wait before making new channel list if no DPP authentication request is received after above two timers.
4	Controller/Proxy Agent	Time	10 Seconds	Time to wait for the DPP authentication response from the Enrollee.
5	Controller	Count	5	Number of times the controller retries for the DPP authentication request before dropping the DPP connection.
6	Enrollee	Time	10 Seconds	Time to wait for the DPP authentication confirmation from the Controller/Proxy agent.

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7	Controller/Proxy Agent	Time	10 Seconds	Time to wait for the DPP configuration request from the enrollee after sending out DPP authentication response.
8	Enrollee	Time	10 Seconds	Time to wait for DPP GAS initial response from the controller/Proxy Agent after configuration request has been sent.

3.2.3.7 DPP User failure Events

SEC_CONNECTION_FAILURE

Description: Event to indicate to use if any failure during the DPP onboarding. Below structure is used to indicate the failure reason to user.

Structure used and parameter:

```
struct mapd_user_conn_fail_event
{
        unsigned char reason[64];
};
```

reason – failure reason.

Usage: This event is used to notify the failure from the DPP handshakes during R3 onboarding. All failures like decryption failures, Key mismatch, missing DPP attributes in messages, etc. are sent. The reason will be mentioned in the user event as a string.

USER_FAIL_EVENTS

Description: Event to indicate to user failure from cmd during the DPP onboarding. Below structure is used to indicate the failure reason to user.

Structure used and parameters:

```
struct mapd_user_fail_event
{
        unsigned char reason_id;
        unsigned char reason[64];
};
```

reason_id:

DPP_QR_PARSING_FAIL – 0x0
DPP_QR_PARSING_FAIL - 0x1
CH_NOT_VALID - 0x2
PEER_DISC_FAIL – 0x3

reason – failure reason

Usage: Below are some failure reason for the user failure events according to the reason ID.

DPP_QR_PARSING_FAIL:

This event is sent to user when there is an issue in the QR code entered by user is wrongly formatted or wrong parameter is passed in the QR code on the controller.

DPP_QR_PARSING_FAIL:

This event is sent to user when any R3 command is given for the non R3(R1/R2) device.

CH_NOT_VALID:

This event is sent to user when an invalid channel is added in the channel list during QR code generation on agent.

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PEER_DISC_FAIL:

This event is sent to user when agent receives the Peer discovery failure from controller or proxy agent during DPP BH connection.

3.2.3.8 DPP State Machine

3.2.3.8.1 DPP Authentication

Spec has defined the default channels for chirping. If there are no valid channels provided (i.e. device not support that channel), corresponding error should be returned to user".

Upon receipt of a DPP Authentication Response frame, the Initiator checks that the DPP Status is set to STATUS_OK and whether a hash of its public bootstrapping key is included in the response. If so, the Initiator performs mutual authentication and should alert the user on the authentication failure of the Responder (STATUS_AUTH_FAILURE)

After the Responder obtains the DPP Authentication Confirm frame, it checks DPP Status and authenticate the Initiator. It should alert the user on the reason for the protocol failure of STATUS_NOT_COMPATIBLE/STATUS_AUTH_FAILURE or authentication failure.

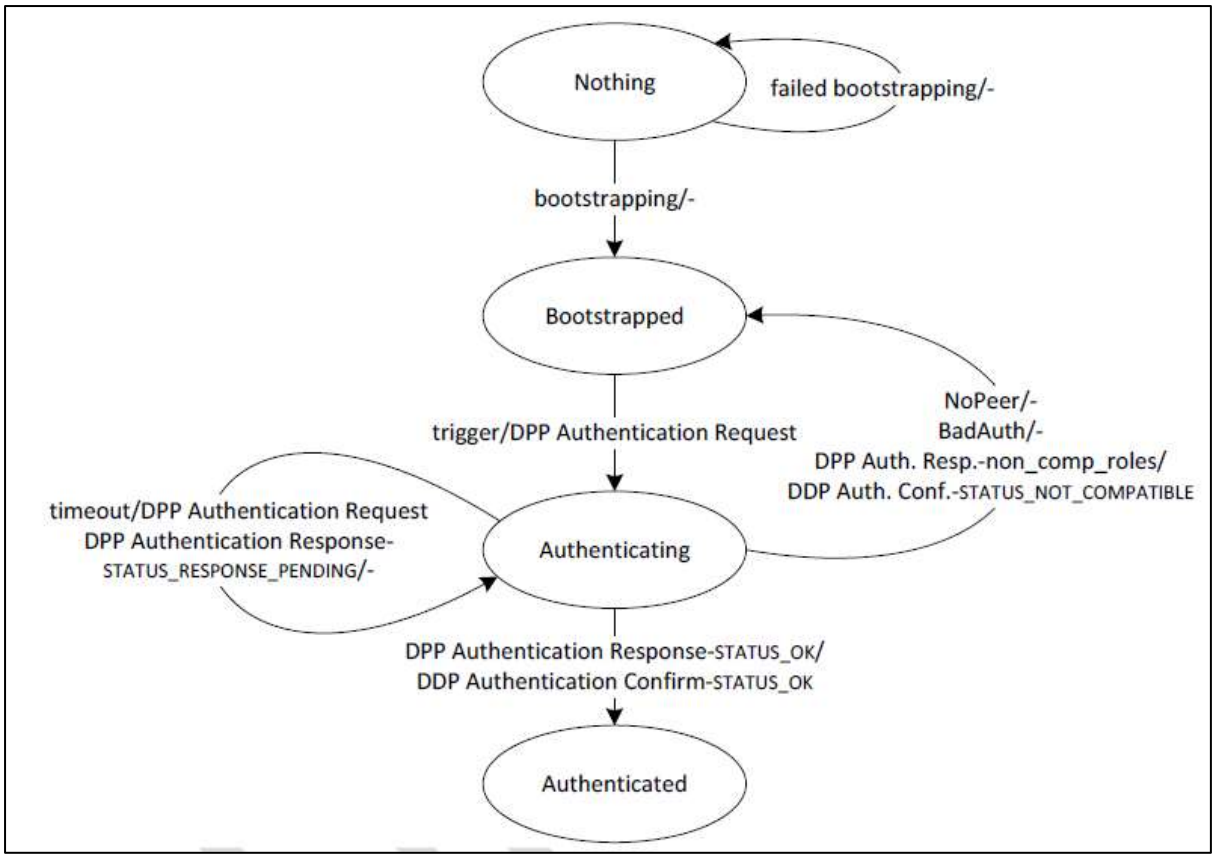


Figure 3- 27 . *DPP initiator state machine*

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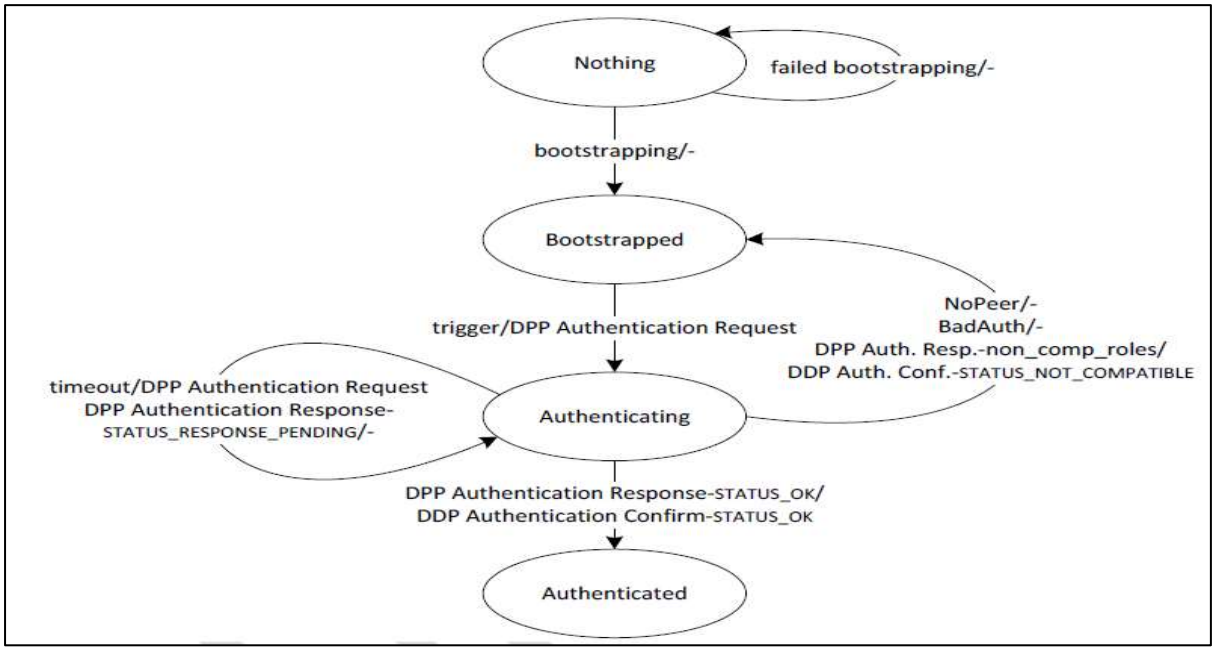


Figure 3- 28. *DPP responder state machine*

3.2.3.8.2 DPP Configuration

When both the Enrollee and the Configurator indicate their protocol version numbers to be 2 or higher, the Enrollee reports the result of configuration processing to the Configurator to allow clear indication of the results on the Configurator's user interface. The result is indicated in the DPP Configuration Result frame sent on the same channel immediately after the final DPP Configuration Response frame.

The expiry defines the expiry date for the DPP Client's Connector information. After the expiry date, the information is no longer valid and may be updated by the Configurator when a further connection between an Enrollee and a Configurator is established through reconfiguration.
Another case to consider is that the reconfiguration should be triggered automatically whenever "PSK or SAE password may have expired or changed.

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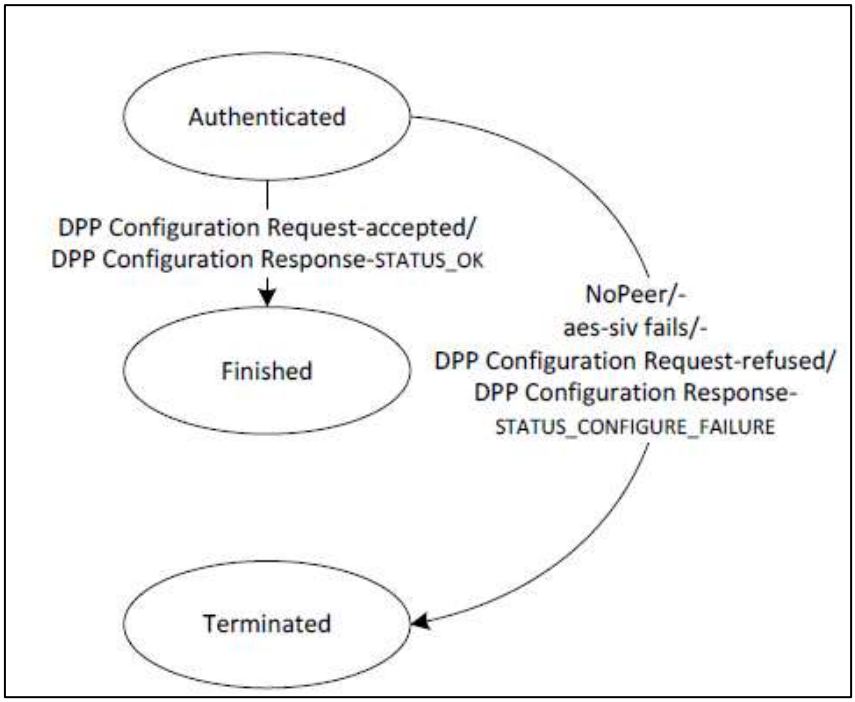


Figure 3- 29 . *DPP configurator state machine*

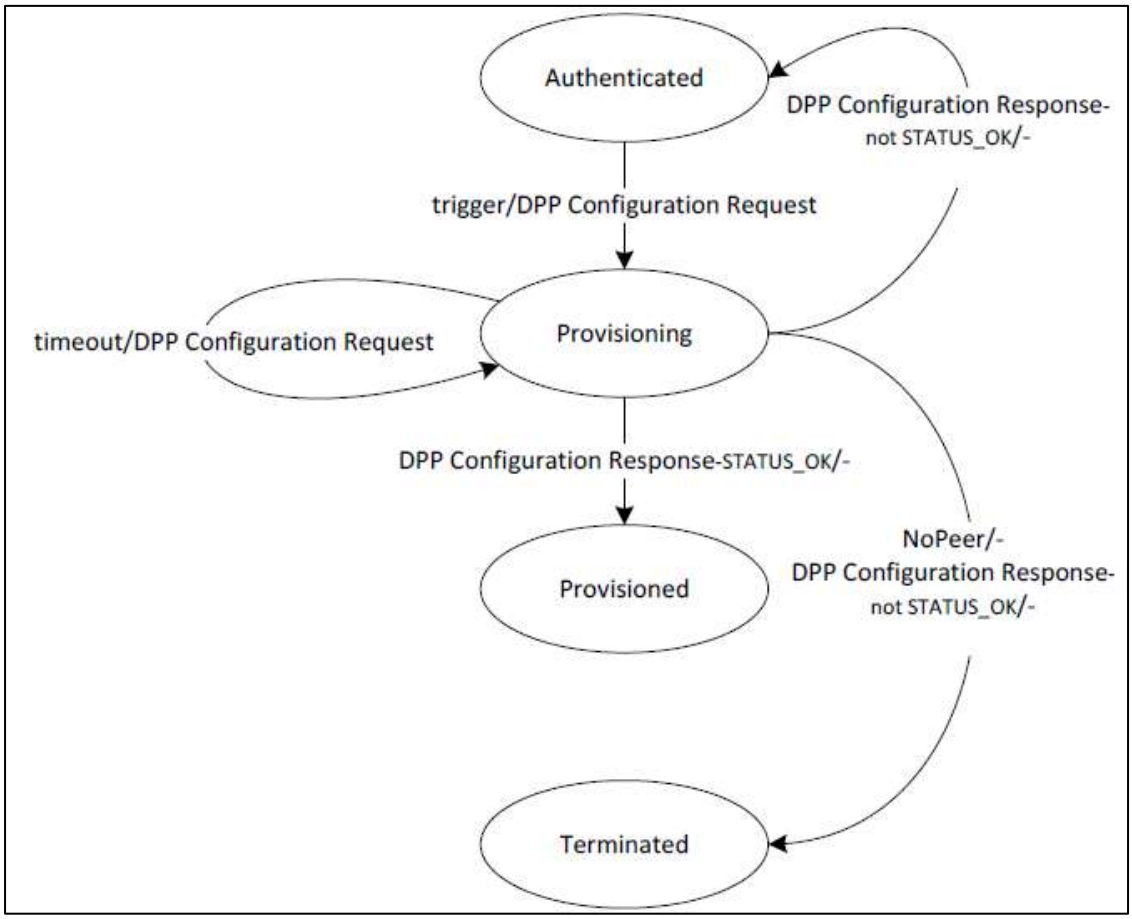


Figure 3- 30. *DPP Enrollee state machine*

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3.2.3.9 DPP Reconfiguration

DPP reconfiguration can be triggered in following ways:

- WPA2 PSK/Passphrase - A STA using a WPA2 PSK or Passphrase follows the standard IEEE 802.11 procedures and an unsuccessful 4-way handshake using the provided PSK or Passphrase may trigger reconfiguration.
- WPA3 Password - A STA using a WPA3 Personal password follows the SAE authentication as per the IEEE 802.11 procedures and an unsuccessful SAE authentication may trigger reconfiguration.
- DPP Connector - A STA using a DPP Connector attempts the Network Introduction Protocol failure with error STATUS_INVALID_CONNECTOR may trigger reconfiguration.
- DPP Client's Connector expiry date - After the expiry date, the information is no longer valid. Enrollee will compare the current time to evaluate the validity of credential, After the credential expired may trigger reconfiguration.
  ap_expiry parameter needs to be added for configurator in file (/etc/dpp_cfg.txt).
  For example:
  ap_expiry=2021-05-25T22:56:22 (YYYY-MM-DDTHH:MM:SS).

3.2.3.10 DPP Onboarding Important Points

This section helps share the DPP Onboarding important details:

- Post onboarding, connector information is saved and maintained on reboot.
- URI will be saved in a dpp configuration file and maintained on reboot.
- Addition of URI is limited to 100, after 100 it will overwrite from 1st.
- If DPP Onboarding Ongoing and DPP Onboarding triggered again then second dpp onboarding trigger will be rejected.
- Default Preferred onboarding method is DPP, and if set to WPS then URI will not be send.
  Preferred onboarding can be modified via API as explained in MAP R3 API section.
- As DPP Spec allows config for DPP/SAE/WPA2-PSK. So make sure authmode is set accordingly in WTS file. DPP+SAE+PSK is suggested combination to be set to cover all cases for Backhaul and for Fronthaul use SAE Transition mode as many Mobile STA don't support DPP and not connect if DPP is set by AP.
- Make sure atleast 1 fronthaul exist as DPP Onboarding uses fronthaul. So minimum of 2 BSS is needed for DPP onboarding, one as Backhaul BSS and one as front haul BSS.

3.2.3.11 DPP Onboarding Procedure

DPP onboarding configuration procedure has been described below parts :
Part 1 : Common for out-of-band bootstrapping (WiFi OR Ethernet) and inband bootstrapping
Part 2a: For out-of-band bootstrapping (WiFi OR Ethernet)
Part 2b: For inband bootstrapping via push button

3.2.3.11.1 Configuration Part 1

This part is common for out-of-band bootstrapping (WiFi OR Ethernet) and inband bootstrapping DPP onboarding.

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Controller: Configure the Device mode as Controller using UI. Set Device Role as Controller and click on Save and Apply button.

- After configuring device as controller, /etc/dpp_cfg.txt will be as below:



Figure 3- 31. Controller's dpp_cfg.txt file example snapshot

Agent: Configure the Device mode as Agent using UI. Set Device Role as Agent and click on Save and Apply button.

- After configuring device as agent, /etc/dpp_cfg.txt will be as below:

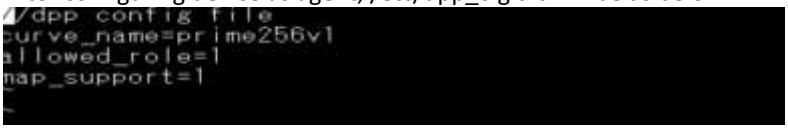


Figure 3- 32. Agent's dpp_cfg.txt file example snapshot

Agent: After configuring the device as an agent, For MAP R3 Onboarding the first step is to generate DPP URI. Click on the button "Generate DPP URI". **Don't click on Save and Apply button.**

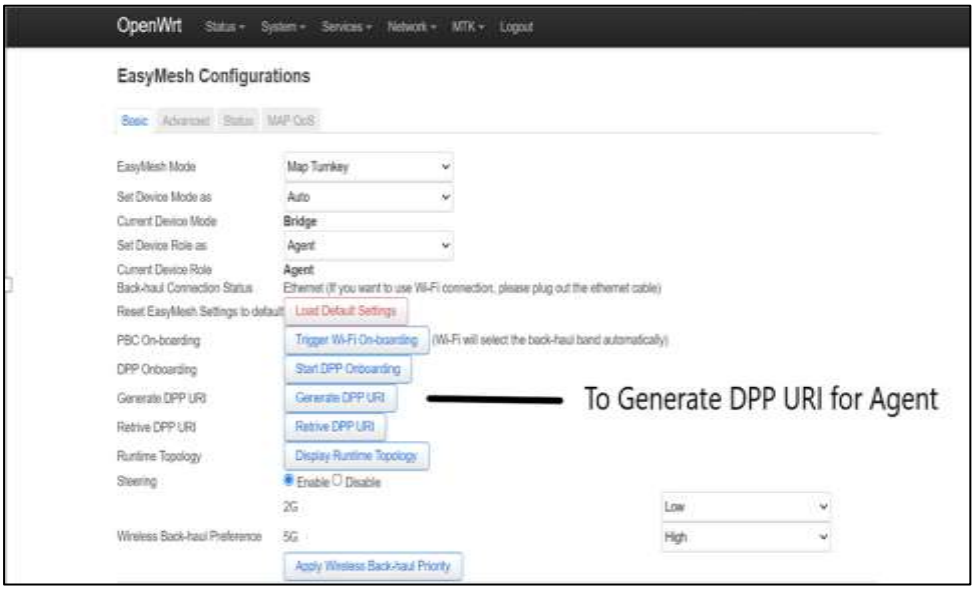


Figure 3- 33. Generate DPP URI for Agent WEBUI snapshot

Command Line Interface

Instead of using Web UI, , the above procedure can also be done as below :

1. Add multiple BSS (At least 2) using commands are as follows for both controller and agents.

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- wificonf -f etc/wireless/mediatek/mt7986-ax6000.dbdc.b0.dat set BssidNum 4
- wificonf -f etc/wireless/mediatek/mt7986-ax6000.dbdc.b1.dat set BssidNum 4
- Here 4 is number of BSSID.

2. Need to reboot both controller and agent manually using “reboot” command.

3. Enable MAP Turnkey on both controller and agent using command as follows:

   - wificonf -f etc/wireless/mediatek/mt7986-ax6000.dbdc.b0.dat set MapMode 1
   - wificonf -f etc/wireless/mediatek/mt7986-ax6000.dbdc.b1.dat set MapMode 1

Note: Above command line interface from step 1 to 3 will not work in UCI supported SKUs . So need to use WEB UI only to configure MBSS and Map Mode.

4. **Controller :**Configure the Device mode as Controller using following command:

   - wificonf -f /etc/map/mapd_user.cfg set DeviceRole 1
   - wificonf -f /etc/dpp_cfg.txt set allowed_role 2

For UCI supported SKUs following command will be used:

- uci set mapd.mapd_user.DeviceRole='1'
- uci set mapd.dpp_cfg.allowed_role='2'
- uci commit mapd

5. **Agent :**Configure the Device mode as Agent using following command:

   - wificonf -f /etc/map/mapd_user.cfg set DeviceRole 2
   - wificonf -f /etc/dpp_cfg.txt set allowed_role 1

For UCI supported SKUs following command will be used:

- uci set mapd.mapd_user.DeviceRole='2’
- uci set mapd.dpp_cfg.allowed_role='1'
- uci commit mapd

6. Set R3 version if not set using following command both on Controller and Agent:
   - wificonf -f /etc/map/1905d.cfg set map_ver R3

For UCI supported SKUs following command will be used:

- uci set 1905d_cfg.map.map_ver='R3'
- uci commit 1905d_cfg

7. Reboot both the controller and agent via “reboot” command, in order to apply above settings.

8. **Agent:** After configuring the device as an agent, using following command generate the DPP URI.
   - wappctrl ra0 dpp dpp_bootstrap_gen type=qrcode

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3.2.3.11.2 Configuration Part 2a

This part 2a is only for out-of-band bootstrapping (WiFi OR Ethernet) DPP onboarding.

Agent: After generating the URI, You can see the Device's URI using the button "Retrieve the DPP URI". **Don't click on Save and Apply button.** You will be able to see the device's URI below

- Retrieve URI button.

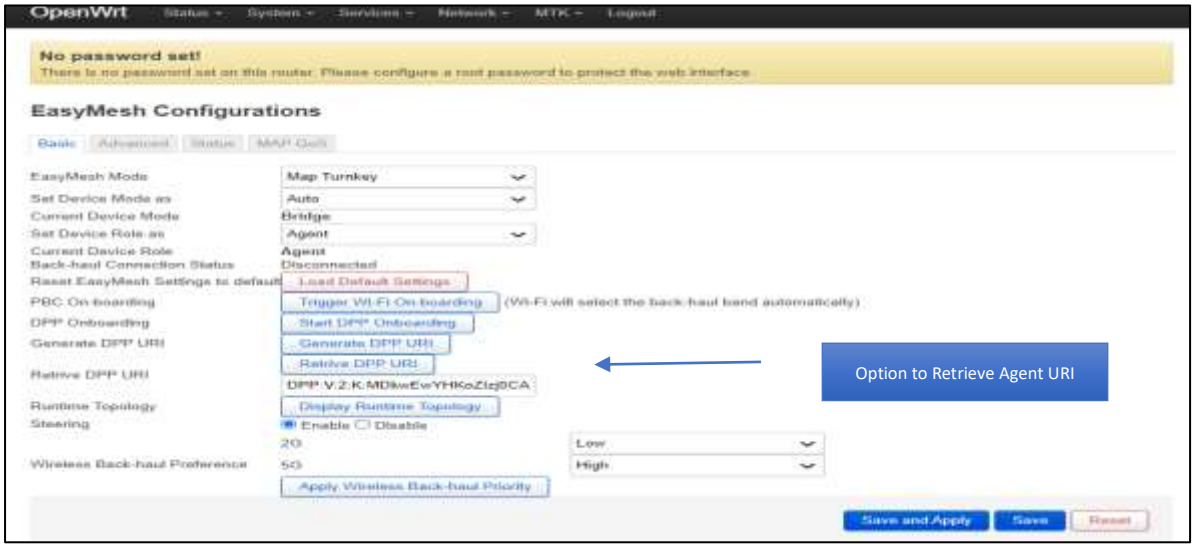


Figure 3- 34 . Retrieve DPP URI of Agent WEBUI snapshot

Copy this URI from retrieve URI, so that can be added on controller side before starting the onboarding procedure.

Example URI format-
DPP:V:2;K:MDkwEwYHKoZIzj0CAQYIKoZIzj0DAQcDIgACPpWgjqvIzFGOs4pF1GaHucnmw9lstQLdVzpDKeEipHo=;;

Controller: In controller device mode, add the DPP URI of the agent device that in the "ADD DPP URI" tab and click on the "Submit DPP URI" Button. **Don't click on Save and Apply button.**

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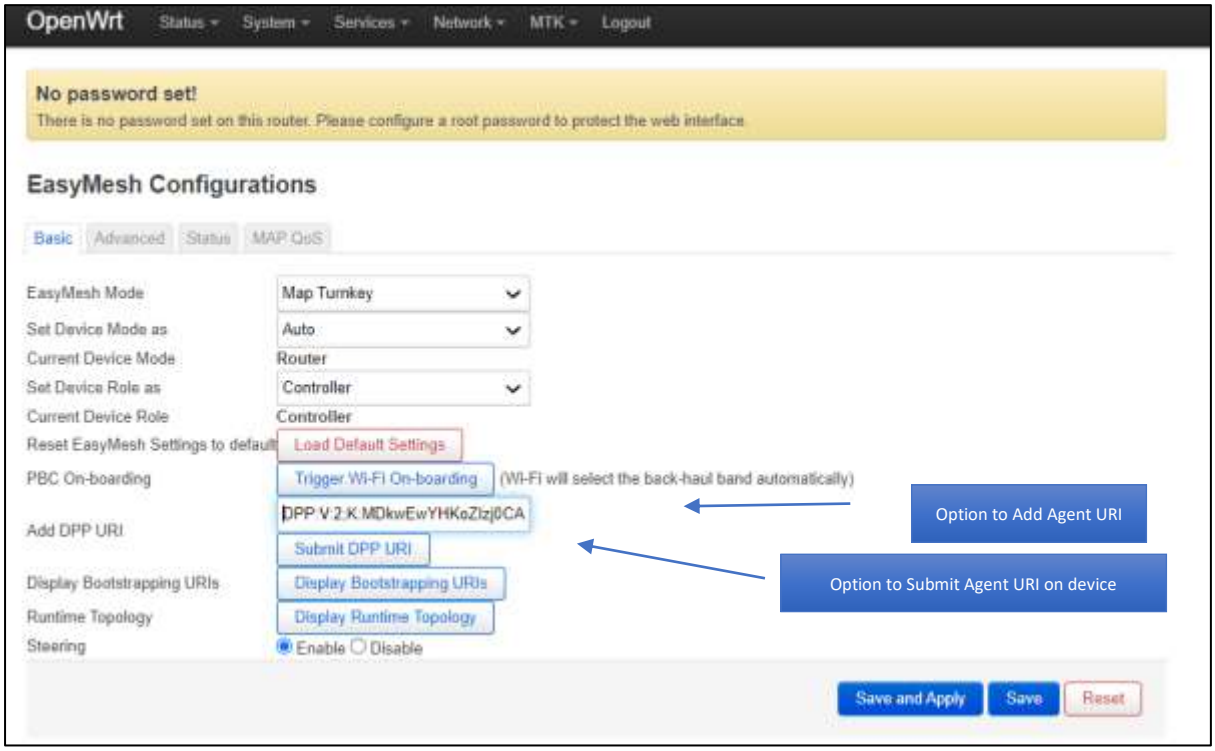


Figure 3- 35. Add and Submit Agent's URI on Controller's WEB UI page snapshot

On submitting URI, QR code will be added to controller /etc/dpp_cfg.txt file for this Agent: See item agt_qr_code_1:

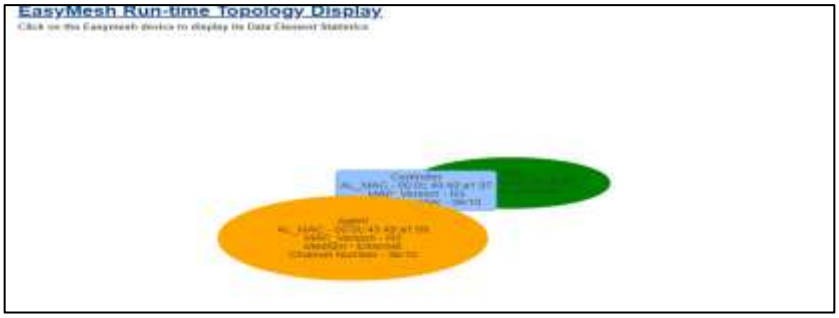
```
# dpp config file
curve_name=prime256v1
allowed_role=2
map_support=1
dpp_private_key=30770201010420985e40f2a0cbae38dade21f6d9a5487ced23c57caf0a4aa895244ec90533e7bda00a06082a8648ce3d030107a144034
akm=1
1905valid=1
1905connector=eyJ0eXAiOiJkcHBDb24iLCJraWQiOiJnV2hPalA1anB1dTRzaFNud1NNdVlsQ1pYakJfQXB3WlBuNEJRNFNaWWJNIiwiYWxnIjoiRVMyNTYifQ
1905netAccessKey=307702010104208bb39089ad7ecb605af0f2eefe12b9146b74dc0f9970a2798d80dacde831d1257a00a06082a8648ce3d030107a1440
1905cSignKey=3059301306072a8648ce3d020106082a8648ce3d03010703420004757294 3f9355f4bdcc562bdf321fae0c84c51d57cc98585c9ae8ab8dc
netKeyExpiry=0
decryptThreshold=5
ppkey=3059301306072a8648ce3d020106082a8648ce3d03010703420004776f261401eb862fb28af9d40e2ec2aaa6d0acbce51b10c6a0ee6b156d3e1c5a
agt_qr_code_1=DPP:V:2;K:MDkwEwYHKoZIzj0CAQYIKoZIzj0DAQcDIgACPpWgjqvlzFGDs4pF1GaHucnmw9IstQLdVzpDKeEipMo=;;
qr_count=1
```

Figure 3- 36. Controller's dpp_cfg.txt file example after addition of agent's URI snapshot

Controller: Multiple URI's can be added one by one with the **Add DPP URI** option in WebUI.

Onboarding Case Ethernet Or Wi-Fi:

a. **Ethernet Onboarding:**
   a. Connect the Ethernet Cable for Ethernet onboarding, it will automatically start to onboard.



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Figure 3- 37. Easymesh topology display after ether onboarding

b. Post Ethernet Onboarding /etc/dpp_cfg.txt will be as follows:

- **Controller:**



Figure 3- 38. Post ethernet onboarding dpp_cfg.txt snapshot at Controller

- **Agent:**



Figure 3- 39. Post ethernet onboarding dpp_cfg.txt snapshot at Agent

b. **Wi-Fi Onboarding:**

Agent: For WIFI Onboarding click on the “START DPP Onboarding” button to initiate R3 WIFI onboarding on Agent as Device Mode. **Don't click on Save and Apply button.**

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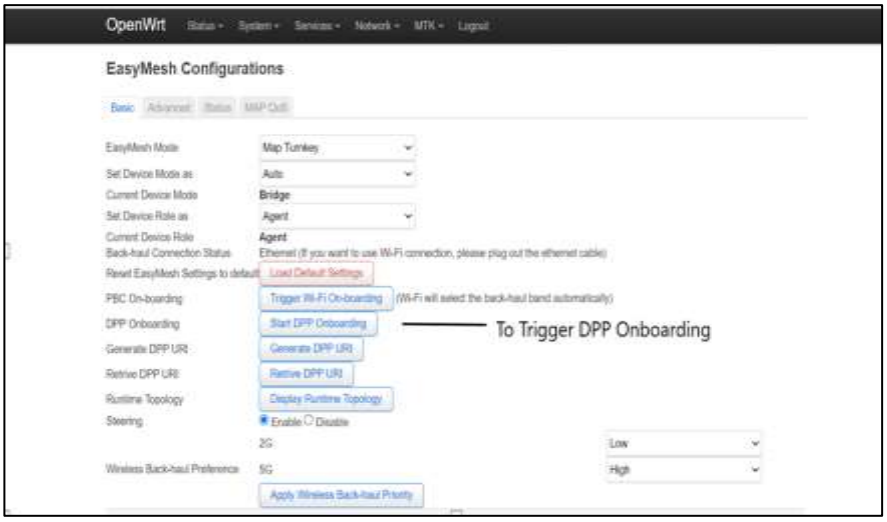


Figure 3- 40. Trigger DPP Onboarding button at Agent's WEBUI page snapshot

a. After following above steps Agent will onboard to controller.

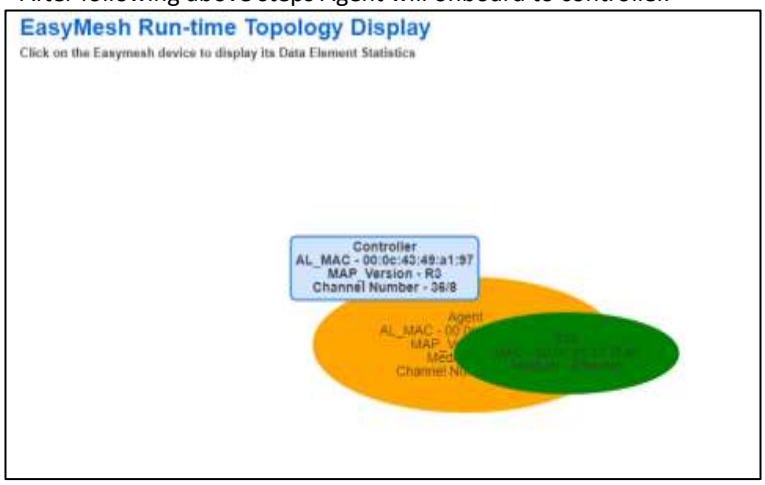


Figure 3- 41. Topology after Agent onboards to controller via WiFi

b. Post Wi-Fi Onboarding /etc/dpp_cfg.txt will be as follows:

- **Agent:**

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Figure 3- 42. Post WiFi onboarding dpp_cfg.txt snapshot at Agent

- **Controller:**



Figure 3- 43. Post ethernet onboarding dpp_cfg.txt snapshot at Controller

Command Line Interface

1. **Agent:** "Retrieve the DPP URI" using following command:
   mapd_cli /tmp/mapd_ctrl get_dpp_uri

2. **Controller:** Submit this URI from retrieve URI, on controller side before starting the onboarding procedure.

   Example URI format-
   "DPP:V:2;K:MDkwEwYHKoZIzj0CAQYIKoZIzj0DAQcDIgADH8ESUBRG2AmjqMnvy2HTnnVOrNVKL5w8D/Jsov60 CZE=;;"

Provide the QR command of the Agent which is to be on boarded using below command in controller.

wappctrl ra0 dpp dpp_qr_code "Generated DPP URI on agent using above example".

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3. **Controller:** Multiple URI's can be added one by one with the above command.
4. **Ethernet Onboarding:** Connect the Ethernet Cable for Ethernet onboarding, it will automatically start to onboard.
5. **Wi-Fi Onboarding:** For WIFI Onboarding give following command to initiate R3 WIFI onboarding on Agent as Device Mode.

   wappctrl ra0 dpp dpp_start

6. After following above steps Agent will onboard to controller.

3.2.3.11.3 Configuration Part 2b

This part is specific for inband bootstrapping via push button.
Controller: To trigger WPS based onboarding, click on button "Trigger Wi-Fi On-boarding".

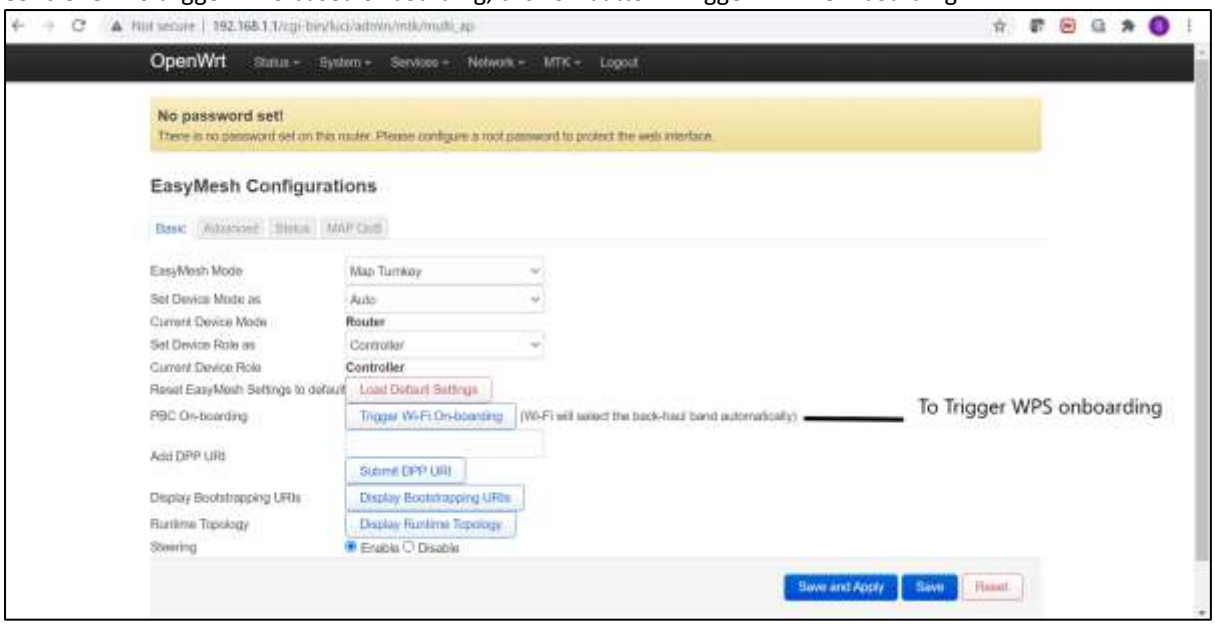


Figure 3- 44. Controller WEB UI page snapshot to trigger WPS onboarding

Agent: Now trigger WPS onboarding on agent, click on button "Trigger Wi-Fi On-boarding".

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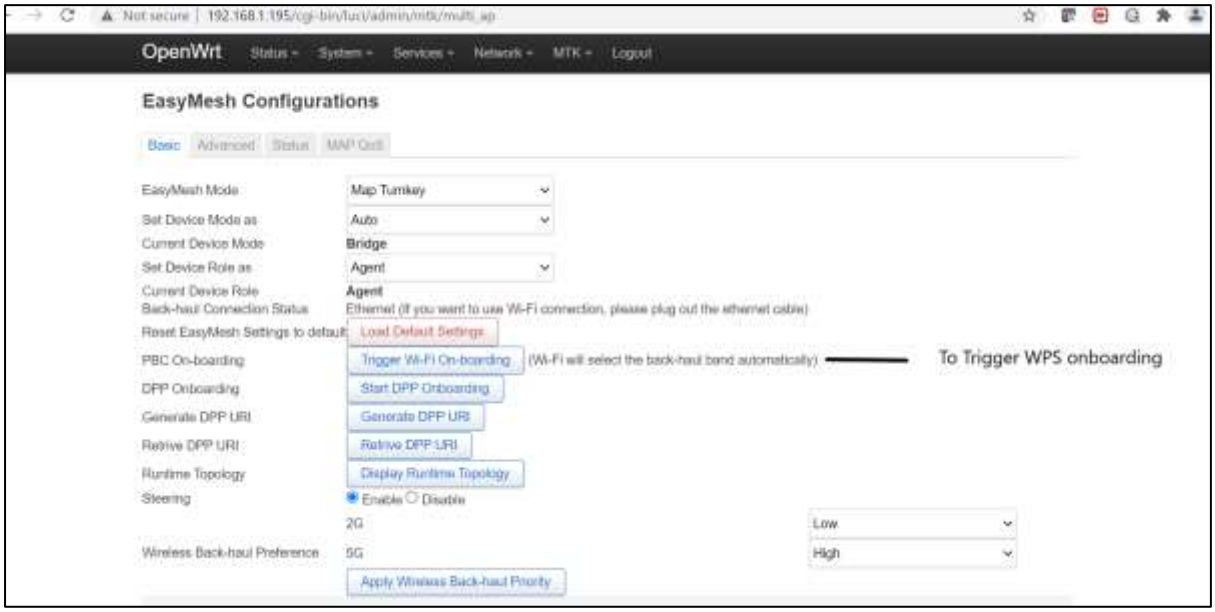


Figure 3- 45. Agent WEB UI page snapshot to trigger WPS onboarding

- After following above steps Agent will onboard to controller.

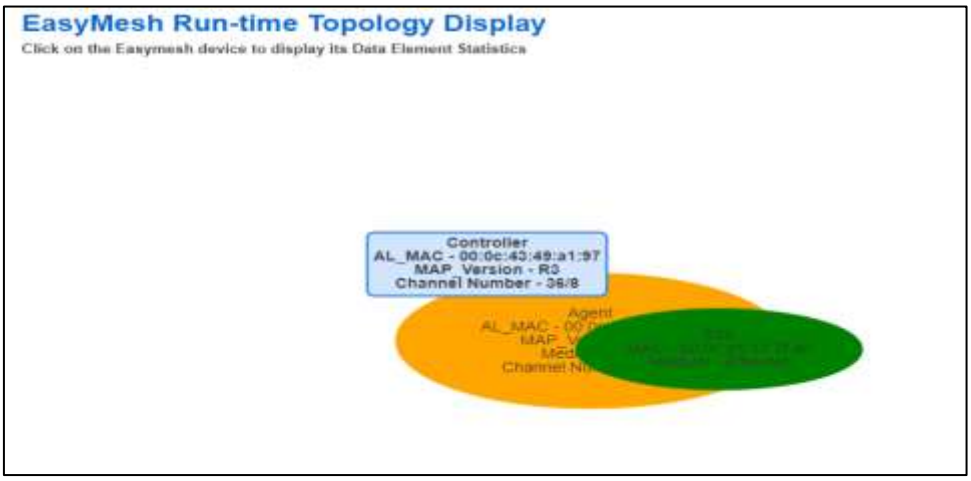


Figure 3- 46. Topology display after WPS onboarding success

- Post Wi-Fi Onboarding /etc/dpp_cfg.txt will be as follows:
- Agent:

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Figure 3- 47. Post WPS onboarding dpp_cfg.txt snapshot at Agent

- Controller:



Figure 3- 48. Post WPS onboarding dpp_cfg.txt snapshot at Controller

Command Line Interface

Instead of using web UI we can trigger WPS based onboarding between the two devices by issuing the following commands on both the devices(First on the controller and then on agent):

"wappctrl ra0 wps_pbc"

Note:-

- If User Set Onboarding type to 1 using dpp_onboard_type type API then it will set the Legacy WPS onboarding, so no URI will be sent in WPS. (wappctrl ra0 dpp dpp_onboard_type type=1)
- Make sure for DPP Push button via WPS, dpp_onboard_type should be 0 then DPP onboarding will happen via Push Button. (wappctrl ra0 dpp dpp_onboard_type type=0)

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3.2.3.12 DPP onboarding after Profile-1 or Profile-2 onboarding

If a Multi-AP R3 Agent has been onboarded using Profile-1 or Profile-2 onboarding procedures (e.g., using the PBC onboarding method or ETH M1/M2 messages). If the Multi-AP R3 agent is onboarded through ETH backhaul, It shall continue sending 1905 AP-Auto configuration Search messages every 30 seconds with a DPP Chirp Value TLV for the DPP onboarding afterwards. And if the Multi-AP R3 Agent onboarded through WiFi backhaul (using legacy WPS onboarding), it will trigger the DPP onboarding afterwards and start sending the DPP presence announcement frames.
Once the DPP URI has been added in the controller then the R3 DPP onboarding will start between agent and the controller.

Command Line Interface

1. Add multiple BSS (At least 2) using commands are as follows for both controller and agents.

   - wificonf -f etc/wireless/mediatek/mt7986-ax6000.dbdc.b0.dat set BssidNum 4
   - wificonf -f etc/wireless/mediatek/mt7986-ax6000.dbdc.b1.dat set BssidNum 4
   - Here 4 is number of BSSID.

2. Need to reboot both controller and agent manually using “reboot” command.

3. Enable MAP Turnkey on both controller and agents using command as follows:

   - wificonf -f etc/wireless/mediatek/mt7986-ax6000.dbdc.b0.dat set MapMode 1
   - wificonf -f etc/wireless/mediatek/mt7986-ax6000.dbdc.b1.dat set MapMode 1

Note: Above command line interface from step 1 to 3 will not work in UCI supported SKUs . So need to use WEB UI only to configure MBSS and Map Mode.

4. **Controller :**Configure the Device mode as Controller using following command:

   - wificonf -f /etc/map/mapd_user.cfg set DeviceRole 1
   - wificonf -f /etc/dpp_cfg.txt set allowed_role 2

   For UCI supported SKUs following command will be used:

   - uci set mapd.mapd_user.DeviceRole='1'
   - uci set mapd.dpp_cfg.allowed_role='2'
   - uci commit mapd

5. **Agent2:** Configure the Device mode as R3 Agent using following command:

   - wificonf -f /etc/map/mapd_user.cfg set DeviceRole 2
   - wificonf -f /etc/dpp_cfg.txt set allowed_role 1

   For UCI supported SKUs following command will be used:

   - uci set mapd.mapd_user.DeviceRole='2’
   - uci set mapd.dpp_cfg.allowed_role='1'
   - uci commit mapd

- **Agent1:** Configure the Device mode as R2 Agent using following command (to be used for wifi onboarding):

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- wificonf -f /etc/map/mapd_user.cfg set DeviceRole 2

6. Set R3 version if not set using following command both on Controller and Agent2:
   - wificonf -f /etc/map/1905d.cfg set map_ver R3

- Set R2 version if not set using following command on agent1 (to be used for wifi onboarding):
  - wificonf -f /etc/map/1905d.cfg set map_ver R2

  For UCI supported SKUs following command will be used:

  - uci set 1905d_cfg.map.map_ver='R3'
  - uci commit 1905d_cfg

7. Reboot both the controller and agents via “reboot” command, in order to apply above settings.

8. **Agent2:** After configuring the device as an agent, using following command generate the DPP URI.
   - wappctrl ra0 dpp dpp_bootstrap_gen type=qrcode

- **ETH onboarding case: (In this example, Controller and Agent 2 is used, both are R3 Device)**

9. Do not give the DPP QR code on the controller before connecting ETH cable to the controller. Connect ETH cable between Agent2 and the controller. Legacy R1/R2 onboarding would happen.
10. Once the onboarding is done between to MAP-R3 devices provide the DPP URI on the controller for the agent2.
    - wappctrl ra0 dpp dpp_qr_code "DPP:V:2;K:MDkwEwYHKoZIzj0CAQYIKoZIzj0DAQcDIgACR6MvlxuYbsBIVGlx3yCfJ9csE0nPWN1v0NiJuePLA80=;;”
    - After giving this command when agent will try the autoconfig search again, then R3 DPP onboarding will be triggered automatically.

- **Wi-Fi onboarding case:**

11. Trigger the WPS onboarding between the R3 controller and R2 Agent (agent1), mechanism to trigger WPS onboarding is same as R1/R2.

12. Once agent1 is onboarded, trigger the WPS onboarding between the R3 agent (Agent2) and R2 agent (Agent1). Once the WPS non R3 onboarding is done, R3 agent will trigger DPP onboarding and start sending the presence announcement frames.

13. Provide DPP URI of the R3 agent to the R3 controller using dpp qr_code command.
    - wappctrl ra0 dpp dpp_qr_code "DPP:V:2;K:MDkwEwYHKoZIzj0CAQYIKoZIzj0DAQcDIgACR6MvlxuYbsBIVGlx3yCfJ9csE0nPWN1v0NiJuePLA80=;;”

14. Whenever the controller receives the Presence announcement corresponding to the agent (for which URI is provided by user) then R3 DPP onboarding will be triggered between Agent2 and the Controller.

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3.2.4 Behavior at boot-up

If previous WiFi BH profile is available at bootup, WiFi connection on BH will start automatically on Bootup. If BH type Ethernet is marked in middle of the connection or after connection is completed, WiFi BH link will be disabled and Ethernet connection will be made.

3.2.5 Backhaul Configuration Save

After onboarding is complete at enroller agent, the configuration of the backhaul BSS is saved in non-volatile memory of the device.

If the device reboots, the same configuration is used for subsequent connections and there is no need to trigger WPS based onboarding again.

Similarly for a MAP R3 device, if the device reboots, the same configuration is used for subsequent connections and there is no need to trigger DPP based onboarding again.

Post DPP Onboarding, when best AP is selected based on best AP selection algorithm and if selected Best AP supports DPP then first DPP Network Introduction Protocol is done and then bSTA connection is formed. Also in R3 device, 11ax capabilities is used using new WiFi6 TLV as defined in R3 spec. In R1/R2 11ax capabilities was shared using HE Capabilities TLV.

DPP Profile is saved in /etc/map/mapd_user.cfg.

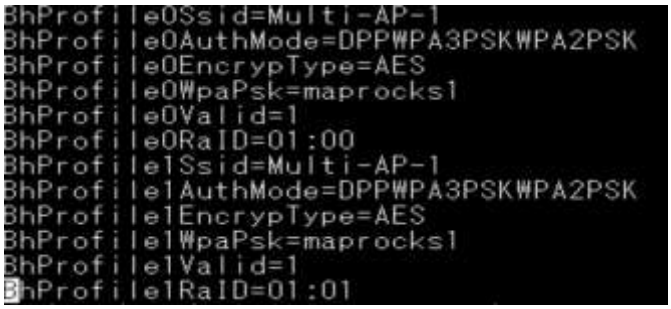


Figure 3- 49. DPP Profile saved in mapd_user.cfg file after successful onboarding snapshot

3.2.6 Re-onboard an already on-boarded device

If a device which was previously on boarded and need to be again DPP onboarding then Easy Mesh default needs to be run on the device first before triggering onboarding or “Load Default Settings” to be done via WEB UI:

1. EashMesh.sh default
   OR
2. Click on “Load Default Settings” Button from Web UI

3.2.7 BSS interface-SSID matching

If user needs to keep the BSS interface-SSID matching, they should add below information in 1905d.cfg file in current device:

bss_config_priority = ra1;ra0;ra3;ra2;rax3;rax1;rax0;rax2

It means the ra1 has the highest priority and rax2 has the lowest priority to be set the wireless setting from the set of configuration which is from MAP controller.

For example, if the wts_bss_info_config in controller is as below:

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Figure 3- 50. "wts_bss_info_config" file present at the controller

When the auto-configuration procedure is completed, the BSS interface-SSID mapping in current device will be as below:

bss interface	SSID
ra1	Multi-AP-24G-1
ra0	Multi-AP-24G-2
ra3	Multi-AP-24G-3
ra2	Multi-AP-24G-4
rax3	Multi-AP-5HG-1
rax1	Multi-AP-5HG-2
rax0	Multi-AP-5HG-3
rax2	Multi-AP-5HG-4

If there's no bss_config_priority information in 1905d.cfg, 1905 daemon would set the wireless setting to the bss interface by the sequence the bss interfaces are initialzed in WAPP. For OpenWRT, the parameter should be present in both 1905d.cfg and mapd_cfg.

3.2.8 Hidden SSID

If user wants to hide some specific ssid, it can set that ssid to a hidden ssid. This ssid will not be shown on the wireless configuration UI or in the beacon of the corresponding BSS.
User needs avoid to set ssid of all fronthaul bss to the hidden ssid, because in some situation, for example DFS, wps will fail.
This is a MTK proprietary solution and only valid with MTK device.
For WPS to be successful, there must be at least one NON-HIDDEN Front haul BSS per radio.
To configure Hidden SSID on a particular BSS , description is present in section 3.2.1 of this document.

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3.2.9 Error handling for Auto-Configuration

Error handling mechanism for auto configuration is used to cover the cases in which the agent could not be configured successfully, for example, when the Controller sync the new configuration to all agents, some agents disconnect from current network, then when these agents reconnect to the MAP network, the configuration will not sync on these agents.

The error handling mechanism is mainly depend on the detection of topology change, when detecting of an agent leaving, the topology service in 1905daemon will delete the leaf that represents the removed agent from the topology tree, and also delete all its child leaves, when these agents reconnect to current network, controller will regard these agents as new agents that are not configured, then if these agents do not start auto configuration procedure by themselves within 30 seconds, the Controller will start auto configuration renew procedure.

For example:

a. The MAP network topology is as below picture, all the agents have been configured by controller.

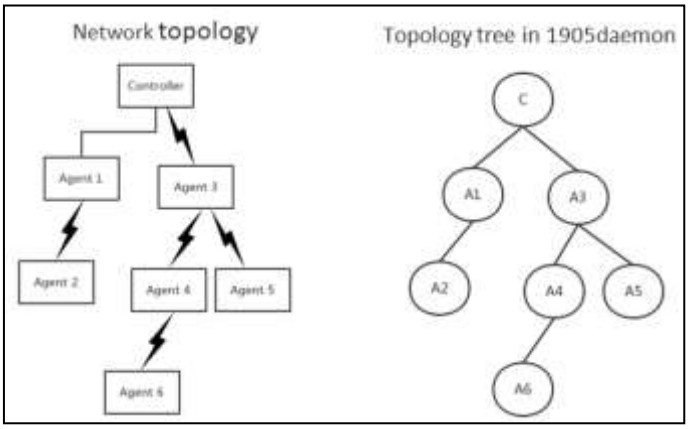


Figure 3- 51. MAP network topology tree

b. Agent3 disconnect from controller, the 1905daemon delete the A3 leaf and all its child leaves A4, A5, A6.

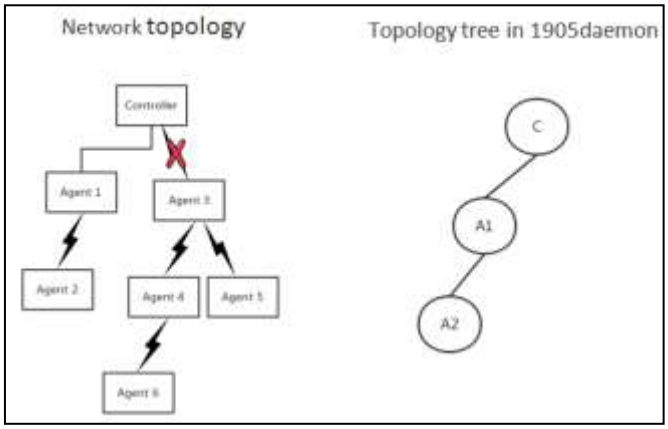


Figure 3- 52. MAP network topology tree A3 disconnects

c. Agent3 reconnect to controller, the controller will regard agent3, 4, 5, 6 as the new agents, if these agents do not start auto configuration procedure within 30 seconds, the controller will start to send renew message to each of them to trigger auto configuration.

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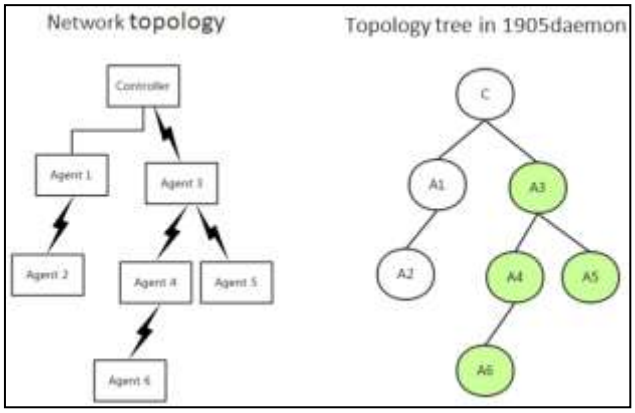


Figure 3- 53. MAP network topology tree A3 reconnects

And also, the error handling mechanism could cover the cases that the existing agent does not start auto configuration procedure within 30 seconds after joining in current MAP network. 1905daemon in Controller would regard an agent as configured when the M2 has been sent to that agent.

3.3 Mediatek Multi-AP Controller Connectivity with Non-MAP devices

Mediatek Multi-AP solution provides provision for the controller to be connected with Non-MAP device through both Ethernet as well as WiFi interfaces. This helps Mediatek Multi-AP network to get WAN connectivity through Non-MAP devices as well.

This feature is also commonly called Third Party connection.

3.3.1 Ethernet BH link for controller with non-MAP uplink device

To get controller connection with non-MAP AP, expectations here is that Ethernet should be plugged on WAN interface from controller to that non-MAP AP.

3.3.2 Wireless BH for controller with non-MAP uplink device:

A Controller's ApCli interface connection is established with the non-MAP AP. This uses the same best AP selection algorithm as used by ApCli interfaces of any other agent. However, a MAP device will never be selected as the target AP for a controller's ApCli interface. If the feature is enabled, and BH configuration profiles are saved for a controller, the Mediatek Multi-AP solution will continue to scan unless a connection is formed with a target AP. The scan process can be stopped by disabling the feature dynamically. If the feature is enabled and profiles were not previously available, controller ApCli interfaces will remain in idle state and scan will not be performed. There are two ways through which the profile can be made available to the controller.

1. Trigger WPS:
    - The ApCli interfaces of controller can connect and get profile by forming a WPS push button connection. The profile thus received will be stored on Non-Volatile memory and will be available after reboot.

2. Store profile directly on non-volatile memory:
    - The profile can be stored directly on Non-volatile memory as detailed in below sections.

The feature needs to be enabled on the controller using additional configurations in configuration file and APIs are required to trigger connection of a MAP controller with a Non-MAP Wireless AP. **By default this feature is disabled.**

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If more than one device has role “Auto” then it is possible that two devices can connect to third party AP simultaneously and get role as controller. In this case these two MAP devices in this case both devices connected to third Party will acts as controller and will be like two separate networks.

3.3.2.1 Procedure

For this feature to work some settings need to be enabled in the Configuration file interface

OpenWRT:

→ Add ThirdPartyConnection=1 in mapd_cfg and run EasyMesh.sh again.

Save the parameters in mapd_cfg as follows:

BhProfile0Ssid=abc5g
BhProfile0AuthMode=OPEN
BhProfile0EncrypType=NONE
BhProfile0WpaPsk=maprocks1
BhProfile0Valid=1
BhProfile1Ssid=abc2g
BhProfile1AuthMode=OPEN
BhProfile1EncrypType=NONE
BhProfile1WpaPsk=maprocks1
BhProfile1Valid=1
BhProfile2Valid=0

Third_party_connection:

Use: Enable/Disable feature dynamically
Usage: /sbin/mapd_cli /tmp/mapd_ctrl set third_party_connection <Enable>
Enable: 1- Feature Enabled, 0- Feature Disabled
Note: if enable dhcp_ctl feature, cmd mode should not use.

cli_wps_pbc:

Use: Force WPS to be triggered on ApCli interfaces
Usage: wappctrl ra0 cli_wps_pbc

3.3.3 Third Party Connection Example

3.3.3.1 Connection by Ethernet

1. Enable flag ThirdPartyConnection=1 in mapd_cfg file.
2. Controller’s WAN port connected to 3rd Party device, which provide internet connection with DHCP server enabled.
3. Make sure controller disable dhcp server and enable dhcp client.

```
cat /etc/config/network
config interface 'lan'
        option device 'br-lan'
        option proto 'dhcp'
```

4. With this featured enabled and BH as ETH nothing else is needed.
5. Rest Easymesh algos will be as applied.

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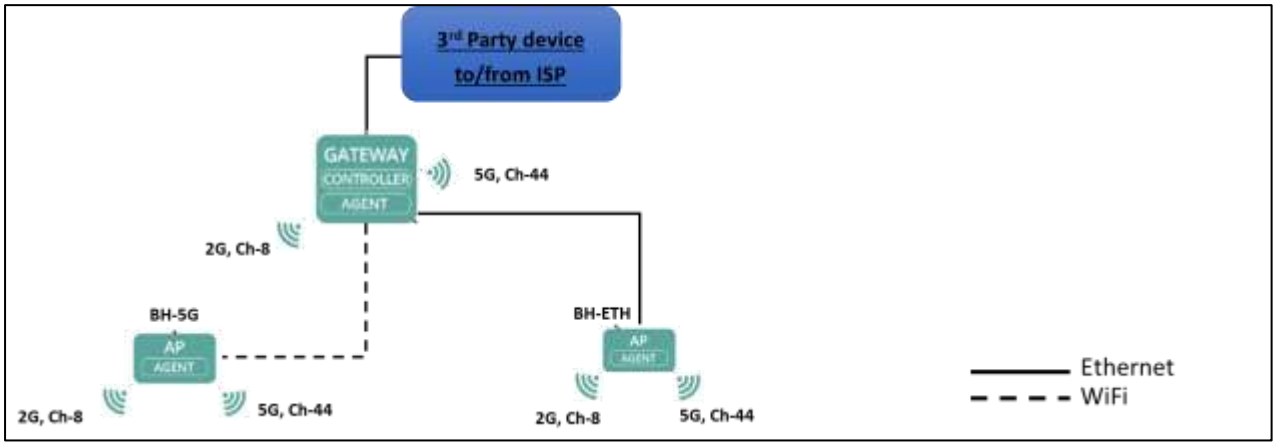


Figure 3- 54. Third Party Connection with Ethernet Example

3.3.3.2 Connection by WiFi

1. Enable flag ThirdPartyConnection=1 in mapd_cfg file.
2. If controller of our MAP network is connected as WiFi. Then APCLI interface of the controller like other MAP devices will connect to 3rd Party device, which provide internet connection with DHCP server enabled.
3. To achieve this either user has to run WPS on both controller and 3rd party device.
4. Or Pre saved config of third party has to be there on controller, in this case at MAPD boot scan will start and connection is made.
5. Rest Easymesh algos will be as applied.

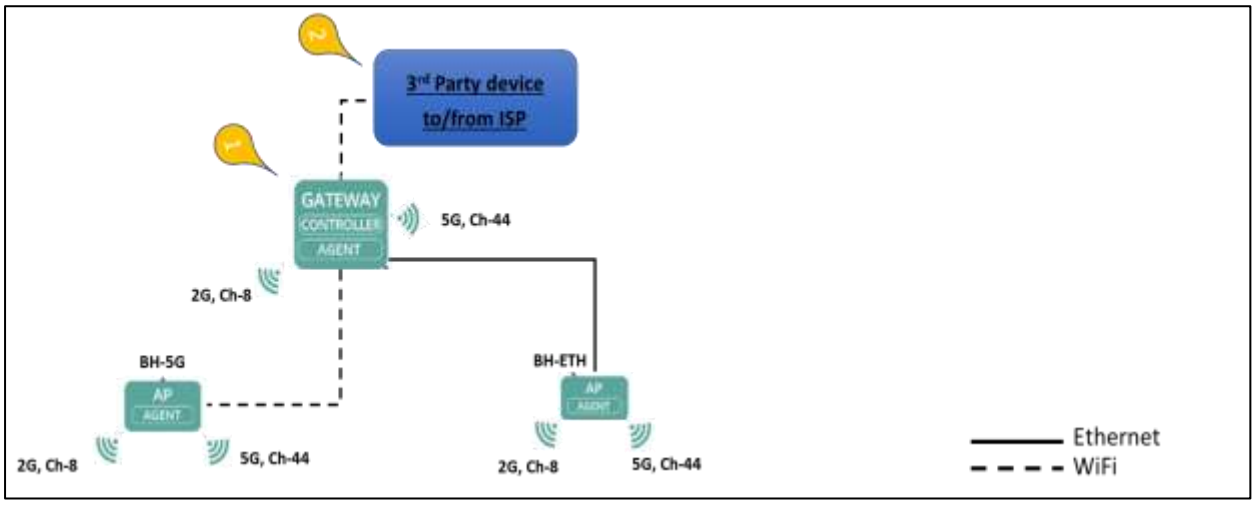


Figure 3- 55. Third Party Connection with WiFi Example

3.3.4 Limitations :

1. If enable dhcp_ctl feature of map and also enable this ThirdPartyConnection feature. Then rerun EasyMesh.sh or reboot in all map devices. And 3rd-AP should enable it's dhcp server. In that case, all STAs connected in the MAP network may have a different subnet ip. So we recommend users reconnect into the MAP network to get a new ip address from 3rd-AP.
2. Automatic Ethernet to WiFi switching between controller and Non-MAP controller is not supported

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3. If radar is detected at any map device, network will disconnect from the Non-MAP controller.
4. If this feature is enabled then it should be done at all the devices in the easymesh network and in same way if it is disabled, then it should be done on all devices.
5. Details of Radar scenario: If Radar is detected at the Non-MAP AP, then it will give CSA and switch to new channel, because of CSA our entire MAP network will also follow the third party AP and switch the channel. However, if RADAR is detected on a MAP device first, then the MAP network will move on to a new channel and controller will continue to scan for Non-MAP AP on all channels except for channels in the maximum BW group (of the network) on which radar was detected. For example: max BW on any device in MAP network is 80M, suppose radar is detected on ch 64 at any one device. controller will block operation on entire channel group from : 52 to 64 , and they will be marked NOP.
6. Channel planning will not be able to converge that radio's channel across the MAP network for which condition exists as below:
    a. Controller is connected to a third party AP via WiFi on the third party AP's radio band say 5G.
    b. There is Ethernet BH between controller and agent on that third party radio band say 5G.
        i. Channel of that radio , say 5G , will be fixed at the controller since it needs to maintain connection with third party AP. So best channel as found by channel planning may or may not be applied to controller's that radio say 5G. For all other devices on all radios best channel will get set by Channel planning. So in this case best channel on that radio, say 5G , may or may not differ between agent and controller.
    c. There is WiFi BH between controller and agent on that third party radio band say 5G.
        i. Channel of that radio , say 5G , will be fixed at the controller since it needs to maintain connection with third party AP. So best channel as found by channel planning may or may not be applied to controller's that radio say 5G. If any other agent is connected to that controller/or in chain via this radio, then similarly for that agent also best channel may or may not be able to apply.
        ii. For all other devices on all radios best channel will get set by Channel planning.

    d. **Channel Overload Condition:** When controller is connected to third party via Wifi . And this third-party radio band channel get's overloaded, then controller will still prefer to remain on this channel in order to maintain the third party connectivity.

3.4 Trigger WPS at front-haul BSS of an Agent

"Trigger WPS at front-haul BSS of an Agent" is a mechanism to trigger WPS at a front-hual BSS of an Agent but command is given on Controller. And this Agent has connected to Controller already.

3.4.1 Main flow

- Configure "Trigger WPS" on controller with MAC address of one front-hual BSS of one Agent
- Controller call _1905_Send_Vendor_Specific_Message to send Vendor Specific Message to the Agent
- Agent do WPS actions after receiving the Vendor Specific Message.
    - Mapd send TRIGGER_WPS event to wapp
    - Wapp send this event to driver
    - Driver do WPS actions.

3.4.2 Procedure

1. Prepare 3 Multi-AP Devices, make sure Agent1 succeed in on-boarding to Controller.
2. Configure, on Controller, "Agent BSS-MAC" as the MAC address of a front-hual BSS, and then push "Trigger WPS at front-haul BSS of an Agent" button.
3. Trigger WPS on WEBUI of Agent2.
4. Agent2 will connet to Agent1 at WPS stage

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5. Finally, Agent2 may connect to either Controller or Agent1 after on-boarding, according to parameters such as: RSSI strength, root distance, etc.

Set Agent AL-MAC as MAC address of one of front-hual BSS interface of an Agent, push “Trigger WPS at Front-haul BSS of an Agent” . This Button is present at the controller WEB UI.

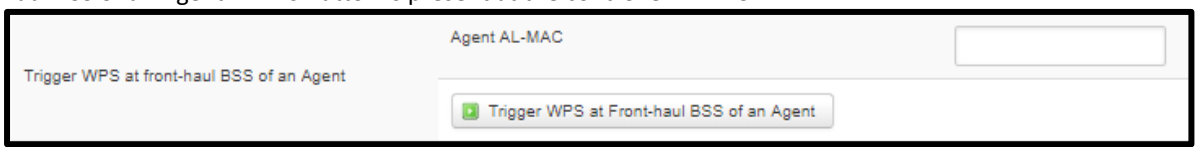


Figure 3- 56. WEB UI button to trigger WPS at fronthaul of an agent

Command Line Interface

Instead of web UI command line can also be used command line for Trigger_map_wps
Use: to trigger wps at the Agent which MAC address of one BSS equals the <bss-mac>
Usage:

mapd_cli /tmp/mapd_ctrl trigger_map_wps <bss-mac>

3.5 Auto Role Selection

This feature is used to set up cable MAP network automatically. We support the following three scenarios and its variation.

1) Chain topology and one MAP device connecting to internet directly.



Figure 3- 57. Chain Topology with direct internet connection to MAP device

2) Chain topology and one MAP device connecting to PON without MAP service.

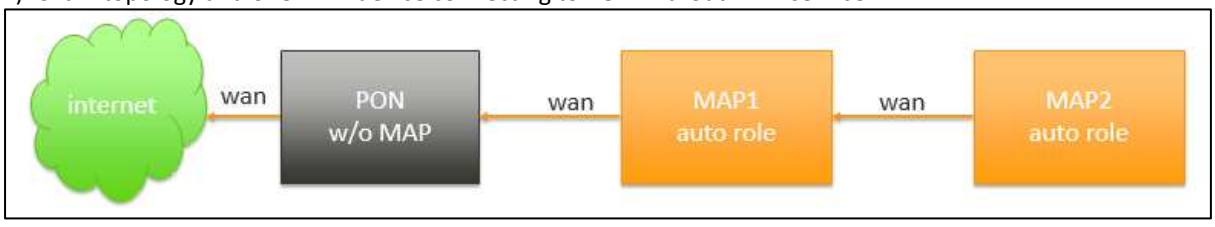


Figure 3- 58. Chain Topology with PON

1) Star topology and all MAP device connecting to PON directly

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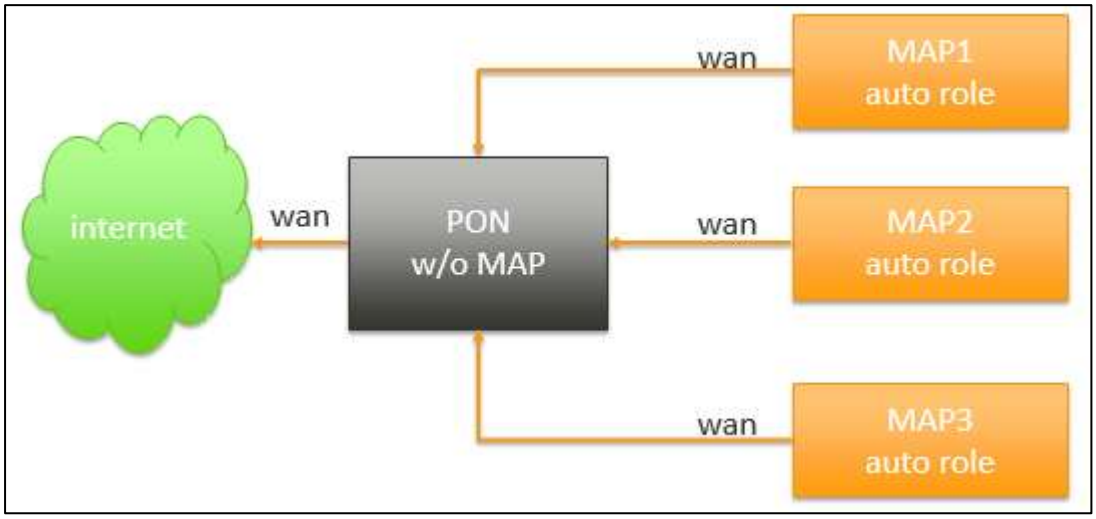


Figure 3- 59. STAR Topology with PON

One of them will be selected as controller and the others will become agent. For scenario one and two, the device has the shortest distance to the internet will become controller. For the scenario 3, as all the device has the same distance to internet, in this case the device with the biggest almac will become controller.

3.5.1 Main flow

- Auto role detection. Check on every device that if it can ping 8.8.8.8/208.67.222.222 with fixed length 88 and block ping with 8.8.8.8/208.67.222.222 from downstream device.
- If ping can success, move to send search state. Send auto-configuration search message on Ethernet and wifi backhaul sta interface every 5s to check if controller is already exist and schedule a 30s timer to decide role. Meanwhile collecting auto configuration search message from the other device. If ping can't success, keep on ping.
- If it receives auto-configuration response message, change own role to agent and cancel timer scheduled in auto role stage and unblock ping with 8.8.8.8/208.67.222.222 from downstream device. If it doesn't receive any auto-configuration response message during auto role detect stage, after timeout, check if its own almac is the biggest among the data base of receiving auto configuration search message, change its role to controller otherwise change role to agent.

3.5.2 Procedure

- Set Device Role as Auto, save and apply

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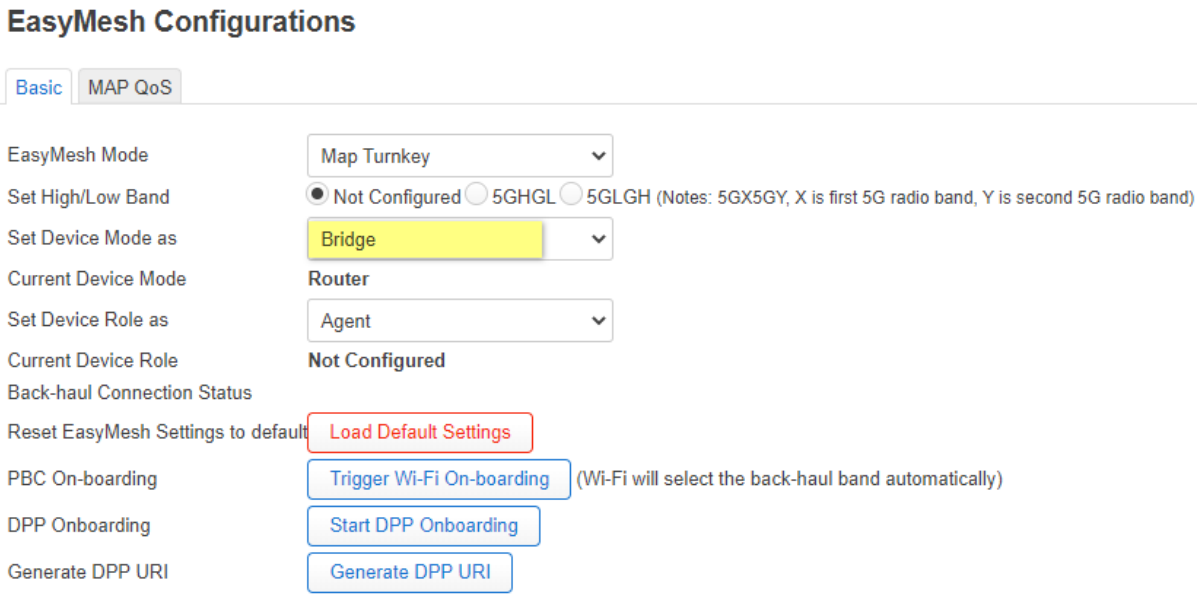


Figure 3- 60. WEB UI button to set MAP device role

For a MAP R3 device whenever device role is set to auto option by UI then UI will update the allowed_role in /etc/dpp_cfg.txt as 0. If modified the role to device role to auto manually then need to manually set the allowed_role to 0 in /etc/dpp_cfg.txt. (allowed_role=0)

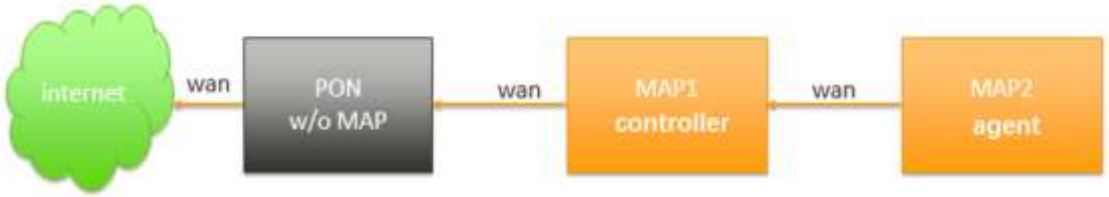
3.5.3 Configure & Test

The Division Rules of controller and agent will be different in different topology.

1) Chain topology and one MAP device connecting to internet directly.

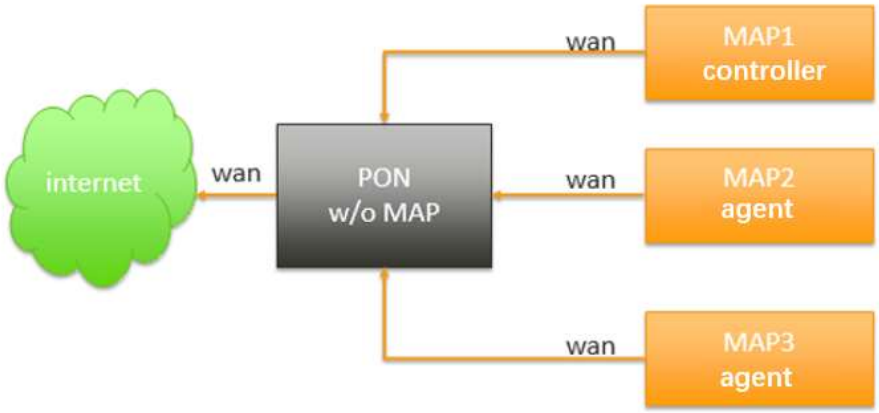


2) Chain topology and one MAP device connecting to PON without MAP service.



3) Star topology and all MAP device connecting to PON directly.

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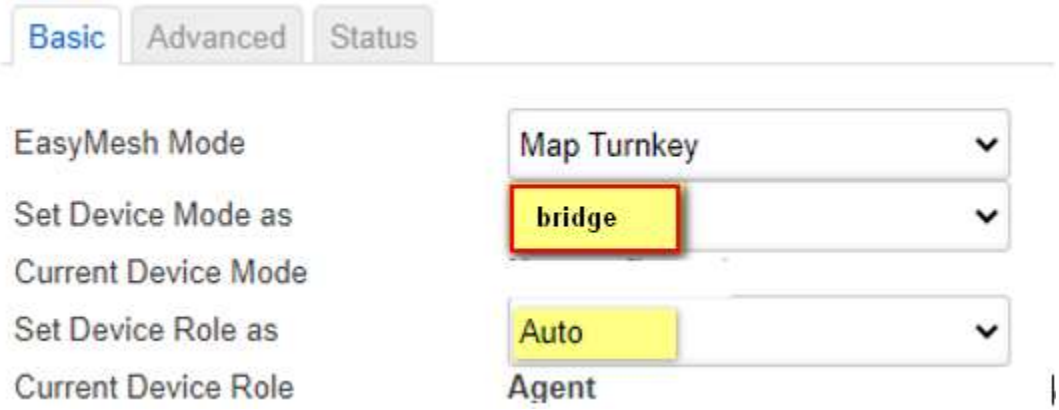


For scenario one and two, the device has the shortest distance to the internet will become controller. For the scenario 3, as all the device has the same distance to internet, in this case the device with the biggest almac will become controller.

After device connect by eth， according to the following settings：

- Set MapMode=1 in the driver dat file
- Set DeviceRole=0 in /etc/map/mapd_user.cfg,
- Set ext_phy_port=<port no.> in /etc/ethernet.txt, refer to chapter 3.7.3
- Reboot

If the above settings are successful，the EasyMesh Mode will be Map Turnkey and the Set Device Role will be auto on UI



After the below setting and connecting devices in different topology, the device will become a controller or agent.

3.6 Multi-AP Renew

Multi-AP renew is a mechanism to update configuration information of operating bss both at controller and agent. It supports both backhaul and fronthaul bss renew.

3.6.1 Main flow of first time set-up

1. Once mapd has started up, it calls lib1905 API _1905_Set_Read_Bss_Conf_and_Renew to trigger local bss setting.
2. 1905 daemon read wts_bss_info_config and send wireless setting event to wapp to set new ssid/authmode/encryption/password/bss role into driver.
3. In wapp,
   a. If the setting is the same with the old one, it just skips it.
   b. If the setting is updated compared to the old one, it applies to the new one.

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c. If the previously configured interface is not configured this time, it disables that interface.

4. Driver just executes the cmd sent by wapp.

3.6.2 Main flow of wireless setting update

1. UI changes bss configuration information and updates wts_bss_info_config file.
2. UI call libmapd API mapd_renew to start renew procedure.
3. mapd calls lib1905 API _1905_Set_Read_Bss_Conf_and_Renew_v2 to start renew state machine.
4. renew state machine in 1905 dump the topology tree into a stack structure with preorder traversal algorithm.
5. renew state machine pops out agent one by one stored in the stack and sends unicast renew message it.
6. Agents do auto-configuration one by one to get and update new wireless setting.
7. After all agents renew done, 1905 sends message to Mapd with MsgId _1905_SET_WIRELESS_SETTING_EVENT.
8. After all agents renew done, Mapd sends command with Cmd ID WAPP_USER_SET_WIRELESS_SETTING to wapp to trigger updating local bss setting.

Note:
The main difference of renew between R1/R2 and R3 is that
1. Controller send AP Autoconfig renew to each agent, then M1 and M2 between Controller and agents for R1/R2 renew
2. Controller send BSS reconfiguration trigger message to each agent, then BSS Configuration Request, BSS configuration response, BSS Configuration Result message between controller and Agents for R3 renew

3.6.3 Procedure

- Change wireless configuration information on UI and push confirm button.
- Wait for renew done and check the topology database to see if all backhaul and fronthaul bss setting are changed.
- MTK->EasyMesh->Advanced->Config BSS->Edit BSS

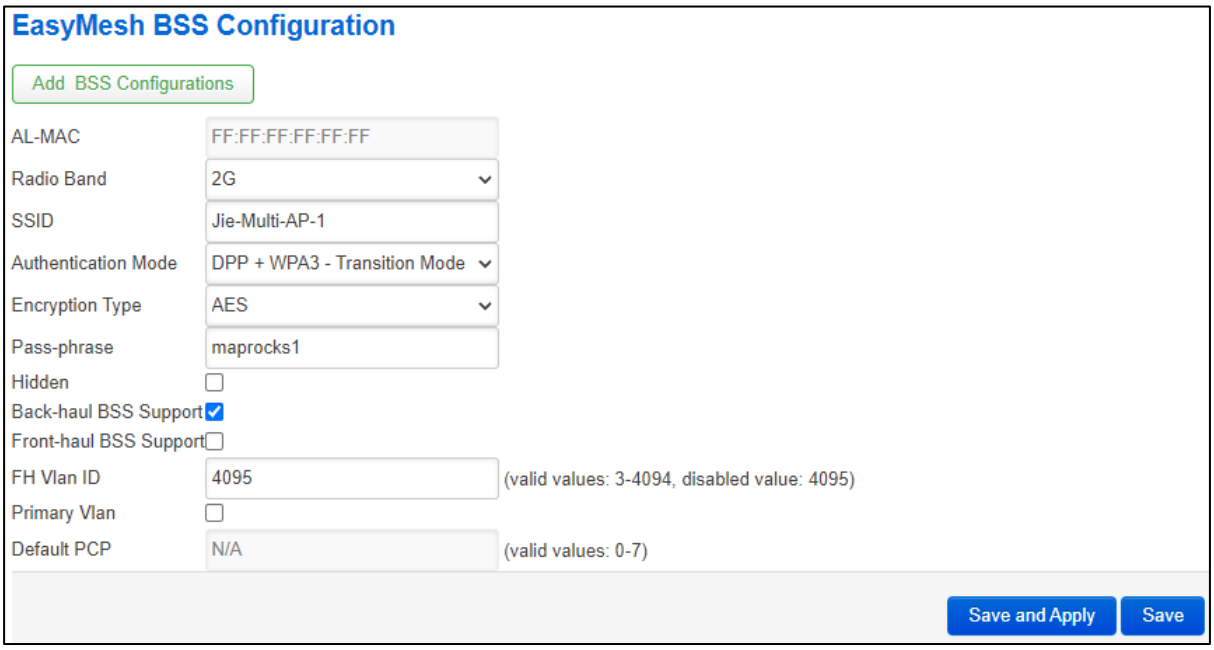


Figure 3- 61. BSS configuration renew Web UI Interface

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- Modify any parameters list above, then push button “Save and Apply”, renew will be triggered, but onboarding(DPP or WPS) should be finished between Controller and Agent as a precondition.
- **For UCI supported SKUs all bss config will update in /etc/config/mapd .**

3.6.4 Limitations

As the controller has to renew all the agents one by one, and it will take a few seconds (8~10 seconds per agent) to renew the whole Multi-AP network. How long it will take depends on the amount of agent in the Multi-AP network. User should not trigger a new renew process before the previous one is done, or else the controller fails to renew all the agents.

3.6.5 Customer Feature to Enable/Disable SSID on Specific Interface

Customer has the requirement to runtime enable or disable SSID on specific interface of the whole mesh network. This is due to customer bind the wireless mode or other Wi-Fi parameter to interface. In this way SSID is bond to interface and they need operate the specific interface.

Take 5G setting for instance, with the first setup, the mesh network is configured as bellow on both Controller and Agents:

Rax0---11ax--- Multi-AP-5LG-1
Rax1--11ac--- Multi-AP-5LG-2
Rax2---11an--- Multi-AP-5LG-3
Rax3---11ax--- Multi-AP-5LG-4

If user wants to disable rax1 on both Controller and Agents, the user could do following step to update the setting:

- Set command on Controller and wait a few seconds to make it done.
  mapd_cli /tmp/mapd_ctrl set_bss_prio
  "ra0;ra1;ra2;ra3;apcli0;**rax0;rax2;rax3;rax1**;apclix0;rai0;rai1;rai2;rai3;apclii0"

or directly

mapd_cli /tmp/mapd_ctrl set_bss_prio "**rax0;rax2;rax3;rax1;**apclix0"

this command will trigger Controller to update local BSS priority and, in the meanwhile, to synchronize the BSS priority setting to Agents by Mediate vendor message.

- Delete the wireless setting by Multi-AP-5LG-2 and do renew
- Once renew is done, the new setting would be:
  Rax0---11ax--- Multi-AP-5LG-1
  Rax1--11ac--- BSS stop
  Rax2---11an--- Multi-AP-5LG-3
  Rax3---11ax--- Multi-AP-5LG-4

If user wants to enable rax1, he could add one wireless setting and do renew. Once the renew procedure is done, the new setting would be:

Rax0---11ax--- Multi-AP-5LG-1
Rax1--11ac--- BSS start with new SSID
Rax2---11an--- Multi-AP-5LG-3
Rax3---11ax--- Multi-AP-5LG-4

3.7 Variations of switch/GMAC combination support

MAP device sometimes needs to get/set information from/to different kind of switch, for example, when the network cable is inserted into switch port, 1905daemon needs to detect this event and send some topology messages to update the topology information. So in 1905daemon, we define a series of Ethernet operations as below:

ops	description

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init()	this operation should guarantee that current device would not forward any 1905 neighbor multicast cmdus(dest mac as 01:80:c2:00:00:13).
get_switch_mode()	get switch mode. 0-GSW, 1-DSA
get_port_type (OUT unsigned int *lport_map, OUT unsigned int *wport_map)	get the types of each switch port @lport_map: indicates the lan port index, e.g. 0x4F=0B01001111 means port 0, 1, 2, 3, 6 are the lan ports @wport_map indicates the wan port index, e.g. 0x30=0B00110000 means port 4, 5 are the wan ports
get_table_entry (OUT struct dl_list *entry_list)	get all the entry from switch table. get all eth 1905 neighbor and none 1905 neighbor imformations @entry_list is the head of the entry list, the element of entry list should be the structure ethernet_client_entry.
search_table (IN unsigned char *mac, OUT struct ethernet_client_entry *entry)	find the specific entry by mac address. get the port number which 1905 packet received and update 1905 topology
get_port_status (IN unsigned char index, OUT int *status)	get the status of a switch port. status should be 1 if Link Up, 0 if Link Down
set_switch_ts_vlan (unsigned short vid[], unsigned short vid_cnt, unsigned char active)	set traffic separation vlans to switch,vid[] include all the ts vlan make sure all the ts vlan packets can be forwarded between lan cpu port and all lan ports
set_switch_transparent_vlan (unsigned short vid[], unsigned short vid_cnt, unsigned char active)	set transparent vlans to switch, vid[] include all the transparent vlan make sure all the transparent vlan packets can be forwarded between lan ports and wan ports
get_port_speed (IN unsigned char index, OUT int *speed)	Get negotiation speed of user specified port
set_port_block (IN unsigned char index, IN int enable)	Block or unblock specified port enable=1, blocking state; enable=0, forwarding state
clear_table()	clear switch mac table and clear fdb table in bridge of lan interfaces
clear_port_table (IN unsigned char port)	clear switch mac table and clear fdb table in bridge of specified lan port
Deinit()	deinitialization procedure

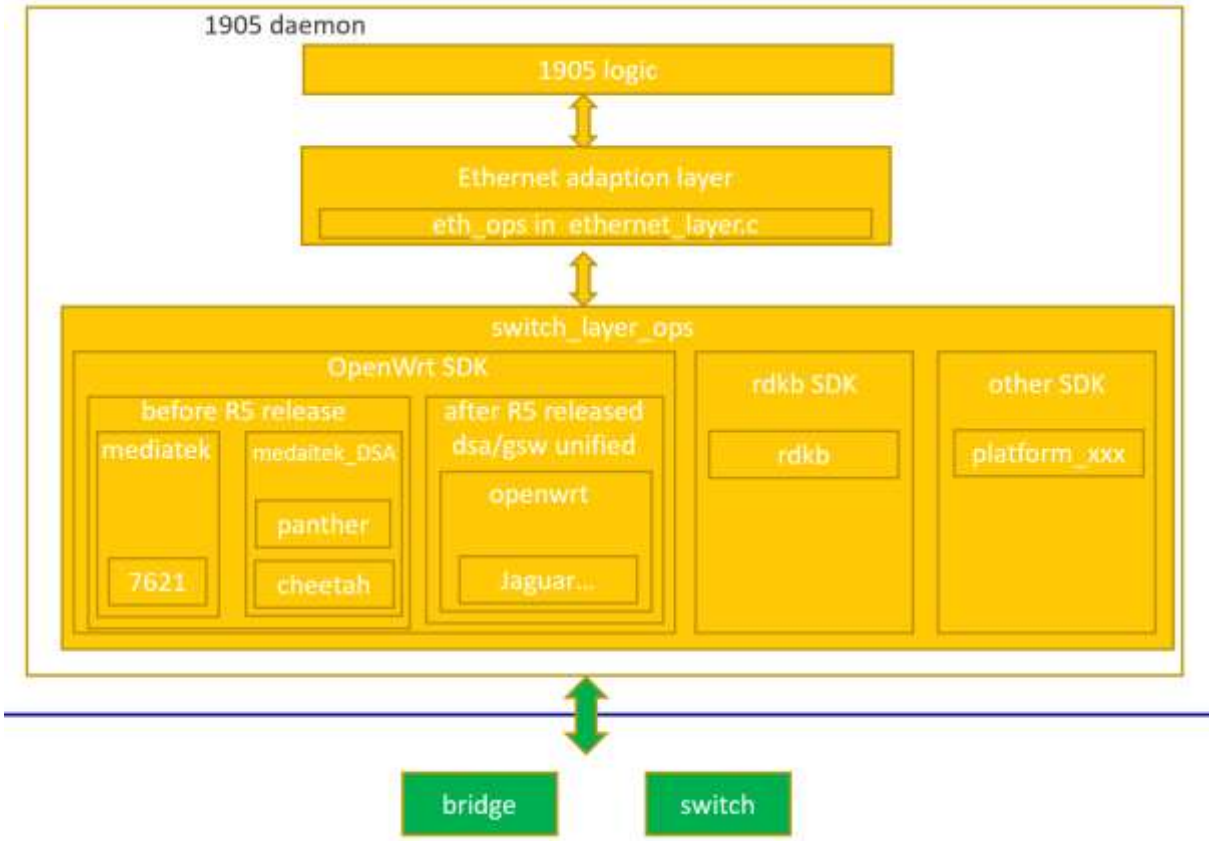
1905daemon would call this Ethernet operation function if the switch can support these operations, now MT7530/MT7531 can fully support these operations.

Make sure SDK_TYPE is correct for your SDK
for openwrt:
SDK_TYPE=openwrt
for rdkb:
SDK_TYPE=rdkb

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3.7.1 Switch OPS

3.7.1.1 Switch OPS arch

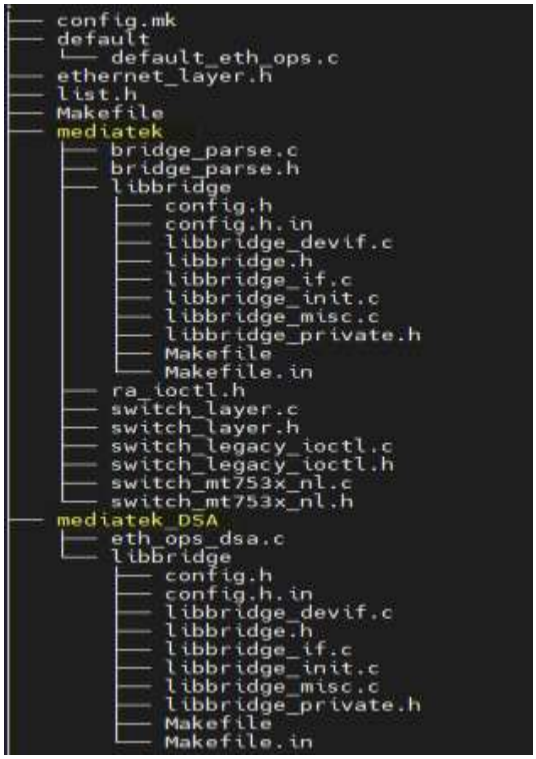


Ops defined in Ethernet adapter layer
Ops registered by switch layer, maybe there are somer different platforms, such as OpenWrt, RDKB, etc.

- Before R5 release

There are two folders for openwrt sdk before R5 release, mediatek is for GSW, mediatek_DSA is for DSA

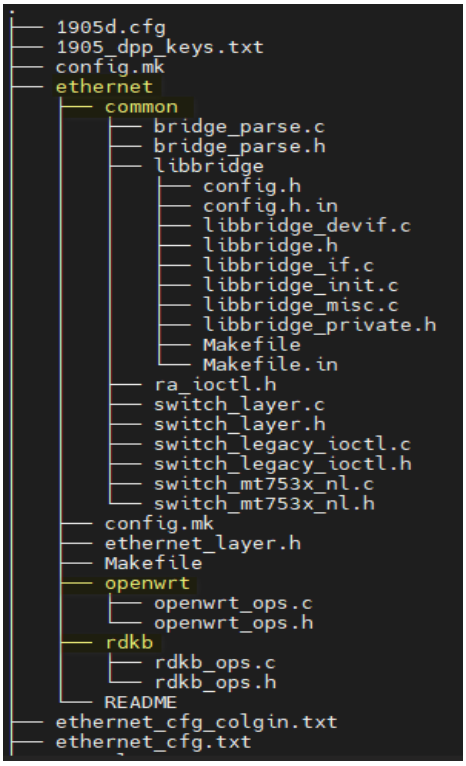
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7621 will use mediatek folder default
Panther, Cheetah will use mediatek_DSA folder by setting SWITCH_TYPE = mediatek_dsa in ethernet/config.mk

➢ After R5 released

folder(openwrt) for openwrt SDK after R5 released, as it's unified switch layer for both DSA and GSW
folder(rdkb) for rdkb SDK

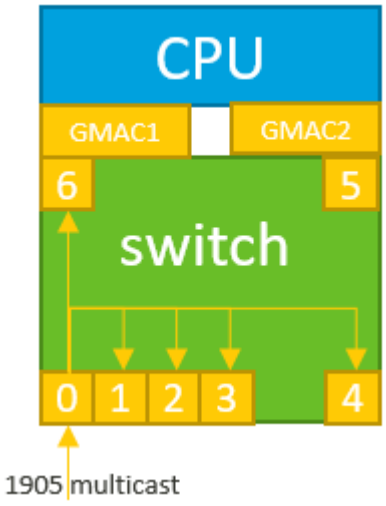


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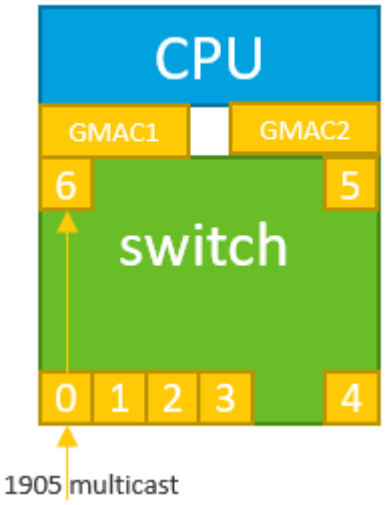
3.7.1.2 Switch OPS description

3.7.1.2.1 Init

1905 multicast cmdu will be flood on each switch port after receiving from one port default. as below:

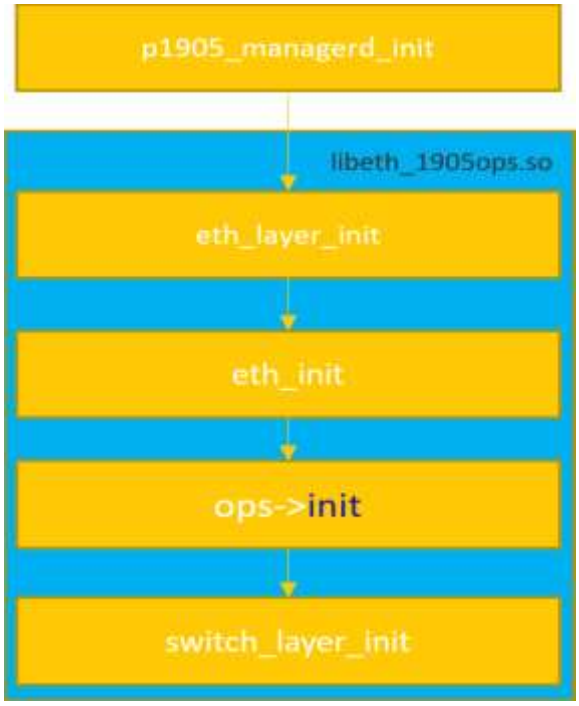


but from MAP spec, 1905 multicast cmdu only can be forward by upper layer 1905 daemon
So in init ops, we prevent flooding 1905 multicast cmdus to each switch port, but only forward to cpu, as below:



Init op can be called while while bringing up 1905 daemon after Easy Mesh enabled

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Call eth_layer_init(void) api provided by libeth_1905ops.so @ 1905 daemon init stadge

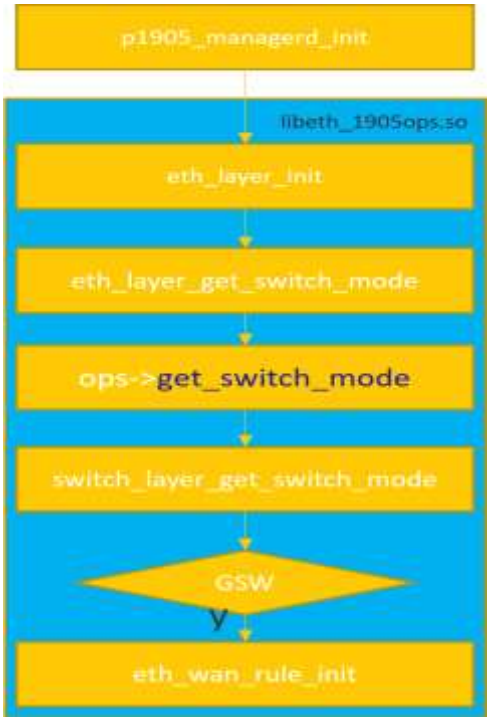
3.7.1.2.2 get_switch_mode

- details

not used after R
need get switch mode(DSA or GSW) of current device

get_switch_mode op can be called while while bringing up 1905 daemon after Easy Mesh enabled

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Call eth_layer_init(void) api provided by libeth_1905ops.so

➢ test cmd

get switch mode of current device, by cmd:

on controller:

$1905ctrl controller get_switch_mode

On agent:

$1905ctrl agent get_switch_mode

Return DSA for DSA switch:

switch mode: DSA

Return GSW for GSW switch:

switch mode: GSW

3.7.1.3 get_port_type

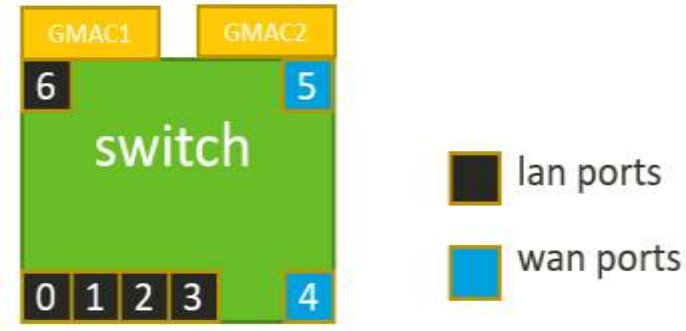
get port bitmap, not used after R5 released.

➢ details

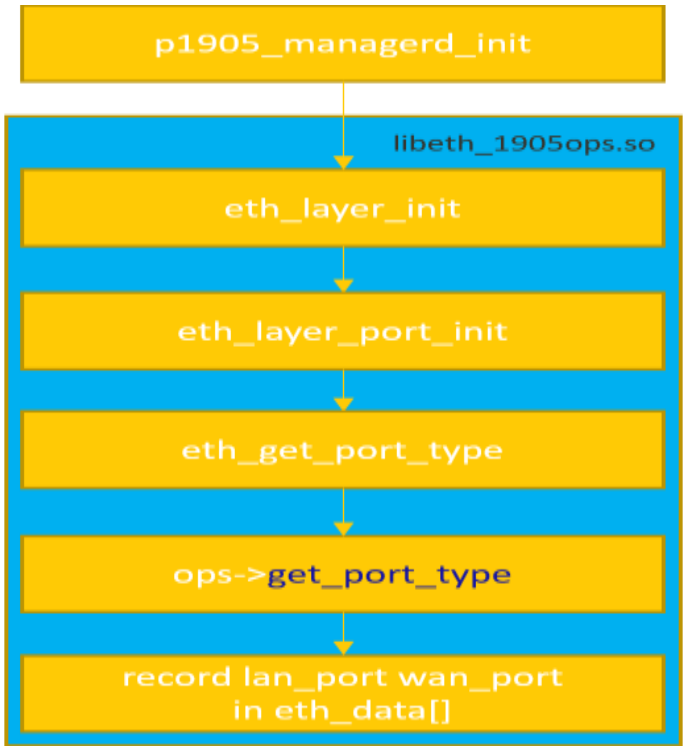
need know how many ports are lan port and which port is wan port

lport_map: indicates the lan port index, e.g. 0x4F=0B01001111 means port 0, 1, 2, 3, 6 are the lan ports
wport_map indicates the wan port index, e.g. 0x30=0B00110000 means port 4, 5 are the wan ports

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- call get_port_type



Call eth_layer_init(void) api provided by libeth_1905ops.so

- test cmd

cmd to get port type
On controller:
1905ctrl controller get_eth_port_type
On agent:
1905ctrl agent get_eth_port_type
Return lport_map 004f and wport_map 0030
lport_map 004f!
wport_map 0030!

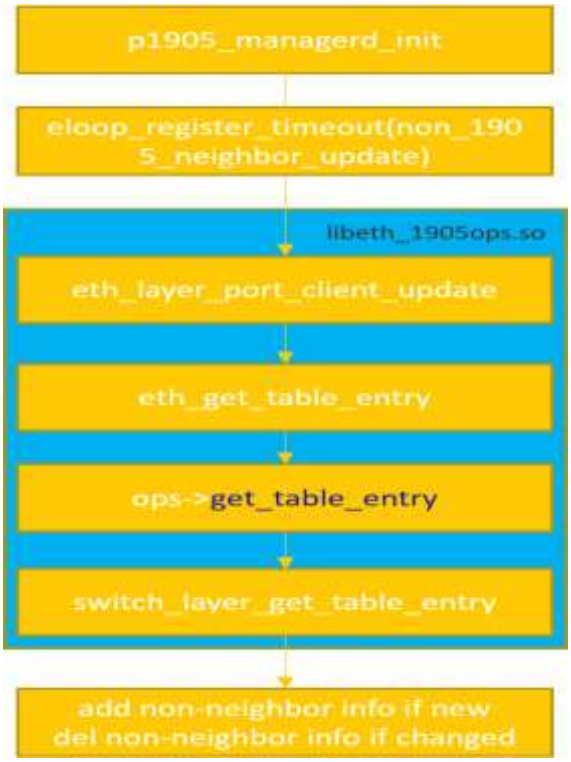
3.7.1.4 get_table_entry

- details

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get all the ethernet entry from switch table.

ethernet entry including connected port no., mac address and age etc.
register a timer to update non-neighbor periodically



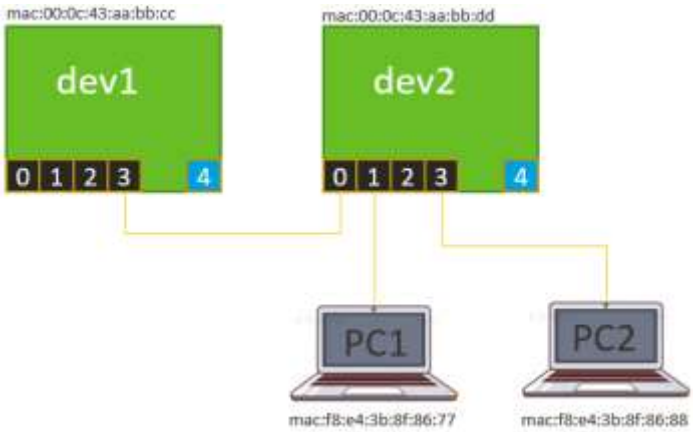
Call eth_layer_port_client_update(void) api provided by libeth_1905ops.so

- test cmd

Cmd to get table entry:
On controller:
1905ctrl controller get_eth_entry
On agent:
1905ctrl agent get_eth_entry
Return all entrys that connect to the device
NO. MAC PORT age
0 xx:xx:xx:xx:xx:xx port-no

Example:

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Run cmd below on dev2

1905ctrl agent get_eth_entry

Return content:

NO.	MAC	PORT	age
0	f8:e4:3b:8f:86:77	1	145
1	f8:e4:3b:8f:86:88	3	143
2	00:0c:43:aa:bb:cc	0	150

3.7.1.5 search_table

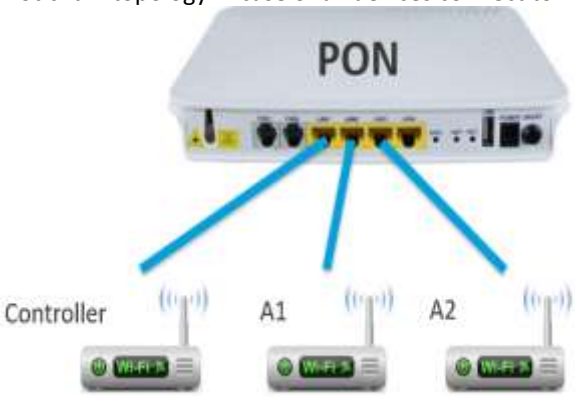
- details

find the specific entry by mac address, and return the port no.

sometimes we need find out which port the ethernet entry is connected to

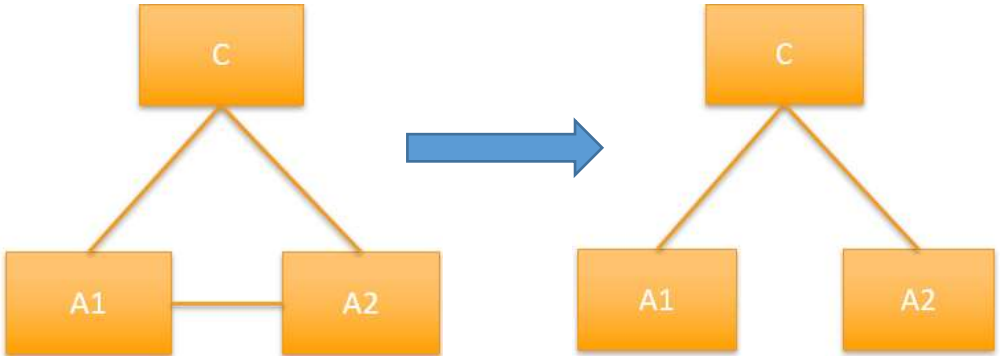
case 1: update port no. after receiving 1905 topology response cmdus

case 2: to form star topology not chain topology in case of all devices connect to PON as below



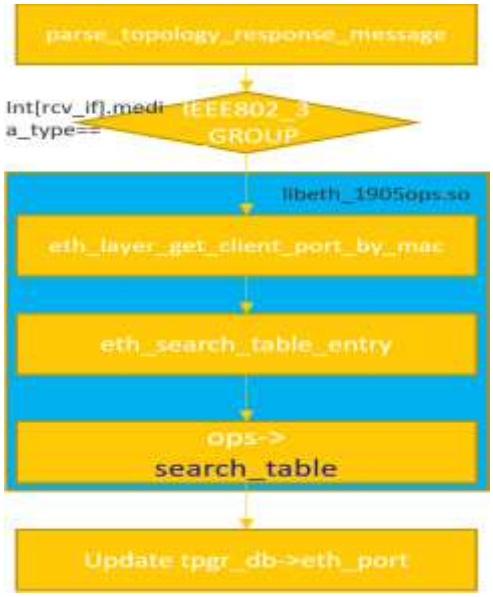
To form topology as below:

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- call search_table op

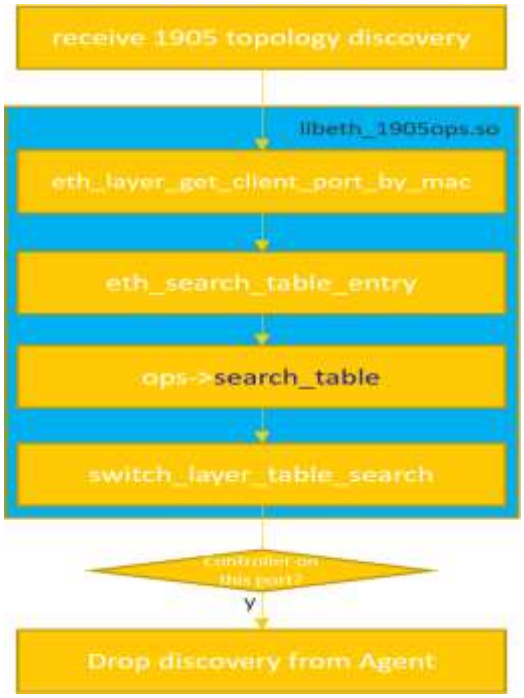
case 1: handle topology response



call eth_layer_get_client_port_by_mac(mac) function to get port no. which cmdu received, after receiving 1905 topology response cmdu from ethernet port, and update eth_port in topology response database

case 2: to form star topology not chain topology in case of all devices connect to PON

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Call eth_layer_get_client_port_by_mac(mac) function to get port no. which cmdu received, after receiving 1905 topology discovery cmdu from ethernet port

Drop 1905 topology discovery cmdu from other agent once controller on the same port.

- test cmd

Cmd to get table entry:

On controller:

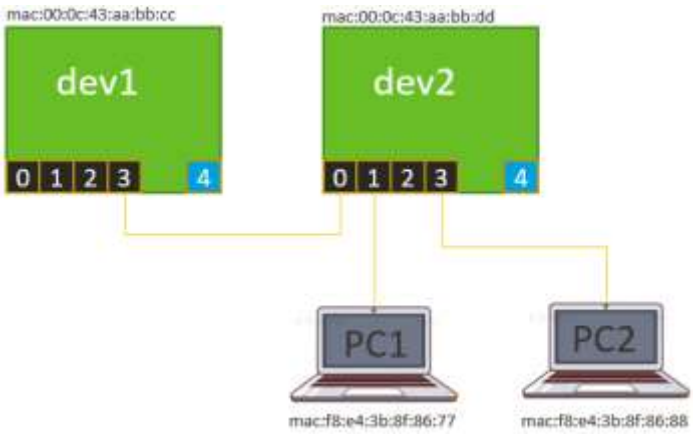
1905ctrl controller find_eth_entry xx:xx:xx:xx:xx:xx

On agent:

1905ctrl agent find_eth_entry xx:xx:xx:xx:xx:xx

Return the port NO. that entry connected

Example:



Run cmd below on dev2

1905ctrl agent find_eth_entry f8:e4:3b:8f:86:88

Return content:

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find f8:e4:3b:8f:86:88 in port(3)

3.7.1.6 get_port_status

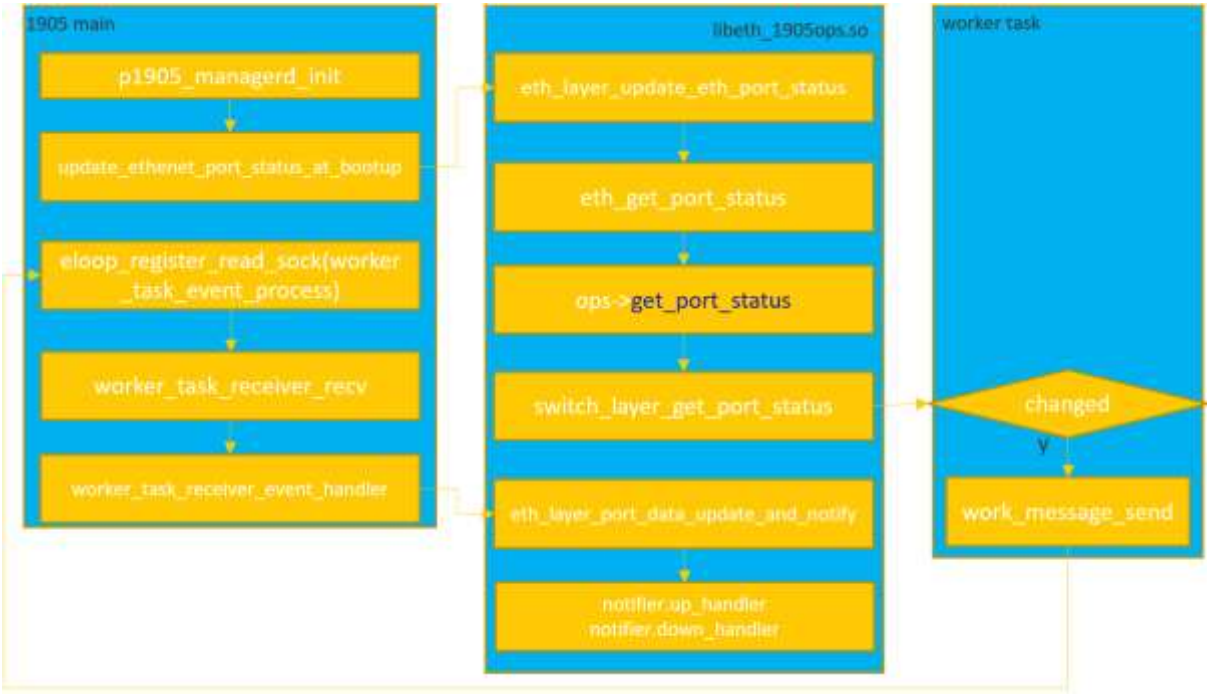
- details

get the status of a switch port. status should be 1 if Link Up, 0 if Link Down

1905 need to know the status of each port on switch.
Maybe do ethernet onboarding after plug-in ethernet cable between two EasyMesh devices
And maybe do wifi onboarding after plug-out ethernet cable between two EasyMesh devices once ethernet onboarding finished.

- call get_port_status op

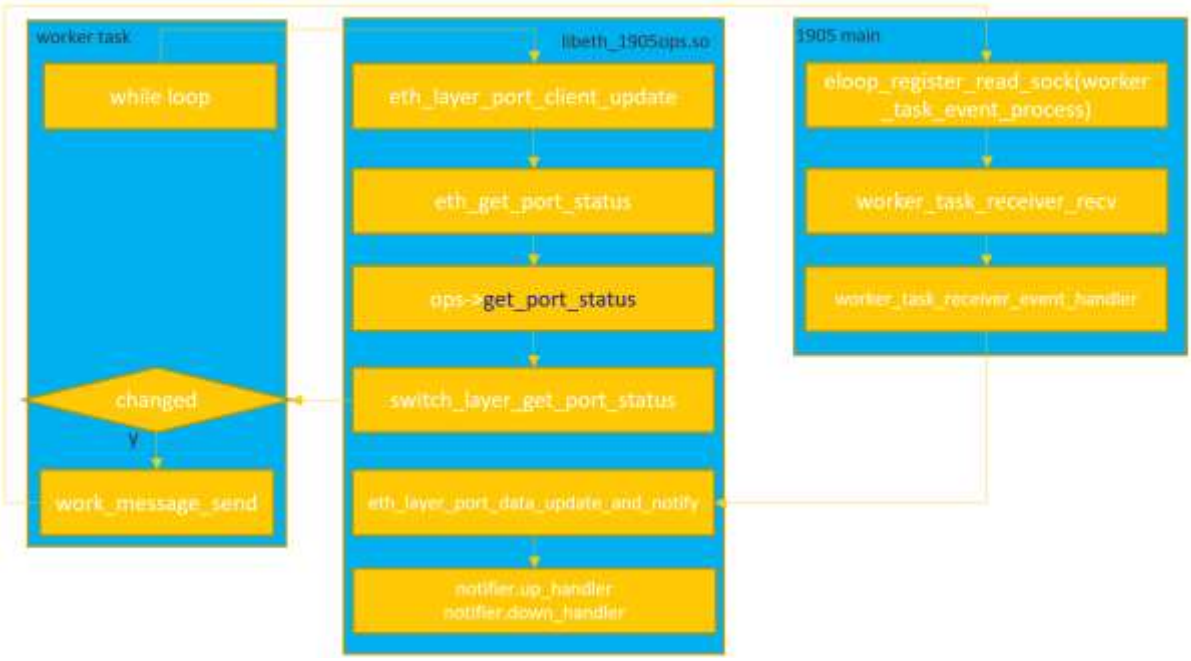
case 1: get port status @ init stadge



Call eth_layer_update_eth_port_status(port_id) @ init stadge of 1905 daemon
if port status changed, will send work message with port no., status to 1905 main process
there is a worker_task_event_process registered by eloop, it will handle the message from worker task, and call eth_layer_port_data_update_and_notify api in libeth_1905ops.so to process up/down handler

case 2: get port status every 1 second in a worker task

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worker task call eth_layer_port_client_update(void) ops get port status every 1 second. if port status changed, will send work message with port no., status to 1905 main process

there is a worker_task_event_process registered by eloop, it will handle the message from worker task, and call eth_layer_port_data_update_and_notify api in libeth_1905ops.so to process up/down handler

- test cmd

Cmd to get port status:

On controller:

1905ctrl controller get_eth_port_status <port-no>

On agent:

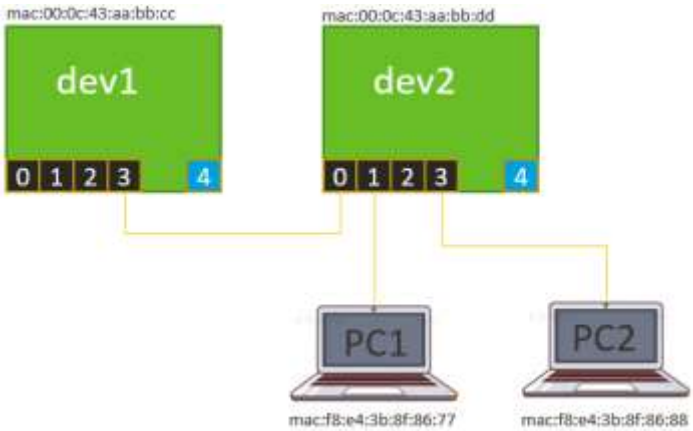
1905ctrl agent get_eth_port_status <port-no>

Return the port status:

Port[x] LINK UP

Port[y] LINK DOWN

Example:



Run cmd below on dev2

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1905ctrl agent get_eth_port_status 0
1905ctrl agent get_eth_port_status 1
1905ctrl agent get_eth_port_status 2
1905ctrl agent get_eth_port_status 3

Return content:
Port[0] LINK UP
Port[1] LINK UP
Port[2] LINK DOWN
Port[3] LINK UP

3.7.1.7 set_switch_ts_vlan

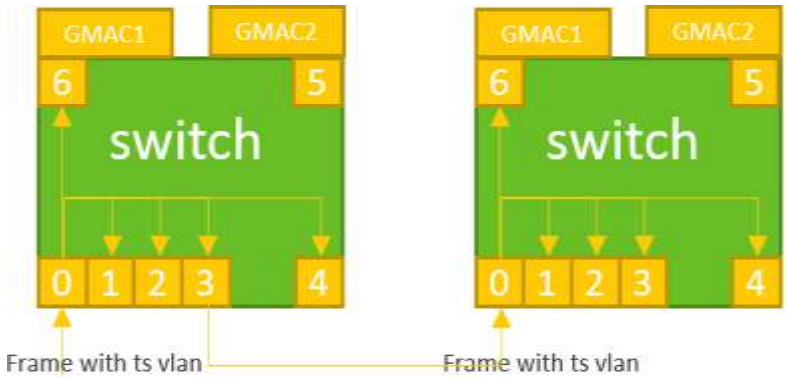
not used after R5 released

- details

set traffic separation vlans to switch,vid[] include all the ts vlans

Reserved ops, not use for now.
As vlan packet can be forward to other ports once received from one port default.

make sure all the ts vlan packets can be forwarded between lan cpu port and all lan ports



- call set_switch_ts_vlan op

Call eth_layer_set_traffic_separation_vlan (vid[], vid_num) once traffic separation vlan configed for EasyMesh.

- Test cmd

Cmd to set ts vlan:
On controller:
1905ctrl controller set_eth_ts_vlan_id <vid1,vid2,…>
On agent:
1905ctrl agent set_eth_ts_vlan_id <vid1,vid2,…>

3.7.1.8 set_switch_transparent_vlan

not used after R5 released

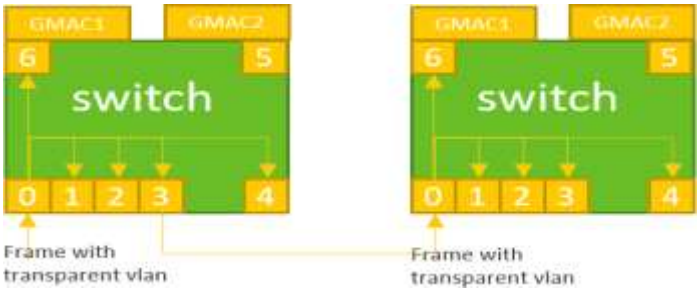
- details

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set transparent vlans to switch,vid[] include all the transparent vlans

Reserved ops, not use for now. for 8.1.3 IPTV service
As vlan packet can be forward to other ports once received from one port default.

make sure all the transparent vlan packets can be forwarded between lan cpu port and all lan ports



- call set_switch_transparent_vlan op

Call eth_layer_set_switch_trasnparent_vlan (vid[], vid_num) once transparent vlan configed for EasyMesh.

- Test cmd

Cmd to set ts vlan:

On controller:

1905ctrl controller set_eth_trans_vlan_id <vid1,vid2,…>

On agent:

1905ctrl agent set_eth_trans_vlan_id <vid1,vid2,…>

3.7.1.9 get_port_speed

get negotiated speed of user specified port
need get_port_speed op once port negotiation speed in link metrics

- call get_port_speed op

once there is a requirement to append switch port negotiation speed in link metrics

call eth_layer_get_eth_port_speed (port_id) with port_id

- test cmd

Cmd to get port negotiation speed:

On controller:

1905ctrl controller get_port_speed <port-no>

On agent:

1905ctrl agent get_port_speed <port-no>

Return the port negotiation speed:

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3.7.1.10 set_port_block

- details

Block or unblock specified port
enable=1, blocking state; enable=0, forwarding state

used in loop case to block and unblock specified port.

- call set_port_block op

Loop occur, need block the port
Loop disappear, unblock the port

Call eth_layer_set_eth_port_block(char index, int enable) with port index and active|disactive status
Block port: eth_layer_set_eth_port_block(port_id, 1)
Unblock port: eth_layer_set_eth_port_block(port_id, 0)

- test cmd

Cmd to block or unblock port:
On controller:
1905ctrl controller set_port_block <port_id, 1>
On agent:
1905ctrl agent get_port_speed <port_id, 1>
Cmd to unblock port:
On controller:
1905ctrl controller set_port_block <port_id, 0>
On agent:
1905ctrl agent get_port_speed <port_id, 0>

3.7.1.11 clear_table

- details

clear switch mac table and clear fdb table in bridge of all lan interfaces
need clear switch mac table and fdb table once ethernet cable plug-out

- call clear_table op

need clear switch mac table and fdb table once ethernet cable plug-out

call eth_layer_clear_switch_table(void) api provided in libeth_1905ops.so
in ethernet_pull_out_handler.

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➢ test cmd

Cmd to clear table:

On controller:

1905ctrl controller clear_table

On agent:

1905ctrl agent clear_table

3.7.1.12 clear_port_table

➢ details

this op is called only by ethernet_layer
clear switch mac table and clear fdb table for specified port once port down

switch mac table can't be clear immediately after port down, but will spend 150 seconds to clear, so need this api to clear switch mac table directly after port down

➢ call clear_port_table op

this op is called in worker_task, when after the port down

call eth_layer_clear_port_table(unsigned char index) api provided in libeth_1905ops.so
in worker_task function.

3.7.1.13 deinit

deinitialization procedure

3.7.2 Default setting

There are some default switch settings which would probably impact the related features:

- lan port vlan id=1
- wan port vlan id=2
- lan cpu port=6

So if customer changes above settings of switch, customer should update the configuration file in /etc/ethernet_cfg.txt as below:

```
#set the vid of the LAN group, the range is 1-4094, and not repeat with wan_vid
#use' ;' as separation character when there are multiple lan vids, for example perportpervlan setting.
#lan_vid=1;2;3;4;

#et the vid of the WAN group, the range is 1-4094, and not repeat with lan_vid
#wan_vid=2;

#et the CPU port number, where the switch and CPU are connected.
# The witch-7530 can be selected as 5 or 6.
```


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```
# The switch-7531 can only set one of 5 and 6, or you can set them at the same time
#cpu_port=6;
#specify external phy port, used as lan port while bridge mode
# panther with one 2.5G phy port, ext_phy_port=6
# panther with two 2.5G phy port, ext_phy_port=5 6
ext_phy_port=
```

For this scenario, switch layer will forward packets received from LAN port to the other LAN port directly based on switch learning table instead of forwarding the packets to CPU port.

The LAN interface shall be bond to the bridge:

```
brctl  addif br-lan eth0
```

The WAN interface shall be bond to the bridge if the current mode is bridge mode:

```
brctl  addif br-lan eth1
```

The LAN interface should be written to /etc/map/1905d.cfg

GSW switch	router mode(default)	lan=eth0
GSW switch	bridge mode	lan=eth0 lan=eth1
DSA switch	router mode(default)	lan=lan0 lan=lan1 lan=lan2 lan=lan3
DSA switch	bridge mode	lan=lan0 lan=lan1 lan=lan2 lan=lan3 lan=eth1

The LAN interface should be written to /etc/ethernet_cfg.txt

Switch mode	Route/Bridge mode	
GSW	Router mode	lan=eth0
GSW	Bridge mode	lan=eth0 eth1
DSA	Router mode	lan=lan0 lan1 lan2 lan3
DSA	Bridge mode	lan=lan0 lan1 lan2 lan3 eth1

The WAN interface should be written to /etc/map/1905d.cfg if the current mode is bridge mode:

```
wan = eth1
```

3.7.2.1 customer setting(wan port 0)

StartMapTurnkey
set_switch_vlan_gmac2_2_7531

call set_switch_vlan_gmac2_2_7531 in function StartMapTurnkey in EasyMesh_openwrt5x.sh

set_switch_vlan_gmac2_2_7531 as below:

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- forward 1905 multicast to cpu port not flooding
  switch reg w 34 8160816
  switch reg w 4 60
  switch reg w 10 ffffffff
  switch vlan clear

- lan port(1234) and cpu port(6), vlan id 1
  switch vlan set 0 1 01110010

- wan port 0 and 5, vlan id 2
  switch vlan set 1 2 10000100

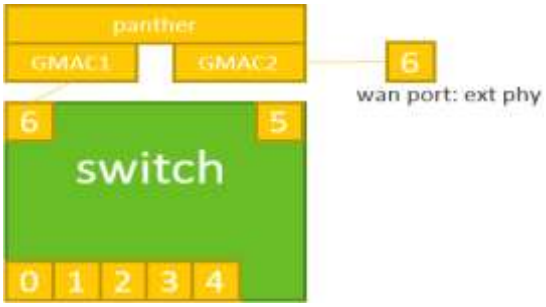
- lan port and cpu port set pvid to 1
  switch vlan pvid 1 1
  switch vlan pvid 2 1
  switch vlan pvid 3 1
  switch vlan pvid 4 1
  switch vlan pvid 6 1

- wan port set pvid to 2
  switch vlan pvid 0 2
  switch vlan pvid 5 2

3.7.3 ext_phy_port setting

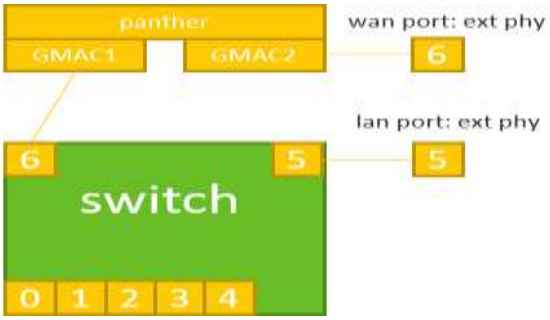
in /etc/ethernet_cfg.txt

- **panther**
  - **case 1: external phy as wan port**
    ext_phy_port=6



    Use external phy as wan port, and port NO. is 6
  - **case 2: one external phy as wan port, another one as lan port**
    ext_phy_port=6
    not support external phy as lan port yet
    only support external phy as wan port

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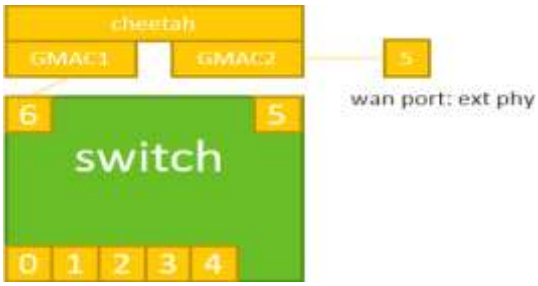


- **cheetah**
  - **case 1: internal phy**
    ext_phy_port=0



Use internal phy(Gephy) as wan port, and port NO. is 0

  - **case 2: external phy**
    ext_phy_port=5



Use external phy as wan port, and port NO. is 5

- **jaguar**
  ext_phy_port=0 8

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Use external phy0 as wan port, and port NO. is 0
Use external phy8 as lan port, and port NO. is 8, need add eth2 to bridge once use phy8 as lan port.

3.7.4 Per-Port-Per-Vlan setting for Non-DSA Architecture

For this setting, customer shall configure the switch setting by their own commands or refer to the setting in perportpervlan_test in EasyMesh_openwrt.sh.
In this setting, every LAN port and WAN port use the different vlan id and every LAN port use the different LAN interface name. Take four LAN port switch for example, the /etc/ethernet_cfg.txt could be configured as below:

```
#set the vid of the LAN group, the range is 1-4094, and not repeat with wan_vid
#use' ;' as separation character when there are multiple lan vids, for example perportpervlan setting.
lan_vid=1;2;3;4;

#et the vid of the WAN group, the range is 1-4094, and not repeat with lan_vid
wan_vid=5;
```

For this scenario, switch layer will forward packets received from LAN port directly to CPU port to bridge layer and the bridge will forward back the packets to other LAN port based on bridge learning table.

ALL the LAN interface shall be bond to the bridge:

```
brctl  addif br-lan eth0.1
brctl  addif br-lan eth0.2
brctl  addif br-lan eth0.3
brctl  addif br-lan eth0.4
```

The WAN interface shall be bond to the bridge if the current mode is bridge mode:

```
brctl  addif br-lan eth1
```

The LAN interface should be written to /etc/map/1905d.cfg

```
lan = eth0.1
lan = eth0.2
lan = eth0.3
lan = eth0.4
```

The WAN interface should be written to /etc/map/1905d.cfg if the current mode is bridge mode:

```
wan = eth1
```

3.7.5 Per-Port-Per-Vlan setting for DSA Architecture

The architecture of the switch with the kernel version higher than 5.x use the DSA architecture.
In this setting all the LAN ports share the same vlan id but their LAN interface name are different. The WAN port use another different vlan id. Take Panther for example, the /etc/ethernet_cfg.txt is configured as below: (add eth1 to lan= if the current mode is bridge mode)

```
br_lan=br-lan
```


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```
lan=lan0 lan1 lan2 lan3 lan4 lan5 eth1
#set the vid of the LAN group, the range is 1-4094, and not repeat with wan_vid
lan_vid=1;

#et the vid of the WAN group, the range is 1-4094, and not repeat with lan_vid
wan_vid=2;

#et the CPU port number, where the switch and CPU are connected.
# The witch-7530 can be selected as 5 or 6.
# The switch-7531 can only set one of 5 and 6, or you can set them at the same time
#cpu_port=6;

#specify external phy port, used as lan port while bridge mode
# panther with one 2.5G phy port, ext_phy_port=6
# panther with two 2.5G phy port, ext_phy_port=5 6
ext_phy_port=
```

For this scenario, switch layer will forward packets received from LAN port directly to CPU port to bridge layer and the bridge will forward back the packets to other LAN port based on bridge learning table.

ALL the LAN interface shall be bond to the bridge:

```
brctl  addif br-lan lan0
brctl  addif br-lan lan1
brctl  addif br-lan lan2
brctl  addif br-lan lan3
brctl  addif br-lan lan4
brctl  addif br-lan lan5
```

The WAN interface shall be bond to the bridge if the current mode is bridge mode:

```
brctl  addif br-lan eth1
```

The LAN interface should be written to /etc/map/1905d.cfg

```
lan = lan0
lan = lan1
lan = lan2
lan = lan3
lan = lan4
lan = lan5
```

The WAN interface should be written to /etc/map/1905d.cfg if the current mode is bridge mode:

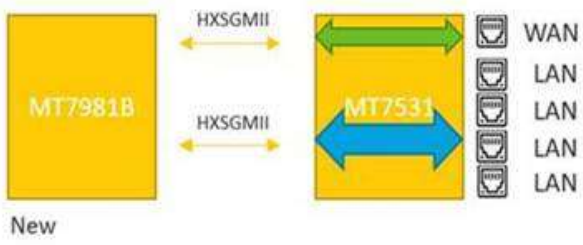
```
wan = eth1
```

3.7.5.1 Customer setting(wan port 0)

Both GMII connect to switch,

wan port: port 0, eth1

lan port: port1, lan1; port2, lan2; port3, lan3; port4, lan4



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3.7.5.1.1 Config port0 (eth1) as the first interface in EasyMesh_openwrt5x.sh

```
prepare_1905_config

if [ $mode = "1" ]; then
      sed -i "1i\lan=$wan_iface $lan_iface" /etc/ethernet_cfg.txt
```

it will generate [lan=**eth1** lan1 lan2 lan3 lan4] to /etc/ethernet_cfg.txt on board

in folder: openwrt\app\wlan_daemon\mapd

3.7.5.1.2 Config vlan setting in EasyMesh_openwrt5x.sh

```
StartMapTurnkey
 set_switch_vlan_gmac2_2_7531
```

change set_switch_vlan to set_switch_vlan_gmac2_2_7531 in EasyMesh_openwrt5x.sh

it will config vlan setting as below:

- forward 1905 multicast to cpu port not flooding
  switch reg w 34 8160816
  switch reg w 4 60
  switch reg w 10 ffffffff
  switch vlan clear

- Set wan port 0 and cpu port 5, vlan id 2
  switch vlan set 1 2 10000100

- wan port set pvid to 2
  switch vlan pvid 0 2
  switch vlan pvid 5 2

in folder: openwrt\app\wlan_daemon\mapd

3.7.5.1.3 Config wan phy port in ethernet_cfg.txt

```
#set the vid of the LAN group, the range is 1-4094, and not repeat with wan_vid
lan_vid=1;

#et the vid of the WAN group, the range is 1-4094, and not repeat with lan_vid
wan_vid=2;

#et the CPU port number, where the switch and CPU are connected.
# The witch-7530 can be selected as 5 or 6.
# The switch-7531 can only set one of 5 and 6, or you can set them at the same time
#cpu_port=6;

#specify external phy port, used as lan port while bridge mode
# panther with one 2.5G phy port, ext_phy_port=6
# panther with two 2.5G phy port, ext_phy_port=5 6
```


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```
ext_phy_port=0
```

In folder: openwrt\app\wlan_daemon\1905daemon

3.7.5.1.4 Change lan port and wan port bit map in eth_ops_dsa.c

```
int get_port_type(OUT unsigned int *lport_map, OUT unsigned int *wport_map)
{
        int ret = ETH_SUCCESS;

        /* GMI <--> 7531 switch port 0 (lan port 1234, wan port 0) */
        *lport_map = 0x0000001E;
        *wport_map = 0x00000001;

        return ret;
}
```

In folder: openwrt\app\wlan_daemon\1905daemon\ethernet\mediatek_DSA

3.7.6 Related features

Below features are related to Variations of switch support, it means if the switch does not support the operations defined by 1905daemon or if the switch configuration is incorrect, all below listed features may not work well.

- Topology information, including non-1905 neighbor, Ethernet 1905 neighbor
- Auto backhaul switching
- Multiple backhaul

3.7.7 Support status

Now in 1905daemon, we have implemented the Ethernet operations for MT7530, MT7531 and switch on MT7628.
If customer needs to implement this part for other vendor's switch, he or she needs to refer to existing solution in ethernet folder in 1905daemon and the 1905daemon Ethernet part porting guide.

3.8 Automatic BH Switching between Ethernet and Wi-Fi

MAP Turnkey provides two variants for BH link switch between Ethernet and Wi-Fi. MAP-Turnkey supports automatic switching between Wi-Fi and Ethernet.
After 2020, the officially released EasyMesh can support automatic Ethernet/Wi-Fi backhaul switching and Ethernet onboarding between two agents.
The flow is as follows:
Switching from Wi-Fi to Ethernet:

1. Agent connects to controller using Wi-Fi BH.
2. Ethernet is plugged between agent and controller.
3. Agent disconnects Wi-Fi BH link and uses Ethernet connection for communication.

Switching from Ethernet to Wi-Fi

1. Initially Agent and Controller are connected via Ethernet link.
2. Ethernet is unplugged between Controller and Agent.
3. Agent detects Ethernet unplug and re-triggers connection on Wi-Fi BH link.

While switching from Ethernet to WiFi, Agent will always attempt to keep the topology unchanged. Hence it will attempt to connect to the controller for three times. If the connection with controller is not restored in three

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attempts, agent can form connection with any other Agent present in the network as per the best AP selection logic.

Note:
The DPP onboarding instance over logical Ethernet instance should take precedence over the one over the Wi-Fi. As a result, we may abort the current DPP onboarding instance over Wi-Fi and start a new one over the Logical Ethernet instance for such a case.

3.8.1.1 Manual BH Switching (without Onboarding)

If the Automatic BH switching is disabled, BH type can be switched manually using Wapp commands. Complete Wi-Fi onboarding (WPS) is not required in this case. Wi-Fi connection will be formed using profile received at the time of previous onboarding.

BH Type Ethernet:

Command:	*wappctrl ra0 map set_bh_type eth*
Behaviour:	Disconnect WiFi BH linkif any. Mark BH type as Ethernet.
Assumption:	Ethernet is already plugged.

BH Type WiFi:

Command:	*wappctrl ra0 map set_bh_type wifi*
Behaviour:	Start connection process on Wi-Fi BH link
Assumption:	Ethernet is already unplugged.

Behaviour at boot-up
If previous WiFi BH profile is available at bootup, WiFi connection on BH will start automatically on Bootup. If BH type Ethernet is marked in middle of the connection or after connection is completed, WiFi BH link will be disabled and Ethernet connection will be made.

Error Handling
If BH type is once configured as Ethernet and onboarding is not complete, use reset_default command before switching back to bh_type_wifi

wappctrl ra0 map reset_default

3.9 DHCP Control

When agents and a controller are in a same LAN, every agent and controller can communication with each other. So every map device should have a same LAN IP segment. When agents onboarding success, it should update IP from the controller. This may need agent open its dhcp client and close its dhcp server. Controler's dhcp server is configured by user. Controller dhcp server default should be enable. When agent disconnected with controller, agent should close dhcp client and enable dhcp server for other device connect in to view agent situation or debug.

3.9.1 Usage

Controller

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- Controller default dhcp server should configured and enable. This dhcp server is used by all agent and device to get its IP.

Agent

- All of the flow agent action is in enable dhcp control situation, if dhcp control is disabled, it's dhcp action is decide by user config in web UI.
- When a agent is power on , dhcp status will flow last configure of dhcp ctl. Then the agent onboarding success or reconnected , it will enable dhcp client to get a ip from the dhcp server which is in a controller. If the agent is disconnected, it will recover its dhcp server for user debug or visit its web ui.

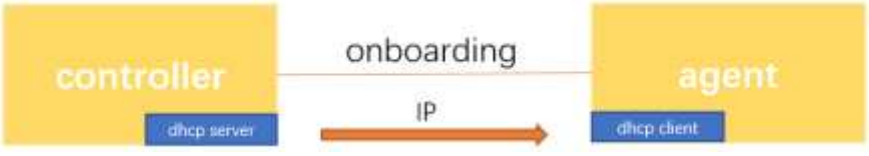
- **FOR PRD1.2(2019-05)**
  Default dhcpctl feature is disable.

  Enable dhcpctl:
  Linux: nvram_set 2860 DhcpCtl 1 , then rerun EasyMesh.h
  OPENWRT: change DhcpCtl=1 in mapd_cfg, then rerun EasyMesh_openwrt.h
  Disable dhcpctl:
  Linux: nvram_set 2860 DhcpCtl 0 , then rerun EasyMesh.h
  OPENWRT: change DhcpCtl=0 in mapd_cfg, then rerun EasyMesh_openwrt.h
- **FOR PRD1.3(2020-01)**
  Default dhcpctl feature is enable. If you want use eth dhcp ctrl feature, you should enable AutoBHSwitching in mapd_default.cfg.

  Enable dhcpctl:
  OPENWRT: change DhcpCtl=1 in mapd_default.cfg, then rerun EasyMesh_openwrt.h
  Disable dhcpctl:
  OPENWRT: change DhcpCtl=0 in mapd_default.cfg, then rerun EasyMesh_openwrt.h
  For UCI supported SKUs following command will be used:
  uci set mapd.mapd_cfg.DhcpCtl='1'
  uci commit mapd
  then rerun EasyMesh_openwrt.h.

Generally, DhcpCtl=1 in mapd_default.cfg During device initialization, so a agent's dhcp server is enable and dhcp client is disable. A controller dhcp server is enable. After they connect, a agent's dhcp server is disable and dhcp client is enable. A controller dhcp server is enable.

1) When agents onboarding with a controller, its dhcp client is enable and its dhcp server disable. Controller 's server is enable. Agents update IP from the controller.



2) When a agent disconnecting with a controller, its dhcp client is disable and its dhcp server enable automatically, Controller 's server is enable. a agent will recover its dhcp server for user debug or visit its web ui.

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According to the following setting, make sure a agent can get IP from controller

- Enter command on the serial port: vi etc/map/mapd_default.cfg , then change DhcpCtl = 1 in a controller
- Reboot controller
- Enter command in on the serial port: vi etc/map/mapd_default.cfg , then change DhcpCtl = 1 in a agent
- Reboot agent

After the below setting, before a agent onboarding success with a controller, the agent's IP is "192.168.1.1"

```
br-lan    Link encap:Ethernet  HWaddr 2E:F3:AE:C2:48:F4
          inet addr:192.168.1.1  Bcast:192.168.1.255  Mask:255.255.255.0
          inet6 addr: fd24:8248:dbe1::1/60 Scope:Global
          inet6 addr: fe80::2cf3:aeff:fec2:48f4/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:2862 errors:0 dropped:0 overruns:0 frame:0
          TX packets:1188 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:450274 (439.7 KiB)  TX bytes:186441 (182.0 KiB)
```

After onboarding, the agent's IP become "192.168.1.224"

```
br-lan    Link encap:Ethernet  HWaddr C6:25:93:63:6B:1F
          inet addr:192.168.1.224  Bcast:192.168.1.255  Mask:255.255.255.0
          inet6 addr: fe80::c425:93ff:fe63:6b1f/64 Scope:Link
          inet6 addr: fde6:10bd:225d::1/60 Scope:Global
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:1603 errors:0 dropped:0 overruns:0 frame:0
          TX packets:322 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:195938 (191.3 KiB)  TX bytes:72936 (71.2 KiB)
```

3.10 Tx/Rx higher layer protocol data

MAP control messages can be used to carry higher layer protocol data over 1905 abstraction layer. A map agent/controller can send/receive higher layer protocol to/from peer map device using this generic mechanism. This feature can be used to send any data/payload specific to customer application over the 1905 abstraction layer. The data will be encapsulated inside a 1905 data frame and sent to the destination ALMAC. At the destination the data can be parsed as per customer defined use case.
The corresponding data messages are send unicast only.

3.10.1 Data flow

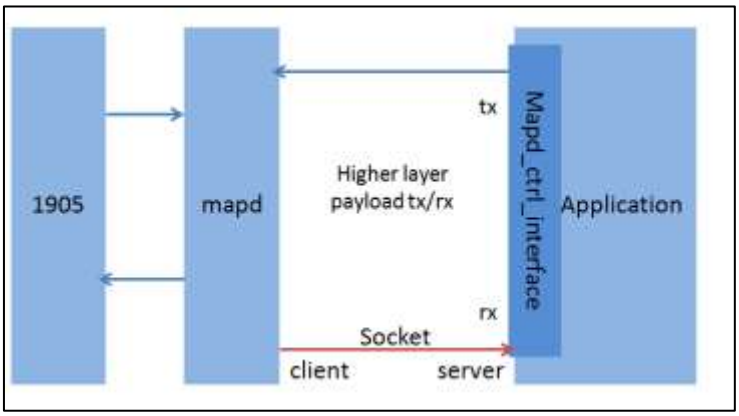


Figure 3- 62. Higher Layer Data flow daemon interaction

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Tx:

- Application uses lib mapd_ctrl_interace to pass higher layer protocol data to mapd
- Mapd ctrl interface passes data payload to 1905 interface and calls _1905_Send_Higher_Layer_Data_Message function.
- 1905 transport payload buffer and protocol value to peer 1905 entity.

Rx:

- Application needs to register a socket interface with mapd interface and implement a mainloop to wait for socket read.
- Once local 1905 receives higher layer protocol data, it generates event _1905_RECV_HIGHER_LAYER_DATA_EVENT to mapd interface.
- Mapd interface then writes this data buffer to socket registered by application to pass this opaque payload data.
- The buffer passed to application is one byte of protocol number followed by actual payload.

3.10.2 Usage

Data TX:
Below mapd_cli command can be used to Tx higher layer data. Payload format depends on customer use case. So MAPD only provides a mechanism (mapdcli API) for customer to use this existing 1905 procedure to transfer higher layer data , encapsulated as a 1905 packet.
Format :
mapd_cli /tmp/mapd_ctrl tx_higher_layer_data <ALMAC> <Protocol> <Payload_len> <Payload>
Example :
mapd_cli /tmp/mapd_ctrl tx_higher_layer_data 00:0c:43:26:60:5a 1 10 aabbccddee

Data RX :
No command is needed, customer can make their own rx packet parser based on use case.
For demo purpose, RX is implemented in application mapd_user_iface. To notify the application about incoming RX data, a new event HIGHER_LAYER_PAYLOAD is implemented in mapd.
Example:
mapd_user_iface /tmp/mapd_ctrl

3.11 WPS Unconfig feature Support

This feature will allow STA or third-party device(Window 10 NB) to setup MAP network's SSID and PSK. In WindowNB from Network and sharing center, we can setup a network.
Network and sharing center -> setup a new connection or network -> setup a new network

In general, when use WPS, AP act as registrar while STA/NB act as enrollee and AP shares the credential while WPS exchanges to configure the STA/NB. But when WPS unconfig feature used, forces AP to act as enrollee and STA/NB act as registrar. STA/NB will share the credential like SSID, PSK to AP while doing WPS exchanges. AP will configure itself as per configuration shared by STA/NB.
In MAP when AP receives credential from STA/NB, this information shared to Wapd through event. Wapp deamon will update wts_bss_config_info with new credential according to AP's index and also send an event to MAP deamon to trigger mapd renew which will sync the BSS of all the devices present on network as per updated credential.

3.11.1 Procedure

1. Set device Role as controller and execute below commands to configure AP in unconfig mode and get pincode:
   iwpriv rax1 set WscConfMode=1

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iwpriv rax1 set WscConfStatus=1
iwpriv rax1 set WscMode=1
iwpriv rax1 set debug=4; iwpriv rax1 set WscGenPinCode=1; iwpriv rax1 set debug=0
iwpriv rax1 set WscGetConf=1

```
root@OpenWrt:/# iwpriv rax1 set WscConfMode=1
root@OpenWrt:/# iwpriv rax1 set WscConfStatus=1
root@OpenWrt:/# iwpriv rax1 set WscMode=1
root@OpenWrt:/# iwpriv rax1 set debug=4; iwpriv rax1 set WscGenPinCode=1; iwpriv
 rax1 set debug=0
[  590.054579] Set_Debug_Proc(): (DebugLevel = 4)
[  590.059958] IF(ra3) Set_WscGenPinCode_Proc:: This command is from ra interface now.
[  590.068036] Set_WscGenPinCode_Proc:: Enrollee PinCode                17312806
[  590.075000] Set_Debug_Proc(): (DebugLevel = 0)
root@OpenWrt:/# iwpriv rax1 set WscGetConf=1
```

Figure 3- 63. Pin Generation

2. Use WIN10 NB or any STA, follow the below steps to configure the AP:
Network and sharing center -> setup a new connection or network -> setup a new network

2.1 Choose **“RalinkAPS_X”**

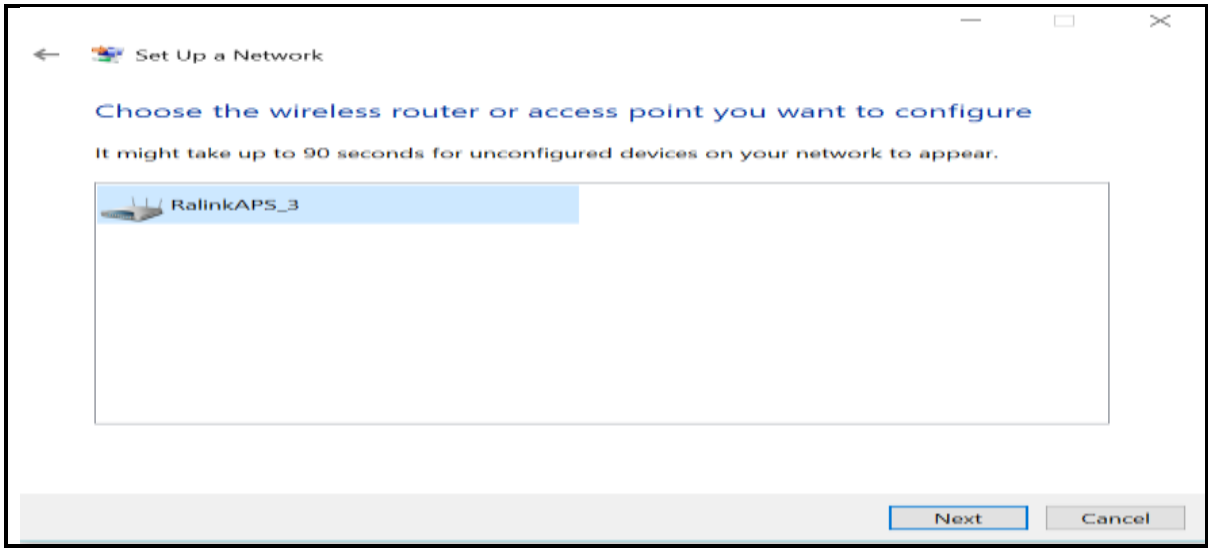


Figure 3- 64. Select AP to configure

2.2 Enter PIN

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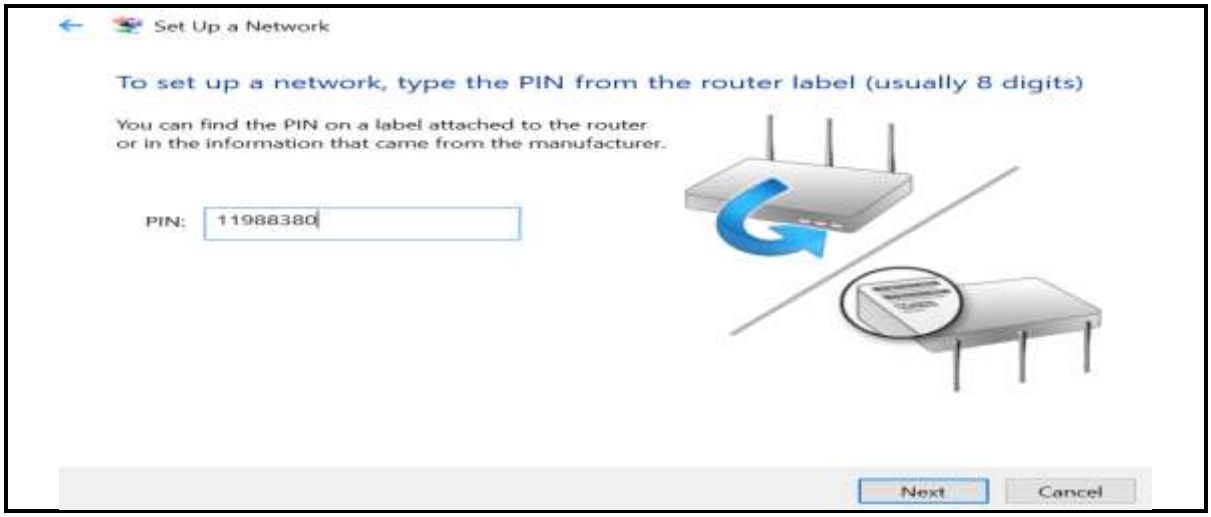


Figure 3- 65. Enter the PIN generated on AP

2.3 Name the SSID

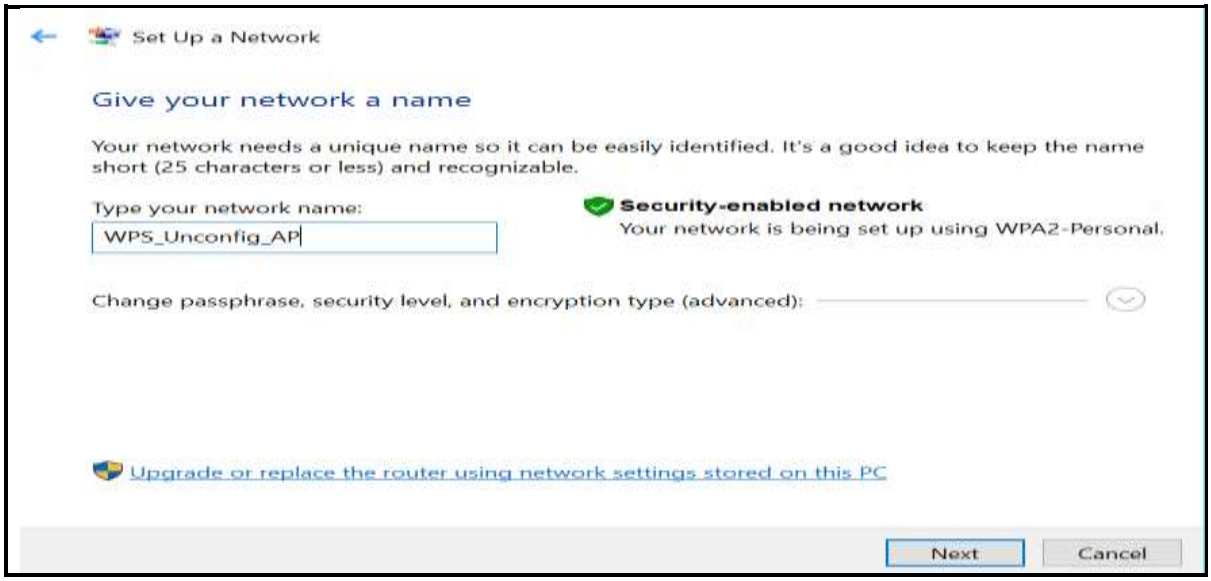


Figure 3- 66. Name the SSID to configure on AP

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2.4, Configuration successful



Figure 3- 67 . Configured the Key & SSID

3. Confirm configuration update in wts_bss_config_info and AP's .dat file

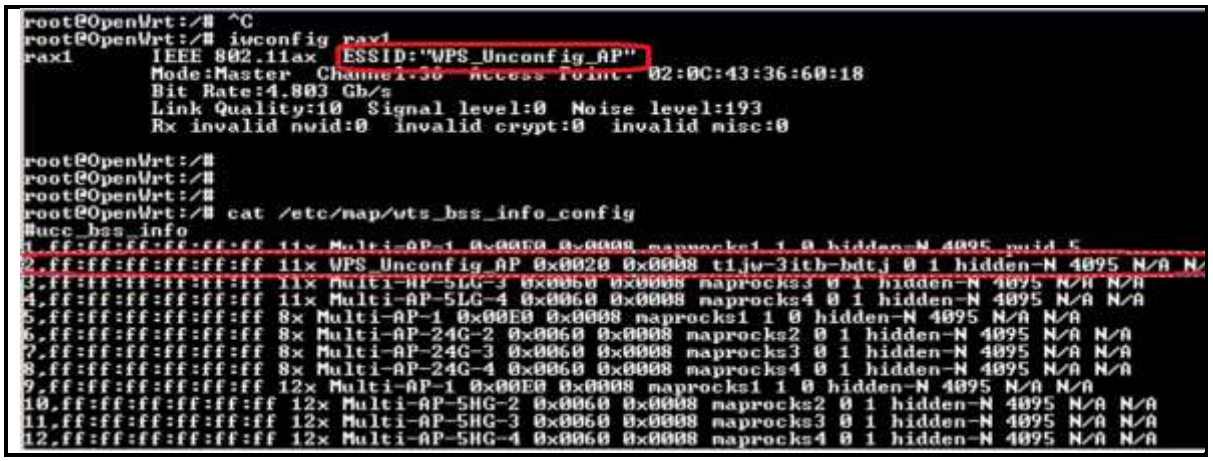


Figure 3- 68. Controller's BSS & WTS_BSS_Config_Info file updated

3.12 Mesh Spatial Reuse

3.12.1 Introduction

Spatial reuse operation addressed in Wi-Fi 6 is to allow the medium to be reused more often between OBSSs in dense deployment scenarios by the early identification of signals from overlapping basic service sets (OBSSs) and interference management.

Currently SR operation is verified only on AX4200 Panther platform.

However, Wi-Fi 6 Spatial Reuse operation can't guarantee the existing link performance while performing parallel transmission, which result in Spatial Reuse is not applicable under Mesh Scenario.
In this section MTK Mesh Spatial Reuse has 2 phases, 1st phase is SRG protect mode, where Mesh APs are prohibited to Spatial Reuse internal traffic. If criteria have met, it can operate as 2nd phase, which is MTK Mesh SR, allow 2 APs' fronthaul traffic open for Spatial Reuse.

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The phase switch between No-SR, SRG Protect and MTK Mesh SR mode will be automatically changed in FW layer according to information brings by MAPD.

3.12.2 Operating Mode

From EasyMesh Turnkey WebGui, select “Mesh SR” on both Controller and Agents to enable this feature.

Table 3- 4. Mesh SR Mode Lists

Mesh SR Mode	SR Available	SR OBSS Traffic	SR Internal Fronthaul
Mesh No SR	N	N	N
Mesh SRG Protect	Y	Y	N
Mesh Spatial Reuse	Y	Y	Y

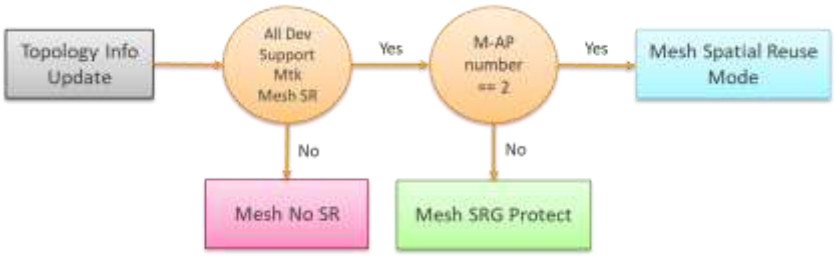


Figure 3- 69. Mesh Mode Switch Criteria

3.12.3 Mesh SRG Protect Mode

While spatial reuse is good for single AP scenario, the reuse behavior may damage the MAP internal traffic when applying to the mesh network. As a result, we would need to configure the SRG info in agents for not reusing the MAP internal traffic but reusing OBSS one.



Figure 3- 70. Mesh SRG SR (not reuse MAP internal traffic, able to reuse OBSS Traffic)

Given that SR group is grouped by BSS color and Partial BSSID, we can have better throughput performance in a dense environment by using different PD thresholds (ex. -110 for SRG and -70 for non-SRG).

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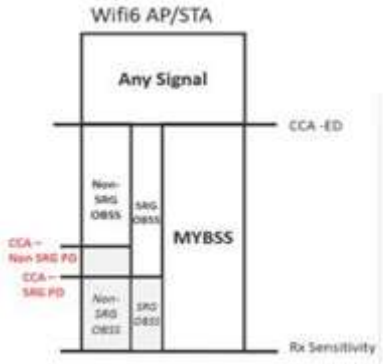


Figure 3- 71. SRG/non-SRG PD threshold

- Upper layer implementation
  1. MAP agent collects BSS Color in mesh
  2. MAP agent reports BSS Color
  3. MAP controller delivers all BSS color to all MAP agents

     The MAP Controller shall send Channel Selection Request message containing one or more Spatial Reuse Request TLVs in order to refresh the SRG info.
     However, the MAP Controller shall clear the “SRG BSS Color bitmap” as 0x0 if one of the MAP Agent does not support “Mesh SRG”
  4. MAP agent sets SRG info - BSS color & partial BSSID to HW
  5. Whenever BSS color collision happens or changes, go to step 2

     MAP agent shall send an unsolicited Operating Channel Report message to the MAP Controller if it autonomously changes the BSS Color of a BSS.
  6. Whenever mesh topology changes, go to step 3

- “SRG BSS Color bitmap” update example
  - For Controller
    1. Whenever Controller driver/fw receives “color change event”, it shall update new self-bitmap to daemon
    2. Whenever Controller driver/fw receives “association/de-association” from STA, it shall update new self-bitmap to daemon
    3. Controller daemon records the bitmap change in Controller’s data structure.
    4. Controller updates new Mesh srg bitmap to Agent1 and Agent2 daemon respectively.
    5. Both Agent1 and Agent2 daemons send “SRG info update” with new Mesh srg bitmap to their own driver/fw
    6. Controller daemon sends “SRG info update” with new Mesh srg bitmap to its driver/fw

  - For Agent
    1. Whenever Agent1 driver/fw receives “color change event”, it shall update new self-bitmap to daemon
    2. Whenever Agent1 driver/fw receives “association/de-association” from STA, it shall update new self-bitmap to daemon
    3. Agent1 daemon sends 1905 msg to controller to update Agent1’s self bitmap
    4. Controller daemon records the bitmap change in Agent1’s data structure.

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5. Controller updates new Mesh srg bitmap to Agent1 and Agent2 daemon respectively.
6. Both Agent1 and Agent2 daemon send “SRG info update” with new Mesh srg bitmap to their own driver/fw
7. Controller daemon sends “SRG info update” with new Mesh srg bitmap to its driver/fw

- Spatial Reuse TLVs defined in MAP r4 draft

  Given that a MAP Agent sets the Spatial Reuse bit to 1 in its AP Wi-Fi 6 Capabilities TLV or other explicit way, the MAP Controller shall start the Spatial Reuse Request procedure to collect and configure the corresponding setting via “Channel Selection Request”, “Channel Selection Response” and Operating Channel Report”.

  Since the BSS Color specified in the Spatial Reuse related TLVs is per RUID basis. As a result, Easymesh script should configure the BSS color usage as per radio one in the profile configuration in advance (ex. 254 for MTK's design).

3.12.4 Mesh Spatial Reuse

While “Mesh SRG Protect Mode” targets the goal for not reusing internal mesh traffic, here we introduce an advanced variant named “Mesh Spatial Reuse” which is built on the basis of “Mesh SRG Protect Mode”. With “Mesh Spatial Reuse”, we can fix the problem that total mesh throughput is limited by airtime sharing among MAP nodes and OBSS CCI interference by gaining additional airtime. From the Proof of Concept, 20% or more total throughput gain can be found.

“Mesh Spatial Reuse” is an exception case of " Mesh SRG Protect Mode ", so they don’t co-work with each other. Here is a brief summary for their effective scopes accordingly.

1. If limitation of “Mesh Spatial Reuse” failed to meet, it will fall back to “Mesh SRG Protect Mode”
2. “Mesh SRG Protect Mode” is targeted for “Reuse OBSS HE PPDU only”
3. “Mesh Spatial Reuse” is targeted for “Reuse OBSS HE”, “OBSS non-HE” and “Internal HE Downlink” traffic.

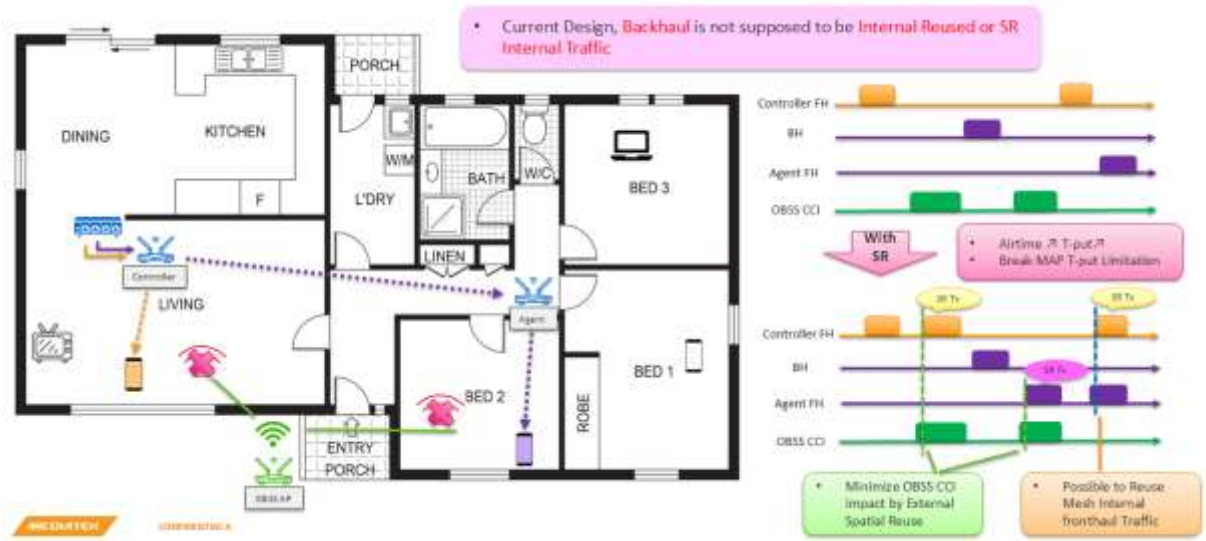


Figure 3- 72. Mesh Spatial Reuse – reuse mesh internal fronthaul and external traffic

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However, to minimize the negative impact accompanied with mesh internal SR, effective SR scope needs to be restricted under certain scenarios. Currently the strategy only applies to the mesh topology with 2 nodes (i.e. 1 MAP controller and 1 agent).

- Limited scenarios for “Mesh Spatial Reuse”
  - Mandatory topology
    - Only 2 AP nodes in mesh network: 1 Controller, 1 Agent
    - All AP nodes are MTK insides support MTK Mesh SR
  - Dynamic control
    - Internal SR HE Downlink PPDU from near-end clients only

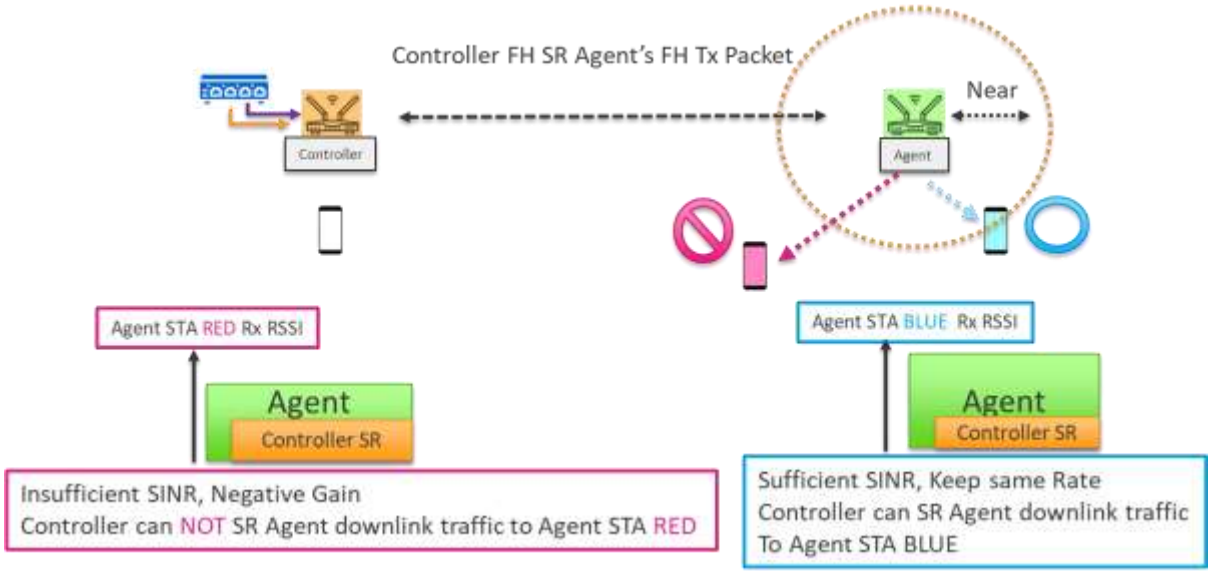


Figure 3- 73. Limited scenario of “Mesh internal SR on only near-end STAs”

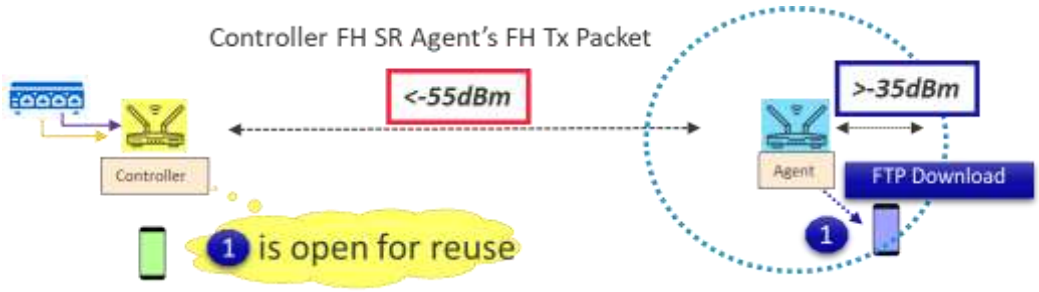


Figure 3- 74. RSSI limitation of “Mesh internal SR on only near-end STAs”

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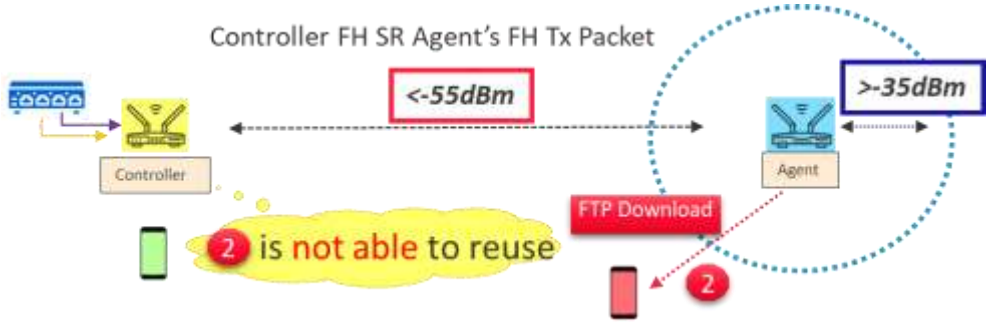


Figure 3- 75. RSSI limitation of "Mesh internal SR not allowed on far-end STAs"

MAP Agents shall send an unsolicited Operating Channel Report message to the MAP Controller if it autonomously changes the BSS Color or BSSID of a BSS. The reported information shall include "SRG BSS Color Bitmap" and "SRG Partial BSSID Bitmap".

On the other hand, a MAP Agent may receive a Channel Selection Request message from the Multi-AP Controller containing one or more Spatial Reuse Request TLVs. For each RUID specified in a Spatial Reuse Request TLV, it shall operate every BSSs and STAs according to the Spatial Reuse Configuration specified in the Spatial Reuse Request TLV, such as "SRG BSS Color Bitmap" and "SRG Partial BSSID Bitmap"

3.12.5 Profile Configuration Related to Mesh SR

Table 3-5. Mesh SR Profile Configuration

Profile	Description
SRDPDEnable = 1	To Enable Dynamic OBSS PD to Increase SR reuse range stronger than ED threshold.
SRMode=1	0: SR in Single AP Mode 1: SR in Mesh Mode
SREnable=1	0: No SR 1: Enable SR

3.12.6 Commands for Debugging

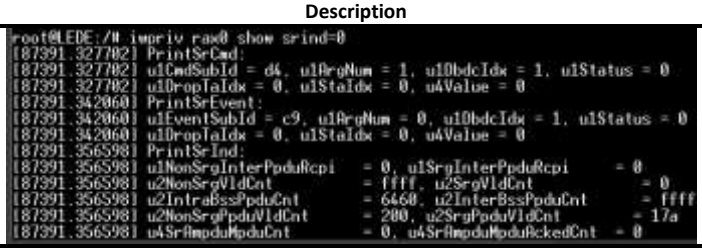
Table 3-6. Mesh SR Debug Commands

Commands	Description
iwpriv <interface> show srmeshtopo=0	Display below 3 parameters UINT_8 map_dev_count; UINT_8 map_dev_sr_support_mode UINT_8 self_role
iwpriv <interface> show srmeshphase=0	Display Mesh SR Phase Decision 0: Mesh No SR

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<table>
<tr><th>Commands</th><th>Description</th></tr>
<tr><td></td><td>1: Mesh SRG Protect
3: Mesh SR</td></tr>
<tr><td>iwpriv <interface> show srmeshremotefhrssi=0</td><td>Display Fronthaul RSSI measurement result
Rssi and FHStat
● Rssi : Signal Level from counterpart AP dbm
● FHStat:<table><tr><th>Status</th><th>Value</th></tr><tr><td>Invalid BSSID</td><td>0</td></tr><tr><td>SCAN SUCCESS</td><td>1</td></tr><tr><td>SCAN FAIL</td><td>2</td></tr></table></td></tr>
<tr><td>iwpriv <interface> show srmeshremotebhinfo=0</td><td>Display Backhaul Associated WCID and Associated Type
● RemoteBhWcid:<table><tr><th>Description</th><th>Value</th></tr><tr><td>Invalid WCID</td><td>0</td></tr><tr><td>Valid WCID</td><td>else</td></tr></table>● RemoteBhType:<table><tr><th>Description</th><th>Value</th></tr><tr><td>Not found wireless BH connection in this radio</td><td>0</td></tr><tr><td>Wireless BH connection in this radio</td><td>1</td></tr></table></td></tr>
<tr><td>iwpriv <interface> show srsrgbitmap=0</td><td>Display SRG Protect Bitmap Information with
Color-31_0, 63_32
pBssid- 31_0, 63_32
in hex format</td></tr>
<tr><td>iwpriv <interface> show srmeshsrgbitmap=0</td><td>Display Virtual SRG Bitmap Information with
Color-31_0, 63_32
pBssid- 31_0, 63_32
in hex format</td></tr>
<tr><td>iwpriv <interface> show srind=0</td><td>Display SR available counter</td></tr>
</table>

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Commands	Description
	

```
root@LEDE:/# iwpriv ra0 show srind=0
[87391.327702] PrintSrCmd:
[87391.327702] u1CmdSubId = d4, u1ArgNum = 1, u10bdcIdx = 1, u1Status = 0
[87391.327702] u1DropTaIdx = 0, u1StaIdx = 0, u4Value = 0
[87391.342060] PrintSrEvent:
[87391.342060] u1EventSubId = c9, u1ArgNum = 0, u10bdcIdx = 1, u1Status = 0
[87391.342060] u1DropTaIdx = 0, u1StaIdx = 0, u4Value = 0
[87391.356598] PrintSrInd:
[87391.356598] u1NonSrgInterPpduRcpi    = 0, u1SrgInterPpduRcpi     = 0
[87391.356598] u2NonSrgVldCnt           = ffff, u2SrgVldCnt         = 0
[87391.356598] u2IntraBssPpduCnt        = 6460, u2InterBssPpduCnt   = ffff
[87391.356598] u2NonSrgPpduVldCnt       = 200, u2SrgPpduVldCnt      = 17a
[87391.356598] u4SrAmpduMpduCnt         = 0, u4SrAmpduMpduAckedCnt  = 0
```

3.13 Single Band Operation

Easymesh operations can be performed with a single active wifi band and the other wifi band in disabled state or both the wifi bands in disabled state with backhaul operating on ethernet.

The following steps can be used to Disable a particular band,

1) Ifconfig “interface” down [For all the interfaces of that particular band]
2) Run “EasyMesh_openwrt.sh” on DUT

The interfaces can be enabled again by using the following steps

1) Ifconfig “interface” up [For all the interfaces of that particular band]
2) Run “EasyMesh_openwrt.sh” on DUT

3.13.1 Limitation

Dfs related functionalities are not expected to work with this feature, due to the following reasons.

1) If 2.4G band is disabled and next best channel is a DFS channel instatenious backhaul switching will not take place as the device would be performing CAC on that particular channel.

3.14 MLD Dynamic Regroup

For an EHT AP with Wifi 7 support all BSS must be affiliated with a MLD group. This can be a Single link MLD group or multi link MLD group, depending on the usecase. We can run time dynamically update the BSS MLD group settings by the help of this feature. This feature is only supported in MLD capable SKU like Eagle and above. The device wifi driver has supporting APIs which can help in BSS Group ID update dynamically.

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Types of MLD groups

The procedure for dynamic regroup is as below:

- By default all the BSS are affiliated with a single MLD groups. All have unique MLD group IDs (for single MLD groups the driver assignes an MLD group from the range 17 to 48). This is before the Easymesh daemons startup.
- Then 1905daemon reads the wts_bss_info_config file (See fig 3.79) for a new parameter the mld_groupID and based on this send some info mainly the number of AP MLD groups and number of affiliated APs in each MLD group to MAPD, MAPD just forwards this event to WAPP where this information is used to call driver APIs to create new mld group or transfer existing mld group, based on the user defined mld configuration. See Figure 3.77.
- User can dynamically update the mld group settings by using the web UI config BSS page present at the controller. Then via the Multi-AP renew feature this setting can be applied. See Figure 3.78.
- Then mld configuration from controller to agent is applied by the help of the MAP R6 spec defined TLV MLO Agent BSS Configuration TLV.
- The MLD grouping has been successful or not can be checked by the generic driver command **" iwpriv ra0 show bssmngr".** See Figure 3.78.

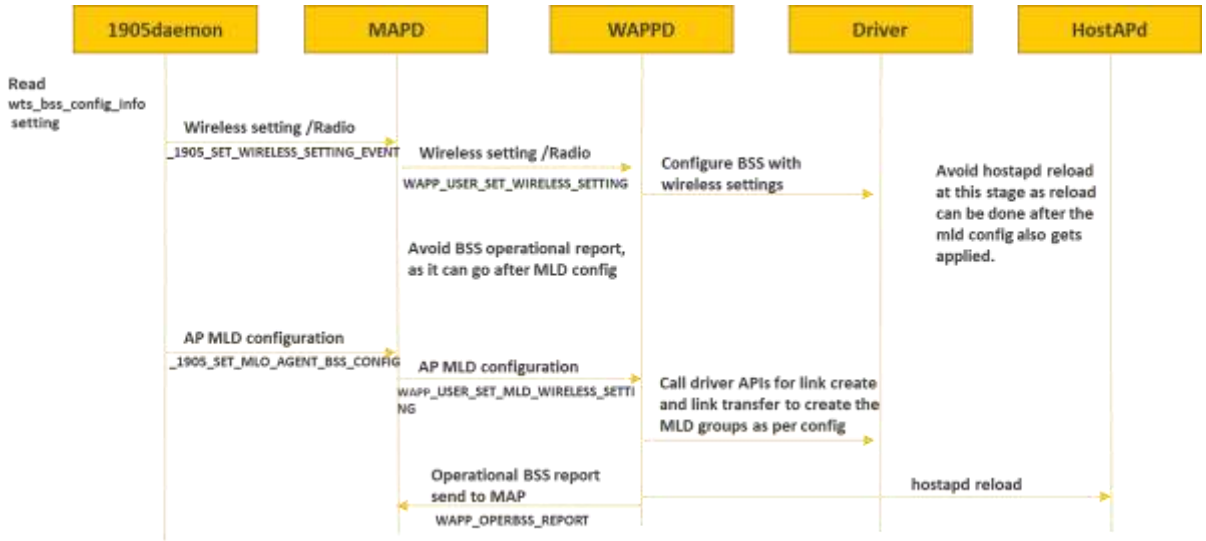


Figure 3- 76. Daemon event interaction diagram for MLD Reconfig/Regroup

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Figure 3- 77. Default MLD grouping with 4 BSS on each radio

Then following wts_bss_info_config file will help to config the mld groups as per the figure 3-78. In this file the triband 11x, 8x and 13x radios are operational. It is made in order to group the backhauls of all bands together , and even the fronthauls of all bands together.

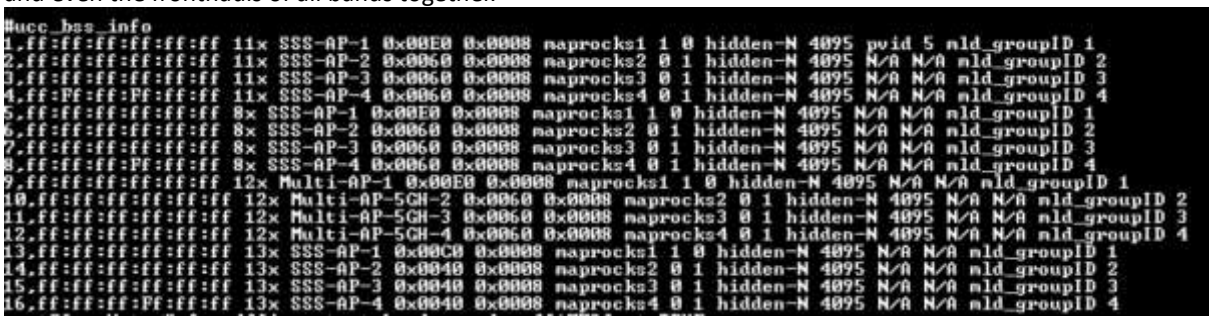


Figure 3- 77. Sample Default wts_bss_info_config file

- **It is important to note that:**
  - When different BSS are configured to be a part of the same 'Multi' MLD group, then SSID, PASSWORD, Auth type, Encrypt type, BH setting, FH setting, mld_groupID <Idx> setting for all these BSS must be kept same.
  - In one band "Idx" value cannot repeat, they should be unique for all BSS in single band.
  - For 'Multi' type MLD groups the MLD group Index can lie from range between 1 to 16.
  - For 'Single' type MLD groups the MLD group Index can lie from range between 17 to 48.
  - When regroup is done then the hostapd reload happens at all mesh devices, so all stations (including agents) will disconnect and connect back again.

3.14.1 Configuring BSS for MLO regroup

For WIFI7 (802.11be) device which supports Multi link operation (MLO) connection we will get the default MLO groups as per the example above. For changing the group dynamically, we can use the web UI new option

Please refer this path in controller WebUI: MTK->Easymesh->Advanced->Configure BSS

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After making the grouping setting as per the important notes in above section, press save and apply.

Example :

- **Typical example of Multi link MLD groups involving all BSSs of all radios**
  - 3 radio bands -- 4 BSS on each band
  - 4 MLD groups -- 3 links in each group

```
iwpriv ra0 show bssmngr
[  175.635982] 7990@ERROR.MLO,ra0,show_bss_mngr_info() 31404: type=0
[  175.642083]
[  175.642083] bss_mngr_con_info_show: type = 0
[  175.647823] bss_mngr_con_info_show: MLO Device, (65)
[  175.652780]  - MLD[01] (active), bmgr_mlo_dev ptr = 00000000574e7f2e
[  175.659167]          MLD Addr:(01:0c:43:26:11:90) Type:(Multi) link_cnt:(3)
[  175.665528]          - Link0: [00] BSS0 2G
[  175.668323]          - Link1: [04] BSS0 5G
[  175.671109]          - Link2: [08] BSS0 6G
[  175.673909]  - MLD[02] (active), bmgr_mlo_dev ptr = 00000000a21a3bc2
[  175.680251]          MLD Addr:(02:0c:43:16:11:90) Type:(Multi) link_cnt:(3)
[  175.686642]          - Link0: [01] BSS1 2G
[  175.689427]          - Link1: [05] BSS1 5G
[  175.692208]          - Link2: [09] BSS1 6G
[  175.694993]  - MLD[03] (active), bmgr_mlo_dev ptr = 00000000023cc084
[  175.701366]          MLD Addr:(03:0c:43:26:11:90) Type:(Multi) link_cnt:(3)
[  175.707722]          - Link0: [02] BSS2 2G
[  175.710529]          - Link1: [06] BSS2 5G
[  175.713313]          - Link2: [10] BSS2 6G
[  175.716095]  - MLD[04] (active), bmgr_mlo_dev ptr = 00000000dcf68b51
[  175.722454]          MLD Addr:(04:0c:43:36:11:90) Type:(Multi) link_cnt:(3)
[  175.728797]          - Link0: [03] BSS3 2G
[  175.731581]          - Link1: [07] BSS3 5G
[  175.734365]          - Link2: [11] BSS3 6G
[  175.737142]
```

Figure 3- 78. Example of the MLO groups

Example move Bss1 of 6G band from MLD group 2 and make it single MLD group

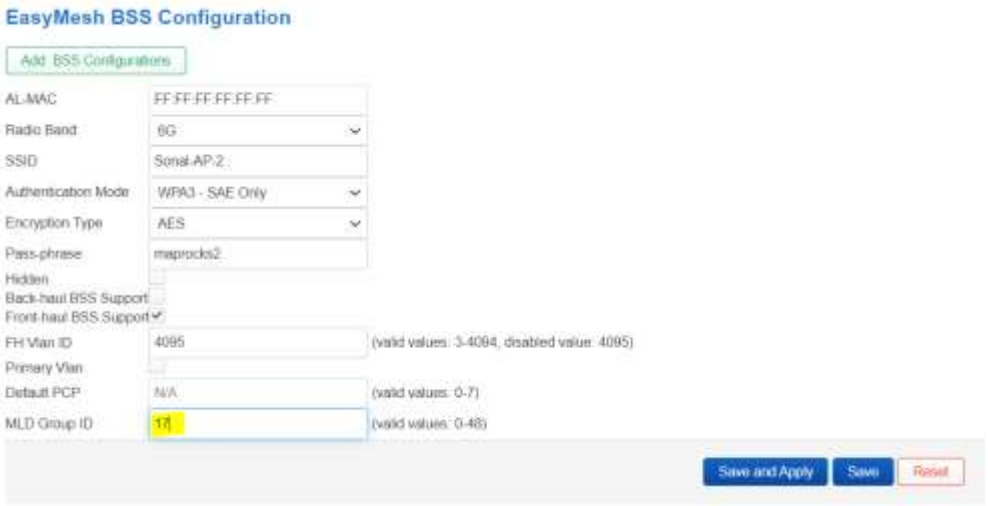


Figure 3- 79. Example how to modify the group

Here we are going to make single MLD group for BSS1 of 6G band. *You can do save and apply to do config renew. This will affectively send out a "renew" command to the controller and modify the wts_bss_info_config as below:*

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```
root@OpenWrt:~# cat /etc/map/wts_bss_info_config
#ucc_bss_info
1,ff:ff:ff:ff:ff:ff 11x SSS-AP-1 0x00E0 0x0008 naprocks1 1 0 hidden-N 4095 pvid 5 mld_groupID 1
2,ff:ff:ff:ff:ff:ff 11x SSS-AP-2 0x0060 0x0008 naprocks2 0 1 hidden-N 4095 N/A N/A mld_groupID 2
3,ff:ff:ff:ff:ff:ff 11x SSS-AP-3 0x0060 0x0008 naprocks3 0 1 hidden-N 4095 N/A N/A mld_groupID 3
4,ff:ff:ff:ff:ff:ff 11x SSS-AP-4 0x0060 0x0008 naprocks4 0 1 hidden-N 4095 N/A N/A mld_groupID 4
5,ff:ff:ff:ff:ff:ff 8x SSS-AP-1 0x00E0 0x0008 naprocks1 1 0 hidden-N 4095 N/A N/A mld_groupID 1
6,ff:ff:ff:ff:ff:ff 8x SSS-AP-2 0x0060 0x0008 naprocks2 0 1 hidden-N 4095 N/A N/A mld_groupID 2
7,ff:ff:ff:ff:ff:ff 8x SSS-AP-3 0x0060 0x0008 naprocks3 0 1 hidden-N 4095 N/A N/A mld_groupID 3
8,ff:ff:ff:ff:ff:ff 8x SSS-AP-4 0x0060 0x0008 naprocks4 0 1 hidden-N 4095 N/A N/A mld_groupID 4
9,ff:ff:ff:ff:ff:ff 12x Multi-AP-1 0x00E0 0x0008 naprocks1 1 0 hidden-N 4095 N/A N/A mld_groupID 1
10,ff:ff:ff:ff:ff:ff 12x Multi-AP-5GH-2 0x0060 0x0008 naprocks2 0 1 hidden-N 4095 N/A N/A mld_groupID 2
11,ff:ff:ff:ff:ff:ff 12x Multi-AP-5GH-3 0x0060 0x0008 naprocks3 0 1 hidden-N 4095 N/A N/A mld_groupID 3
12,ff:ff:ff:ff:ff:ff 12x Multi-AP-5GH-4 0x0060 0x0008 naprocks4 0 1 hidden-N 4095 N/A N/A mld_groupID 4
13,ff:ff:ff:ff:ff:ff 13x SSS-AP-1 0x00C0 0x0008 naprocks1 1 0 hidden-N 4095 N/A N/A mld_groupID 1
14,ff:ff:ff:ff:ff:ff 13x SSS-AP-2 0x0040 0x0008 naprocks2 0 1 hidden-N 4095 N/A N/A mld_groupID 17
15,ff:ff:ff:ff:ff:ff 13x SSS-AP-3 0x0040 0x0008 naprocks3 0 1 hidden-N 4095 N/A N/A mld_groupID 3
16,ff:ff:ff:FF:ff:ff 13x SSS-AP-4 0x0040 0x0008 naprocks4 0 1 hidden-N 4095 N/A N/A mld_groupID 4
```

Figure 3- 80. Modified wts_bss_info_config_file for BSS1 of 6G band

After renew the Bss1 of 6G will leave the Multi group 2 and form a single MLD group 17.

```
root@OpenWrt:/# iwpriv ra0 show bssmngr
[ 2887.847035] 7990@ERROR.MLO,ra0,show_bss_mngr_info() 31404: type=0
[ 2887.853140]
[ 2887.853140] bss_mngr_con_info_show: type = 0
[ 2887.858880] bss_mngr_con_info_show: MLO Device, (65)
[ 2887.863837]   - MLD[01] (active), bmgr_mlo_dev ptr = 00000000574e7f2e
[ 2887.870180]         MLD Addr:(01:0c:43:26:11:90) Type:(Multi) link_cnt:(3)
[ 2887.876524]         - Link0: [00]
[ 2887.879311]         - Link1: [04]
[ 2887.882096]         - Link2: [08]
[ 2887.884878]   - MLD[02] (active), bmgr_mlo_dev ptr = 00000000a21a3bc2
[ 2887.891220]         MLD Addr:(02:0c:43:16:11:90) Type:(Multi) link_cnt:(2)
[ 2887.897563]         - Link0: [01]
[ 2887.900343]         - Link1: [05]
[ 2887.903127]   - MLD[03] (active), bmgr_mlo_dev ptr = 00000000023cc084
[ 2887.909469]         MLD Addr:(03:0c:43:26:11:90) Type:(Multi) link_cnt:(3)
[ 2887.915809]         - Link0: [02]
[ 2887.918593]         - Link1: [06]
[ 2887.921380]         - Link2: [10]
[ 2887.924161]   - MLD[04] (active), bmgr_mlo_dev ptr = 00000000dcf68b51
[ 2887.930502]         MLD Addr:(04:0c:43:36:11:90) Type:(Multi) link_cnt:(3)
[ 2887.936844]         - Link0: [03]
[ 2887.939628]         - Link1: [07]
[ 2887.942415]         - Link2: [11]
[ 2887.945202]   - MLD[17] (active), bmgr_mlo_dev ptr = 00000000141902e0
[ 2887.951544]         MLD Addr:(02:0c:43:16:33:90) Type:(Single) link_cnt:(1)
[ 2887.957971]         - Link2: [09]
[ 2887.960753]
```

Figure 3- 81. Single Group made for BSS 1 of 6G band

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4 MAP Network Maintenance

4.1 Introduction

All the Turnkey features for network maintenance and stability form a part of this chapter.
Mainly they are:

1. **Network Healing:** which aims to restore network connectivity
2. **Network optimization:** which is a mechanism to automatically form the best topology based on the current conditions.
3. **BH link steering and prioritizations:** which gives flexibility to the user to give more priority to a particular BH.
4. **Loop detection and avoidance:** which is introduced to solve circle topology

4.2 Network Healing

Network healing feature aims to restore network connectivity if one of the Agents goes down or backhaul link is broken.
If backhaul link of an Agent is broken either due to its upstream Multi-AP device going down or due to any other reason, then the Agent will try to search for other Multi-AP devices in the network to which it can connect.
Mediatek solution for network healing provides an efficient mechanism to restore network connectivity with minimum disruptions.
The link failover feature developed by Mediatek considers the following parameters for making new connection:

- Best Uplink AP selection based on RSSI, number of hops, uplink rate etc.
- Avoid Repeater to Repeater looping.

To achieve this, Mediatek Multi-AP devices add a vendor specific IE in the beacon and probe response to show:

- Distance from root
- Connectivity to controller

Based on these parameters, the agent tries to connect to the best available Agent in the network.

Steps followed by the Multi-AP device for network healing:

- Multi-AP device gets a trigger of backhaul link disconnected
- It will trigger a scan to find out the backhaul bss available in the network
- Based on the scan results and received RSSI of the probe response/beacons of the backhaul bss, Multi-AP device will find the best candidate backhaul BSS to connect to using a proprietary method.
- After it has found the best backhaul bss, it will attempt connection.
- If connection fail, it will move to the next best bss
- If connection success, then it will update the vendor specific IE and data will be resumed.

Note: In case of 3rd Party device, Mediatek Multi-AP devices will not consider uplink rate as there will be no Mediatek MAP vendor specific IE in the beacon and probe response. So in that case the best uplink ap selection will not consider uplink rate.

For MAP R3 device, in network healing if DPP connection has to be formed then first DPP Network Introduction Protocol is done and then bSTA connection is formed.

4.2.1 Procedure

If original topology was as a daisy chain by default and due to some reason agent 1 got powered off, then agent 2 will realize this firstly by receiving a backhaul link disconnection indication, then agent 2 will use network

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healing (link failover feature) to find its way back in the network, by selecting Best Uplink AP based on various factors as mentioned.

This can be confirmed from:

1. The before and after Map network topology as described in image below.
2. It can also be checked from Agent 2's iwconfig command, BSSID of apclii0 at agent 2 will be changed after agent 1 power off is done.

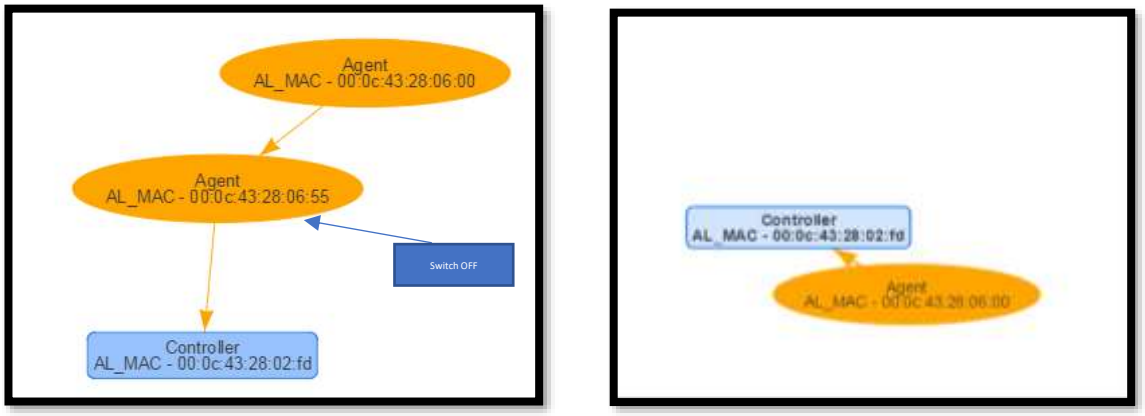


Figure 4- 1. Link Fail over example Image 1 and Image 2

When Agent 1 is powered ON again, then it will also join the best possible Easymesh node based on Network healing. Final topology will become as below:

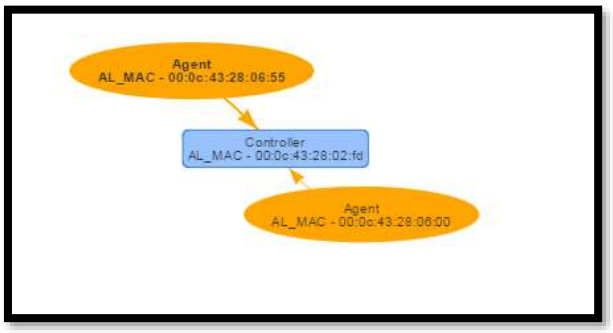


Figure 4- 2. Link failover example Image 3

4.2.2 Feature Management Interfaces

4.2.2.1 WEB UI Interface

You can also trigger Best Uplink AP selection manually via WEBUI, by clicking on the "Trigger Up-link AP Selection".



Figure 4- 3. WEB UI button to trigger Best AP selection at the controller

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4.3 Backhaul Link Steering and Prioritization

Backhaul link is defined as the link which one agent forms with the uplink MAP device which may be a controller or another agent. The backhaul link can be made using ethernet or wireless. The below mechanism can be use for steering and prioritizing only if the BH link is a wireless link. For ethernet BH link , always highest priority is given and also its steering is not possible.

4.3.1 Backhaul Link Steering

Mediatek Turnkey solution provides and API for user to steer the BH link by use of MAP specification define BH steer procedure.
An agent remembers the target AP to which it has to steer only until the connection is formed. Hence an agent will go through best AP selection process after it gets disconnected post completion of backhaul steering process. So this command has just a one time effect.

Network optimization and BH steering features are conflicting in nature. If BH steering feature need to be used, then automatic network optimization feature need to be disabled and vice versa. If BH steer is given via mapd_cli command then Network optimization will get disabled internally.

4.3.1.1 Feature Management Interfaces

4.3.1.1.1 Command Line Interface

Command line interface can also be used to trigger BH steering from console. Command is as below :

BH steer command	To change the current BH of an agent	mapd_cli /tmp/mapd_ctrl steerbh < APCLI MAC Address of current connected BH> <target BSSID of other AP BH, may or may not be of same band as current Backhaul>

For example :

- Initially if we have a Star topology:
  - Agent 1 → Controller ← Agent 2

- We can use Backhaul steering to convert star to daisy chain:
  - Controller← Agent 2← Agent 1

For this you need to feed the MAC address of BackHaul MAC and the target BSSID in the BH steer command.

- 00:0C:43:28:02:87 is MAC Address of Controller at interface rai0



Figure 4- 4. BH Steer example Image 1

- 00:0C:43:28:07:0A is MAC Address of Agent 2 at interface rai0 which is Target BSSID

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Figure 4- 5. BH Steer example Image 2

- iwconfig at Agent 1 shows that it is connected to Controller's rai0 interface :

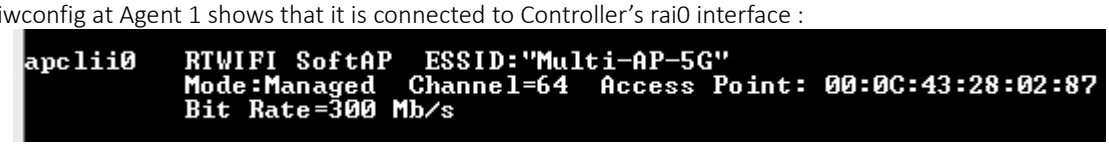


Figure 4- 6. BH Steer example Image 3

- 06:0C:43:28:07:1C is MAC Address of Agent 1 at interface apclii0 which is BH address



Figure 4- 7. BH Steer example Image 4

Command given at the console :

- mapd_cli /tmp/mapd_ctrl steerbh 06:0C:43:28:07:1C 00:0C:43:28:07:0A

After backhaul steering complete, you can check connection status on console and UI to confirm that BH steer has actually happened.

- iwconfig at Agent 1 shows that it is connected to Agent2's rai0 interface :

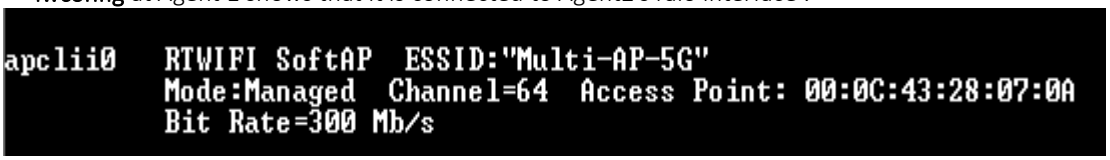


Figure 4- 8. BH Steer example Image 5

4.3.1.1.2 WEB UI Interface

If you want to steer any backhaul link in the network then you can use the following UI button on the Controller Web Page:

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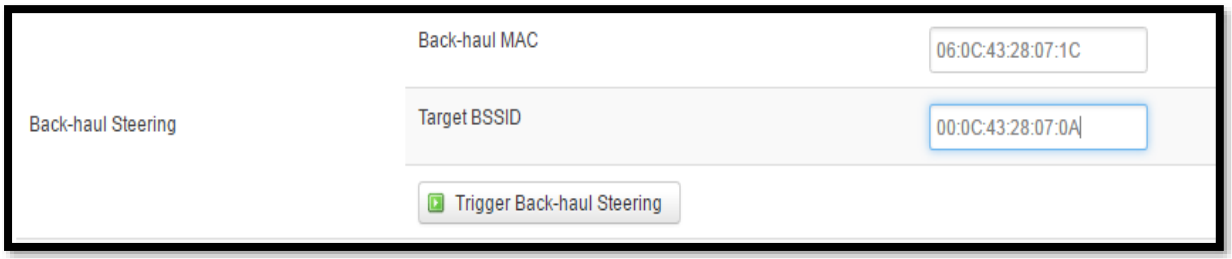


Figure 4- 9. BH steer command Controller WEB UI button snapshot

"Back-hual MAC" is the MAC address of current back-hual interface of the Agent which want to steer.
"Target BSSID" is the MAC address of the back-hual interface of the Controller or Configured Agent to which the want to steer. This BSSID radio may or may not be of the same radio band as current Backhaul.

4.3.2 Backhaul Link Prioritization

The Backhaul link selection at the agents happens via MediaTek's Multi-AP Best AP selection algorithm.
When WPS connection is triggered on enrollee device, it will scan on all the bands and find out all the channels on which WPS registrar is present (with same UUID). Based on this scan result, device will select an interface on which WPS handshake will happen.

WPS registrar provides profiles of all the available Backhaul BSS on each of the bands during the WPS handshake. Based on these profiles, the enrollee agent selects the best suited profile based on the available bands.

But it is also possible for user to give more preference to a particular Backhaul link using the below prioritization techniques. Ethernet is always given the highest priority and this cannot be altered. So if Ethernet connection exists between two devices then BH will always be ethernet and not wifi.

But priority can be set for other bands. Like for example if there are 3 bands 2G , 5GL and 5GH, then by default priority setting in config file is kept as 3 for 2G , 1 for 5GL and 1 for 5GH. This means that 5GH and 5GL are given highest and equal priority i.e. 1 and 2G is given lowest priority i.e. 3. The interface on which backhaul connection will be made depends on this priority. If an interface is programmed with priority 0, we will not attempt connection on that interface ever.

Note that just by setting the priority for a radio band as the highest, does not guarantee that the BH link will be formed at that band. Since the best AP selection also considers other factors also. Please see the main feature flow in below section.
It should also be noted that for proper DFS functionality we expect that at least one BH on non-DFS band is always present. Hence it is discouraged to program priority of 2.4G band as 0.

Mediatek solution provides Automatic Backhaul link selection mechanism. A backhaul link priority is configured: Ethernet > 5G > 2.4G as per default. This can be changed dynamically as well. It should be noted that changing backhaul priority will impact the existing connection and hence should be triggered only on stable networks. Backhaul switching should be avoided immediately after connection and should be triggered after auto configuration is complete. Also triggering backhaul preference in quick succession will impact connection process of previously triggered command to change priority hence it should be avoided.

- BH Link prioritization is a mediatek proprietary feature. So only a mediatek agent can support it. Most of the logic is present agent side.
- Network Optimization should be disabled via cli command before using Backhaul steer or band switching.

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- It should be noted that changing backhaul priority will impact the existing connection and hence should be triggered only on stable networks. Backhaul switching should be avoided immediately after connection and should be triggered after auto configuration is complete. Also triggering backhaul preference in quick succession will impact connection process of previously triggered command to change priority hence it should be avoided.
- While setting the BH priority, it is recommended to set at least one band as the highest priority i.e. 1

NOTE for SKU with MLO enabled:

- BH Priority have no role for MLO, infact all the BH will have the highest priority i.e 1
- WPS registrar provides profiles for the band on which the WPS handshakes had happened.

4.3.2.1 Main Flow

Flow at the Enrollee Agent

1. When WPS is triggered on the device, it will start scan and find out all the bands on which WPS registrar are active and have the same UUID
2. If it finds registrars with more than 1 UUID, then it stops WPS on all interfaces due to PBC overlap.
3. If it finds multiple WPS PBC registrars with same UUID, then it triggers WPS handshake on one band.
4. It gets multiple Backhaul BSS profiles in M8 of WPS handshake. The backhaul profiles contains the following parameters :
    a. SSID
    b. AuthMode
    c. EncryptType
    d. PSK
5. Above configurations are saved in nvram/config file for future use. These parameters are not band specific. Agents need to scan for these configs on all the bands/channels
6. After the device gets multiple Backhaul BSS profiles and finds them in scan results, it applies the best Backhaul BSS selection procedure and finds the best BSS to connect to.
7. According to the algorithm, 5G band is given preference over 2.4G band for backhaul link creation.
8. 1 being the highest priority and 3 being the lowest.
9. Additionally we have provision for configuring minimum RSSI values which will be considered at the time of best AP selection.
10. If the minimum RSSI Threshold value for a band is configured, the best AP calculation can be performed thrice, as explained below:
    a. First calculations will consider only the higher priority band and APs with considerably good RSSI.
    b. If a best AP is not found in above step, we will consider APs on second priority band and with considerably good RSSI.
    c. If an AP is found in second step, by tuning the RSSI correctly, we can safely assume that AP on second priority band will provide better “t-put” than first priority band.
    d. If the AP is not found in second step as well, we will consider all the APs on all priority bands and pick the AP with maximum “t-put” as candidate AP without considering the RSSI threshold programmed.
11. It should be noted that if connection is not completed in 30 seconds after triggering connection, or if the connection is rejected by the peer AP, the peer AP will blacklisted for further connections for 5mins. This means that the AP will not be considered in subsequent connection attempts for next 5 mins.
12. For 6E case, Agent will scan only on PSC channels. If Controller is setup on non-PSC channel then there are chances for BH to be on 5G/24G. In case of non-PSC it is advised to set either 24G or 5G BH priority as 1, so that first scan always happens on 24G/5G and bestAP will use RNR to get non-PSC channel of 6G. For PSC case and using default BH priority, BH might get formed on either 5G or 6G depending on Score.

Flow at the Controller/Existing Agent

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1. When WPS is triggered, WPS registrar is enabled on first BSS of all the bands with the same UUID.
2. When an enrollee agent attempts connection to the fronthaul BSS using WPS on one band :
    a. WPS is stopped on the other band.
    b. Backhaul BSS credentials of all the backhaul BSS on all the bands is given to the enrollee agent in M8 during WPS 8-way handshake.

3. In case when Backhaul priority for 2G and 5G are equal, WPS will be performed on 5G interface.

4.3.2.2 Main flow for Hostapd and MLO supported SKU

Flow at Enrollee Agent (except below points all other points mentioned above are valid for MLO as well):
1. When WPS is triggered on the device, it will start scan on 5G first and if it finds UUID then proceed with WPS connection else starts WPS on another remaining band.
2. Post WPS, Hostapd/Supplicant will handover the credentials to WAPP daemon. To handover the credentials "Wps_cred_processing" is set to 1 in wps_supplicant.conf files.
3. Supplicant will provide credentials only for the band on which WPS handshakes had happened. There will be only one set of credentials.
4. According to the algorithm all bands will be scanned starting from 5G, 6G and 24G, all the bands have equal priority in case of MLO.
5. The bestAP scan and score calculation will be performed on all bands as explain below:
    a. Score calculation will happen on 5G first followed by 6G and 24G.
    b. Apart from invidual score all bands will have MLO score as well. This MLO score will be the score combining other MLD group BSS. For example: 5G, 6G and 24G will have individual scores as X, Y and Z. and MLO scores X+Y+Z for all bands. Based on above score the band which have highest individual and MLO score will be selected for primary link.
    C. Since there are lots of DFS channels on 5G and there are high chances that because RADAR and CAC 5G BH will be affected if Primary link is 5G. So, the current algorithm will prefer 6G to form primary link to avoid disconnection.

4.3.2.3 Feature Management Interfaces

4.3.2.3.1 Configuration File Interface

For openwrt , please update the /etc/map/mapd_cfg as below and run EasyMesh_openwrt.sh again. This setting will remain persistent in the text file after multiple reboots in openwrt platforms.

- BhPriority2G=3
- BhPriority5GL=1
- BhPriority5GH=1
- BhPriority6G=1
- ScanThreshold2g=-88
- ScanThreshold5g=-88
- ScanThreshold6g=-88

4.3.2.3.2 Command Line Interface

There is a mapd_cli command, to change the BH priorities run time. APIs are provided to switch the band of operation of the backhaul STA from one band to another. This is achieved by dynamically changing the priority of each band using the following command:

mapd_cli /tmp/mapd_ctrl bh_priority <2G_priority> <5GL_priority > <5GH_priority> <6G_priority>

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Once this command is given, BHSTA will disconnect and start the best AP selection procedure based on the new priority. If it is required that STA should not connect on a particular band, then the priority of that band should be made 0.

For example, the current Band priority is as follows:
5GH: 1
5GL: 2
2.4G: 3
and the BHSTA connection is on 5GH.
Now if the user want that the connection be made on 5GL, then there are 2 options

Option 1:
Set the priority of all other bands to 0.
mapd_cli /tmp/mapd_ctrl bh_priority 0 1 0 0
In this way, connection will be made only of 5GL. Connection will not be attempted on other bands.

Option 2:
Give the highest priority to 5GL and lower priority to other bands:
mapd_cli /tmp/mapd_ctrl bh_priority 3 1 2 3
In this way, highest priority is given to 5GL, but if due to some reason, connection is not made on 5GL, then connection is attempted on other bands also.

NOTE:
For 6E MAP device, if user want connection to be made on 6G then user can follow either of the two methods:
1st method:
Set the priority of all other bands to 0.
mapd_cli /tmp/mapd_ctrl bh_priority 0 0 0 1
In this way, connection will be made only of 6G. Connection will not be attempted on other bands.

2nd method:
Give the highest priority to 6G and lower priority to other bands:
mapd_cli /tmp/mapd_ctrl bh_priority 3 3 3 1
Set “NetOptUserSetPriority” as 1 to consider BH priority on Network Optimzation.
NetOptUserSetPriority=1 (mapd_default.cfg)
In this way, highest priority is given to 6G, but if due to some reason, connection is not made on 6G, then connection is attempted on other bands also.

- There is a mapd_cli command, to change the RSSI threshold for Scan results run time :
  Example:
  mapd_cli /tmp/mapd_ctrl scan_thresh_2g -75
  mapd_cli /tmp/mapd_ctrl scan_thresh_5g -75

- The blacklisting duration can be configured dynamically also. For changing the value of black list timeout dynamically:
  mapd_cli /tmp/mapd_ctrl bl_timeout <secs>

4.3.2.3.3 WEBUI Interface

- from Web UI , you can set Backhaul priority as follows :

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Figure 4- 10. WEB UI Button for changing BH priority/preference

4.4 Network Optimization

Network optimization is a mechanism to automatically form the best topology based on the current conditions. This mechanism is completely controlled and implemented at the controller. Hence, it can also be referred to as centralized network optimization.

- Controller will contain the entire information about the network.
- Controller will gain this network information by making its agents do an off-channel scan.
- Using all the scan results the controller will decide the final optimized network topology.
- The controller will realize this new topology by sending out backhaul steer command to all the agents whose BH is changed.

For network optimization in MAP R2, Scan Request (to collect scan results from the agents in the network) will be sent only if that MAP R2 agent supports fresh scan in their scan capabilities as per MAP R2 specification. Otherwise for MAP R1 device Network optimization scan will be done using proprietary off channel scan mechanism. Network optimization relies on present devices and environment only and cannot use old scan results.

4.4.1 Network Optimization for MLDs in mesh network

For Mesh devices which are 802.11be MLO capable, the network optimization internal algorithm will consider the information for all the MLO links which one mesh device can make with the other using cumulative MLD scoring.

- Using this MLD score final optimized network topology will be decided. Hence there will be more chances of an MLD agent to make connection with another MLD agent/controller as compared to a non-MLD device when in a mixed network.
- Finally, the BH steer command will be sent to all the Agents whose backhaul needs to be changed or primary backhaul needs to be changed.
- In this command the target BSSID will be the best out of all the MLD links possible with the uplink device . This BSSID will be chosen so that the primary link in the MLO connection is formed on this BSS.
- However in current design 5G BSS is never given to form the primary link, even if its score is coming as the best, for avoiding traffic interruption because of DFS channels CAC time.

4.4.2 Trigger conditions

Network optimization is triggered in the following scenarios:

- Uplink Rate variation
  - Controller will trigger network optimization if the uplink rate of each device varies a lot from the previous uplink rate, in the current topology.
- Topology change:
  - Controller triggers network opt. with any change in topology
    - Any new device joins the network
    - Any device leaves the network
- Post DFS channel selection:

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- Controller will trigger network optimization, once CAC gets completed on a DFS channel as chosen by the controller. This is done to bring back the network on 5G, if CAC is performed by the main radio.

- Post Channel planning:
  - Controller will trigger network optimization post channel planning, because when network moves to a new channel , there is possibility of network optimization as well.

4.4.3 Fail Safe Mechanism for network optimization

There is a failsafe mechanism that will trigger network optimization in every 2 minutes interval in case BH stays on 2G only. As soon as BH is switched to 2G, it will set a timer function, that will check if the BH is still present on 2G, it will trigger network optimization. This mechanism is disabled by default. To enable this mechanism, below mapd config parameter 'NetworkOptPrefer5Gover2G' need to be set to '1'. The number of times this periodic mechanism need to be executed can also be configured using another parameter 'NetworkOptPrefer5Gover2GRetryCnt'.

NetworkOptPrefer5Gover2G=<0/1>
NetworkOptPrefer5Gover2GRetryCnt=<1/2/3/....so on>

4.4.4 Important points

1. Network Optimization gets automatically disabled when a manual BH steer command is given at the controller to change the network. So in this case if user wants to run network opt again then you can enable it dynamically via the cli command

2. In order to run network optimization, BSS with BH Enable must have separate SSID than BSS with only FH Enabled on any device in the network.

3. Network optimization does not run for devices with ETH backhaul. However, other devices with Wi-Fi BH can choose to connect with ETH backhaul device on the basis of best score.

4. Network Optimization should be disabled via cli command before using Backhaul steer or band switching.

5. Network optimization and BH priority coexistence:
   - By default the BH priority is set as 3(2G) , 1 (5G or 5GL if applicable), 1(5GH if applicable), 1 (6G if applicable), but it will not be considered by network optimization algorithm. So algorithm decides best backhaul by internal logic.
   - If user changes the bh priority dynamically via command then bh priority setting will be taken into consideration. Network optimization will also give priority to whatever user has set
   - A new flag is added in the mapd_default: NetOptUserSetPriority. By default value of this flag is 0. If this flag is set then also network optimization will consider the BH priorities set by the user in default file. Means by enabling this flag "NetOptUserSetPriority", user bh priority will be given more preference over network optimization logic for selecting the best BH per Agent.

6. Network optimization and BH priority coexistence for MLD agents:
   - As BH Priority will not be considered in MLD Agent as per section 4.3.2 Notes. Netowrk optimization of MLD Controller can choose the best Primary backhaul for each connected MLD agent as per section 4.4.1.
   - But choosing of the best primary BH can be overridden by setting:
     1. Set BH_priority for each radio to 1(best priority) or 3(last priority) in MLD agent and enable NetOpt and BH priority coexistence in MLD agent by enabling flag "NetOptUserSetPriority".
     2. Set "NtwrkOptMldBhPriority" to 1 in MLD controller.
   - Example:

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1. Customer set BH_Priority of 6G radio as highest. Now network Optimization will consider results of 6G only and select the best score device in 6G radio. Also, in this example if you have 2G+5G MLD link that MLD device will not be considered because the best possible primary Backhaul is 6G.
2. Customer set the BH_priority of 5G/2G radio as highest. same as example 1.
3. Suppose customer set BH_priority of 5G and 6G as high. Now network optimization will not consider 5G as high priority (to avoid disconnection when choose BH as 5G) for 2G+5G+6G MLD Link MLD device. So, the Primary Backhaul will be on 6G.
4. In continuation of point 3, for example if NetOpt find that the MLD device(A1) has 2G+5G MLD Link, then in that case MLD controller will choose 5G primary backhaul, if the connection is happening with A1.
5. Similarly, if the BH_priority of 2G and 5G is high. For 3Link MLD device primary backhaul connection will happen on 2G, but for 2 Link MLD device 5G+6G MLD Link primary backhaul will be chosen as 5G.
6. Assume Customer set BH_Priority of 2G and 6G is high. For 2G+5G+6G MLD Link MLD device MLD Controller will choose 6G as Primary Backhaul Link. But in case if MLD Device assume A1 is having MLD link only 2G+5G then for A1 2G will be considered as primary backhaul link. For this case we choose 6G first because we give preference to 6G.

4.4.5 Procedure

a. Onboard the agent with the controller. With networkopt feature enabled(by default it is enabled).
b. add one more agent to MAP Network and onboard this with controller

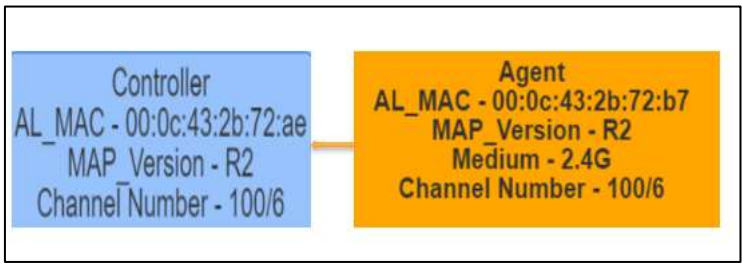


Figure 4- 11. Network optimization example Image 1

c. Add one more agent to MAP Network and onboard this with controller first agent onboard with controller on backhaul 2.4 Ghz and second agent onboard with backhaul 5Ghz .

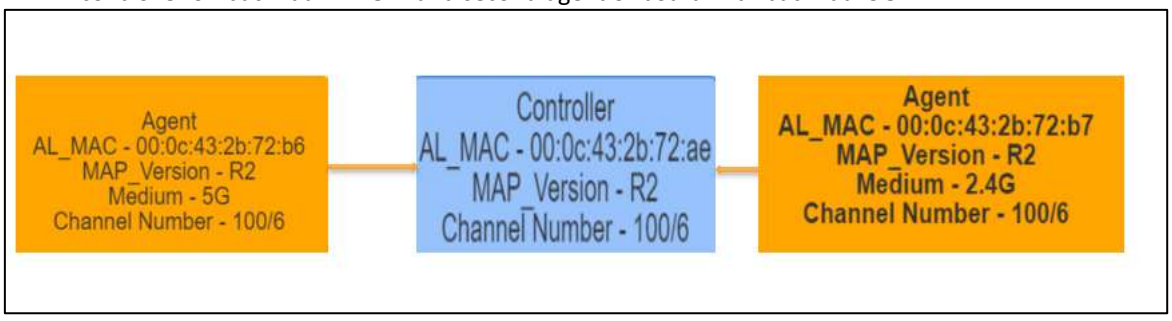


Figure 4- 12. Network optimization example Image 2

d. Network opt will triggerd due to new device added in MAP Network after 45 second default timer and both agent try to make a connection with best link .so both agent will make a link with 5Ghz backhaul .as shown in below image.

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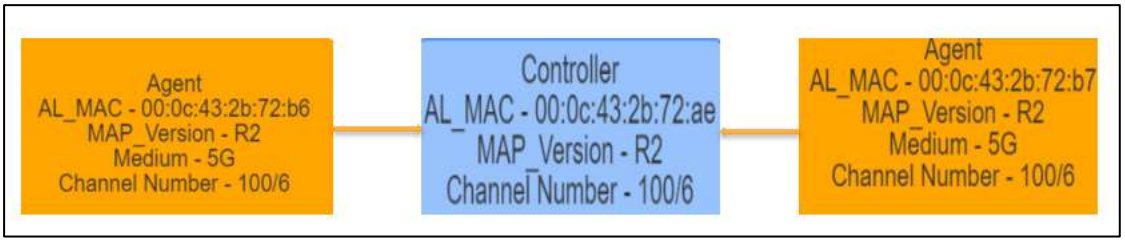


Figure 4- 13. Network optimization example Image 3

4.4.6 Command Line Interface

Different parameters of network optimization can be configured via mapd_cfg for OpenWRT

- **NetworkOptimizationEnabled 1**
  - Enable/disable the feature
- **NtwrkOptConnectWaitTime 45**
  - The time (seconds) for which controller will wait after a new connection to trigger network optimization.
- **NtwrkOptDisconnectWaitTime 45**
  - The time (seconds) for which controller will wait after a disconneciton to trigger network optimization
- **mapd_cli /tmp/mapd_ctrl set Network_Opt_Enable 1**
  - Network optimization enabled/disable cli command to enable/ disable network optimization dynamically.
  - Network Optimization should be disabled via this command before using Backhaul steer or band switching.

For UCI supported SKUs following command will be used:

- uci set mapd.mapd_cfg.NetworkOptimizationEnabled='1'
- uci set mapd.mapd_cfg.NtwrkOptConnectWaitTime='45'
- uci set apd.mapd_cfg.NtwrkOptDisconnectWaitTime='45'
- uci commit mapd

```
root@OpenWrt:/#
root@OpenWrt:/# uci set mapd.mapd_cfg.NetworkOptimizationEnabled='1'
root@OpenWrt:/# uci set mapd.mapd_cfg.NtwrkOptConnectWaitTime='45'
root@OpenWrt:/# uci set mapd.mapd_cfg.NtwrkOptDisconnectWaitTime='45'
root@OpenWrt:/# uci commit mapd
```

4.5 ETH Backhaul between Agents (Loop detection and avoidance)

This feature is used to let agent access to agent via Ethernet and do configuration automatically. Loop detection and avoidance mechanism is introduced to solve circle topology. When ETH plug in, WIFI backhaul connecting to

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the same upstream device will be broken. When ETH plug out, WIFI backhaul connection will be triggered immediately. This feature is mainly based on topology information.

4.5.1 Main Flow

- When ETH plug in, the agent will check if it can access controller based on topology database. If it is, the agent will trigger auto-configuration by itself.
- When ETH plug out, the agent will trigger scan and find the best AP to connect.
- When ETH plug in, the device will do loop detection to find if there is a ring in topology (two or more paths to controller). The loop detection rules are
    - Count up the current hopcnt (WIFI hopcnt) to controller from the WIFI backhaul and the hopcnt (ETH hopcnt) to controller from the ETH backhaul. Take WIFI backhaul connection as 1 hop and take ETH connection as 0 hop.
    - If ETH hopcnt > WIFI hopcnt, do nothing. Let the peer device do the decision.
    - If ETH hopcnt < wifi hopcnt, break WIFI backhaul connection.
    - If ETH hopcnt == WIFI hopcnt, the device with smaller almac breaks WIFI backhaul connection. The compared device locates in its eth neighbor side and it has different WIFI backhaul neighbor and ETH neighbor.

Take an example as topology as below:

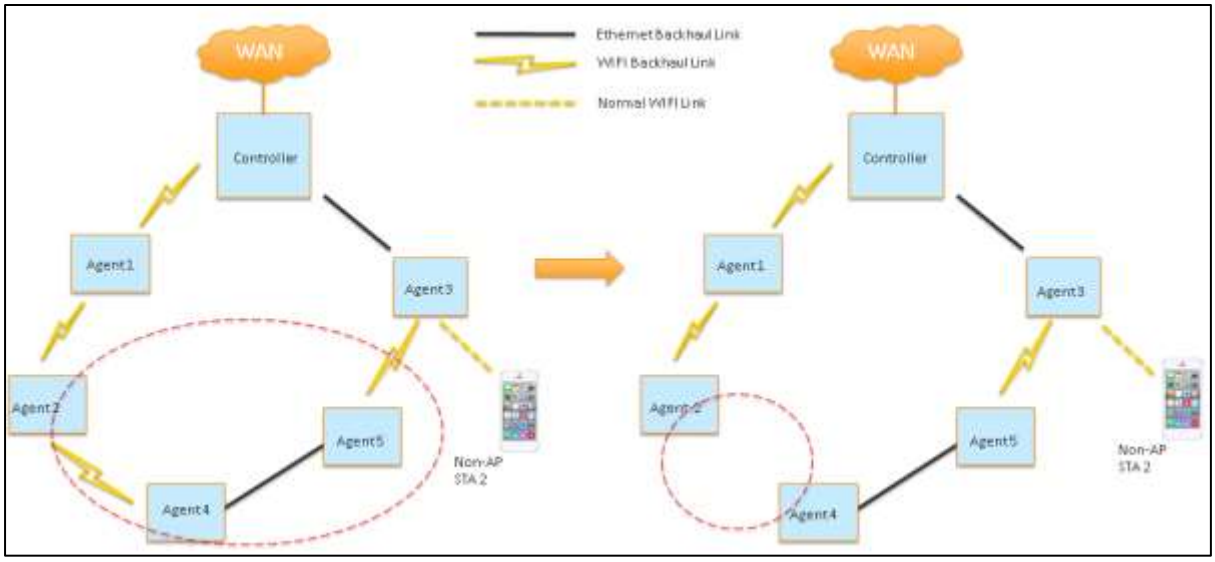


Figure 4- 14. Loop Detection

- For Agent4:
    - A4: has 1 apcli0 neighbor(A2) and 1 ETH neighbor(A5), start ring detection.
    - Apcli0 can access controller via A2 and eth0 can access controller via A5 .Ring exists.
    - WiFi hopcnt=3, ETH hopcnt = 1
    - WiFi hopcnt > ETH hopcnt
    - Do decision, disconnect apcli0.

- For Agent5:
    - A5: has 1 apcli0 neighbor(C) and 1 ETH neighbor(A4), start ring detection.
    - Apclii0 can access controller via A3 and ETH can access controller via A4. Ring exists.
    - WiFi hopcnt=1 of apcli, ETH hopcnt = 3
    - WiFi hopcnt < ETH hopcnt
    - Do nothing(let the peer A4 do the decision)

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Configuration Interface

Enable AutoBHSwitching in mapd_default.cfg on the agent side.

AutoBHSwitching =1

For UCI supported SKUs following command will be used:

- uci set mapd.mapd_cfg.AutoBHSwitching='1'
- uci commit mapd

4.5.2 Limitations

- This feature depends on switch. As agents need detect ETH link up and link down states, so different SWITCH needs different operation.
- Agents need send topology discovery message with vendor specific tlv to notify the peer device that ETH link up. So it is not compatible with other vendor.
- This feature cannot support the topology that all ETH exist a ring.

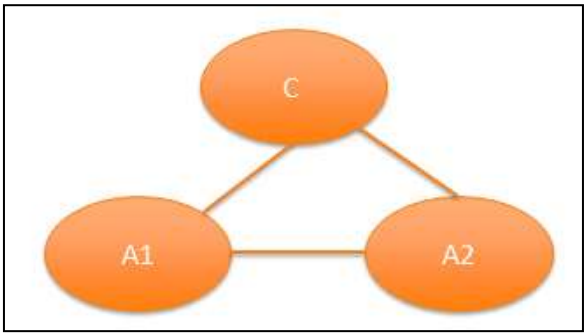


Figure 4- 15. Un-supported Topology

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5 Database and Diagnostics

5.1 Introduction

Database maintenance and diagnostics are an integral part of MediaTek's Turnkey solution. This chapter explains about all of them.
Databases supported are:

1. Topology Database: Information about all the MAP supported nodes in the network.
2. Client Database: Information about all the clients present in the network.
3. Wi-Fi 6 Support: Information of the Wi-Fi 6 capable Agents and Wi Fi 6 capable associated STAs.
4. DPP configurations: Information used to form DPP connection.

Diagnostics supported are:

5. Data elements: Set of diagnostic parameters about radio/BSS/clients etc. which are mandatory as per MAP R2 specification. Some new parameters were also added in MAP R3 version.

5.2 Topology Database

5.2.1 Feature Details

1905 Topology messages are used to learn about network topology by all nodes in the network. Following are topology messages:

1. Topology Discovery: Neighbor-Multicast (ieee-1905 std)
   a. Sent by each node on all interfaces periodically
   b. Consumed by direct neighbor, never forwarded
   c. Helps build DIRECT NEIGHBORS
2. Topology Notification: Multicast-Relay (MAP extension 17.1.5 1905 Topology Notification message)
   a. Sent whenever a node detects change on its topology or interface i.e. new link up/down, STA associated/disconnected etc.
   b. Relayed by all nodes, nodes discard previously seen Notifications to prevent loop
3. Topology Query: Unicast (ieee-1905 std)
   a. Sent by a node to a specific destination to query of its current state and interfaces
   b. Periodically sent for each entry in Database, age out the unresponsive nodes
4. Topology Response: Unicast (MAP extension 17.1.4 1905 Topology Response message)
   a. Sent in response to Topology Query to querying node
   b. Provides its current state
   c. Used by both Direct and Distant Neighbors
   d. Has following TLVs
      (a) Device Info
      (b) Capabilities
      (c) List of Legacy Neighbors (for each of its interfaces)
      (d) 1905 Device Neighbors on its interfaces
      (e) Attached WLAN Stations

The following example shows the topology message exchange example when RE2 joins RE1 and how these messages are used to update current topology at Root.

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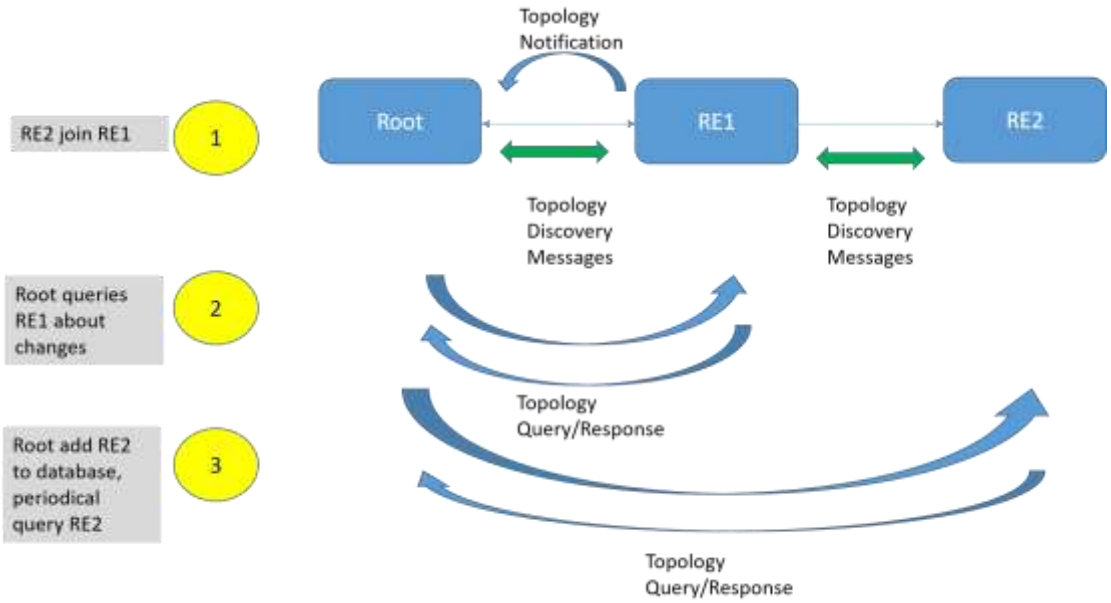


Figure 5- 1. Topology Message exchange example

1) Adding Distant Neighbor Device

Topology Response message from a node contains IEEE1905_TLV_TYPE_NEIGHBOR_DEVICE, which lists 1905 Neighbor devices for each of its interfaces. While processing Topology Response NEIGHBOR_DEVICE TLV from a direct neighbor, if a particular 1905 Device is not present in Database, a new DB entry is created and this neighbor is added as DISTANT NEIGHBOR. A topology Query message is immediately scheduled to get more information directly from this Distant Neighbor. Periodic Topology Query message is also scheduled for this Distant Neighbor entry.

2) Ageout for DB entry

Periodic Query and Response between neighbors (direct or distant) will act like heartbeat or hello. With reception of Topology Response message, its DB entry is refreshed.

5.2.2 Feature Management Interfaces

5.2.2.1 Command line Interface

There are below command line interface commands to fetch the topology data at each Multi-AP device :

1. mapd_cli /tmp/mapd_ctrl dump_topology
2. mapd_cli /tmp/mapd_ctrl dump_topology_v1
3. mapd_cli /tmp/mapd_ctrl show_bh_link_metric
4. mapd_cli /tmp/mapd_ctrl show_sta_metric

On giving these commands on the console following output is obtained:

1. mapd_cli /tmp/mapd_ctrl dump_topology

Following information is shown on console:

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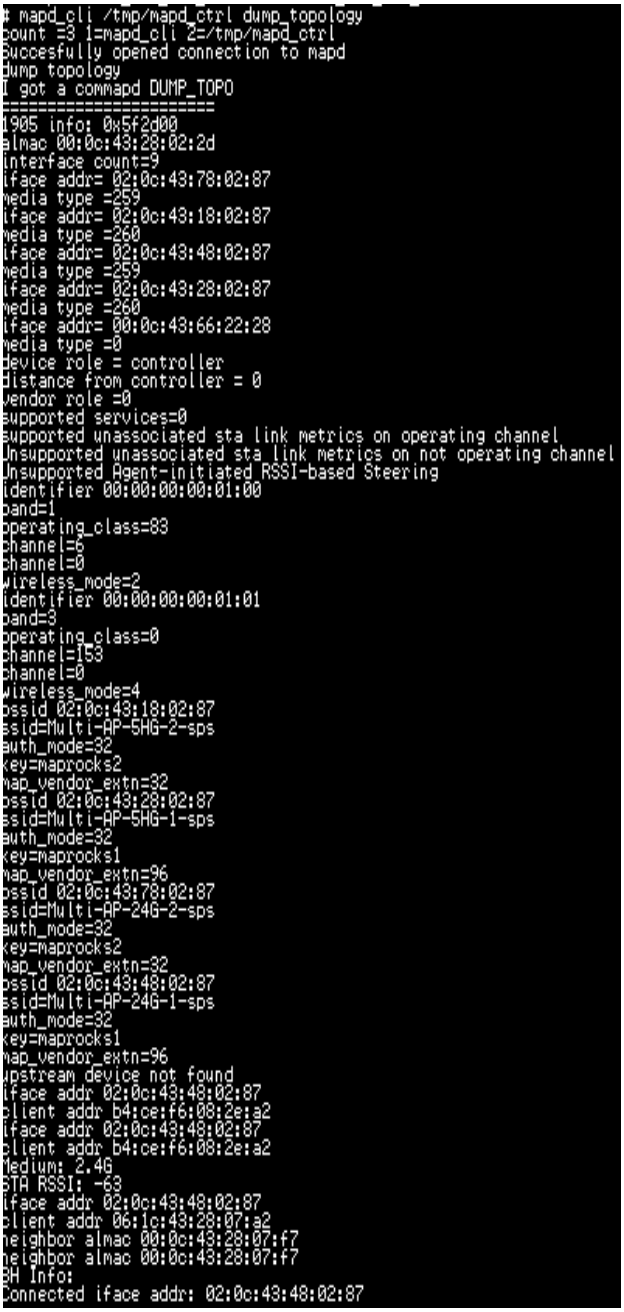
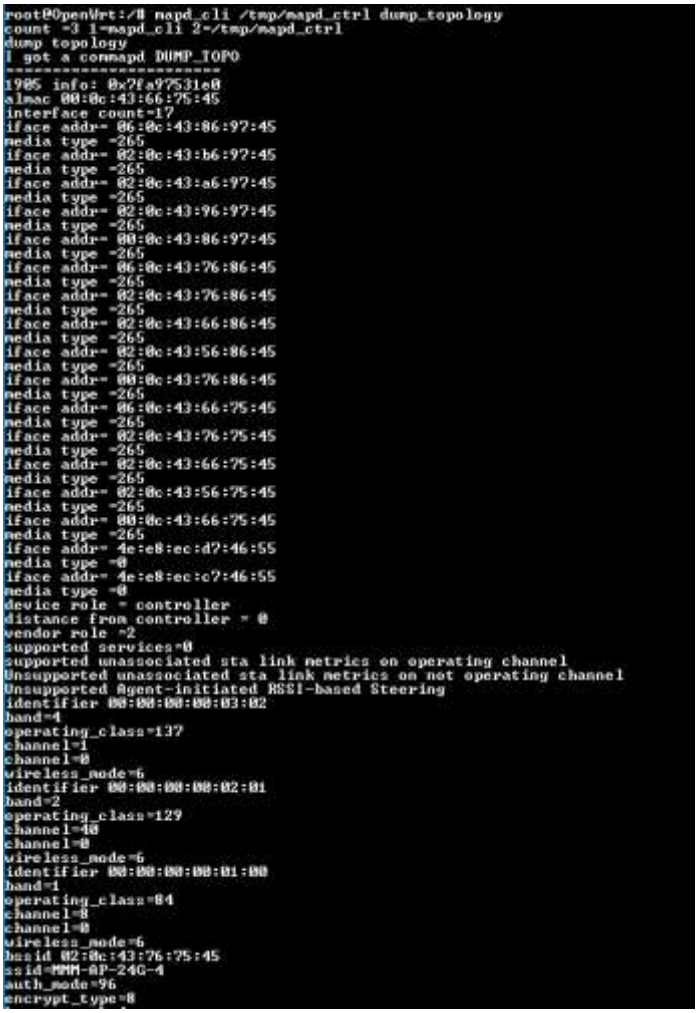
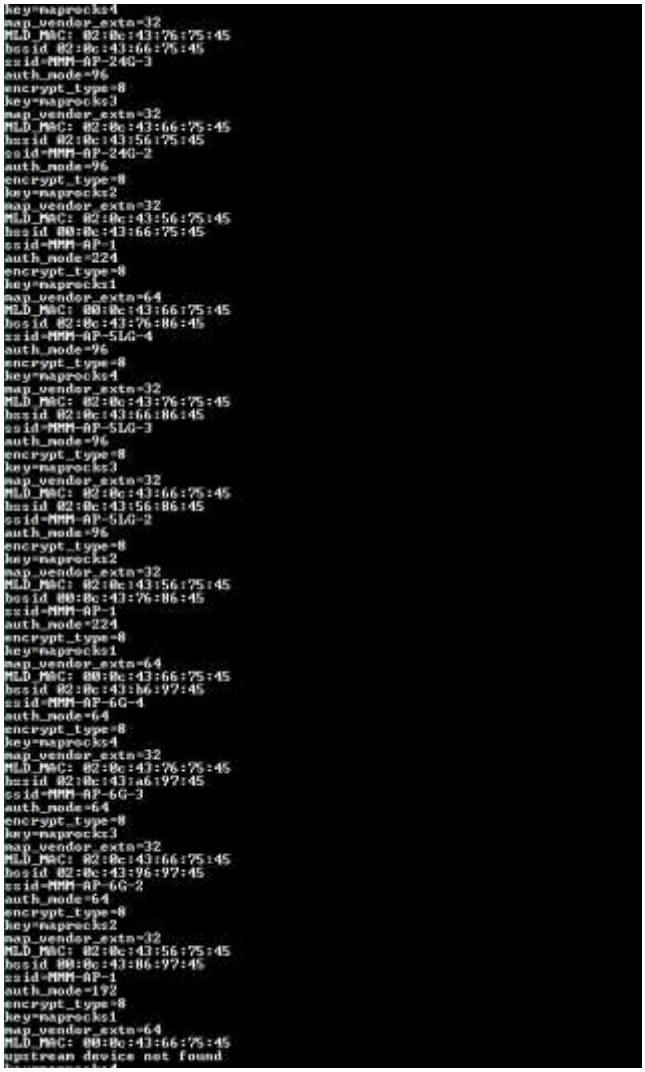


Figure 5- 2. Dump topology command console log snapshot

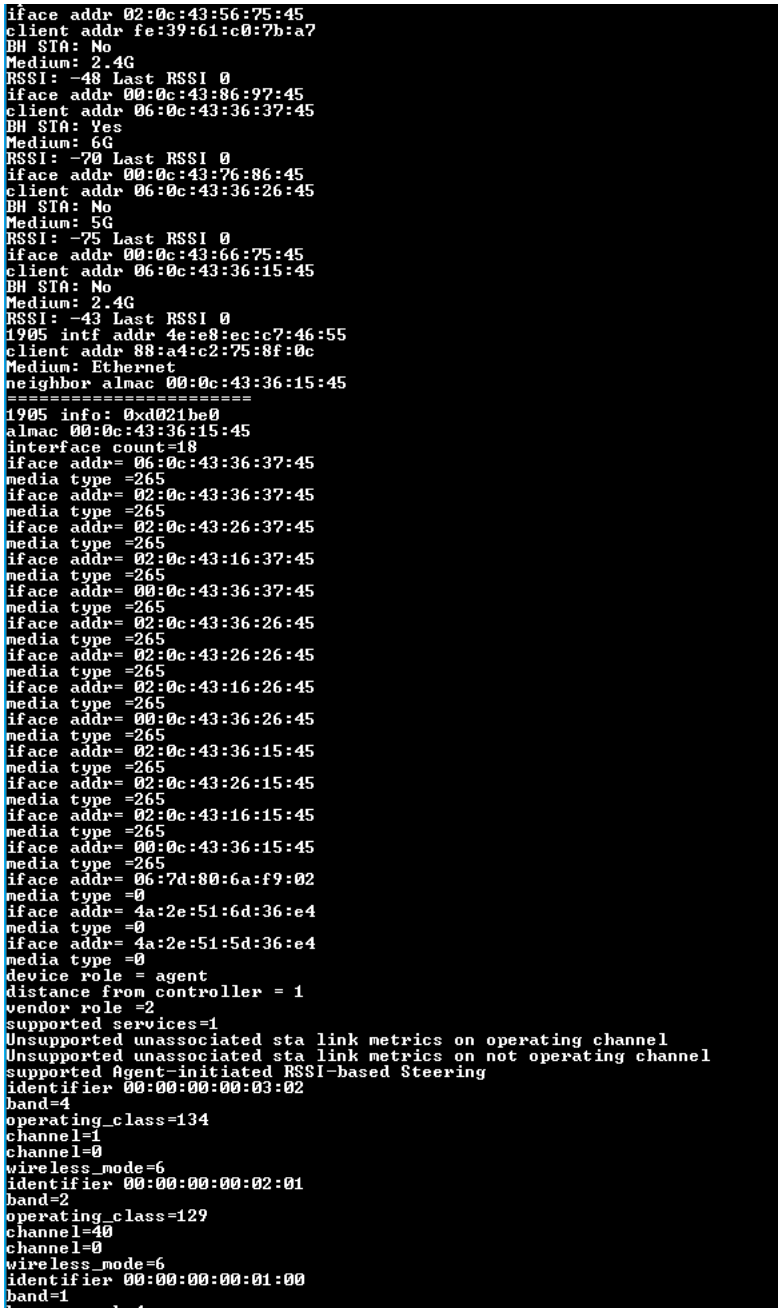
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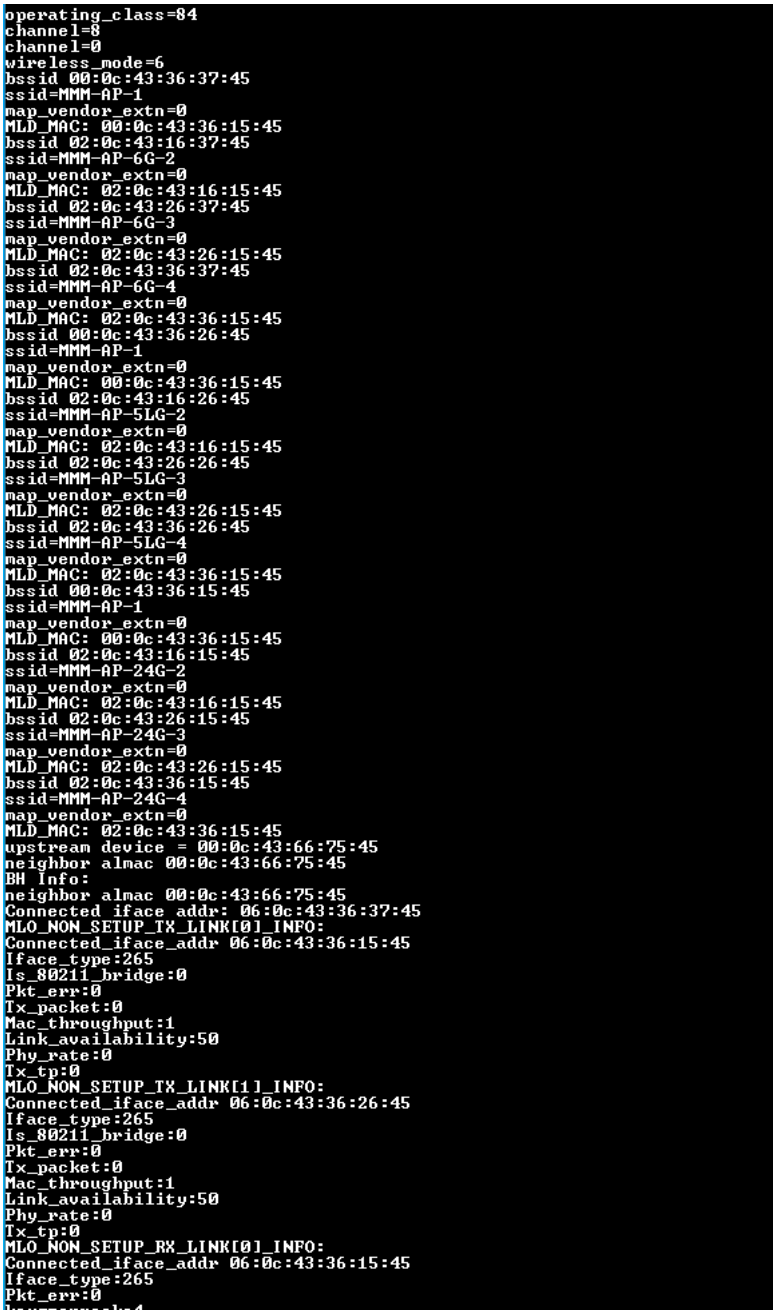
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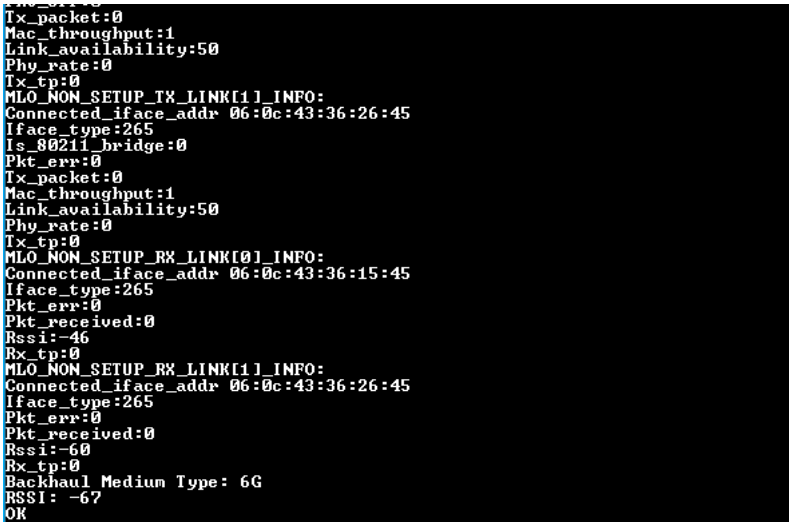
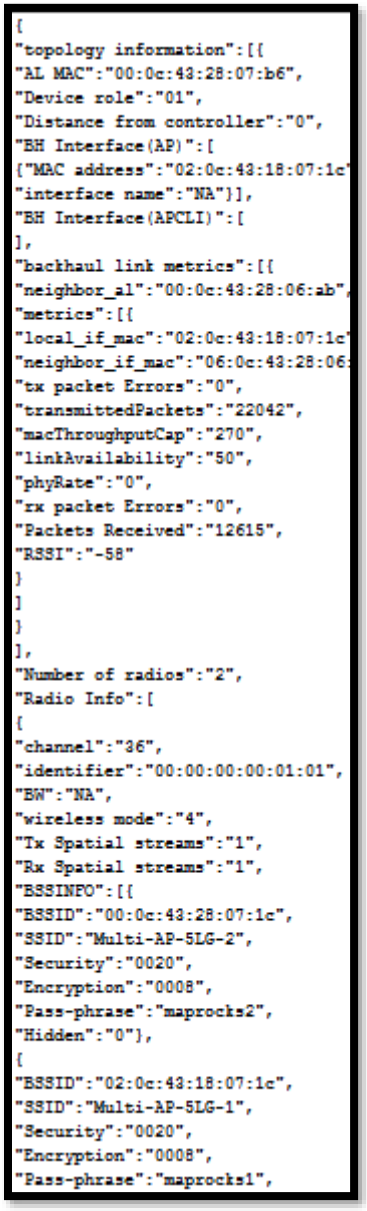


***Figure 5- 3 . Dump* topology command console log with MLD parameters**

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2. mapd_cli /tmp/mapd_ctrl dump_topology_v1

Following information is shown on console in JSON format (used for UI display):




```
{
"topology information":[{
"AL MAC":"00:0c:43:28:07:b6",
"Device role":"01",
"Distance from controller":"0",
"BH Interface(AP)":[
{"MAC address":"02:0c:43:18:07:1c"
"interface name":"NA"}],
"BH Interface(APCLI)":[
],
"backhaul link metrics":[{
"neighbor_al":"00:0c:43:28:06:ab",
"metrics":[{
"local_if_mac":"02:0c:43:18:07:1c"
"neighbor_if_mac":"06:0c:43:28:06:
"tx packet Errors":"0",
"transmittedPackets":"22042",
"macThroughputCap":"270",
"linkAvailability":"50",
"phyRate":"0",
"rx packet Errors":"0",
"Packets Received":"12615",
"RSSI":"-58"
}
]
}
],
"Number of radios":"2",
"Radio Info":[
{
"channel":"36",
"identifier":"00:00:00:00:01:01",
"BW":"NA",
"wireless mode":"4",
"Tx Spatial streams":"1",
"Rx Spatial streams":"1",
"BSSINFO":[{
"BSSID":"00:0c:43:28:07:1c",
"SSID":"Multi-AP-5LG-2",
"Security":"0020",
"Encryption":"0008",
"Pass-phrase":"maprocks2",
"Hidden":"0"},
{
"BSSID":"02:0c:43:18:07:1c",
"SSID":"Multi-AP-5LG-1",
"Security":"0020",
"Encryption":"0008",
"Pass-phrase":"maprocks1",
```


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```
"Hidden":"0",
"connected sta info":[
{
"STA MAC address":"06:0c:43:28:06:fe",
"last assoc time":"0",
"BH STA":"Yes",
"Medium":"5G",
"uplink rssi":"-60"}
]]]
},
{
"channel":"6",
"identifier":"00:00:00:00:01:00",
"BW":"NA",
"wireless mode":"2",
"Tx Spatial streams":"1",
"Rx Spatial streams":"1",
"BSSINFO":[{
"BSSID":"02:0c:43:38:07:1c",
"SSID":"Multi-AP-24G-2",
"Security":"0020",
"Encryption":"0008",
"Pass-phrase":"maprocks2",
"Hidden":"0"},
{
"BSSID":"02:0c:43:28:07:1c",
"SSID":"Multi-AP-24G-1",
"Security":"0020",
"Encryption":"0008",
"Pass-phrase":"maprocks1",
"Hidden":"0"}]
}],
"BH Info":[

],
"Other Clients Info":[

]
},
{
"AL MAC":"00:0c:43:28:06:ab",
"Device role":"02",
"Distance from controller":"1",
"Upstream 1905 device":"00:0c:43:28:07:b6",
"BH Interface(AP)":[
],
"BH Interface(APCLI)":[
{"MAC address":"06:0c:43:28:06:fe",
"interface name":"N/A"}],
```


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```
"backhaul link metrics":[{
"neighbor_al":"00:0c:43:28:07:b6",
"metrics":[{
"local_if_mac":"06:0c:43:28:06:fe",
"neighbor_if_mac":"02:0c:43:18:07:1c",
"tx packet Errors":"0",
"transmittedPackets":"28286",
"macThroughputCap":"270",
"linkAvailability":"50",
"phyRate":"0",
"rx packet Errors":"0",
"Packets Received":"11181",
"RSSI":"-62"
}]},
"Number of radios":"2",
"Radio Info":[
{
"channel":"36",
"identifier":"00:00:00:00:01:01",
"BW":"NA",
"wireless mode":"4",
"Tx Spatial streams":"0",
"Rx Spatial streams":"0",
"BSSINFO":[{
"BSSID":"02:0c:43:18:06:fe",
"SSID":"Multi-AP-5LG-1",
"connected sta info":[
]},
{
"BSSID":"00:0c:43:28:06:fe",
"SSID":"Multi-AP-5LG-2"}]
},
{
"channel":"6",
"identifier":"00:00:00:00:01:00",
"BW":"NA",
"wireless mode":"4",
"Tx Spatial streams":"0",
"Rx Spatial streams":"0",
"BSSINFO":[{
"BSSID":"02:0c:43:28:06:fe",
"SSID":"Multi-AP-24G-1"},
{
"BSSID":"02:0c:43:38:06:fe",
"SSID":"Multi-AP-24G-2"}]
}],
"BH Info":[
{
"neighbor almac addr":"00:0c:43:28:07:b6",
"connected iface addr":"06:0c:43:28:06:fe",
"Backhaul Medium Type":"5G",
"RSSI":"-62"}
],
"Other Clients Info":[
{
"1905 Intf address":"00:0c:43:28:80:5f",
"Client Address":"68:f7:28:21:02:09",
"Medium": "Ethernet"}
]
}]
```


Figure 5- 4. Dump_topology_v1 command JSON file snapshot

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```
{
"topology information":[
{
"AL MAC":"00:0c:43:66:15:45",
"MAP Version":"R3",
"Device role":"01",
"Distance from controller":"0",
"BH Interface(AP)":[
{"MAC address":"00:0c:43:86:37:45",
"interface name":"NA"}],
"BH Interface(APCLI)":[
],
"backhaul link metrics":[{
"neighbor_al":"00:0c:43:36:15:45",
"metrics":[{
"local_if_mac":"00:0c:43:86:37:45",
"neighbor_if_mac":"06:0c:43:36:37:45",
"tx packet Errors":"20",
"transmittedPackets":"19235",
"macThroughputCap":"6483",
"linkAvailability":"50",
"phyRate":"0",
"rx packet Errors":"0",
"Packets Received":"7569",
"RSSI":"-51"
}
]
}
],
"Number of radios":"3",
"Radio Info":[
{
"channel":"9",
"identifier":"00:00:00:00:03:02",
"BW":"NA",
"wireless mode":"6",
"Tx Spatial streams":"4",
"Rx Spatial streams":"4",
"BSSINFO":[{
"BSSID":"02:0c:43:b6:37:45",
```


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```
"MLD_MAC":"02:0c:43:36:15:45",
"SSID":"MMM-AP-6G-4",
"Security":"0040",
"Encryption":"0008",
"Pass-phrase":"maprocks4",
"Hidden":"0"},
{
"BSSID":"02:0c:43:a6:37:45",
"MLD_MAC":"02:0c:43:26:15:45",
"SSID":"MMM-AP-6G-3",
"Security":"0040",
"Encryption":"0008",
"Pass-phrase":"maprocks3",
"Hidden":"0"},
{
"BSSID":"02:0c:43:96:37:45",
"MLD_MAC":"02:0c:43:16:15:45",
"SSID":"MMM-AP-2",
"Security":"0040",
"Encryption":"0008",
"Pass-phrase":"maprocks2",
"Hidden":"0"},
{
"BSSID":"00:0c:43:86:37:45",
"MLD_MAC":"00:0c:43:66:15:45",
"SSID":"MMM-AP-1",
"Security":"00c0",
"Encryption":"0008",
"Pass-phrase":"maprocks1",
"Hidden":"0",
"connected sta info":[
{
"STA MAC address":"06:0c:43:36:37:45",
"last assoc time":"65535",
"BH STA":"Yes",
"Medium":"6G",
"uplink rssi":"-51"}
]}]
},
{
```


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```
"channel":"40",
"identifier":"00:00:00:00:02:01",
"BW":"NA",
"wireless mode":"6",
"Tx Spatial streams":"4",
"Rx Spatial streams":"4",
"BSSINFO":[{
"BSSID":"02:0c:43:76:26:45",
"MLD_MAC":"02:0c:43:36:15:45",
"SSID":"MMM-AP-5LG-4",
"Security":"0060",
"Encryption":"0008",
"Pass-phrase":"maprocks4",
"Hidden":"0",
"connected sta info":[
{
"STA MAC address":"8a:73:26:83:d4:68",
"MLD_MAC":"00:00:00:00:00:00",
"last assoc time":"0",
"BH STA":"No",
"Medium":"5G",
"uplink rssi":"-50"}
]},
{
"BSSID":"02:0c:43:66:26:45",
"MLD_MAC":"02:0c:43:26:15:45",
"SSID":"MMM-AP-5LG-3",
"Security":"0060",
"Encryption":"0008",
"Pass-phrase":"maprocks3",
"Hidden":"0"},
{
"BSSID":"02:0c:43:56:26:45",
"MLD_MAC":"02:0c:43:16:15:45",
"SSID":"MMM-AP-2",
"Security":"0060",
"Encryption":"0008",
"Pass-phrase":"maprocks2",
"Hidden":"0"},
{
```


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```
"BSSID":"00:0c:43:76:26:45",
"MLD_MAC":"00:0c:43:66:15:45",
"SSID":"MMM-AP-1",
"Security":"00e0",
"Encryption":"0008",
"Pass-phrase":"maprocks1",
"Hidden":"0",
"connected sta info":[
{
"STA MAC address":"06:0c:43:36:26:45",
"last assoc time":"65535",
"BH STA":"No",
"Medium":"5G",
"uplink rssi":"-71"}
]}]
},
{
"channel":"3",
"identifier":"00:00:00:00:01:00",
"BW":"NA",
"wireless mode":"6",
"Tx Spatial streams":"4",
"Rx Spatial streams":"4",
"BSSINFO":[{
"BSSID":"02:0c:43:36:15:45",
"MLD_MAC":"02:0c:43:36:15:45",
"SSID":"MMM-AP-24G-4",
"Security":"0060",
"Encryption":"0008",
"Pass-phrase":"maprocks4",
"Hidden":"0"},
{
"BSSID":"02:0c:43:26:15:45",
"MLD_MAC":"02:0c:43:26:15:45",
"SSID":"MMM-AP-24G-3",
"Security":"0060",
"Encryption":"0008",
"Pass-phrase":"maprocks3",
"Hidden":"0"},
{
```


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```
"BSSID":"02:0c:43:16:15:45",
"MLD_MAC":"02:0c:43:16:15:45",
"SSID":"MMM-AP-2",
"Security":"0060",
"Encryption":"0008",
"Pass-phrase":"maprocks2",
"Hidden":"0"},
{
"BSSID":"00:0c:43:66:15:45",
"MLD_MAC":"00:0c:43:66:15:45",
"SSID":"MMM-AP-1",
"Security":"00e0",
"Encryption":"0008",
"Pass-phrase":"maprocks1",
"Hidden":"0",
"connected sta info":[
{
"STA MAC address":"06:0c:43:36:15:45",
"last assoc time":"65535",
"BH STA":"No",
"Medium":"2.4G",
"uplink rssi":"-61"}
]}]
}],
"BH Info":[

],
"Other Clients Info":[
{
"1905 Intf address":"02:31:8e:4a:f7:cf",
"Client Address":"88:a4:c2:75:8f:0c",
"Medium": "Ethernet"}
]
},
{
"AL MAC":"00:0c:43:36:15:45",
"MAP Version":"R3",
"Device role":"02",
"Distance from controller":"1",
"Upstream 1905 device":"00:0c:43:66:15:45",
```


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```
"BH Interface(AP)":[
],
"BH Interface(APCLI)":[
{"MAC address":"06:0c:43:36:37:45",
"interface name":"N/A"}],
"backhaul link metrics":[{
"neighbor_al":"00:0c:43:66:15:45",
"metrics":[{
"local_if_mac":"06:0c:43:36:37:45",
"neighbor_if_mac":"00:0c:43:86:37:45",
"tx packet Errors":"17",
"transmittedPackets":"19411",
"macThroughputCap":"1",
"linkAvailability":"50",
"phyRate":"0",
"rx packet Errors":"0",
"Packets Received":"298",
"RSSI":"-51"
}
],
"non setup metrics 0":[{
"Tx_Link":"0",
"connected_iface_addr":"06:0c:43:36:15:45",
"Iface_type":"265",
"Pkt_err":"23",
"Is_80211_bridge":"0",
"Tx_packet":"0",
"Mac_throughput":"1",
"Link_availability":"50",
"Phy_rate":"0",
"Tx_tp":"0",
"Rx_Link":"0",
"connected_iface_addr":"06:0c:43:36:15:45",
"Iface_type":"265",
"Pkt_err":"0",
"Pkt_received":"0",
"Rssi":"-60",
"Rx_tp":"0"
}
],
```

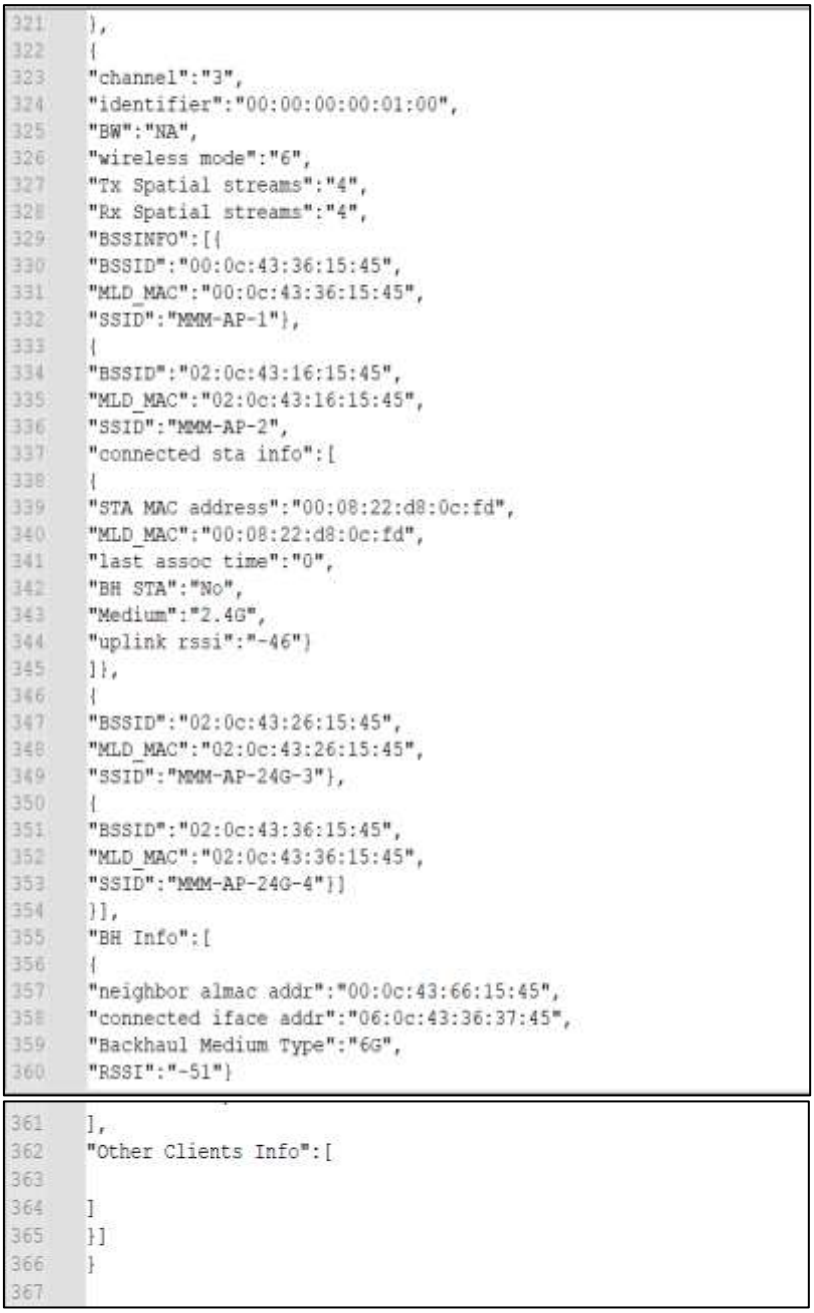

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```
"non setup metrics 1":[{
"Tx_Link":"1",
"connected_iface_addr":"06:0c:43:36:26:45",
"Iface_type":"265",
"Pkt_err":"6",
"Is_80211_bridge":"0",
"Tx_packet":"0",
"Mac_throughput":"1",
"Link_availability":"50",
"Phy_rate":"0",
"Tx_tp":"0",
"Rx_Link":"1",
"connected_iface_addr":"06:0c:43:36:26:45",
"Iface_type":"265",
"Pkt_err":"0",
"Pkt_received":"0",
"Rssi":"-65",
"Rx_tp":"0"
}
]
}
],
"Number of radios":"3",
"Radio Info":[
{
"channel":"9",
"identifier":"00:00:00:00:03:02",
"BW":"NA",
"wireless mode":"6",
"Tx Spatial streams":"4",
"Rx Spatial streams":"4",
"BSSINFO":[{
"BSSID":"00:0c:43:36:37:45",
"MLD_MAC":"00:0c:43:36:15:45",
"SSID":"MMM-AP-1"},
{
"BSSID":"02:0c:43:16:37:45",
"MLD_MAC":"02:0c:43:16:15:45",
"SSID":"MMM-AP-2"},
{
```


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```
"BSSID":"02:0c:43:26:37:45",
"MLD_MAC":"02:0c:43:26:15:45",
"SSID":"MMM-AP-6G-3"},
{
"BSSID":"02:0c:43:36:37:45",
"MLD_MAC":"02:0c:43:36:15:45",
"SSID":"MMM-AP-6G-4"}]
},
{
"channel":"40",
"identifier":"00:00:00:00:02:01",
"BW":"NA",
"wireless mode":"6",
"Tx Spatial streams":"4",
"Rx Spatial streams":"4",
"BSSINFO":[{
"BSSID":"00:0c:43:36:26:45",
"MLD_MAC":"00:0c:43:36:15:45",
"SSID":"MMM-AP-1"},
{
"BSSID":"02:0c:43:16:26:45",
"MLD_MAC":"02:0c:43:16:15:45",
"SSID":"MMM-AP-2",
"connected sta info":[
{
"STA MAC address":"00:08:22:d8:0c:fc",
"MLD_MAC":"00:08:22:d8:0c:fd",
"last assoc time":"65535",
"BH STA":"No",
"Medium":"5G",
"uplink rssi":"-54"}
]},
{
"BSSID":"02:0c:43:26:26:45",
"MLD_MAC":"02:0c:43:26:15:45",
"SSID":"MMM-AP-5LG-3"},
{
"BSSID":"02:0c:43:36:26:45",
"MLD_MAC":"02:0c:43:36:15:45",
"SSID":"MMM-AP-5LG-4"}]
```


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```
},
{
"channel":"3",
"identifier":"00:00:00:00:01:00",
"BW":"NA",
"wireless mode":"6",
"Tx Spatial streams":"4",
"Rx Spatial streams":"4",
"BSSINFO":[{
"BSSID":"00:0c:43:36:15:45",
"MLD_MAC":"00:0c:43:36:15:45",
"SSID":"MMM-AP-1"},
{
"BSSID":"02:0c:43:16:15:45",
"MLD_MAC":"02:0c:43:16:15:45",
"SSID":"MMM-AP-2",
"connected sta info":[
{
"STA MAC address":"00:08:22:d8:0c:fd",
"MLD_MAC":"00:08:22:d8:0c:fd",
"last assoc time":"0",
"BH STA":"No",
"Medium":"2.4G",
"uplink rssi":"-46"}
]},
{
"BSSID":"02:0c:43:26:15:45",
"MLD_MAC":"02:0c:43:26:15:45",
"SSID":"MMM-AP-24G-3"},
{
"BSSID":"02:0c:43:36:15:45",
"MLD_MAC":"02:0c:43:36:15:45",
"SSID":"MMM-AP-24G-4"}]
}],
"BH Info":[
{
"neighbor almac addr":"00:0c:43:66:15:45",
"connected iface addr":"06:0c:43:36:37:45",
"Backhaul Medium Type":"6G",
"RSSI":"-51"}
],
"Other Clients Info":[

]
}]
}
```

Figure 5- 5***. Dump_topology_v1 command JSON file snapshot with MLD parameters***

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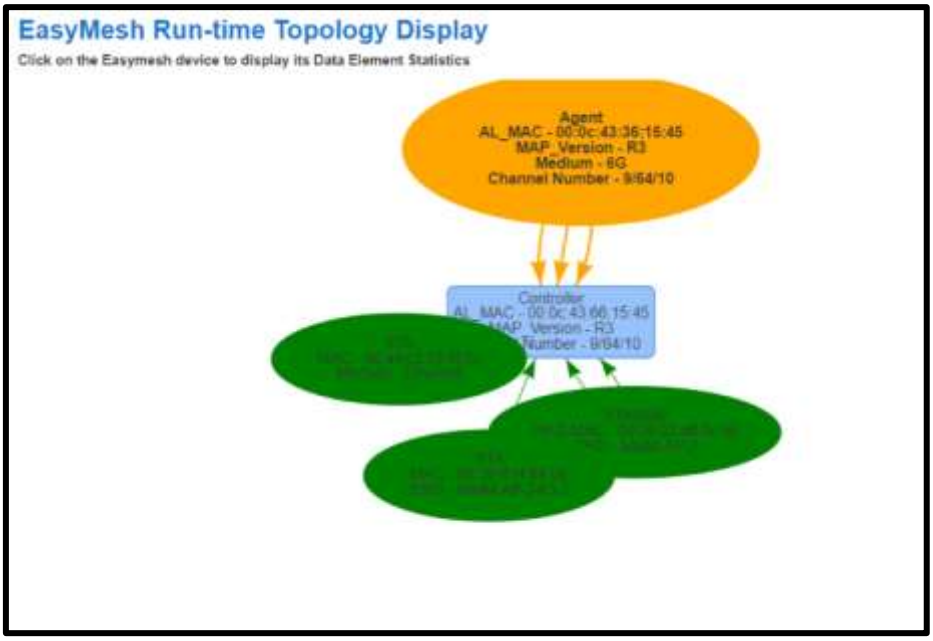


Figure 5- 6. runtime topology with NON-MLD STA, MLD STA and MLD Agent

Details of topology related parameters present in JSON format are present in the below sections.
Parameter Description for JSON Output:

Table 5- 1. Dump Topology Parameter Description

Parameter name	Description	Example	Mandatory
AL MAC	1905.1 abstraction layer (AL) MAC address: A locally administered Extended Unique Identifier-48(EUI-48) value that uniquely identifies a 1905.1 abstraction layer.	AL MAC 00:0c:43:28:07:08	yes
Device role	This refers to the Multi-AP Device Role. It can be an “Agent” or “Controller” or “Auto”. Auto means role has not been decided yet.	0- Auto 1- Controller 2- Agent	yes
Distance from controller	Distance of the MAP device from the controller.	Distance of controller from itself is = 0 Distance of agent from the controller is >= 1 If agent is not connected to the controller then its distance fron controller = 255 (maximum)	yes
Upstream 1905 device	This field comes only if an upstream device exists for the given MAP device	Upstream 1905 device":"00:0c:43:28:07:08",	no
BH Interface(AP)	Own Backhaul AP interfaces at the MAP device. It contains two sub fields: -MAC address (6-Byte) -interface name (eg. ra0/rai0/rai1 etc) , not being filled in current code reserved for later use	BH Interface(AP): MAC address:"00:0c:43:28:07:a2" interface name:"N/A"	no
BH Interface(APCLI)	Backhaul APCLI interfaces at the MAP device. It contains two sub fields: -MAC address (6-Byte) -interface name (eg. apcli0/apclii0 etc) , not being filled in current code reserved for later use	BH Interface(AP): MACaddress:"06:0c:43:28:07:30" interface name:"N/A"	no
backhaul link metrics	Backhaul link metric information provides the link statistics between the device and all its 1905 neighbor, only if there is a neighbor.	backhaul link metrics	Yes

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	This field is present per 1905 device in the network for each of its 1905 neighbors. It contains multiple sub-fields. Please refer to Table-2 for details about the sub-fields.		
Number of radios	Number of bands supported by the MAP device (24G , 5GL , 5GH)	Values can be 0 to 3	yes
Radio Info	This parameter contains lot of details for each radio band supported by the MAP device. Please refer to Table-4 for details regarding the sub-fields.	Radio Info	yes
BH Info	This field will only come in an agent's topology dump. It gives the own uplink backhaul link information about uplink neighbour. It contains multiple sub fields. Please refer Table-5 for the details.	BH Info	no

Table 5- 2. Backhaul Link Metric description

Backhaul Link Metric Parameters	Description	Example
neighbor_al	the ALMAC address of the neighbor	"neighbor_al":"00:0c:43:28:07:65",
metric	link metric structure , it contains the multiple subfields as described in Table-3	"metrics":[{

Table 5- 3. Per BH metric information description

Backhaul Link Metric Parameters	Description	Example
local_if_mac	MAC address of local interface connected to the neighbour	"local_if_mac":"00:0c:43:28:07:a2",
neighbor_if_mac	MAC address of neighbour's interface connected to our local interface	"neighbor_if_mac":"06:0c:43:28:07 :30",
tx packet Errors	Estimated number of lost packets on the transmit side of the link during the measurement period. It is a 4-octets (any integer value).	"tx packet Errors":"0",
transmittedPackets	Estimated number of packets transmitted by the Transmitter of the link on the same measurement period used to estimate packetErrors. It is a 4-octets (any integer value).	"transmittedPackets":"895",
macThroughputCap	The maximum MAC throughput of the Link estimated at the transmitter and expressed in Mb/s. It is a 2-octets (any integer value).	"macThroughputCap":"270",
linkAvailability	The estimated average percentage of time that the link is available for data transmissions. It is a 2-octets value between 0 and 100.	"linkAvailability":"50",
phyRate	Last Tx Rate	"phyRate":"0",
rx packet Errors	Estimated number of lost packets during the measurement period. It is a 4-octets (any integer value).	"rx packet Errors":"0",
Packets Received	Number of packets received at the interface during the same measurement period used to count packet errors. It is a 4-octets (any integer value).	"Packets Received":"547",
RSSI	Estimated RSSI in dB at the receive side of the Link expressed in dB. It is a 1 octet (signed value).	"RSSI":"-63"

Table 5- 4. Radio Information description

Radio Info Parameter	Description	Example
Channel	Channel number of the radio band	channel:36

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Identifier	Unique implementation defined Radio Identifier of the radio band.	identifier:00:00:00:00:01:01
BW	This field is reserved for the Bandwidth of the radio band for future use. Current value of BW is not applicable.	BW: NA
wireless mode	This shows the wireless mode(phy mode) of the current radio band as per the below table: MODE_CCK=0 MODE_OFDM=1 MODE_HTMIX=2 MODE_HTGREENFIELD=3 MODE_VHT=4	wireless mode:4
Tx Spatial streams	shows the HT capability Tx Spatial Stream, value can vary from 0 to 4	Tx Spartial streams:1
Rx Spatial streams	shows the HT capability Rx Spatial Stream, value can vary from 0 to 4	Rx Spartial streams:1
BSSINFO	This parameter is contains the following sub-fields for each BSS for this radio band	BSSINFO
BSSID	The MAC address of the BSS interface	BSSID:00:0c:43:28:07:a2
SSID	The SSID of the BSS interface	SSID:Multi-AP-5LG-1
Security	The Auth mode settings of the BSS interface: 0x0001 - Open 0x0002 - WPA - Personal 0x0004 - Shared 0x0008 - WPA - Enterprise 0x0010 - WPA2 - Enterprise 0x0020- WPA2- Personal	Security:0020
Encryption	The Encryption type settings of the BSS interface 0x0001 - NONE 0x0002 - WEP 0x0004 - TKIP 0x0008- AES	Encryption:0008
Pass-phrase	PASSPHRASE for this BSS (MAX 64 byte)	Pass-phrase:maprocks1
Hidden	The field shows that the BSS is hidden or not	Hidden : 0
connected sta info	If any station is connected to the BSS then the below sub-fields are present for all connected stations	connected sta info
STA MAC address	MAC address of the connected phone, station or client	STA MAC address :9c:35:eb:f0:34:0c
MLD_MAC	MLD_MAC address of connected phone if connection established is MLO connection otherwise it would be ZERO for non-mlo STA	MLD_MAC:00:08:22:d8:0c:fd
last assoc time	This field is reserved for the time since the station got associated to the BSS in seconds. It is a two byte value. Right now the value is always 0 as it is invalid and we do not require it	last assoc time : 0
BH STA	This gives info about the Station, whether it is a 1905 device or not Yes: if client is a 1905 device; No : if client is not a 1905 device.	BH STA : No
Medium : Ethernet/ 2.4G/ 5G	This field tells about the Backhaul connection by means of which the client is connected to this BSS	Medium : 5G
uplink rssi	This is the connected stations RSSI in dBm. At the controller this field is valid for all the connected station in the entire MAP network, but for the Agent dump this field is valid only if the station is connected to the agent on which topology dump is taken	uplink rssi :-88

Table 5- 5. Backhaul Information description

Backhaul Information	Description	Example
neighbor almac	This is a 6 byte value which gives the neighbours AL-MAC	neighbor almac addr:"00:0c:43:28:80:f3"
Connected iface addr	This is a 6 byte MAC address of the interface at the 1905 device by which it is connected to its neighbour.	connected iface addr:"06:0c:43:28:a3:2f"

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Backhaul Medium Type	This gives info about the Backhaul link connection medium (2.4G/5G or Ethernet)	Backhaul Medium Type:"5G"
RSSI	If the BH medium is Ethernet, then this field is N.A. otherwise it gives the RSSI value of the Backhaul link in dBm.	RSSI:"-82"

Table 5- 6. Non-Setup Metrics when device is Agent and MLD-ENABLED

Metric Info Parameter	Description	Example
Tx_Link	Just a additional parameter to tell about which Tx Link's information is to be displayed. As of now value for Tx_Link can be 0 or 1 as there will be only 2 non-setup link and other is primary. The parameters till the parameter «Rx_Link» are associated with this Tx_Link.	"Tx_Link":"0",
Connected_iface_addr	This is a 6 byte MAC address of the interface at the 1905 device which is a non-setup interface address.	"connected_iface_addr":"06:0c:43:36:26:45",
Iface_type	This field shows whether the non-setup link is wireless or Ethernet. In Case of Ethernet: IEEE_802_3_u = 0x0000 IEEE_802_3_ab = 0x0001 If it is wireless then it also specifies its mode also. IEEE_802_11_b = 0x0100 IEEE_802_11_g = 0x0101 IEEE_802_11_a = 0x0102 IEEE_802_11_n_2_4G = 0x0103 IEEE_802_11_n_5G = 0x0104 IEEE_802_11_ac = 0x0105 IEEE_802_11_ad = 0x0106 IEEE_802_11_af = 0x0107 IEEE_802_11_ax = 0x0108 IEEE_802_11_be = 0x0109	"Iface_type":"265",
Pkt_err	Estimated number of lost packets during the measurement period. It is a 4-octets (any integer value).	"Pkt_err":"0",
Is_80211_bridge	It is a boolean type parameter. It signifies whether the current non-setup link is part of the bridge or not.	"Is_80211_bridge":"0",
Tx_packet	Number of packets sent by the interface during the same measurement period used to count packet errors. It is a 4-octets (any integer value).	"Tx_packet":"33",
Mac_throughput	The maximum MAC throughput of the Link estimated at the transmitter and expressed in Mb/s. It is a 2-octets(any integer value).	"Mac_throughput":"1",
Link_availability	The estimated average percentage of time that the link is available for data transmissions. It is a 2-octet value between 0 and 100.	"Link_availability":"50",
Phy_rate	Last Tx Rate	"Phy_rate":"0",
Tx_tp	Transmission Throughput	"Tx_tp":"0",
Rx_Link	Just a additional parameter to tell about which Tx Link's information is to be displayed. As of now value for Tx_Link can be 0 or 1 as there will be only 2 non-setup link and other is primary. All the parameters below this parameter are associated with this Rx Link.	"Rx_Link":"1",
Pkt_received	Number of packets received at the interface during the same measurement period used to count packet erros. It is a 4-octets (any integer value).	"Pkt_received":"128214",
Rssi	If the BH medium is Ethernet, then this field is N.A. otherwise it gives the RSSI value of the Backhaul link in dBm. It is a 1 octet (signed value).	"Rssi":"-57",

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Rx_tp	Receiver Throughput	"Rx_tp":"0",

3. mapd_cli /tmp/mapd_ctrl show_bh_link_metric

Following information is shown on console:

```
root@OpenWrt:/# mapd_cli /tmp/mapd_ctrl show_bh_link_metric
count =3 1=mapd_cli 2=/tmp/mapd_ctrl
I got a commapd SHOW_BH_LINK_METRIC
BH Link Info for Agent 1 with AL-MAC: 00:0c:43:36:15:45
        Agent Mesh Version : 3
        Number of connected clients : 1
        Backhaul BSSID: 00:0c:43:86:37:45
        Setup Link APCLI-MAC : 06:0c:43:36:37:45
                Number of Packet Errors: 6
                Backhaul uplink traffic volume(count of packets): 1937
                Backhaul downlink traffic volume(count of packets): 89
                Backhaul RSSI: -49
                Packet Retransmission count: 122
                Backhaul Link Rate(downlink/uplink)(Mb/s): 6483/1152
                Radio: 6G
                        CH: 9 Op Class: 137 BW : 320
        Non-Setup Link 1 APCLI-MAC : 06:0c:43:36:15:45
                Number of Packet Errors: 5
                Backhaul uplink traffic volume(count of packets): 0
                Backhaul downlink traffic volume(count of packets): 0
                Backhaul RSSI: -58
                Packet Retransmission count: 0
                Backhaul Link Rate(downlink/uplink)(Mb/s): 102/412
                Radio: 2G
                        CH: 3 Op Class: 81 BW : 20
        Non-Setup Link 2 APCLI-MAC : 06:0c:43:36:26:45
                Number of Packet Errors: 0
                Backhaul uplink traffic volume(count of packets): 0
                Backhaul downlink traffic volume(count of packets): 0
                Backhaul RSSI: -66
                Packet Retransmission count: 0
                Backhaul Link Rate(downlink/uplink)(Mb/s): 648/6
                Radio: 5G
                        CH: 40 Op Class: 129 BW : 160
reply=0
root@OpenWrt:/# [mapd][parse_bss_config_report_tlv][7786][TOPO]bss mapg
```

Figure 5- 7. show_bh_link_metric command console log

Table 5-7. Parameter information for show_bh_link_metric command

Metric Info Parameter	Description	Example
Agent Mesh Version	It depicts which MAP Version this ALMAC agent supports	Agent Mesh Version : 3
Number of connected clients	It depicts the count of connected clients to this agent	Number of connected clients : 1
Backhaul BSSID	The backaul BSSID of uplink device to which APCLI interface of this agent is connected to	00:0c:43:36:15:45
APCLI-MAC	The 48 bit MAC address of APCLI interface	Setup Link APCLI-MAC : 06:0c:43:36:37:45
Number of Packet Errors	Estimated number of lost packets during the measurement period. It is a 4-octets (any integer value).	Number of Packet Errors: 6
Backhaul uplink traffic volume	Estimated uplink traffic volume in the count of packet format during the measurement period. It is a 4-octets (any integer value).	Backhaul uplink traffic volume(count of packets): 1937
Backhaul downlink traffic volume	Estimated downlink traffic volume in the count of packet format during the measurement period. It is a 4-octets (any integer value).	Backhaul downlink traffic volume(count of packets): 89
Backhaul RSSI	If the BH medium is Ethernet, then this field is N.A. otherwise it gives the RSSI value of the Backhaul link in dBm. It is a 1 octet (signed value).	Backhaul RSSI: -49
Packet Retransmission count	Estimated number of packets re-transmitted by the Transmitter of the link on the same measurement period used to estimate number of packet errors. It is a 1-octet (any integer value).	Packet Retransmission count: 122
Backhaul Link Rate	It displays the downlink/uplink rate for this apcli link. It's units is Mb/s	Backhaul Link Rate(downlink/uplink)(Mb/s): 5763/1152
Radio	Depicts the radio type(2g/5g/6g) information comprising of: 1. CH : operating channel number 2. Op Class : Operating class number	Radio: 6G CH: 9 Op Class: 137 BW : 320

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		3. Bw : the BW associated to this radio	

4. mapd_cli /tmp/mapd_ctrl show_sta_metric

Following information is shown on console:

```
root@OpenWrt:/# mapd_cli /tmp/mapd_ctrl show_sta_metric
count =3 1=mapd_cli 2=/tmp/mapd_ctrl
I got a commapd SHOW_STA_METRIC
STA INFO for sta : 36:53:89:c7:f3:e8
        Uplink Speed(Mb/s): 263
        Downlink Speed(Mb/s): 780
        STA uplink traffic volume(count of packets): 188
        STA downlink traffic volume(count of packets): 0
        Packet Retransimission count: 25
        Uplink RSSI: -48
        BH STA : NO
        Medium: 5G
STA INFO for sta : 6e:1f:13:d6:06:ba
        Uplink Speed(Mb/s): 6
        Downlink Speed(Mb/s): 468
        STA uplink traffic volume(count of packets): 1160
        STA downlink traffic volume(count of packets): 0
        Packet Retransimission count: 137
        Uplink RSSI: -55
        BH STA : NO
        Medium: 5G
reply=0
root@OpenWrt:/#
```

Figure 5- 8. show_sta_metric command console log

Table 5-8. Parameter information for show_sta_metric command

Metric Info Parameter	Description	Example
Uplink speed	The uplink speed of this STA in Mb/s	Uplink Speed(Mb/s): 263
Downlink speed	The downlink speed of this STA in Mb/s	Downlink Speed(Mb/s): 780
STA uplink Traffic Volume	Number of packets sent at the interface during the measurement period. It is a 1-octet(any integer value).	STA uplink traffic volume(count of packets): 188
STA downlink traffic volume	Number of packets received at the interface during the measurement period. It is a 1-octet(any integer value).	STA downlink traffic volume(count of packets): 0
Packet Retransimission count	Estimated number of packets re-transmitted by the Transmitter of the link on the measurement period. It is a 1-octet (any integer value).	Packet Retransimission count: 25
Uplink RSSI	This is the connected stations RSSI in dBm.	Uplink RSSI: -48
BH STA	This gives information about the Station, whether it is a 1905 device or not. YES : if client is a 1905 device. NO: if client is not a 1905 device.	BH STA : NO
Medium	Shows the connectd clients and the band on which this connected STA is present	Medium: 5G

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5.2.2.2 WEBUI Interface

You can see the Topology details in the UI as well. Click on Show all device info

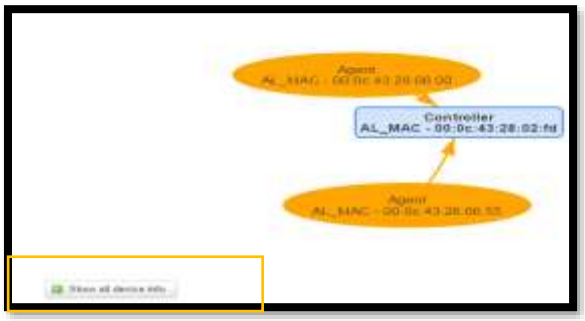


Figure 5- 9. Topology display WEB UI page snapshot 1

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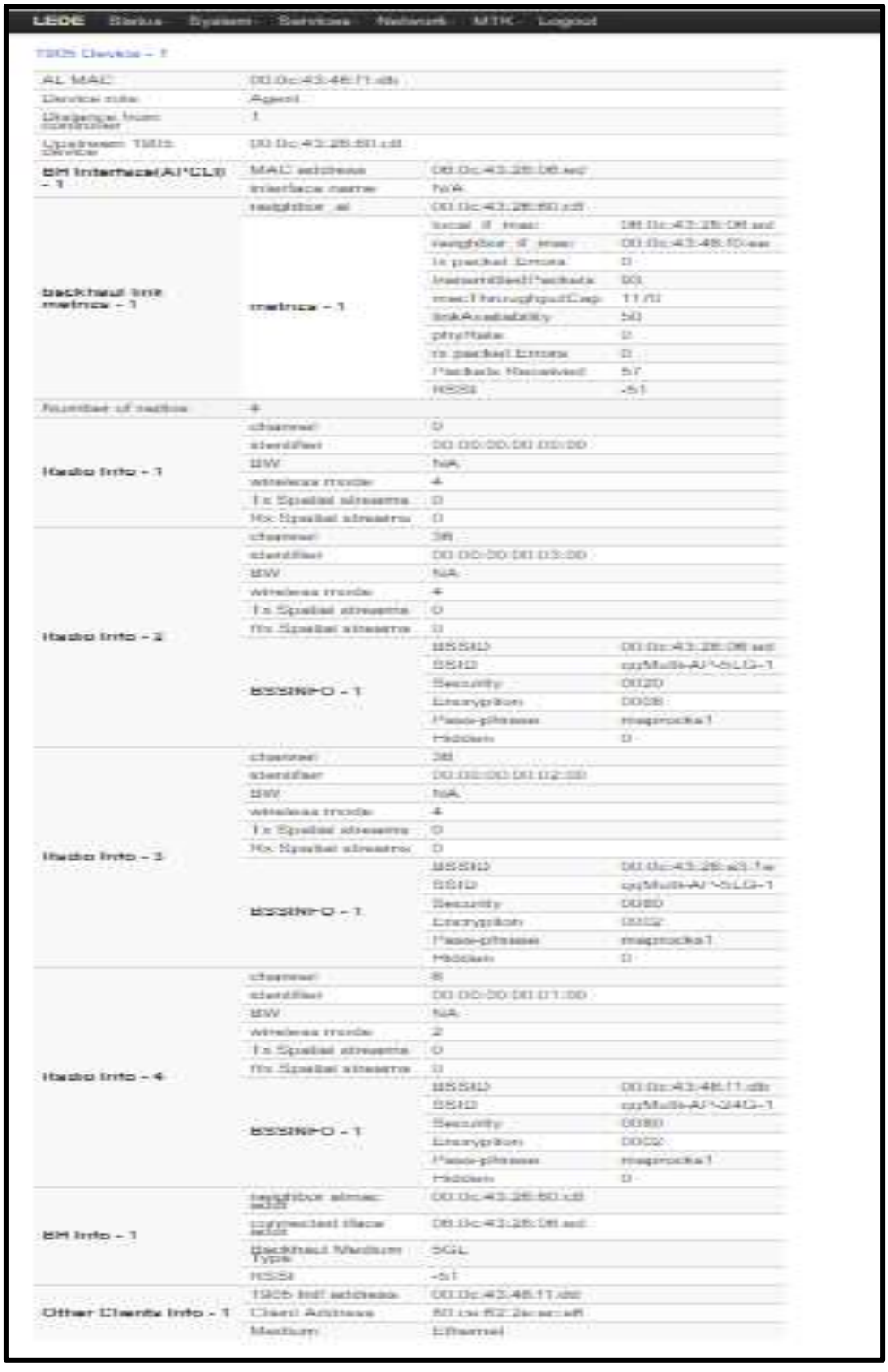


Figure 5- 10. Topology display WEB UI page snapshot 2

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Similarly if you scroll down the page, information for other devices in the network will also come.

5.3 Client Database

5.3.1 Feature details

An AP must be able to handle various clients. Legacy clients do not support 802.11v BTM transition and therefore 11v BTM request is not applicable. Some newer clients support 802.11v BTM but their behaviors vary depending on the implementation.
Per the experiments, some 11v-capable clients always followed BTM requests from the AP and others only accepted the requests conditionally. Therefore, the AP must find a suitable steering method for each client adaptively.
An algorithm runs to learn client capabilities from each steering result and becomes better as more steering occurs. Algorithm adapts the 11k, 11v characteristics based on previous results and takes better steer decisions next time

The learnings from this algorithm are stored in a client database and are synchronized among the Multi-AP devices to take better steering decisions.

A total of 128 client entries are maintained by each Multi-AP device. If the client database is full and a new entry needs to be added, a client entry which was least recently used in the client database is removed.

5.3.2 Feature Management Interfaces

5.3.2.1 Command line Interface

1. Run the following command to get the client list and Client MAC to Client ID mapping
   mapd_cli /tmp/mapd_ctrl mib sta_seen_list

```
root@OpenWrt:/#
root@OpenWrt:/# mapd_cli /tmp/mapd_ctrl mib sta_seen_list
count =4 1=mapd_cli 2=/tmp/mapd_ctrl
Succesfully opened connection to mapd
I got a commapd MIB sta_seen_list
MIB cmd
reply=730
STA SEEN List:
        d4:8a:39:15:e9:26       id = 0
        2e:a2:5b:07:66:aa       id = 1
        4a:86:f9:26:07:77       id = 2
        d6:a5:9b:5c:42:1f       id = 4
        08:25:25:b3:c9:f0       id = 3
        16:6b:eb:1f:15:04       id = 6
        de:44:e5:62:38:44       id = 7
        82:68:6c:5c:fb:53       id = 8
        66:39:e9:fb:d6:ff       id = 9
        b6:de:6e:48:a6:13       id = 5
        7c:78:7e:b0:5d:41       id = 11
        32:6b:f2:20:64:a4       id = 12
        06:2f:d5:73:dc:c1       id = 13
        0a:c6:5e:40:37:2c       id = 14
        2e:d2:fb:2a:32:63       id = 15
        08:25:25:a1:f1:d4       id = 16
        6e:00:af:34:a3:a8       id = 17
        7a:70:40:a5:59:d6       id = 18
        1e:ed:44:22:fd:69       id = 19
        82:6d:93:43:71:bf       id = 20
        32:83:2f:35:eb:ac       id = 21
        9a:25:16:6f:4b:3a       id = 10
        8c:b8:4a:67:a7:2d       id = 22
        b4:c4:fc:9d:be:ec       id = 24
        90:78:b2:be:dd:71       id = 23
```

Figure 5- 11. MIB sta_seen_list command console logs snapshot

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1. Select the client ID for which information need to be shown and issue the following command to get the information
    a. mapd_cli /tmp/mapd_ctrl mib sta <client ID>
    Eg 1 : mapd_cli /tmp/mapd_ctrl mib sta 23



Figure 5- 12. MIB sta <id> command console logs snapshot

Eg 2: mapd_cli /tmp/mapd_ctrl mib sta 57

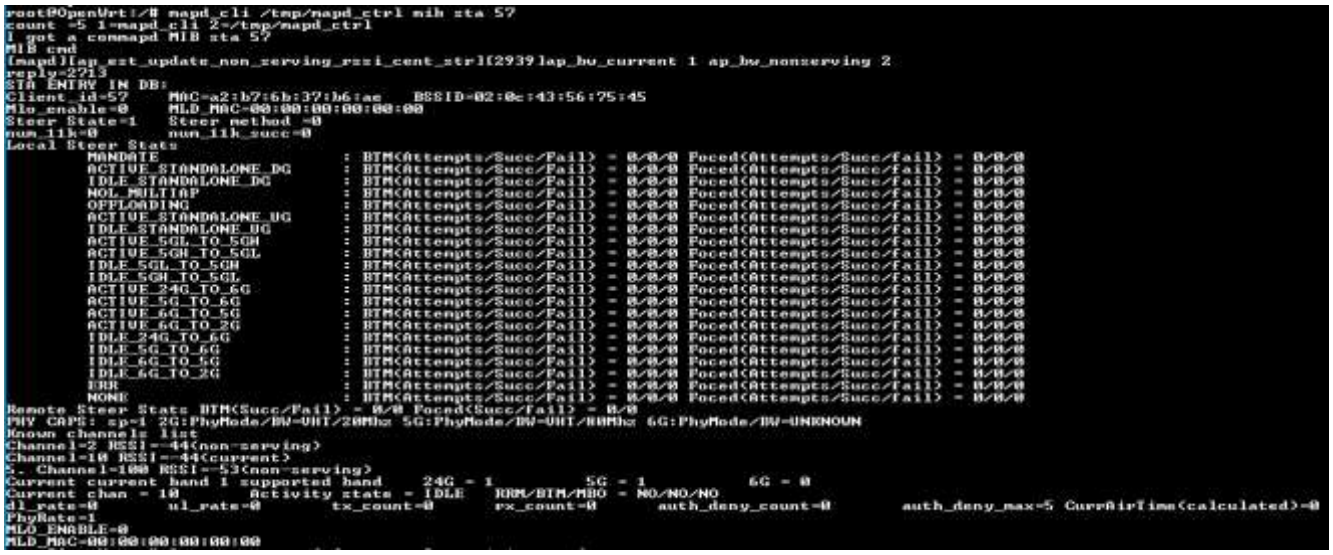


Figure 5- 13. MIB sta <id> command console logs snapshot with MLD parameters

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5.3.2.2 WEBUI Interface

To see the information of all the clients connected in the network, click on “Display Client Capabilities”

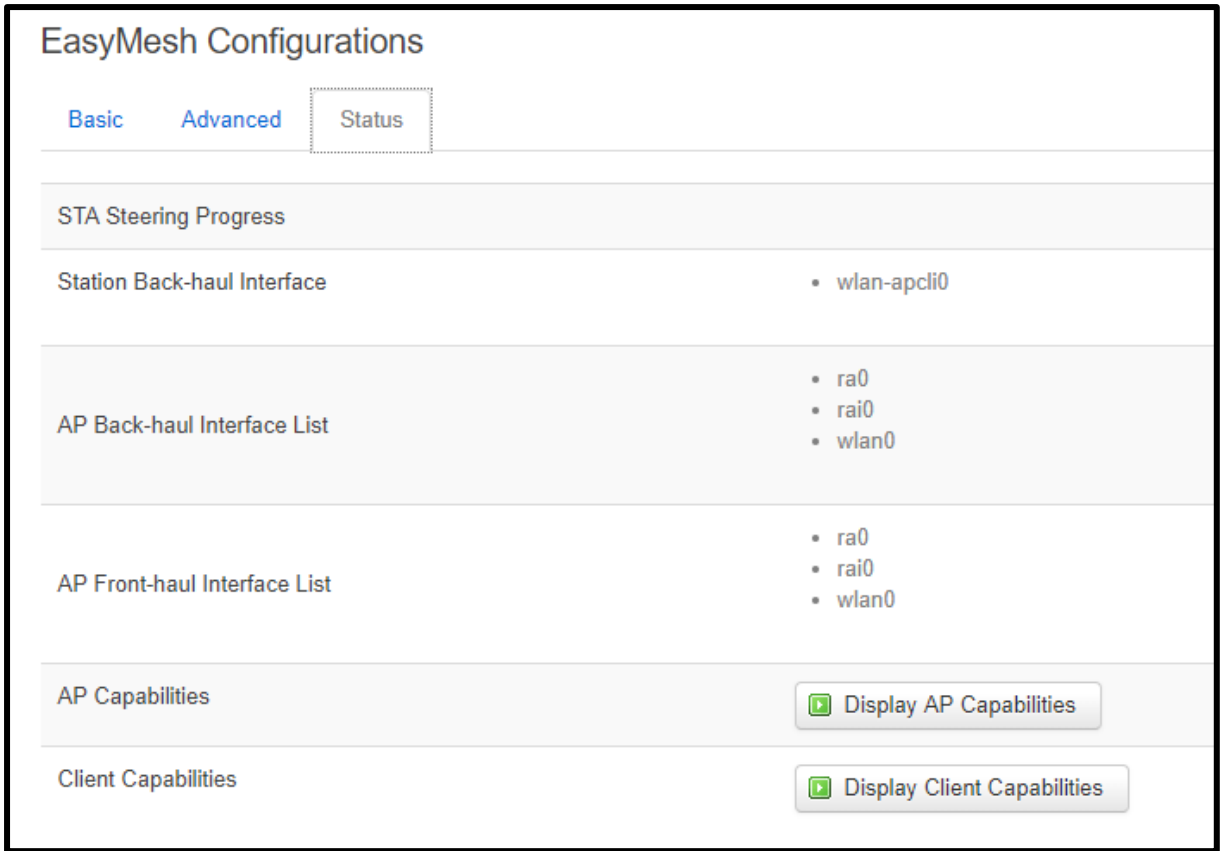


Figure 5- 14. WEB UI Client capability display button



Figure 5- 15. WEB UI Client capability page snapshot



Figure 5- 16. WEB UI Client capability page snapshot with MLD parameters

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5.4 Data Elements

5.4.1 MAP R2 DE Support

We support all the main data elements displays as supported by the Easymesh version R2 specification. The significance of some of these statistics is as follows:

Table 5- 7. MAP R2 Data elements

S. no.	Parameter Name	Significance
Radio Info - 1		
1	identifier	This is a Radio ID which indicates that band is 5G or 2.4G. For example on AX1800 SKU : 00:00:00:00:01:01 stands for 5G Radio and 00:00:00:00:01:00 stands for 2.4G
2	Radio Noise	This field is currently not operational due to HW limitation of 7915 chip.
3	Radio Transmit	The percentage of time (linearly scaled with 255 representing 100%) the radio has spent on individually or group addressed transmissions by the AP. When more than one channel is in use by BSS operating on the radio, then the Transmit value is calculated only for the primary channel.
4	Radio Receive Self	The percentage of time (linearly scaled with 255 representing 100%) the radio has spent on receiving individually or group addressed transmissions from any STA associated with any BSS operating on this radio. When more than one channel is in use by BSS operating on the radio, then the ReceiveSelf value is calculated only for the primary channel.
5	Radio Receive Other	The percentage of time (linearly scaled with 255 representing 100%) the radio has spent on receiving valid IEEE 802.11 PPDUs that are not associated with any BSS operating on this radio. When more than one channel is in use by BSS operating on the radio, then the ReceiveOther value is calculated only for the primary channel.
BSSInfo		
1	UnicastBytesSent	BSS wide statistics for total unicast bytes transmitted.
2	UnicastBytesReceived	BSS wide statistics for total unicast bytes received.
3	MulticastBytesSent	BSS wide statistics for total multicast bytes transmitted.
4	MulticastBytesReceived	BSS wide statistics for total multicast bytes received.

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5	BroadcastBytesSent	BSS wide statistics for total broadcast bytes transmitted.
6	BroadcastBytesReceived	BSS wide statistics for total broadcast bytes received.
7	AssociationStatus	Does this BSS allow association of any client or not
8	Number of Clients	The number of STAs currently associated to this BSS.
Client Extended Info - 1		
1	LastDataDownlinkRate	The data transmit rate in Kbps that was most recently used for transmission of data PPDUs from the access point to the associated STA.
2	LastDataUplinkRate	The data transmit rate in Kbps that was most recently used for transmission of data PPDUs from the associated STA to the AP.
3	UtilizationReceive	The amount of time the radio has spent on the channel receiving data from this STA in milliseconds.
4	UtilizationTransmit	The amount of time the radio has spent on the channel transmitting data to this STA in milliseconds.

5.4.2 MAP R3 DE Support

- Multi-AP R3 improving its comprehensive diagnostic information, increasing the scope of metrics covered in line with their availability from enhanced Wi-Fi protocols such as Data Elements support of these metrics.
- R3 DE feature includes the following :
  - Controller use AP Capability Query to get Diagnostics Information.
  - Agent sends Device Inventory TLV information using AP Capability report message.

5.4.2.1 R3 DE Spec Details

- **AP Capability Report Message:**
  - If a Multi-AP Agent that implements Profile-3 sends an AP Capability Report message, it shall also perform the following:
    - Include one Device Inventory TLV in the AP Capability Report message.
  - If a Multi-AP device that implements Profile-3 sends an AP Capability Report message, then for the following fields in the Device Inventory TLV:
    - the Serial Number string shall remain fixed over the lifetime of the device, including across firmware updates
    - if a Multi-AP Agent supports multiple firmware images, the Software Version string shall be the software version of the active firmware image
    - to allow version comparisons, the Software Version string should be in the form of dot-delimited integers, where each successive integer represents a more minor category of variation. For example, 3.0.21 where the components mean: Major.Minor.Build
    - if a Multi-AP Agent supports multiple execution environments, the ExecutionEnv string shall be the active execution environment

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5.4.2.2 R3 DE design

- **Controller Scope:**
  - Controller of MAP R3 will send the Capability Query to Agent.
  - This inform MAP R3 Agents to respond the Capability report with Device Inventory TLV.
  - AP Capability Query message format
    - No TLVs are required in this message.
- **Agent Scope:**
  - On receiving Capability Query message from Controller, agent will respond with Capability Report message.
  - Capability report message includes:

AP Capability Report message format

The following TLVs shall be included in this message:

- One AP Capability TLV (see section 17.2.6) [Profile-1].
- One or more AP Radio Basic Capabilities TLV (see section 17.2.7) [Profile-1].
- Zero or more AP HT Capabilities TLV (see section 17.2.8) [Profile-1].
- Zero or more AP VHT Capabilities TLV (see section 17.2.9) [Profile-1].
- Zero or more AP HE Capabilities TLV (see section 17.2.10) [Profile-1].
- Zero or more AP Wi-Fi 6 Capabilities TLV (see section 17.2.72) [Profile-3].
- One Channel Scan Capabilities TLV (see section 17.2.38) [Profile-2].
- One Device 1905 Layer Security Capability TLV (see section 17.2.67) [Profile-3].
- One CAC Capabilities TLV (see section 17.2.46) [Profile-2].
- One Profile-2 AP Capability TLV (see section 17.2.48) [Profile-2].
- One Metric Collection Interval TLV (see section 17.2.59) [Profile-2].
- **One Device Inventory TLV (see section 17.2.76) [Profile-3].**

5.4.2.3 R3 DE SW Stack

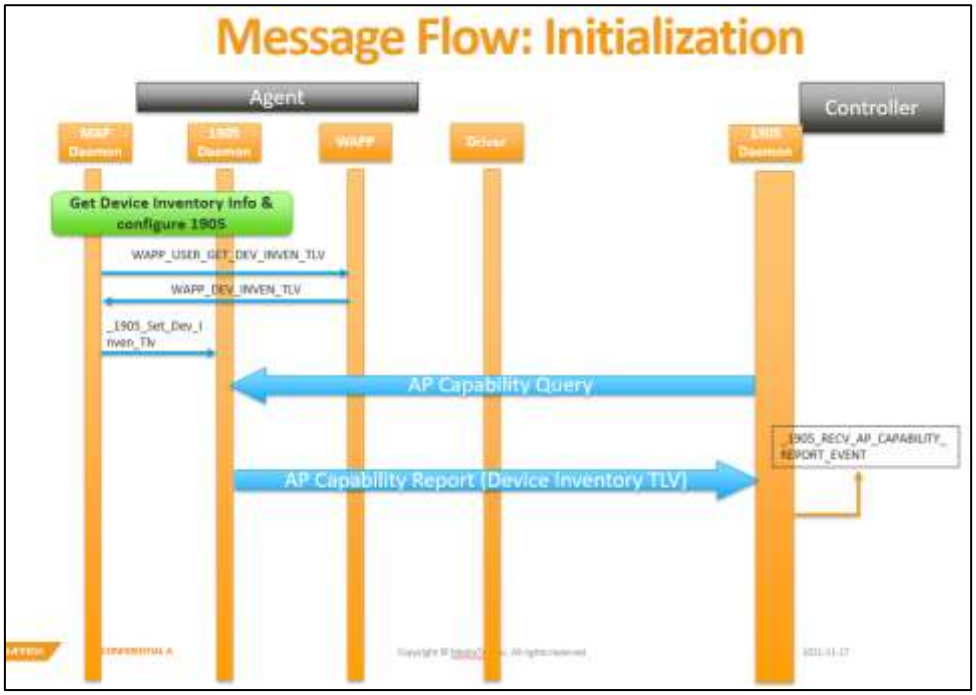


Figure 5- 17. R3 DE SW Stack Msg flow diagram

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- MAPD gets it data from WAPP by using WAPP_USER_GET_DEV_INVEN_TLV

5.4.3 Feature Management Interfaces

5.4.3.1 MAP R3 DE value changes are per SKU

In mapd_default.cfg, there are DE related fields mentioned, ADMIN can change these fields as per there requirement and SKU.

Example:-
/*this field is to tell the serial number of the board.*/
DESerialNumber=v3.0

/*this field is to be used as the SW/SDK version running on the board*/
DESoftwareVersion=v3.0

/*this field is to be used as the EXECUTION env. */
DEExecutionEnv=Linux-Openwrt

/*this filed is to be used as vendor using this chip*/
DEChipsetVendor=MediaTek

5.4.3.2 Command Line Interface

The following commands can be used for getting DE statistics of a particular device.

- Command to refresh the DE stats of a particular device. Input the ALMAC of the device.
- Eg: mapd_cli /tmp/mapd_ctrl get_de_stats 00:0c:43:48:f4:f0
- Command to dump the stats from the variable to a file (file path: /tmp/dump_de.txt)

  Eg: mapd_cli /tmp/mapd_ctrl get_de_stats_dump

5.4.3.3 WEB UI Interfaces

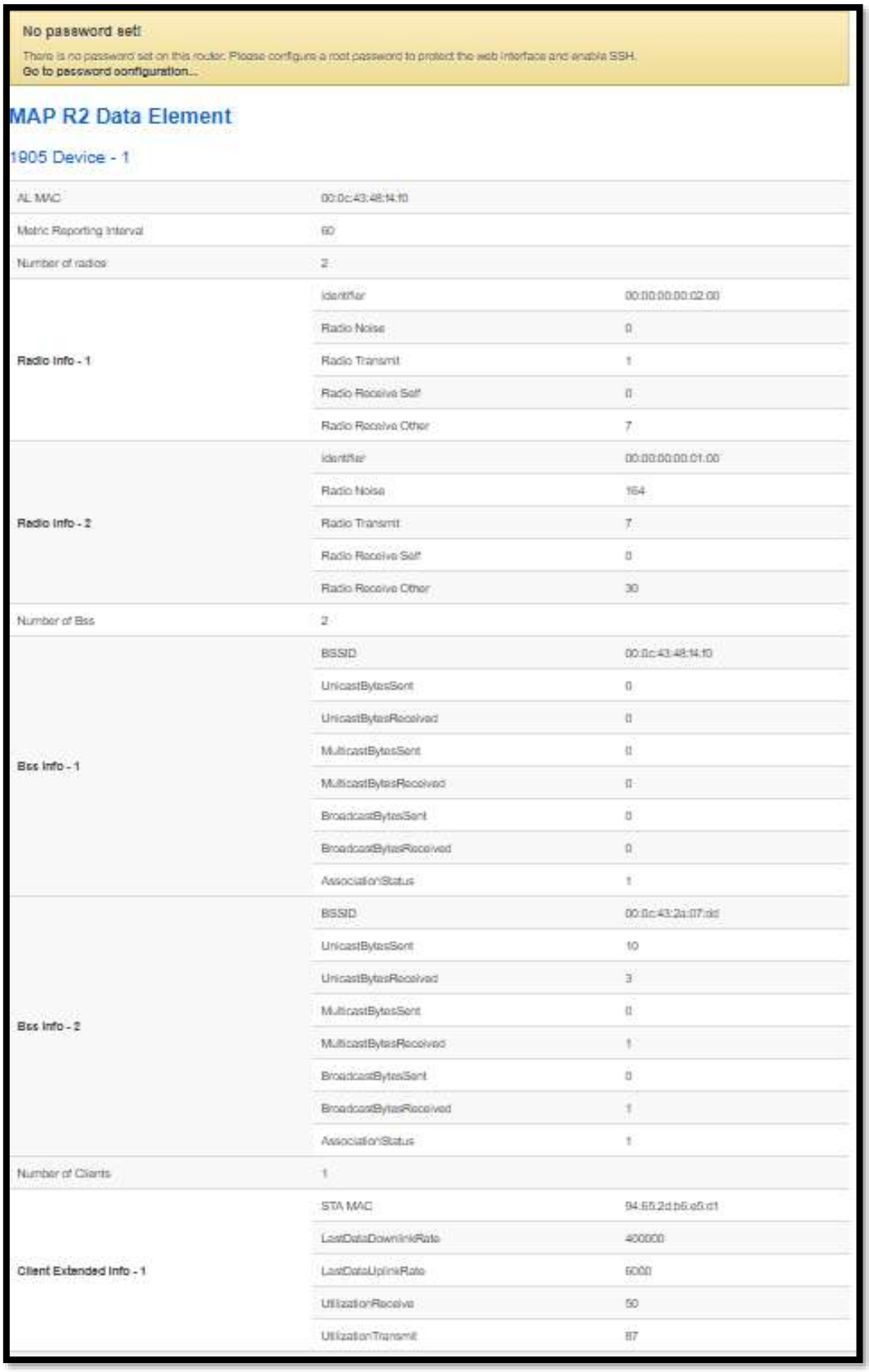
- Open the Runtime Topology Display page
- Choose a MAP device by clicking on the bubble for that MAP dev
- A new page opens on which the Data element statistics of that device are visible.

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Figure 5- 18. Click on topology display bubble for DE display page

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No password set!

There is no password set on this router. Please configure a root password to protect the web interface and enable SSH.

Go to password configuration...

MAP R2 Data Element

1905 Device - 1

AL MAC	00:0c:43:48:f4:f0	
Metric Reporting Interval	60	
Number of radios	2	
Radio Info - 1	Identifier	00:00:00:00:02:00
	Radio Noise	0
	Radio Transmit	1
	Radio Receive Self	0
	Radio Receive Other	7
Radio Info - 2	Identifier	00:00:00:00:01:00
	Radio Noise	164
	Radio Transmit	7
	Radio Receive Self	0
	Radio Receive Other	30
Number of Bss	2	
Bss Info - 1	BSSID	00:0c:43:48:f4:f0
	UnicastBytesSent	0
	UnicastBytesReceived	0
	MulticastBytesSent	0
	MulticastBytesReceived	0
	BroadcastBytesSent	0
	BroadcastBytesReceived	0
	AssociationStatus	1
Bss Info - 2	BSSID	00:0c:43:2a:07:dd
	UnicastBytesSent	10
	UnicastBytesReceived	3
	MulticastBytesSent	0
	MulticastBytesReceived	1
	BroadcastBytesSent	0
	BroadcastBytesReceived	1
	AssociationStatus	1
Number of Clients	1	
Client Extended Info - 1	STA MAC	94:65:2d:b6:e5:d1
	LastDataDownlinkRate	400000
	LastDataUplinkRate	6000
	UtilizationReceive	50
	UtilizationTransmit	87

Figure 5- 19. Data Elements (DE) Display Page WEB UI snapshot

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5.5 Wi-Fi6 Support

- Supporting Wi-Fi 6 in MAP R3 means that the Controller can learn the capabilities of the Wi-Fi 6 capable Agents and Wi Fi 6 capable associated STAs so that it can better utilize Wi-Fi 6 operations in the MAP network.
- In R3, new WiFi6 Capability TLV has been added.
- For R1, R2 HE Capability TLV was provided for WiFi 6 Capability information sharing to controller by Agent.
- Controller use AP Capability Query to get WiFi6 Capability Information, and Agent sends WiFi6 Capability information using AP Capability response message.

5.5.2 Wi-Fi 6 Details

- **Capability Information**
  - Agent informs controller about WiFi6 capability in AP capability report message:
    - Role: AP/Non-AP STA role
    - Support/No Support Parameters :
      - HE 160
      - HE 80+80
      - MU Beamformer
      - SU Beamformee/Beamformer
      - Beamformee STS Less 80/Greater 80
      - UL MU-MIMO
      - UL/DL OFDMA
      - Max DL MU-MIMO TX/ Max UL MU-MIMO RX
      - Max DL OFDMA TX/ Max UL OFDMA RX
      - RTS/MU RTS
      - Multi- BSSID
      - MU EDCS
      - TWT Requester/TWT Responder
    - MCS/NSS
- **Policy configuration**
  - Controller configures the policy for WiFi6 STA Status Report using Metric Policy Reporting TLV
    - 0: Do not include Associated Wi-Fi6 STA Status TLV in AP Metrics Response
    - 1: Include Associated Wi-Fi6 STA Status TLV in AP Metrics Response Associated Wi-Fi6 STA Status Inclusion Policy.

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5.5.3 Wi-Fi 6 SW Stack

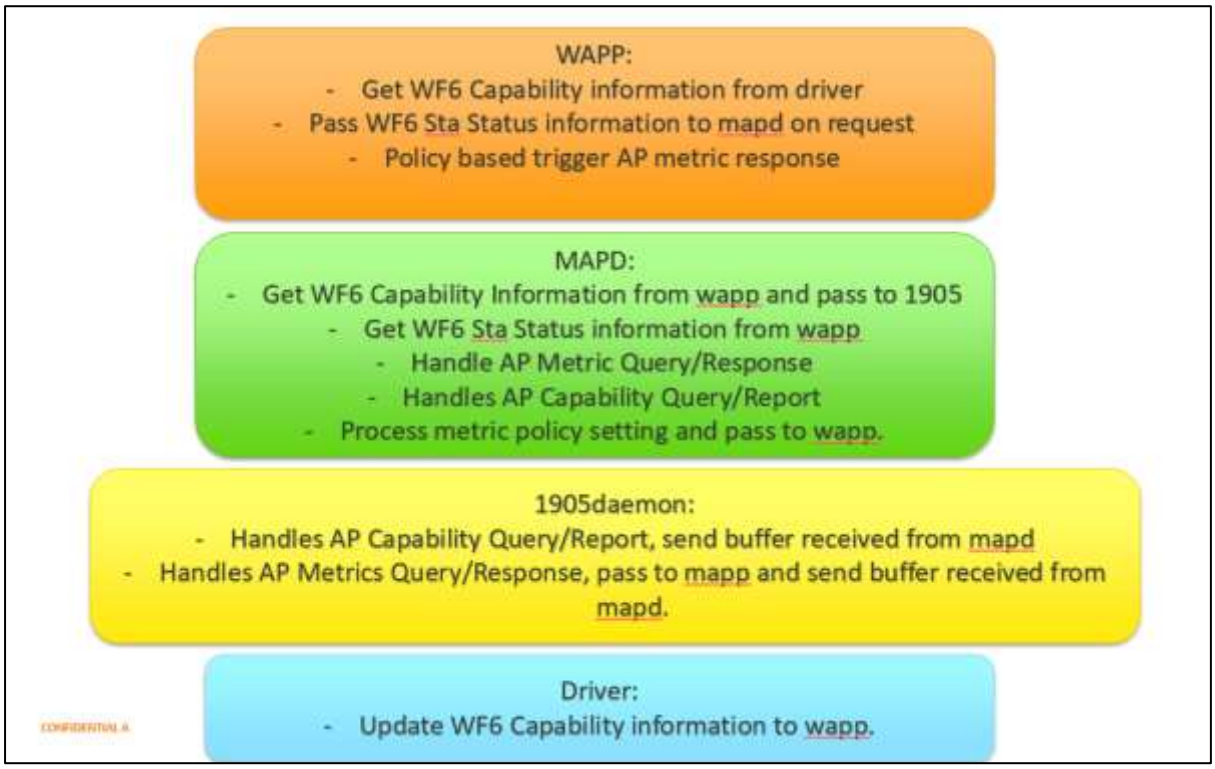


Figure 5- 20. WiFi 6 SW Stack

- To get the WiFi6 capability from Driver to MAPD
- MAPD gets wifi6 capability at boot time from WAPP by using the "WAPP_USER_GET_AP_WF6_CAPABILITY".
- Response of the same is received by WAPP on event name "WAPP_AP_WF6_CAPABILITY".
- WF6 capabilities are taken from Driver by WAPP by OID "OID_WF6_CAPABILITY".

- "WAPP_USER_GET_ASSOC_WIFI6_STA_STATUS" this is to get the associated WiFi6 client as per the spec from WAPPD.
- Response will come "WAPP_ASSOC_WIFI6_STA_STATUS" in this format from WAPPD.

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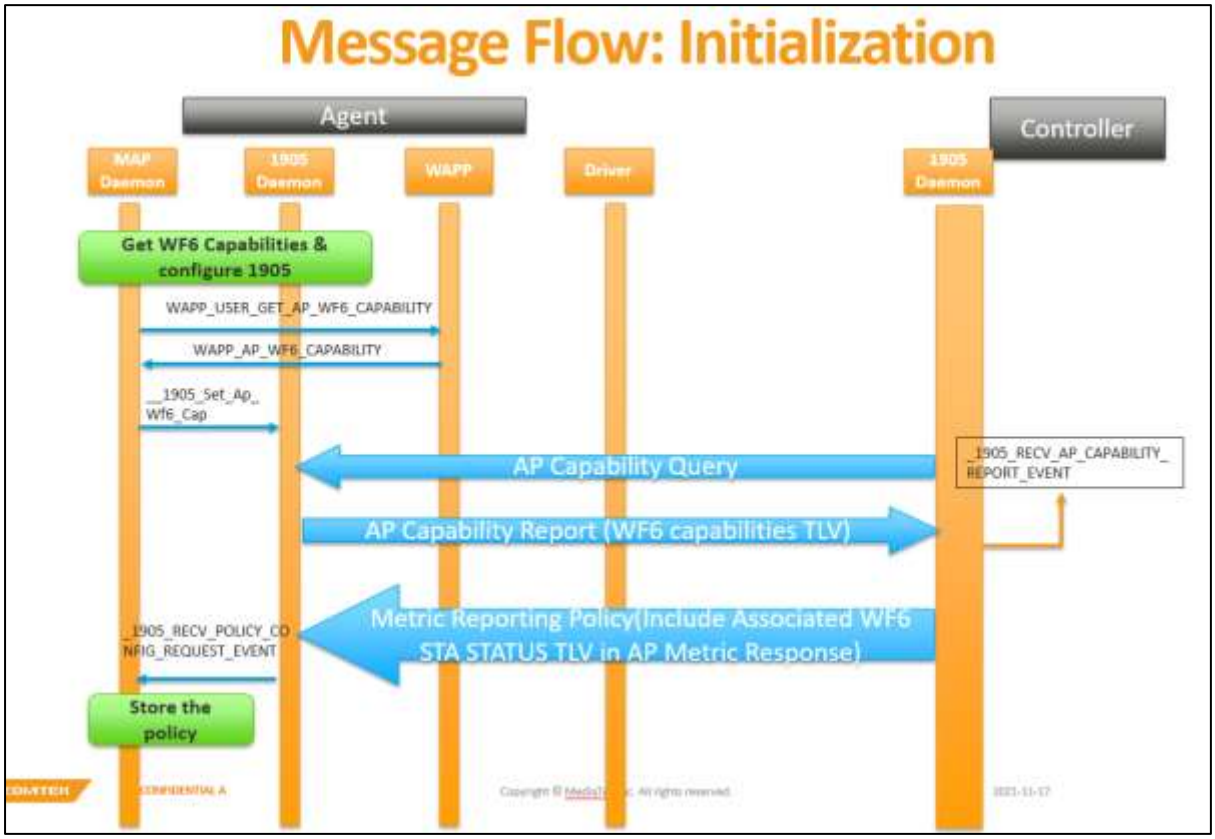


Figure 5- 21. WiFi 6 Msg Flow Image 1

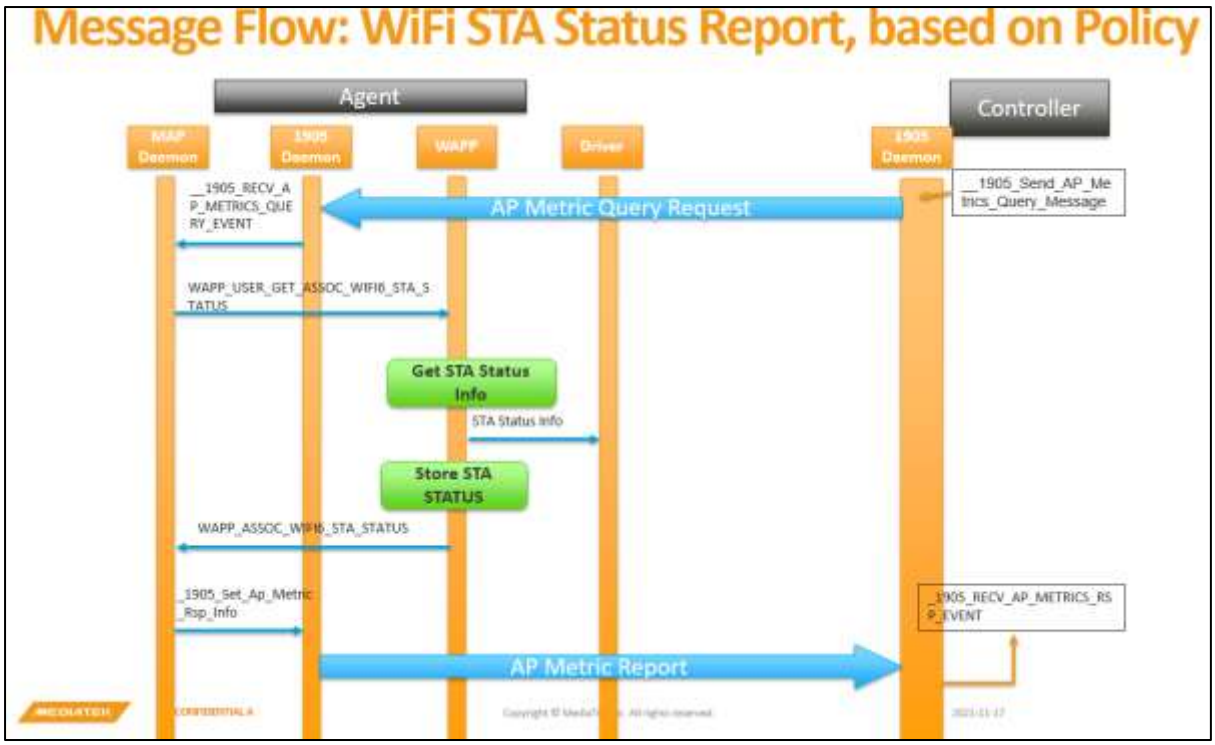


Figure 5- 22. WiFi 6 Msg Flow Image 2

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5.6 DPP Configuration file

MAP R3 controls the DPP connection using DPP Configuration file present in /etc/dpp_cfg.txt folder.

Compile Time Parameters:

//dpp config file
curve_name=prime256v1
allowed_role=1 → By default setting is agent
map_support=1

Run Time Parameters:

//Default dpp config file on Image flash
curve_name=prime256v1
allowed_role=1 → Setting will be changed based on controller/Agent/Auto Role setting. If done from WEB UI, it will automatically update the settings. If done manually then need to modify the settings manually.
map_support=1



Figure 5- 23. dpp config file after device is configured as Controller



Figure 5- 24. dpp config file after device is configured as Agent



Figure 5- 25. dpp config file of Controller device when Agent onboard to Controller

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Figure 5- 26. dpp config file of Agent device when Agent onboard to Controller

Parameter details maintained in /etc/dpp_cfg.txt:

- curve_name = Defines the Elliptic curve and prime256v1 is supported.
- allowed_role = Defines the device role for DPP:
  - 1 – Agent/Enrollee
  - 2 – Controller/Configurator
- map_support = For MAP it needs to be set as 1.
- dpp_private_key = Private Protocol Key. This will be AutoSaved upon bootstrap key generate.
- _akm = This variable used only on Agent for connection.
  - 1 – DPP, 2- PSK, 3-SAE, 4- SAE+PSK, 5-PSK+DPP, 6-SAE+DPP, 7 – DPP+SAE+PSK
- _1905connector, _1905cSignKey, _1905netAccessKey explained in section 5.7
- _netKeyExpiry not used in Turnkey.
- _decrypThreshold used by 1905.
- _ppKey = Public Configurator privacy protection key
- agt_qr_code_1 = Agent QR code added on Controller.
- qr_count = Shows number of qr code saved on controller.
- bSTA Configuration response parameters saved on Agent:
  - _ssid = Backhaul AP SSID to connect.
  - _passPhrase = Key for Backhaul AP connection received in configuration response.
  - _connector = It's verified by Csign key on each device. And used to generate PMK on each device.
  - _netAccessKey= Is used during Network Introduction Protocol.
  - _cSignKey= Csign key is used to verify the signed connector on all Agents. Need save unique Csign key on all devices to check the signature.
  - _Valid = 1 → means profile save for AP is valid and need to be used for connection.
  - presence_band_priority - DPP chirping will be started based on priority flag present in /etc/dpp_cfg.txt i.e presence_band_priority. If presence_band_priority=5g, then presence announcement will be started on 5G channels first then on 2G channels. EasyMesh Daemons needs to be restarted post this change or need to reboot the device in order to apply this change.
    - **Note : By default presence_band_priority variable not set in /etc/dpp_cfg.txt, so default value will be used as 5g.**

For 1905 parameters maintained during reboot refer section 5.7

For Boot Up and link failover support, all connectors and URI will be stored in this file. And System will use these connectors and URI to form connection on boot up.

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5.7 1905 DPP parameters stored

Need to save necessary parameters on each device once any device leave, reboot. Then the device will recovery with these parameters.

All the parameters will be kept in /etc/map/1905_dpp_keys.txt



Figure 5- 27. 1905_dpp_keys.txt file snapshot

5.7.1 csign key

csign key is used to sign the connector of 1905 layer on controller.
All the connectors on each device are signed by the same csign key by controller in the MAP network.

5.7.2 Csign key

Csign key is used to verify the signed connector on all Agents
Need save unique Csign key on all devices to check the signature

5.7.3 private net-access-key(pk)

pk and NK(Net-access-key) are pair of net-access-key
They are used to generate N.x to negotiate PMK on two devices.
pk need to be stored on each device.
NK don't need to be stored as it is stored in connector, so don't need to store again.
We need make sure N.x = pk * NK is the same on two peer devices by DH algorithm

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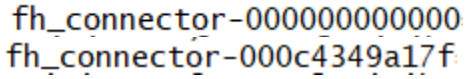
5.7.4 Connector

Connector of FH BSS on each device

It’s verified by Csign key on each device.
And used to generate PMK on each device.

5.7.5 fh_connector

FH connector of FH BSS for each device on Controller.
fh_connector-<almac>:



if almac all zero, indicate it’s fh_connector of Controller itself;
if almac non-zero, indicate it's fh_connector of one agent

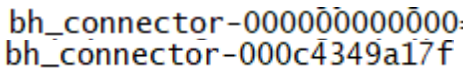
Save fh_connector for each device on Controller after onboarding;
Read and apply self fh connector to driver after reboot for controller
Read and save the fh connector for each agent in memory for each, will give fh connector to each agent after bh connection.

FH connector of FH BSS for each agent on Controller.

Save fh_connector for each device on Controller, read and apply after reboot, will

5.7.6 bh_connector

BH connector of BH BSS for each device on Controller.
bh_connector-<almac>:



if almac all zero, indicate it’s bh_connector of Controller itself;
if almac non-zero, indicate it's bh_connector of one agent

Save bh_connector for each device on Controller after onboarding;
Read and apply self bh connector to driver after reboot for controller
Read and save the bh connector for each agent in memory for each, will give bh connector to each agent after bh connection.

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6 Steering

6.1 Introduction

Mediatek's Multi-AP Turnkey solution provides a centralized approach in which the controller is responsible for the client steering decisions for all the clients connected to any device of the entire multi-ap network.

Centralized steering includes:

1. **Multi -AP Steering**: Steering client from one MAP device to another
2. **Band Steering**: Steering client from one radio band to another on same MAP device
3. **Client DB:** Connected client database management at Controller

6.2 Centralized Steering feature details

Centralized Steering requires the following mechanisms:

1. Maintenance of topology database:

Topology database is maintain other AP's BSSID, channel info, PHY capability etc. It will be used by algorithm to select the threshold (on the Root, the threshold is lower to delay AP steering), initiate coordination among APs, to estimate MCS rate and airtime for best AP selection etc.

2. Coordination between MAP devices:

Mechanism between Aps to do the RSSI measurement and setup blacklist for best possibility steering

3. Centralized client server model:

Controller is the server and the agents the client. Controller being responsible for steering decisions requires the multi-ap agents to send the metrics information periodically. It also requires the agent to follow client steering request triggered by the controller after a steering decision has been made at the controller. The controller chooses a new MAP agent periodically for load balancing and upgrade steering. However, the controller performs multi-ap steering or band steering (performance of STA downgraded) when the STA Metrics Reporting RCPI threshold trigger is received from the agent to which the client is connected. Load balancing includes channel overload and moving STA to a better band (2.4G --> 5G). To achieve this using MAP specification, following messages are used:

a. **Steering Policy and AP Metric Reporting**: Controller send Policy Config Request message (17.1.8 Multi-AP Policy Config Request message) to configure the new detected agent (when new agent join).
   i. Steering policy: Set Agent Initiated RSSI-based Steering to disallowed to disable steering decisions being taken at the agent.
   ii. Channel Utilization Threshold: align single AP band steering parameters.
   iii. RSSI Steering Threshold: depends on role (LowRSSIAPSteerEdge_Root/ LowRSSIAPSteerEdge_RE), it will be different threshold settings.
   iv. AP Metrics Reporting Interval: The interval at which the agent will send ap metrics report periodically. Configured via MetricRepIntv in the /etc/mapd_strng.conf file(set to 10 seconds default).
   v. STA Metrics Reporting RCPI threshold**:** A non zero value will enable the agent rcpi monitor mechanism. When the client RCPI crosses the threshold the agent will send an unsolicited sta link metrics response to the controller.
   vi. STA Metrics Reporting RCPI Hysteresis Margin Override: The purpose of the hysteresis margin is to prevent an agent from sending the unsolicited sta link metrics response

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frequently after the Sta's RCPI crosses the STA Metrics Reporting RCPI threshold. A non-zero value overrides the default STA Hysteresis margin(default value is 5). Reducing the value increases the sensitivity to client RCPI changes.

For example,
if the value is 2 and when the client RCPI crosses the STA Metrics Reporting RCPI threshold then if the client RCPI changes by 2 an unsolicited sta link metrics response will be sent by the agent to the controller.

vii. Associated STA Link Metrics Inclusion Policy: Set to 1 include sta link metrics stats in th periodic ap metrics response.
viii. Associated STA Traffic Stats Inclusion Policy: Set to 1 to include sta traffic stats in the periodic ap metrics response

Note:
It is mandatory to set Associated STA Traffic Stats Inclusion Policy and Associated STA Link Metrics Inclusion Policy to 1 for Centralized Steering.

b. **Beacon metrics query/response –** In centralized steering the controller triggers 11k beacon request to get the STA's RCPI. When the STA is connected to an agent,the controller sends the Beacon metrics query message to trigger the 802.11K beacon request at the agent .The agent receiving the 802.11k beacon report from the STA will send a beacon metrics report to the controller.
c. **Unassociated sta link metrics query/response** – The controller sends the unassociated sta link metrics query to the agent when the STA doesn't support 802.11k. The agent after measurement will respond with unassociated sta link metrics response.

6.2.1 Decision for Multi-AP Steering or Standalone Band Steering

1. On 5GHz, if RSSI (or rate for active client) is lower than standalone downgrade threshold, attempt standalone downgrade steering
2. For idle client, RSSI < RSSISteeringEdge_DG
3. For active client, RSSI < RSSICrossingThreshold_DG || MCS < MCSCrossingThreshold_DG
4. On 2.4GHz, if RSSI (and rate for active client) are higher than standalone upgrade threshold, attempt standalone upgrade steering
5. For idle client, RSSI > RSSISteeringEdge_UG
6. For active client, RSSI > RSSICrossingThreshold_UG && MCS > MCSCrossingThreshold_UG
7. Others, go through Multi-AP steering procedures

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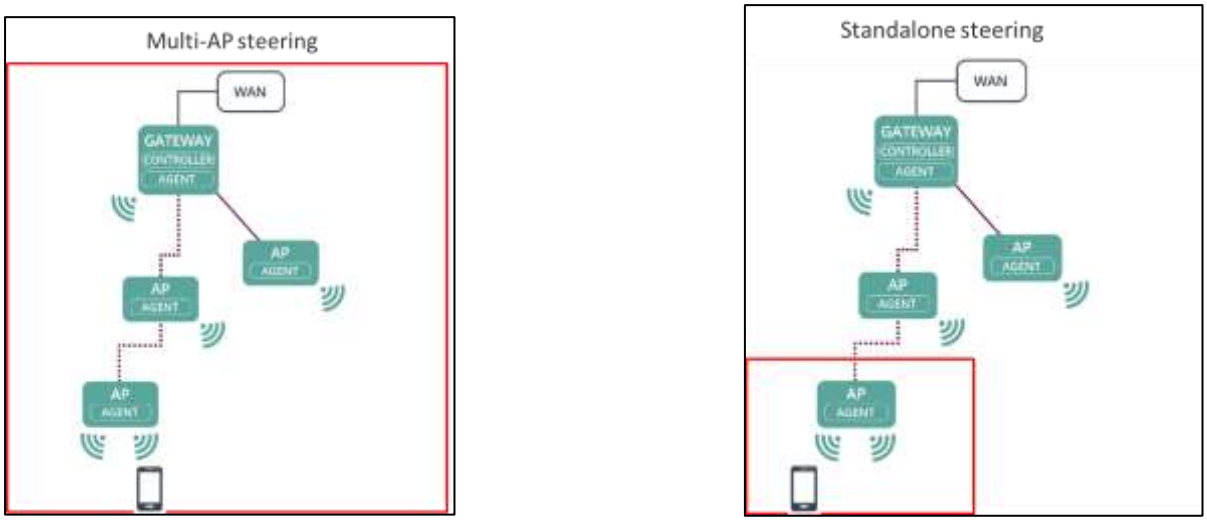


Figure 6- 1. Pictorial description of scope of STA steering(red highlight) in MultiAP steering vs Standalone Band Steering

6.2.2 Band Steering

Band steering as the name implies means that client is steered from one band to the other on the same MAP device. Hence it is also known as Standalone Steering.
There are two primary types of steering that are supported: -

- Idle post-association steering
- Active post-association steering.
  - Active Non-Overloaded Steering
  - Active Overloaded Steering

6.2.2.1 Band steering for MLD clients

Band steering is not supported for a multi link device (MLD) client. For the MLD client Band steering is not applicable, as multi link device is expected to be already connected with multi links, so switching the primary link on the same device via band steering will not give much advantage.

6.2.2.2 Idle post-association steering

- An idle client is defined as a client for which the sum of Avg Tx Bytes/sec and Avg Rx Bytes/sec is less that *Activitythreshold*
- An idle client can be steered by either of the two methods:-
  - Forced Steering: Disconnect + Deauth
  - BTM Steering: IEEE80211v BSS Transition Management
- IDLE Clients with recent enough RSSI on 5G band, whose estimated non-serving RSSI(2.4Ghz) has fallen below *RSSISteeringEdge_DG* are forced steered/ BTM steered to 2.4G band if the 2.4G channel is non-overloaded. If the RSSI is quite old, the RSSI is refreshed by triggering a series of NULL frames to the IDLE client. If IDLE client is known to fail BTM steering, forced steering is attempted.
- IDLE Clients with recent enough RSSI on 2.4Ghz band, whose non-serving RSSI(RSSI on 5G) has increased above *RSSISteeringEdge_UG* are forced steered/ BTM steered to 5G band if the 5G channel is non-overloaded. If the RSSI is quite old, the RSSI is refreshed by triggering a series of NULL frames to the IDLE client. If IDLE client is known to fail BTM steering, forced steering is attempted.

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Table 6- 1. Band Steering Steering decision based on client action

Client condition	• 2.4G Channel Condition	• 5GL Channel Condition	• 5GH channel Condition	• PHYBasedSelection	• Steering Action
Client Connected on 2.4G with Estimated 5G RSSI > *RSSISteeringEdge_UG*	Don't care	NOL	NOL	ON	If the client is VHT steer it to 5GH otherwise steer to 5GL
	Don't' care	NOL	NOL	OFF	Steer to least utilized 5G Channel
	Don't care	NOL	OL	Don't care	Steer to 5GL
	Don't Care	OL	OL	Don't' care	Do not Steer
	Don't Care	OL	NOL	Don't' care	Steer to 5GH channel
Client Connected on 2.4G with Estimated 5G RSSI < *RSSISteeringEdge_UG*	Don't' care	Don't' care	Don't' care	Don't' care	Do not Steer
Client Connected on 5GL/5GH, with Estimated 2.4G RSSI < *RSSISteeringEdge_DG*	OL	Don't' care	Don't' care	Don't' care	Do not steer
	NOL	Don't' care	Don't' care	Don't' care	Steer to 2.4 channel
Client Connected on 5GL with Estimated 2.4G RSSI > *RSSISteeringEdge_DG*	Don't' care	Don't' care	OL	Don't Care	Do not steer
	Don't' care	Don't' care	NOL	OFF	Do not steer
	Don't Care	Don't Care	NOL	ON	If the client is VHT, steer it to 5GH, otherwise do nothing
Client Connected on 5GH with Estimated 2.4G RSSI > *RSSISteeringEdge_DG*	Don't Care	OL	Don't Care	Don't Care	Do not steer
	Don't Care	NOL	Don't Care	OFF	Do not steer
	Don't Care	NOL	Don't Care	ON	If the client is Non-VHT, steer it to 5GL, otherwise do nothing

6.2.2.3 Active post-association steering.

- An ACTIVE client is defined as a client for which the sum of Avg Tx Bytes/sec and Avg Rx Bytes/sec is greater than *ActivityThreshold*.
- Only ACTIVE clients that are both 802.11K and 802.11V capable are eligible for steering
- Only BTM steering is used for ACTIVE clients to cause minimum service disruption.
- There are two types of steering that are defined for Active Clients:-
  - Active Non-Overloaded Steering
  - Active Overloaded Steering[Offloading]

6.2.2.3.1 Active Non-Overloaded Steering

- Active clients whose serving RSSI and DL MCS have increased above specific thresholds, are BTM steered to 5G band provided that the 5G channel is not overloaded and the steering of Active client

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doesn't make the 5G channel as overloaded. 11k helps to predict the impact of moving the client to the target channel. If the Active client is known to fail BTM steering, then no steering is attempted.

- Active clients whose serving RSSI and DL MCS have increased above specific thresholds, are BTM steered to 5G band provided that the 5G channel is not overloaded and the steering of Active client doesn't make the 5G channel as overloaded. 11k helps to predict the impact of moving the client to the target channel. If the Active client is known to fail BTM steering, then no steering is attempted

6.2.2.3.2 Active Overloaded Steering [Offloading]

Active clients connected to an Overloaded channel are BTM steered to a non-overloaded channel provided that the steering of Active client doesn't make the target channel as overloaded. 11k helps to predict the impact of moving the client to the target channel. If the Active client is known to fail BTM steering, then no steering is attempted

6.2.2.4 Band Steering Examples

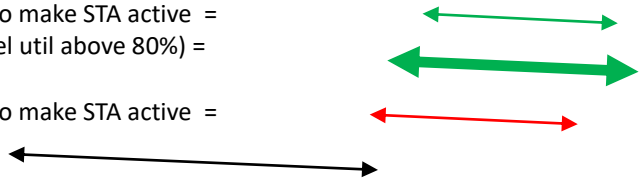
1: agent_post_assoc_ol_active_dg:

Steps:
a. Configure MAP agent and controller in HE mode wpa2_psk_aes and keep centralized steering enabled, phy based steering and nol_multiap steering disabled on both.
b. Onboard the agent with the controller.
c. Connect STA1(which supports 5G only) and 11KV dual band STA2 both to 5G interface of the agent.
d. Run traffic with STA2, just to make it active and make 5G OL with the help of heavy traffic with STA1 and keep 2G NOL.

Expected result:
Check the dual band STA2 get steered to 2G interface of the agent.

Legends used in below diagrams:
5G connection with just nominal traffic to make STA active =
5G connection with heavy traffic (channel util above 80%) =

2G connection with just nominal traffic to make STA active =
MAP onboarding =



Initial setup for the test:

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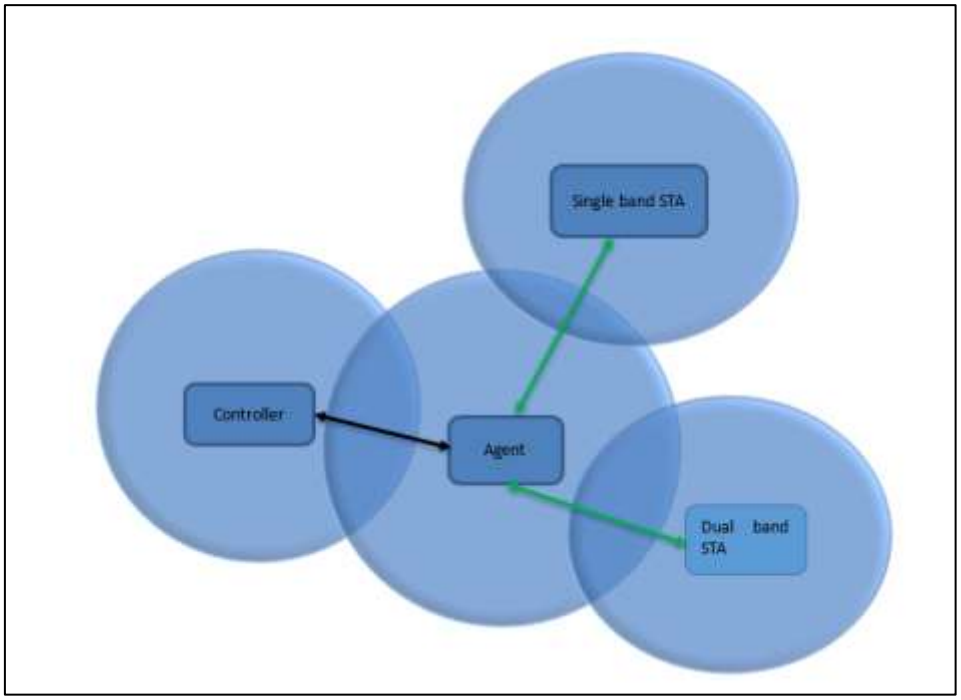


Figure 6- 2. Band Steering example Image 1

Just before band steering supposed to happen:

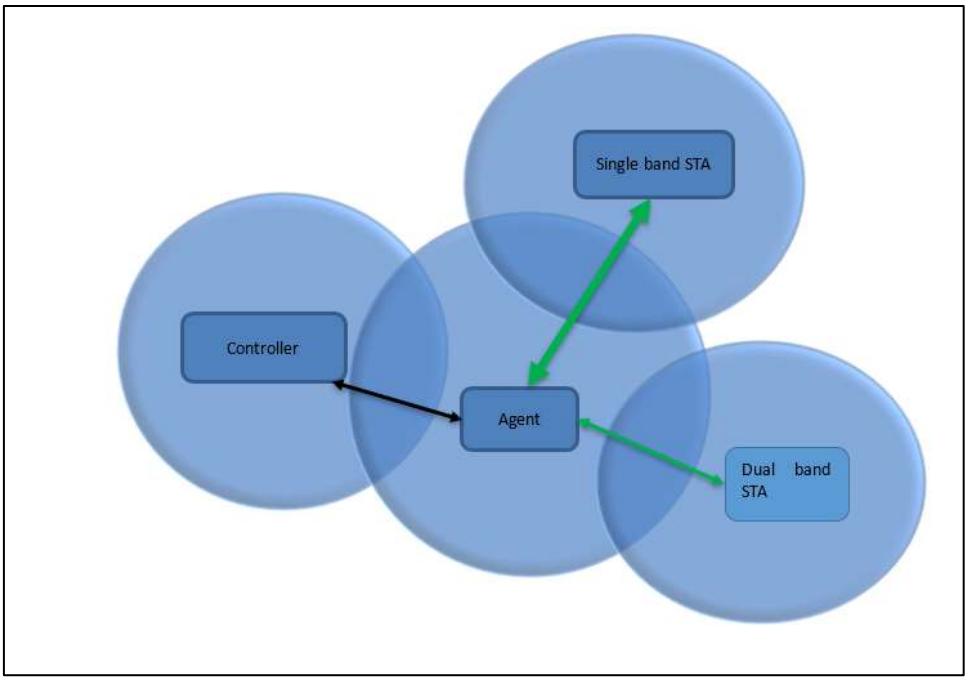


Figure 6- 3. Band Steering example Image 2

After Band steering is performed (the STA got steered to 2G interface of the Agent):

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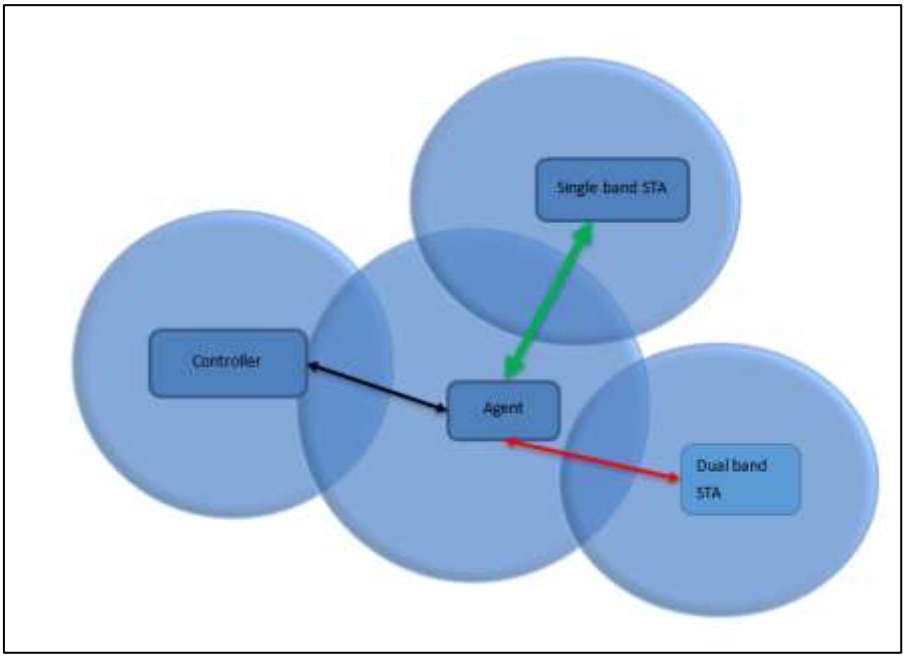


Figure 6- 4. Band Steering example Image 3

2: agent_post_assoc_rssi_active_ug:
Steps:
1. Configure MAP agent and controller in HE mode wpa2_psk_aes and keep centralized steering enabled, phy based steering and nol_multiap steering disabled on both.
2. Onboard the agent with the controller.
3. Blacklist the 11KV dual band STA for some certain time on 5G interface of the agent.
4. Connect the STA to 2G interface of the agent.
5. Run traffic with STA, just to make it active
6. Wait for the blacklist to be over.
7. Now make RSSI good in 5G.

Expected result:
Check the dual band STA get steered to 5G interface of the agent.

Legends used in below diagrams:
5G connection with just nominal traffic to make STA active =
2G connection with just nominal traffic to make STA active =

MAP onboarding =

When the STA is out of Agent's 5G range and within Agent's 2G range:

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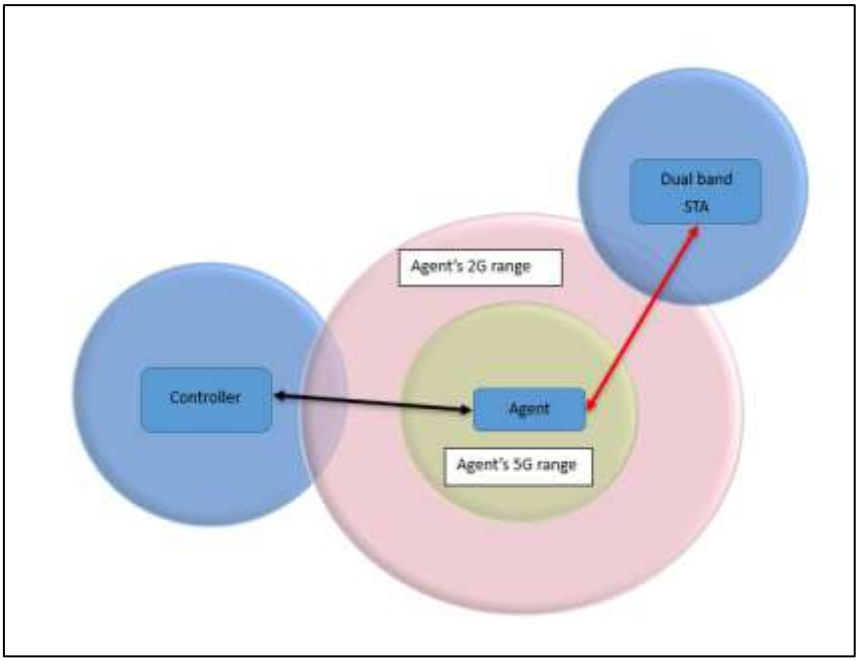


Figure 6- 5. Band Steering example Image 4

When the STA is within Agent's 5G range (the STA got steered to 5G interface of the Agent):

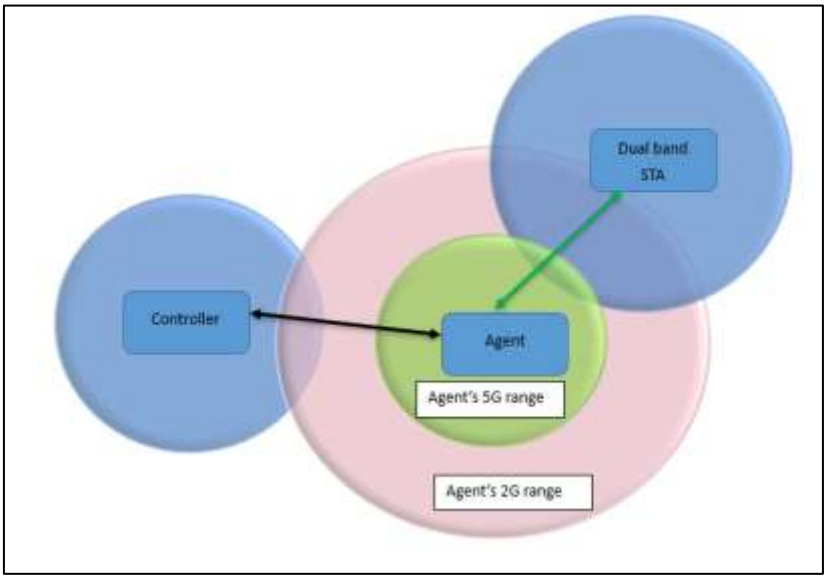


Figure 6- 6. Band Steering example Image 5

6.2.3 Multi-AP Steering

Multi-AP steering is a mechanism to attempt to move a client from 1 AP to another in a Multi-AP Network environment. Best BSS selection for steering client to another MAP dev firstly requires channel measurement. Based on Channel measurement (Radio Resource measurement /Beacon measurement) score is calculated for

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each BSS found in measurement result. If it validated the steering criteria and has highest score then that BSS is selected as the best BSS for that client and steering is initiated.

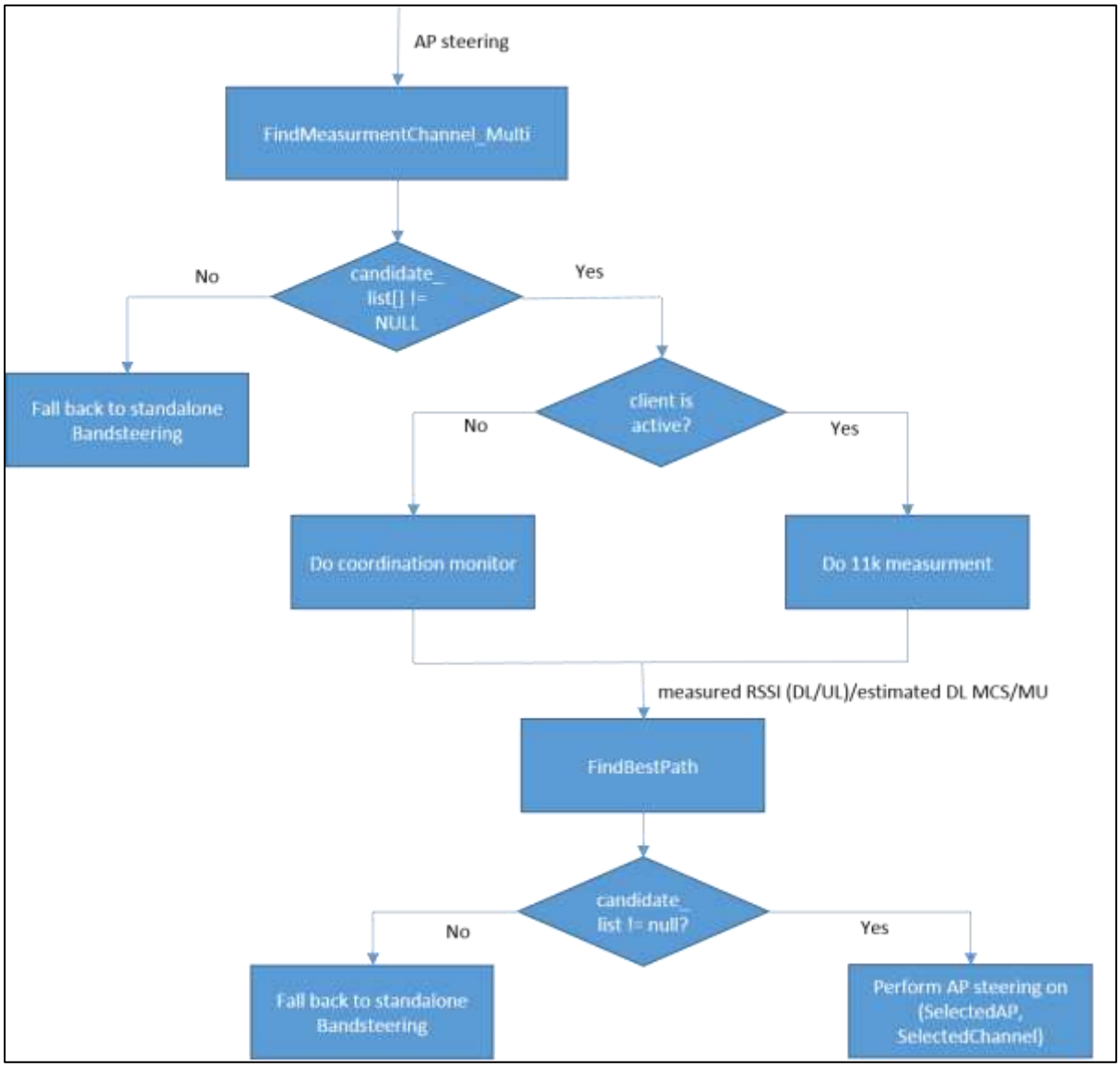


Figure 6- 7. Flow Chart for AP Steering Procedure

6.2.3.1 Selection of Measurement Channels

In multi-AP steering we need to perform the measurement to find out the AP which can provide best performance to client. Firstly, we need to decide the channel that wants to measure. It will consist two cases.

- Case 1: if client cannot support RRM, then the measurement channel will be same as current serving channel (to perform Airmonitor)
- Case 2: if the client can support RRM, it will follow the filters to determine the channel.
  - Filter 1: same channel as serving channel
  - Filter 2: 6G channel first
  - Filter 3: 5G channel second
  - Filter 4: lowest channel utilization

For active clients, all overloaded channels will be excluded.

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6.2.3.2 Selection of Best BSS

Based on measurement results, select the BSS based on the following rules:

1. RSSI difference between the current BSS and the target BSS and Link Capacity is better than the current link.
    i. The difference should be greater than a threshold
    ii. The threshold is different for upstream devices and downstream devices
2. Link Capacity between the current BSS Link capacity threshold and target BSS.
    i. Link capacity of the target BSS should be greater than the current BSS link capacity threshold.
    ii. Link capacity threshold is link capacity on current RSSI + RSSI margin.
3. Upstream BSS is preferred over downstream and peer BSS (same distance from root)
4. 6G band BSS is preferred over 5G BSS. 5G band BSS is preferred over 2.4G band BSS.
5. BSS on channel with lowest channel utilization is preferred.

6.2.3.3 Selection of Best BSS for MLD Clients

1. All the above point in section 6.2.3.2 are also valid in case if clients are MLO capable.
2. For a MLD client the steering decision is taken after considering the cumulative score of all the MLD links it is cable of making, hence for this reason 11k measurement/air monitoring and score calculation etc need to consider the channels for all multi links.
   For the MLD client Band steering is not applicable, as multi link device is expected to be already connected with multi links, so switching the primary link on the same device via band steering will not give much advantage.

6.2.3.4 Steer coordination in Multi-AP environment

There will be two reasons (at least) we need the coordination among APs.

1. For the best probability of steering success, it is necessary that the controller initiates the steering coordinate with the other nodes in the network so that they can install blacklists and withhold probe responses as appropriate (MAP 17.2.31 Client Association Control Request message)

2. Without this coordination, the legacy steering mechanism would often lead to the client associating on a different BSS than intended in cases where signal strength is similar between Root and RE or the client's own BSS selection algorithm prefers a different BSS than the target one. Even steering using 802.11v BSS Transition Management Request (at least with current clients) is not as reliable without the steering coordination.

3. To position the client with improved WiFi Network performance by steering the clients to a better-serving AP through a better RSSI. If the client does not support 802.11k beacon measurement (or a client that fails for consecutive attempts of 802.11k beacon report), we need AP coordination to monitor the client RSSI and report to initiating AP (MAP 17.1.20 Unassociated STA Link Metrics Query message)

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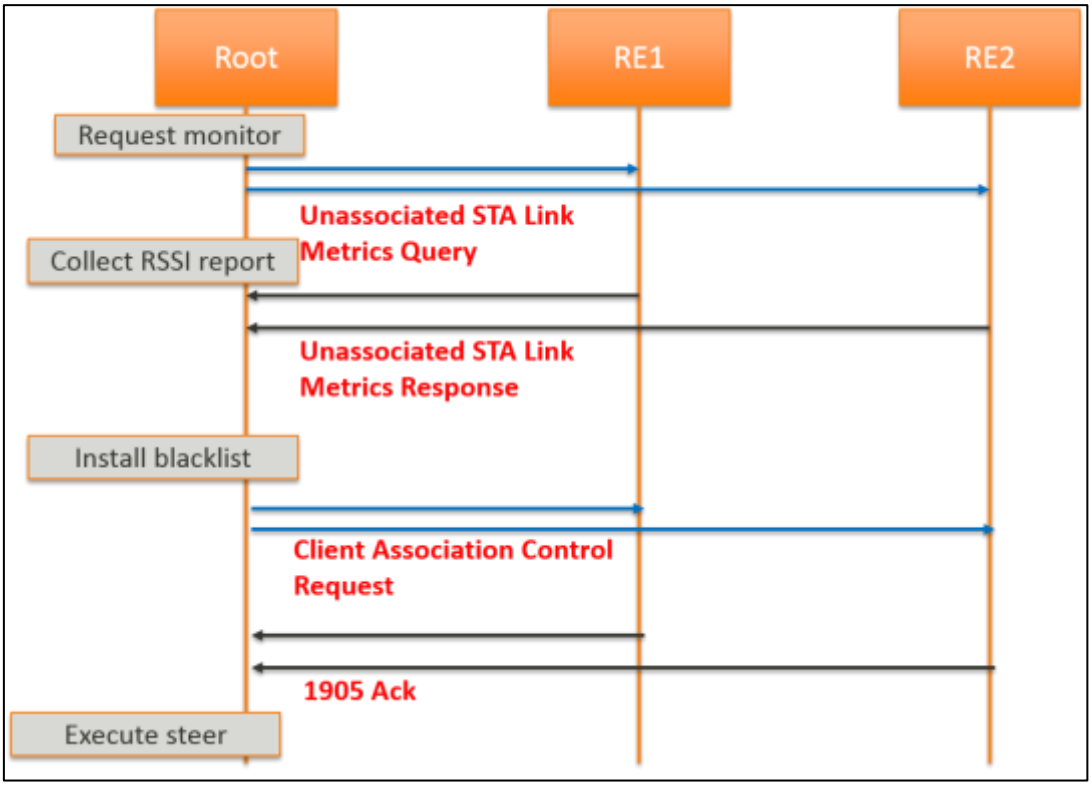


Figure 6- 8. Steering coordination message exchange between different MAP devices

6.2.3.5 Tri-band detection

Since the controller is responsible for taking steering decisions, the controller will maintain a centralized client database.
We currently rely on probe request and association to determine if the client is tri band capable.
In the centralized steering design, the client database will be updated based on

- Probe Requests received by Controller

However not all clients can be identified as dual band based on the above method as some clients may not be in vicinity of the Controller for it to receive their probe requests.

For all such cases, we try to detect whether the client is dual band capable or not by utilizing the supported operating classes element during the client association phase which is mandatory if the client is MBO capable. If operating class IE is not found in association request, then we send rrm request. We exclude the currently operated band for rrm. If client send proper rrm response on another band, controller will update database for the client.
If there is no supported operating classes element, we initially assume that all the clients are tri-band capable and allow the tri-band steering for all clients. Based on the steering attempts failures beyond a configured threshold (default is 5), we mark it as single band or dual band.

The threshold can be configured via the **CentSteerMaxBSFail** parameter in /etc/mapd_strng.conf.

6.2.3.6 Multi AP Steering example

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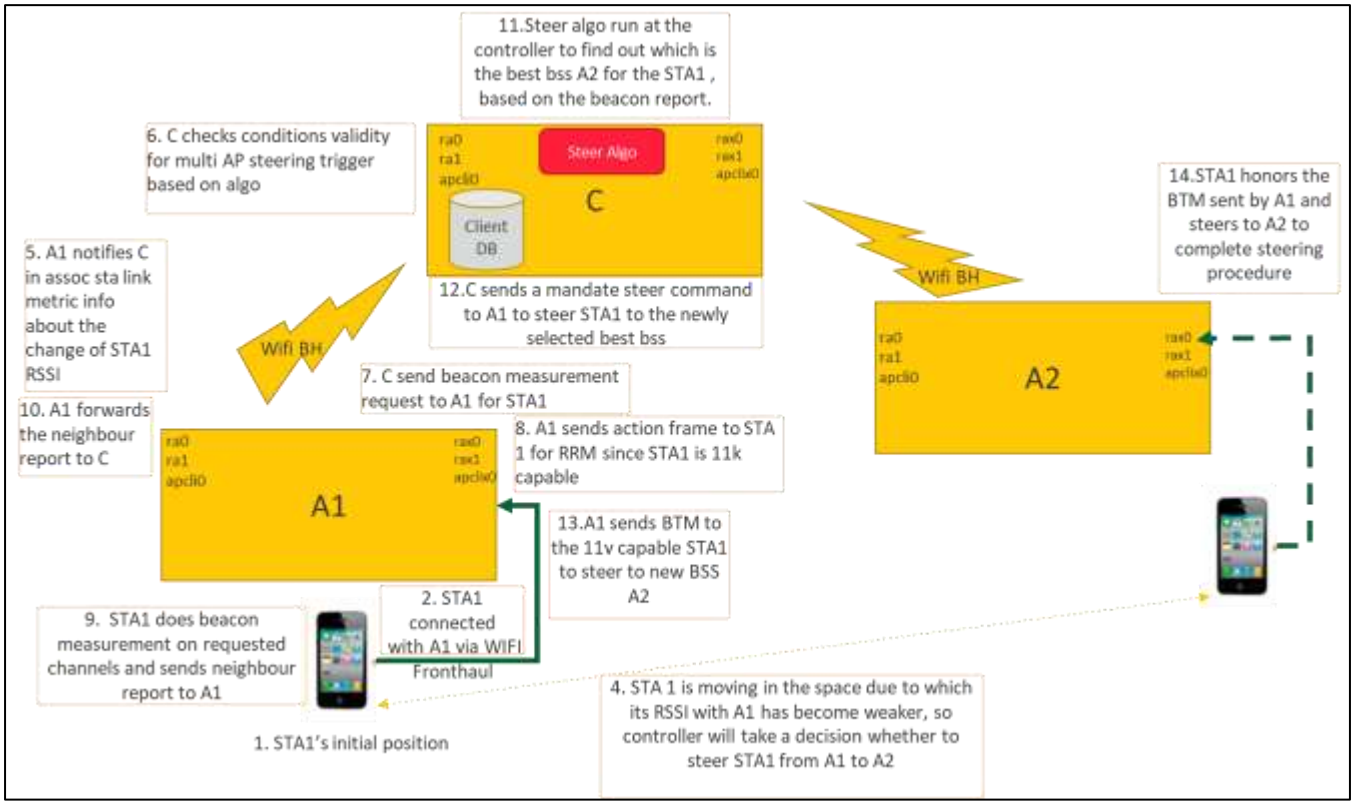


Figure 6- 9. Multi AP centralized steering procedure deployment example

In the above Figure, the Easymesh network has one controller 'C' and 2 Agents 'A1' and 'A2' connected with 'C' in a tree topology via wifi backhaul. The client 'STA1'(11kv capable) is initially connected to 'A1'. The controller mainitains all the client database in the easymesh network. 'STA1' is moving in space due to which its RSSI degrades with 'A1'. So, the centralized steering alogorithm running at the controller helps to steer this client to a better BSS which is 'A2', as explained step by step in the image.

6.2.4 Feature Management Interfaces

6.2.4.1 Compile Time Options

Mapd:

Enable CONFIG_AP_CENT_STR=y in mapd_config

1905daemon:

Enable CENT_STR_SUPPORT=y in config.mk

6.2.4.2 Configuration file interface

Enable/Disable Centralized steering:

Configure CentralizedSteering =1 in /etc/map/mapd_default.cfg.

For Map Turnkey we cannot disable thie Cnetealized steeringfeature as it is mandatory.

6.2.4.3 Configuration Parameters for Band Steering

The steering configuration parameters can be over-ridden by specifying them in /etc/mapd_strng.conf. The parameters and their default values are as below:-

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Table 6- 2. Band Steering Configurable Parameters

Parameter	Description	Default Value
ActivityThreshold	The Avg sum of Tx Bytes and Rx Bytes per second above which the client is considered as ACTIVE.	3000 Bps
RSSISteeringEdge_UG	The point at which the measured or estimated RSSI on 5 GHz dictates a node associated on 2.4G should be steered to 5 GHz	20db
RSSISteeringEdge_DG	The point at which the measured or estimated RSSI on 2.4 GHz dictates a node associated on 5 GHz should be steered to 2.4 GHz	20db
MinRSSIOverloaded	Uplink RSSI (in dB) above which pre-association steering and post-association offloading is allowed.	20db
PHYBasedSelection	In pre-association, put non HT client to 2.4G channel. In post-association, to prioritize VHT capable clients to high power 5G channel, and put non-VHT clients to other channels.	0
MaxClientOverloaded	The maximum connected client number to indicate potential overloaded.	100
ForcedRssiUpdate	Force RSSI Update enable/disable	0(disabled)
RSSIAgeLim	Number of seconds to check if last record RSSI is fresh enough.	5s
RSSIMeasureSamples	Number of NULL frames to be sent for client uplink RSSI measurement.[Only applicable when [ForcedRssiUpdate=1]	5
MCSCrossingThreshold_UG	The rate (in Kbps) above which an active client is a potential candidate for upgrade to 5 GHz.	50000
RSSICrossingThreshold_UG	The value (in dB) the uplink RSSI must be above to be considered for active steering to 5 GHz. This threshold is in conjunction with the MCSCrossingThreshold_UG.	30db
MCSCrossingThreshold_DG	The rate (in Kbps) at which a rate crossing event should be generated for a potential active client downgrade to 2.4 GHz. The default basically disable downgrade steering.	6000
RSSICrossingThreshold_DG	The value (in dB) the uplink RSSI must be below to be considered for active steering to 2.4 GHz.	15db

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	This threshold is an additional trigger (an OR condition) for downgrade steering.	
CUOverloadThreshold_2G	Medium utilization threshold (in percentage %) for an overload condition on 2.4 GHz	80
CUOverloadThreshold_5G_L	Medium utilization threshold (in percentage %) for an overload condition on the first 5G radio	70
CUOverloadThreshold_5G_H	Medium utilization threshold (in percentage %) for an overload condition on the second 5G radio	70

6.2.4.4 Configuration Parameters for Multi AP steering

Required Parameter Configuration at controller:
For the controller, the steering policy of the controller contains the Metric Reporting Policy TLV. This has different parameters that can be modified by writing them in the "mapd_strng.conf" file.

Table 6- 3. Multi AP Steering Configurable Parameters

Parameter type	mapd_strng.conf file field/token	Example value	Default value
AP Metrics Reporting Interval	MetricRepIntv	10 (In seconds)	10
Max number of Band steering fail count for dual-band detection	CentSteerMaxBSFail	3 (After 3 failures of client band steering with no band steering success previously the client is marked as Single band device)	5
STA Metrics Reporting RCPI Threshold	MetricPolicyRcpi_24G	150 (RCPI threshold values can range from 0 to 220, if 0 then do not report STA Metrics based on RCPI threshold.)	100
	MetricPolicyRcpi_5GL		
	MetricPolicyRcpi_5GH		
	MetricPolicyRcpi_6G		
STA Metrics Reporting RCPI Hysteresis Margin Override	MetricPolicyHys_24G	2 (Hysteresis can range from 0 to 5, if 0 is sent then code will use 5 by default)	0
	MetricPolicyHys_5GL		
	MetricPolicyHys_5GH		
	MetricPolicyHys_6G		
AP Metrics Channel	MetricPolicyChUtilThres_24G	150 (Channel Utilization threshold values	0

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Utilization Reporting Threshold	MetricPolicyChUtilThres_5GL	can range from 0 to 255, if 0 then do not report AP Metrics based on Channel utilization threshold)	
	MetricPolicyChUtilThres_5GH		
	MetricPolicyChUtilThres_6G		
Associated STA Traffic Stats Inclusion Policy.	MetricPolicyTrafficInclusion_24G	0: Do not include Associated STA Traffic Stats TLV in AP Metrics Response 1: Include Associated STA Traffic Stats TLV in AP Metrics Response	1
	MetricPolicyTrafficInclusion_5GL		
	MetricPolicyTrafficInclusion_5GH		
	MetricPolicyTrafficInclusion_6G		
Associated STA Link Metrics Inclusion Policy.	MetricPolicyMetricsInclusion_24G	0: Do not include Associated STA Link Metrics TLV in AP Metrics Response 1: Include Associated STA Link Metrics TLV in AP Metrics Response	1
	MetricPolicyMetricsInclusion_5GL		
	MetricPolicyMetricsInclusion_5GH		
	MetricPolicyMetricsInclusion_6G		
RSSI threshold for MultiAp steer	LowRSSIAPSteerEdge_root	The value (in dB) the uplinkRSSI must be below to trigger AP steering for a STA associated on the Root device. To convert this value in RSSI, noise floor is needs to be added. EX. To compare with RSSI, noise floor(-94) is needs to be add in LowRSSIAPSteerEdge_root.	20
	LowRSSIAPSteerEdge_RE	The value (in dB) the uplink RSSI must be below to trigger AP steering for a STA associated on the RE device. To convert this value in RSSI, noise floor is needs to be added. EX. Same as above.	40
Minimum Rssi to steer from one bss to another bss.	MinRssiIncTh_Root	The amount the RSSI of the upstream AP must be better than the RE to cause AP steering to occur.	5
	MinRssiIncTh_RE	The amount the RSSI of a downstream AP must be greater than the connected AP to cause AP steering to occur.	10
	MinRssiIncTh_Peer	The amount the RSSI of a RE must be greater than that of another RE to cause AP steering to occur.	10
Threshold to decide channel overload	CUOverloadTh_2G	It is a percentage value of threshold for channel overloading. If channel utilization is more than CUOverloadTh_2G then sta will not steer at that channel.	80
	CUOverloadTh_5G_L		70
	CUOverloadTh_5G_H		
	CUOverloadTh_6G		

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To disable any type of steering	disable_pre_assoc_strng	By using these parameter, any steering can be disabled. 0 means steering is enabled.	0
	disable_post_assoc_strng		
	disable_offloading		
	disable_nolmultiap		
	disable_active_ug		
	disable_active_dg		
	disable_idle_dg		
	disable_idle_ug		

6.2.4.5 Command Line Interface

- Set metric policy parameter format:
  metric_policy_set <radio band 0/1/2> <parameter ID 0 to 4> <value>

  radio band 0/1/2
  BAND_2G 0x01
  BAND_5GL 0x02
  BAND_5GH 0x03
  parameter ID 0 to 4
  PARAM_ID_RCPI_THRESH 0
  PARAM_ID_HYS 1
  PARAM_ID_METRIC_INCLUSION 2
  PARAM_ID_TRAFFIC_INCLUSION 3
  PARAM_ID_CH_UTIL_THRESH 4

Example :
To set the 2G band's RCPI threshold as 150, you can give the below command:
  mapd_cli /tmp/mapd_ctrl metric_policy_set 1 0 150
To set the 5GH band's Channel utilization threshold as 150, you can give the below command:
  mapd_cli /tmp/mapd_ctrl metric_policy_set 3 4 150

- Command to trigger mandate steering from CLI:
  mapd_cli /tmp/mapd_ctrl steermand <client mac> <Target mac>

Example:
  mapd_cli /tmp/mapd_ctrl steermand be:d2:fb:39:7a:d7 02:0c:43:59:a0:81

Controller side log:

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Figure 6- 10. Controller side console logs for Mandate Steering

Agent side log:



Figure 6- 11. Agent side console logs for Mandate Steering

1. How to check Channel Utilization on each Radio and BSS info.

mapd_cli /tmp/mapd_ctrl mib

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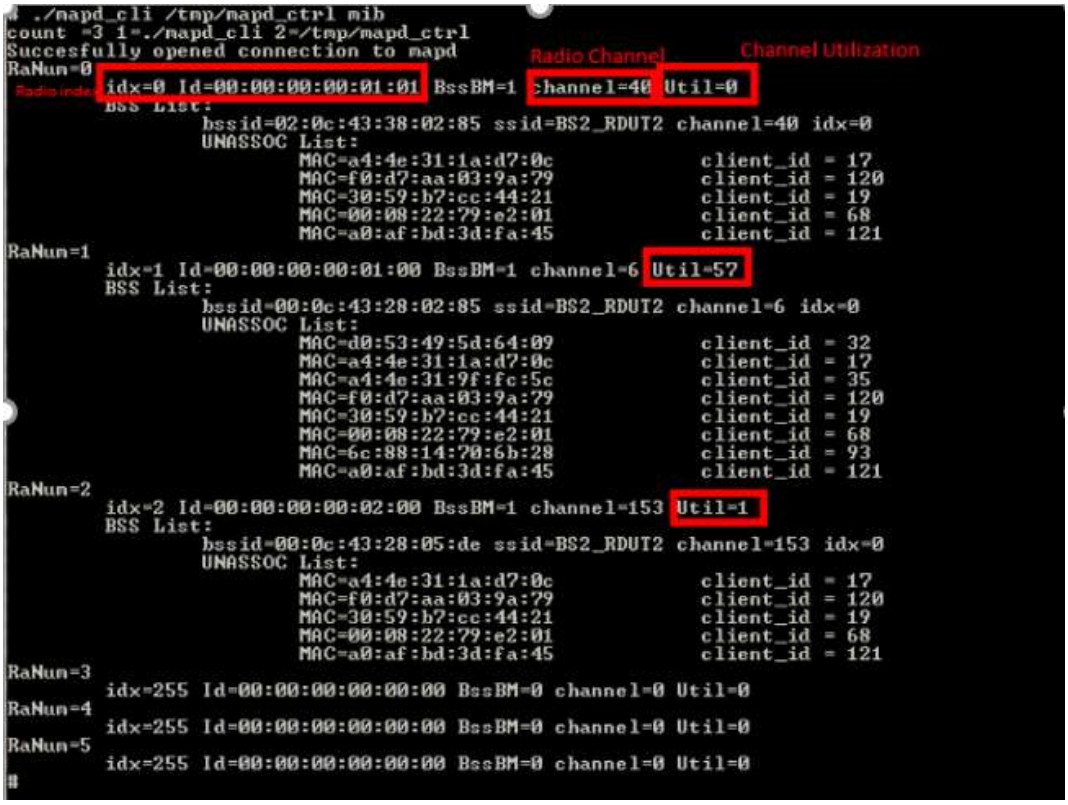


Figure 6- 12. Mapd_cli mib command console logs

RaNum	Radio number
Idx	Radio Index
Id	Radio Identifier
BssBM	BSS bitmap
channel	Channel Number
Util	Channel utilization
Bss List	List of all MBSS for this radio

2. How to check the clients that are monitored by the mapd for steering

mapd_cli/tmp/mapd_ctrl mib sta_seen_list

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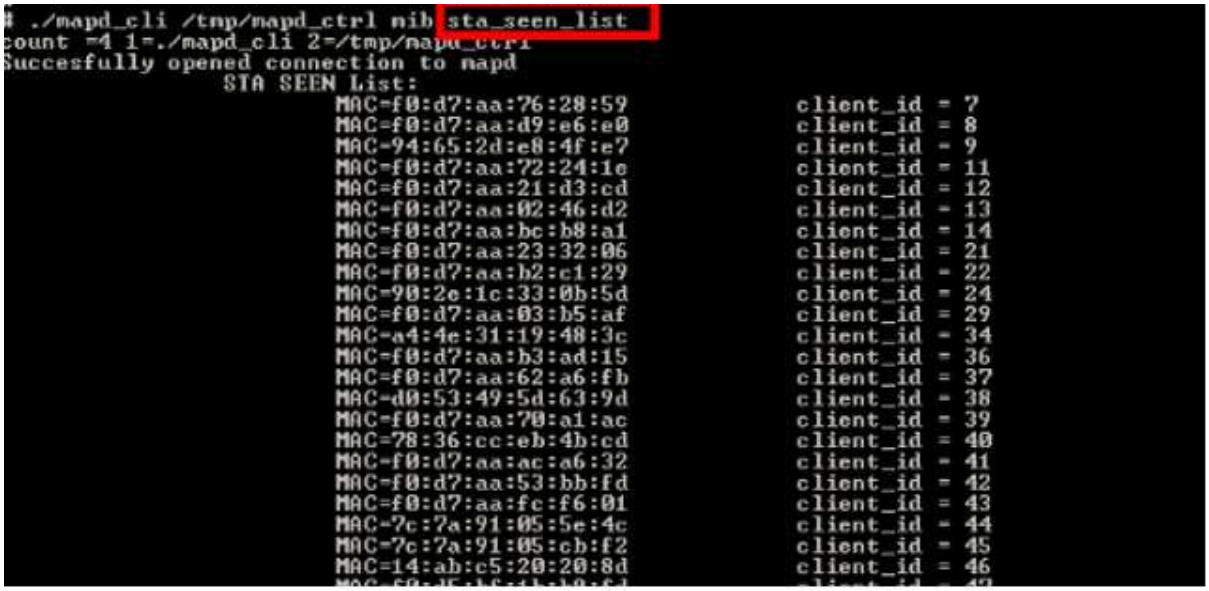


Figure 6- 13. . Mapd_cli mib sta_seen_list command console logs



Figure 6- 14. Mapd_cli mib sta <id> command console logs

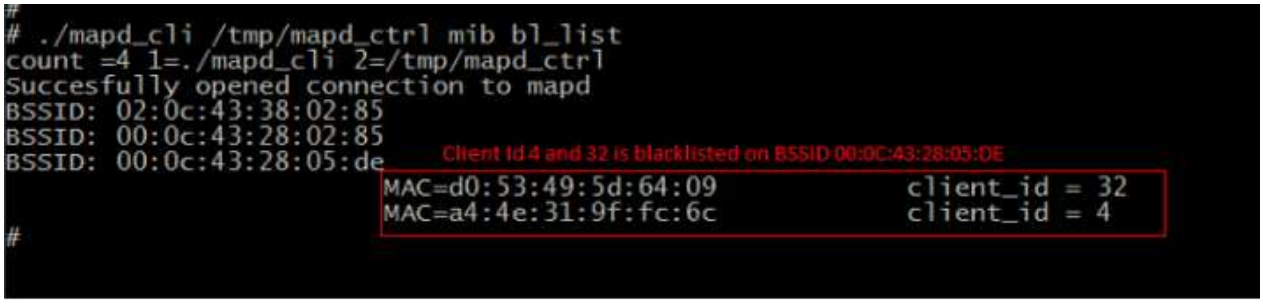


Figure 6- 15. Mapd_cli mib bl_list command console logs

3. How to check the clients that are associated with the MAP device

mapd_cli /tmp/mapd_ctrl mib assoc_sta_list

This command shows the following info :

- **sta_mac:**
  - client's mac address
- **mld_mac** :

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- Client's multi link device mac address , which is same for all band interfaces of same client.
- If mld_mac is all 0 BSSID then it means that the device is not an MLO capable device or the connection made by the device is non-MLO .

- **id:**
  - Client ID as seen in the sta seen list
- **is_setup_link :**
  - This flag indicates which sta mac is the primary MLD link.
  - This field is relevant only if the connection with station is an MLo connection.

```
root@OpenWrt:/# mapd_cli /tmp/mapd_ctrl mib assoc_sta_list
count =4 1=mapd_cli 2=/tmp/mapd_ctrl
I got a commapd MIB assoc_sta_list
MIB cmd
reply=354
ASSOC STA List:
sta_mac         06:0c:43:26:17:17       mld_mac         06:0c:43:26:17:17       id = 36         is_setup_link = 1
sta_mac         06:0c:43:26:19:19       mld_mac         06:0c:43:26:17:17       id = 303        is_setup_link = 0
sta_mac         06:0c:43:26:18:18       mld_mac         06:0c:43:26:17:17       id = 33         is_setup_link = 0
sta_mac         02:12:19:83:10:e6       mld_mac         00:00:00:00:00:00       id = 235        is_setup_link = 0
```

Figure 6- 16. Mapd_cli mib assoc_sta_list concole logs

6.2.4.6 WEBUI Interface

- If in case client is not able to steer, then you can adjust the RSSI Threshold of all the agents via controller UI.

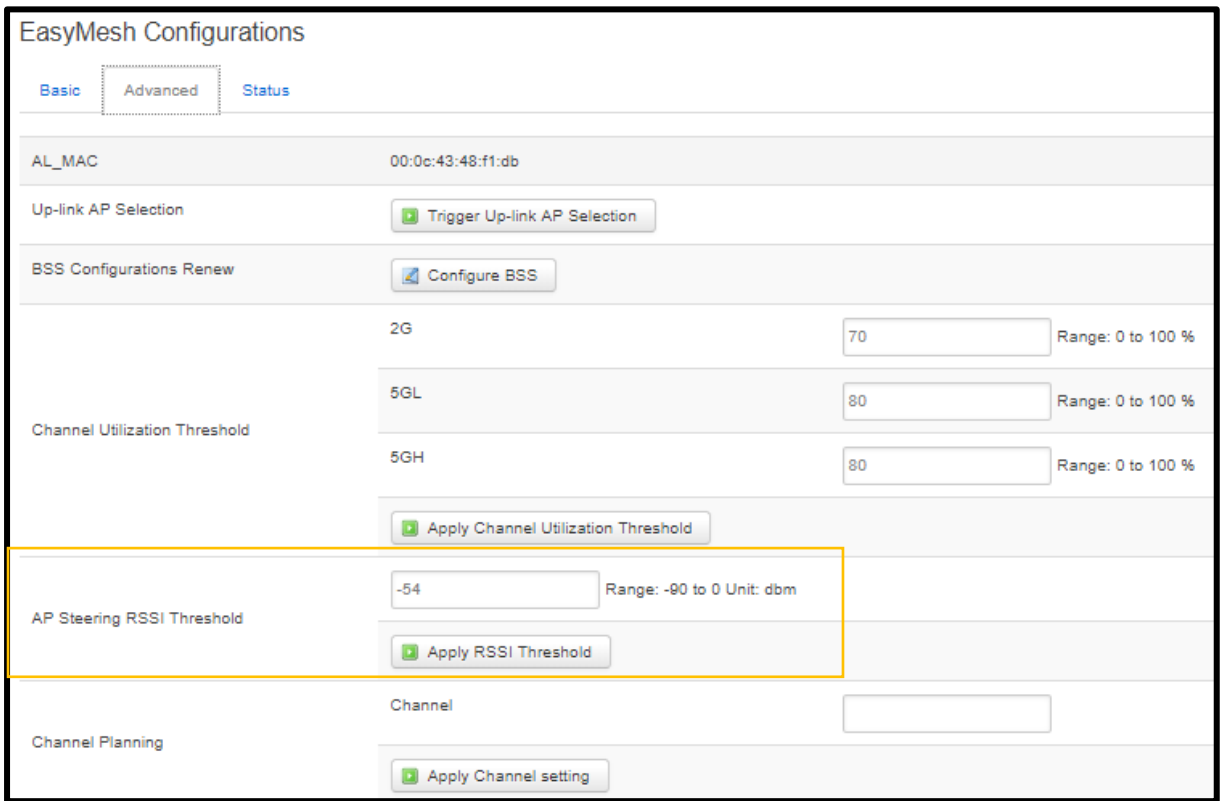


Figure 6- 17. WEB UI method to change the AP Steering RSSI Threshold

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- Observe the steering progress on the Agent 1 in following display:

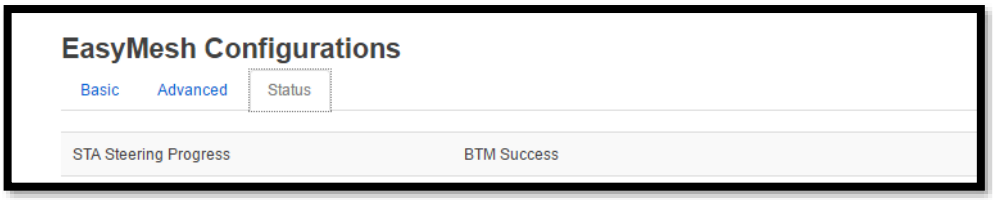


Figure 6- 18. WEB UI display parameter to show Steering status

- Mandate Steering

To move a client mandatorily from one map device to another, you can use the Trigger Mandate Steering on Agent using the button on the web page:

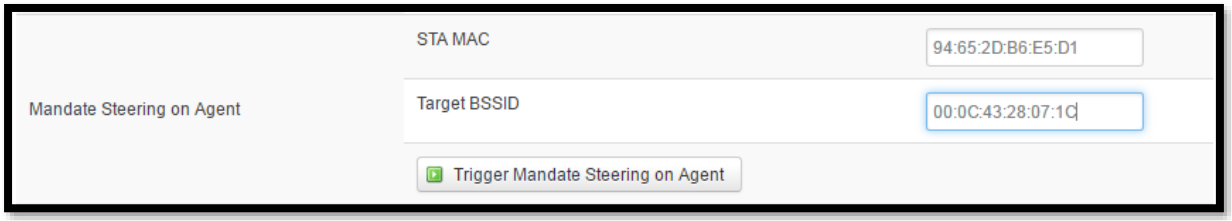


Figure 6- 19. WEB UI button to give mandate steering command

6.2.5 Limitations

1. Since the centralized steering design relies on the controller to take the steering decisions for the entire network, the STA steering time will be impacted by the processing delay taken at the controller (slower then distributed steering).
2. Pre-assoc Band Steering is not supported with centralized steering in MAP. It cannot be supported because there is no mechanism provided by the MAP specification for the agents to share probe requests with the controller before association attempt is made to take any pre-assoc decisions.
3. Band Steering during CAC cannot be handled.
4. 802.11r based fast transition not supported across bands. If FT is enabled on the DBDC AP, FT capable station, when band steered, would fall back on non-FT Association.

6.2.6 Known IOT Issues

Table 6- 4. Steering known IOT issues

Issue	Impact	Seen with	Root Cause	Resolution
802.11k failure with some phones when PMF is enabled	MAP Steering; Post Association Active Steering; Offloading	• Iphone 6s: MN8X2HN/A; IOS Version 12.1.4 • Samsung Galaxy S8: SM-G950FD,	Problem with third party STA wherein they do not encrypt the RRM response frames •	Disable PMF on the APUT

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		Android 9.0(Feb 15 2019) • Samsung Galaxy S9		
BTM Failure: Some phones reject BTM requests most of the times	BTM Steering(and hence Active Steering) not possible	• Samsung Galaxy S8: SM-G950FD, Android 9.0(Feb 15 2019) • Samsung Galaxy S9. • Samsung Galaxy S10.	Unknown	None; If the Phone is rejecting BTM request, it will not be active steered
Some STAs advertise their capability to support DFS channel (through 11k), but never honor the BTM request to move to DFS Channel BSS.	BTM Steering(and hence Active Steering) not possible to the DFS channel	• Nokia 5.1: TA-1102; Android 9.0	Problem with third party STA	None; If the Phone is rejecting BTM request, it will not be active steered
BTM Failure: Some stations respond to BTM requests with BTM response success but do not steer		• Intel AX 200		None; If the stations is not steering after BTM Response, it will not be active steered
No BTM response for some STA	Impact on BTM steering	• Realme XT when MBO enabled • Oneplus when PMF enabled	Problem with third party STA	None; If the stations is not responding steering cannot be done
Connect issue with with Eagle having hostapd support	Impact on connection	• Redmi Note 5 Pro	Hostapd rejects the association with this phone as on sniffer phone is showing MBO capable, but not PMF capable, which is an invalid case	None, station is not following the MBO spec, so hostapd is behaving

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6.3 User Configurable Target BSS of Client Steering

In continuation of Multi-AP Centralized Steering Section 4.20.
If user wants that no station will move to a particular BSS, user can add BSS-id of that BSS in restrict BSS list in the controller.

6.3.1 Feature Details

This Feature is controller specific as this run on centralized steering.

We are maintaining a permanent record across reboots of mapd's local DB of added/deleted restricted BSS-ids.
"/etc/map/restirct_steer_config" is the file MAPD used to have a permanent record.
User can edit this file but has to mention one BSSID per line in the file. This is only applied at boot time.

While mapd is running user can add the BSSID to restrict BSS-id list by using mapd_cli command. Commands for these are available in later portions.

The flow chart of the restricted BSSid feature:

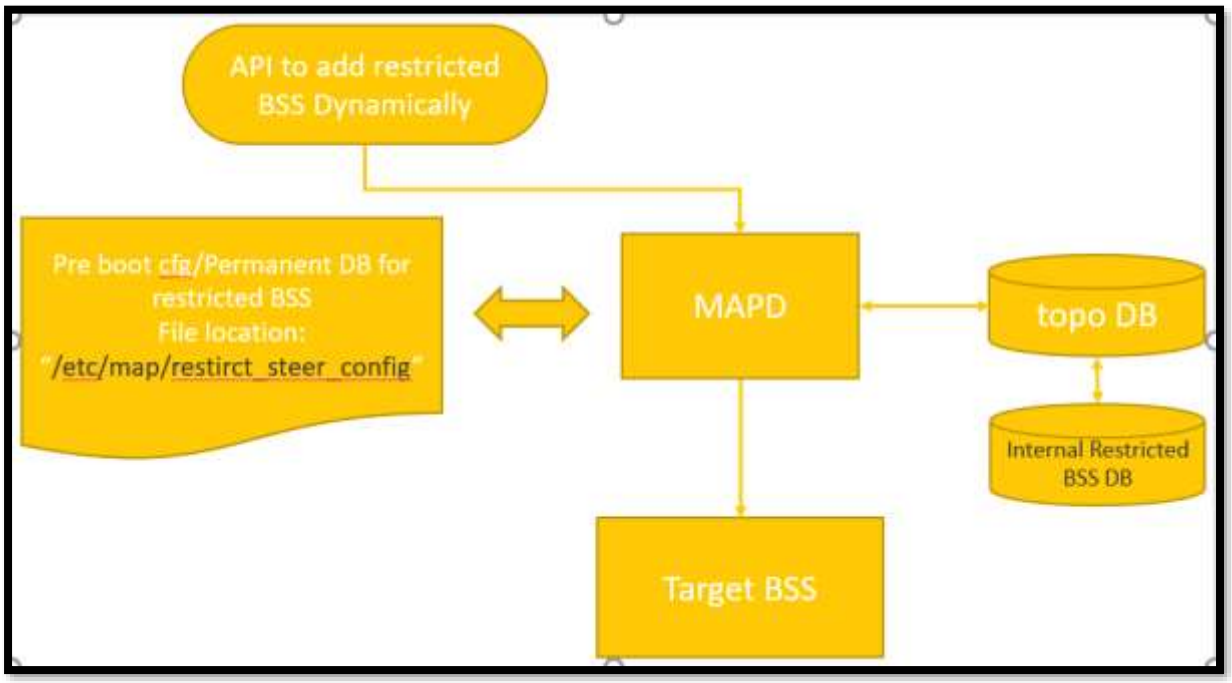


Figure 6- 20. Flow chart for restricted BSSID feature

While selecting the best BSS for STA to move controller look into its local DB of restricted BSS-ids. And if it found that the best BSS is in restrict list, it will skip that BSS and look for second best BSS.

But if the controller finds that the selected best BSS is not in the restricted BSS list it will use that for steering.

6.3.2 Feature Management Interfaces

6.3.2.1 Configuration file interface

"/etc/map/restirct_steer_config" is the file MAPD used to have a permanent record.

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User can edit this file but has to mention one BSSID per line in the file. This is only applied at boot time.
Example:

vim /etc/map/restirct_steer_config
A0:b0:c0:d0:e0:f0
11:22:33:44:55:66

6.3.2.2 Command Line Interface

- To Add the bssid in the restrict bssid list:
  mapd_cli /tmp/mapd_ctrl steer_restrict_bss ADD <BSSID>
  Example:
  mapd_cli /tmp/mapd_ctrl steer_restrict_bss ADD 11:22:33:44:55:66

- To Remove the bssid from the restrict bssid list:
  mapd_cli /tmp/mapd_ctrl steer_restrict_bss REMOVE <BSSID>
  Example:
  mapd_cli /tmp/mapd_ctrl steer_restrict_bss REMOVE 11:22:33:44:55:66

6.4 User configured ACL

ACL stands for Access control list. In MAP turnkey solution we have provided a mapd_Cli API to configure ACL at specific MAP device for specific station , to allow/disallow its connection with the MAP device.
For this purpose internally whitelisting and blacklisting is implemented.

6.4.1 Feature Management Interfaces

6.4.1.1 Command Line Interface

Following is generic command format to configure ACL list on BSS in network:

- STA add/delete to ACL list:
  mapd_cli /tmp/mapd_ctrl acl_ctrl <type> <acl_cmd> <sta_mac> <bssid/almac>

- Configure ACL list:
  mapd_cli /tmp/mapd_ctrl acl_ctrl <type> <acl_cmd> <bssid/almac>

The <type> and <acl_cmd> has following values:

Table 6- 5. ACL command allowed values

<type>	Description
0	acl_cmd for all dev in n/w
1	acl_cmd for dev
2	acl_cmd for bss

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<acl_cmd>	Description
0	Add entry to list
1	Del entry from list
2	Flush list
3	Set ACL policy to 0
4	Set ACL policy to 1 (Whitelist)
5	Set ACL policy to 2 (Blacklist)
6	Show ACL list in mapd

Example

- <type> 0: Operation on all dev
  - To add/del sta(06:0c:43:28:a1:63) on all bss of dev

    mapd_cli /tmp/mapd_ctrl acl_ctrl 0 0 06:0c:43:28:a1:63 add

    mapd_cli /tmp/mapd_ctrl acl_ctrl 0 1 06:0c:43:28:a1:63 delete
  - For flush/policy/show on all dev

    mapd_cli /tmp/mapd_ctrl acl_ctrl 0 2 flush list on all dev

    mapd_cli /tmp/mapd_ctrl acl_ctrl 0 4 set whitelist on all dev

- <type> 1: Operation on dev (agent/controller)
  - To add/del sta(06:0c:43:28:a1:63) on dev (almac: 00:0c:43:4a:20:e5)

    mapd_cli /tmp/mapd_ctrl acl_ctrl 1 0 06:0c:43:28:a1:63 00:0c:43:4a:20:e5 add

    mapd_cli /tmp/mapd_ctrl acl_ctrl 1 1 06:0c:43:28:a1:63 00:0c:43:4a:20:e5 delete
  - For flush/policy/show on dev

    mapd_cli /tmp/mapd_ctrl acl_ctrl 1 2 00:0c:43:4a:20:e5 flush list on dev

    mapd_cli /tmp/mapd_ctrl acl_ctrl 1 4 00:0c:43:4a:20:e5 set whitelist on dev

- <type> 2: Operation on bss
  - To add/del sta(06:0c:43:28:a1:63) on bss (00:0c:43:4a:20:e7) in n/w

    mapd_cli /tmp/mapd_ctrl acl_ctrl 2 0 06:0c:43:28:a1:63 00:0c:43:4a:20:e7 add

    mapd_cli /tmp/mapd_ctrl acl_ctrl 2 1 06:0c:43:28:a1:63 00:0c:43:4a:20:e7 del
  - For flush/policy/show on all dev

    mapd_cli /tmp/mapd_ctrl acl_ctrl 2 2 00:0c:43:4a:20:e7 flush list on bss

    mapd_cli /tmp/mapd_ctrl acl_ctrl 2 4 00:0c:43:4a:20:e7 set whitelist bss

    mapd_cli /tmp/mapd_ctrl acl_ctrl 2 6 00:0c:43:4a:20:e7 show map acl list for bss

- Example for setting up whitelist for bss (00:0c:43:4a:20:e7)
  - Configure whitelist for bss

    mapd_cli /tmp/mapd_ctrl acl_ctrl 2 4 00:0c:43:4a:20:e7
  - Add sta(06:0c:43:28:a1:63) to whitelist

    mapd_cli /tmp/mapd_ctrl acl_ctrl 2 0 06:0c:43:28:a1:63 00:0c:43:4a:20:e7
  - Del sta(06:0c:43:28:a1:63) from whitelist

    mapd_cli /tmp/mapd_ctrl acl_ctrl 2 1 06:0c:43:28:a1:63 00:0c:43:4a:20:e7
  - Flush whitelist for bss

    mapd_cli /tmp/mapd_ctrl acl_ctrl 2 2 00:0c:43:4a:20:e7
  - Show map whitelist for bss

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```
mapd_cli /tmp/mapd_ctrl acl_ctrl 2 6 00:0c:43:4a:20:e7
```


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7 Multi-AP Channel planning

7.1 Introduction

Mediatek Multi-AP Channel Planning is a centralized feature which runs at the controller. The aim of this feature is that controller converges the entire MAP network devices and all their radios on a common channel.

Channel planning is a feature which was available in map R1 version also, but it has been enhanced in MAP R2 release by making use of information that we can gather in MAP R2 version.

In MAP R1, aim was to converge the entire map network on a common channel, at each radio. But in MAP R2, first we try to find the best channel amongst all the preferred channels based on channel conditions, and then converge all devices to use this best channel for each radio. In MAP R3 and R4 , all existing features of R1 and R2 are available.

Table 7- 1. MAP R1 vs. MAP R2 Channel planning differences

MAP R1	MAP R2
Aim is to converge the entire map network on a common channel, at each radio.	Aim is to converge the entire map network on a common **BEST** channel, at each radio.
The new channel selection is from intersection list based only on the fact that maximum devices are currently operating on that channel.	The new channel selection is from the intersection list based on the environment conditions of all channels
This runs only when new dev enters network.	This also runs when the environment conditions vary for current channel and MAP network is in idle state.
To run this controller needs: • channel preference list • current operating channel	To run this controller needs: • channel preference list • current operating channel • All channels scan results indicating the channel conditions

This convergent channel planning approach helps in ensuring that a MAP device is able to monitor the transmissions by a station which is connected to a different MAP device present in the network. It is focused on enhancing the efficiency of client steering. AP Steering of legacy clients (devices which don't support 802.11k) can happen only if all devices are on same channel. AP Steering utilizes air monitor feature for monitoring Unassociated Client RSSI on same channel for handoff procedures. Placing Fronthaul BSSs on same channel will also provide faster handoff of roaming clients.

Mt7603 can't support channel planning R2 feature, as it can't get the accurate parameters needed for choosing the best channel. Mt7603 will report zero for edcca, my tx air time, my rx air time, obss time and noise to mapd, thus causing channel planning follow the R1 logic.

7.2 Channel Planning feature details

7.2.1 Message flow

When a new device is detected in the network, controller will send it a Channel Preference Query. After receiving a Channel Preference Report, controller will try to find the best possible channel as per our algorithm.
The best possible channel for a radio as per MAP R1 channel planning is:

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- A channel which is present in agent's Channel Preference report and not mark as un-operable.
- The channel which has been announced as operable by maximum number of agents in the network.
- The channel on which maximum number of radios are currently operating.

The best possible channel for a radio as per MAP R2 channel planning is:

- A channel which is present in agent's Channel Preference report and not mark as un-operable.
- A channel which is selected from the intersection list of channels supported by all devices based on the environment conditions of all channels and has the best score among them.

After selecting the best channel, controller will send the Channel Selection Request to the agent. Agent reconfigures the radio to the new channel and respond with Channel Operating report.

If the channel preference list provided by the agent does not contain any channel which preferred other agents, then it will continue to operate on its existing channel.

All logic to do channel planning is present at controller side. Agent if present in network, helps controller to gather channel condition information. Agent acts as a slave here and only responds to controller's requests. Agent listens and responds to controller's below requests:

- Do scan and sent back scan report of particular channel list
- Switch to channel to do CAC and send back CAC response
- Ch selection request to switch to a particular channel

Both agent and controller perform channel scan at each radio to gather all channels condition information. If no agent is present in network, then also controller is capable of doing channel planning by doing all of the above processes by itself.

Please Note:
Channel configuration in MAP is taken care by Channel planning feature. Hence, boot up channel selected by AutoChannel in any case is not used. So AutoChannel feature should be kept disabled when MAP is enabled.

7.2.2 When is channel planning run

- **Bootup Run for channel planning:**

After "**ChPlanningInitTimeout",** controller will run channel planning once for each radio. At this time if there is any agent already connected to controller it will also help controller to do channel planning. So after channel planning runs once, all radios of all devices will come to the best channel selected by controller ch planning logic. If any new devices joins the network after bootup run is complete then after its "**ChPlanningInitTimeout"**, its radio's will also sync to this bootup time best channel.

- **Current channel overload triggered channel planning:**

After bootup run network will stay at current best channel under normal circumstances.
If due to some traffic the conditions of current best channels on any radio vary adversely, then controller can go for monitoring the current channel and later on to run channel planning if all conditions are met.

- **User trigger channel planning:**

Channel planning can be forcefully run via mapd_cli command. This will not do any channel monitoring, but start searching for a better channel than the current one by instantly running channel planning.

- **When a new device enters the network:**

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If any new devices joins the network its radios will also sync to the controller's best channel which was chosen at boot-up run. Here all radios converge to the best channel based on existing database. There is no need for channel planning scan to run again.

7.2.3 Channel planning stages

These are the stages as defined by Map R2 channel planning algorithm.

7.2.3.1 Channel Monitoring

Channel Monitoring is important because sometimes high utilization surges can occur, which may not last for long durations, or they are like a false alarm. So it is important that the current channel be monitored for at least "**ChPlanningR2MonitorTimeoutSecs"** duration. At the end of this duration, the averaged data collected during the monitor interval will be analyzed and compared with the overload thresholds again. For threshold details you can refer to the section 7.3.4. If now also it exceeds thresholds then it means channel planning need to run.

- **Channel monitoring prohibit interval:**

  We also have a prohibit timer "**ChPlanningR2MonitorProhibitSecs"** which is configurable so that we can have some gap between two monitoring runs. So we must stay on a particular channel for at least the prohibit duration before we can decide to run monitoring/channel planning again.

- **Idle MAP network:**

  Before channel monitoring starts, it is mandatory that "MAP network is in idle state" otherwise results will not be consistent.

- **Limitation with channel monitoring**

  When overlapping BSS is using only secondary channels for uplink and downlink traffic, then if this traffic is overloading the secondary channel then also channel planning monitoring will not get triggered. This is because it is a hardware limitation to detect high utilization on secondary channels. Eg: If a MAP network is on channel 36 with 160MHz bandwidth (occupying channel 36 to 64) and OBSS traffic is running on channel 52 with 80MHz bandwidth, then channel utilization will not show high utilization on channel 36 due to this HW limitation. Hence the channel monitoring cannot be triggered.

7.2.3.2 Channel Scanning

After the monitor duration is over controller decides whether to go to the next step for channel planning which is to do channel scan. If channel planning is required then, controller gathers the information about all the channels.

- Each channel's score and rank is calculated internally by the channel planning algorithm which makes use of the channel conditions found by the controller.
- The higher the rank the better the channel. It is possible that two or more channels can have same rank/score. In this case any one of them can be chosen.
- In 5G case it is also possible that you find that rank of a DFS channel is the highest. If the algo decides to move to that DFS channel, then user may have to wait for at least 65sec to see its effect on each device. This is because first DFS CAC will be done on this new channel and only if no radar was found on it , then the command to switch to this channel will be given by the controller to all devices and itself. Please refer to the DFS CAC R2 section for more details.

Channel scanning is an integral part of channel planning. So its message flow between the various deamons can be described in two parts Initialization and requested scan :

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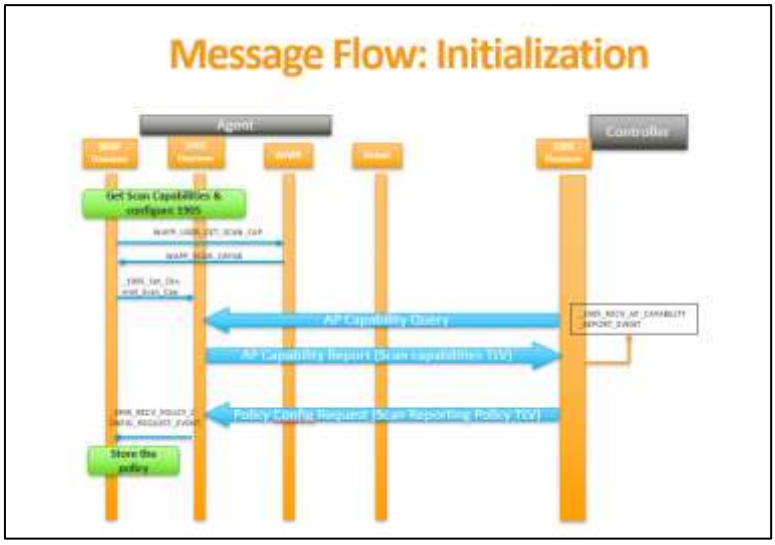


Figure 7- 1. Channel scan initialization message flow

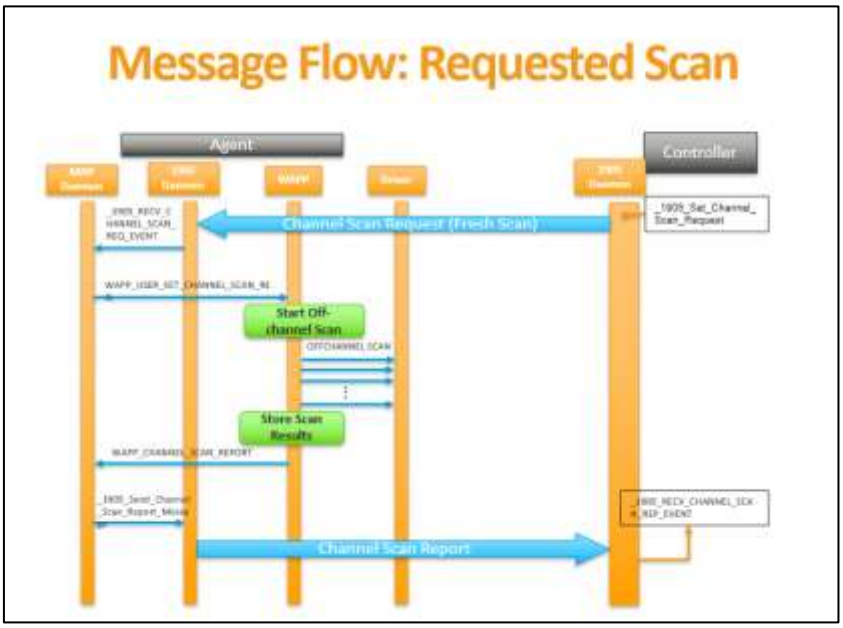


Figure 7- 2. Channel scan request/report message flow

7.2.3.3 Channel Switching

If channel planning finds a channel which has higher rank that existing channel then channel switch will be initiated by the controller to all devices including itself by using channel selection request.

- If switch has to be done to a DFS channel, then in R2 case mostly CAC would have already be done first on a dedicated radio or by a main radio of some agent in network (if dedicated radio is not available on any dev) . In this case controller will switch the channel of all devices one by one. First the device which is having max hop count will be switched and likewise the others and thus, channel switching can be efficient.

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7.2.4 Channel Planning with BW consideration

If this feature is enabled, then the channel planning best channel selection decision, will not just be based on individual channel ranks, but also on the ranks of the complete bandwidth group to which this channel belongs to.
Example if controller is currently operating on bandwidth 80MHz (channel 36), that means controller can occupy all channels from 36 to 48. If 36 is found to be the worst channel and 48 is found to be the best channel by the algo, still overall group rank of Band 36 to 48 can fall due to the score of channel 36.In this case if another band say 149 to 161 is having better rank than the band 36 to 48 by a certain margin (ChPlanningR2MinScoreMargin) then algo will choose band 149 to 161 for its operation.
This feature is used in 5G and 6G bands. Based on controller's BW the groups are formed with available channels as per preference of all devices. Then group score determins the best channel group as per the existing BW.

Note: This feature should be kept off if the channel score display on web UI is used for demonstration.

7.2.5 User Preferred Channel Configuration

Mediatek Multi-AP turnkey solution provides a mechanism to configure the whole network on a user defined channel. If a channel is marked as user preferred, best channel selection logic mentioned in 4.8.2 will not be used. Instead use defined channel will carry the maximum weightage and controller will attempt to configure all agents on user preferred channel.
If the user preferred channel is not present in an agent's Channel Preference Report, channel to this particular agent will not be assigned. In all other cases user preferred channel will always be the first preference. The configuration interfaces are mentioned in below sections.

7.2.6 RADAR Handling in channel planning

MAP Turnkey feature uses channel planning feature to handle RADAR detection on any agent in the network. An agent leverages dual backhaul system to convey RADAR detection to the controller.
Flow is as follows:

1. Any agent which detects RADAR on 5G BH will form link on 6G BH or 2.4G BH or other 5G BH if tri-band according to priority.
2. Agent will select a new random channel for 5G Hz from the valid channels in that band.
    a. The channel might be DFS or non-DFS channel.
3. Agent will add this channel info in its CSA IE and move to this channel after CSA time.
    b. Connected clients should automatically move with it.
4. Agent will send new channel preference report to controller using 6G or 2.4G BH informing presence of RADAR on a channel.
5. Controller detects RADAR from channel preference report and triggers channel planning again to find an operable channel from channel preference report.
6. Controller will send channel selection request to an agent using 6G BH or 2.4G BH or other 5G BH.
7. Agent configures new channel. They will do a CSA first and set this new channel's information in CSA IE before moving to new channel. Connected clients should automatically move to new channel.
8. If the channel selected is non-DFS then Network Opt will be re-triggered after 45 secs (As configured default value in variable – NtwrkOptConnectWaitTime, NtwrkOptDisconnectWaitTime) on 2.4G or 5G (Tri-Band) connection.
9. Network opt will switch the Agent BH channel to 5G once it finds out 5G channel is better.
10. If Channel Planning choose DFS channel post radar detection, Agent will switch to 5G after performing CAC on Controller and Agent as shown below:

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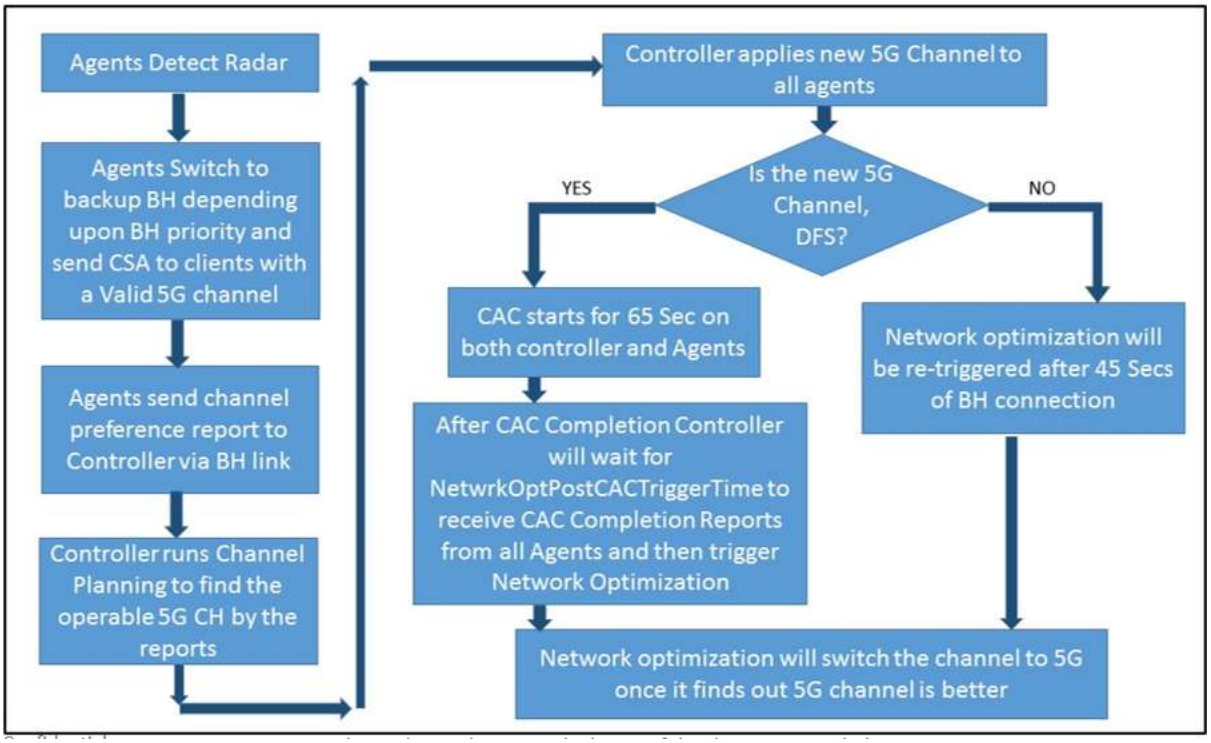


Figure 7- 3. Radar handling by Channel planning flow when Radar is detected at the agent Image 1

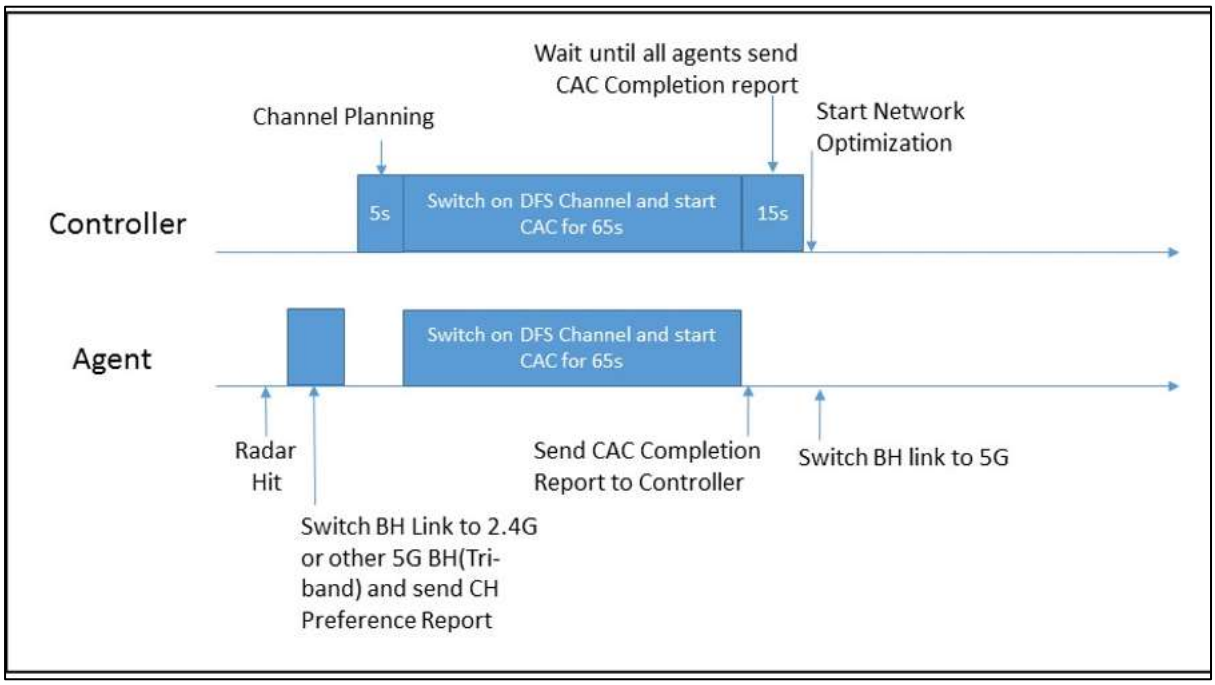


Figure 7- 4. Radar handling by Channel planning flow when Radar is detected at the agent Image 2

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7.2.7 Channel Planning examples

7.2.7.1 Channel Planning run at Bootup

Controller after boot-up scan channel, rerun the channel planning and try to make the channel of all devices same. If divergent channel planning is disabled.

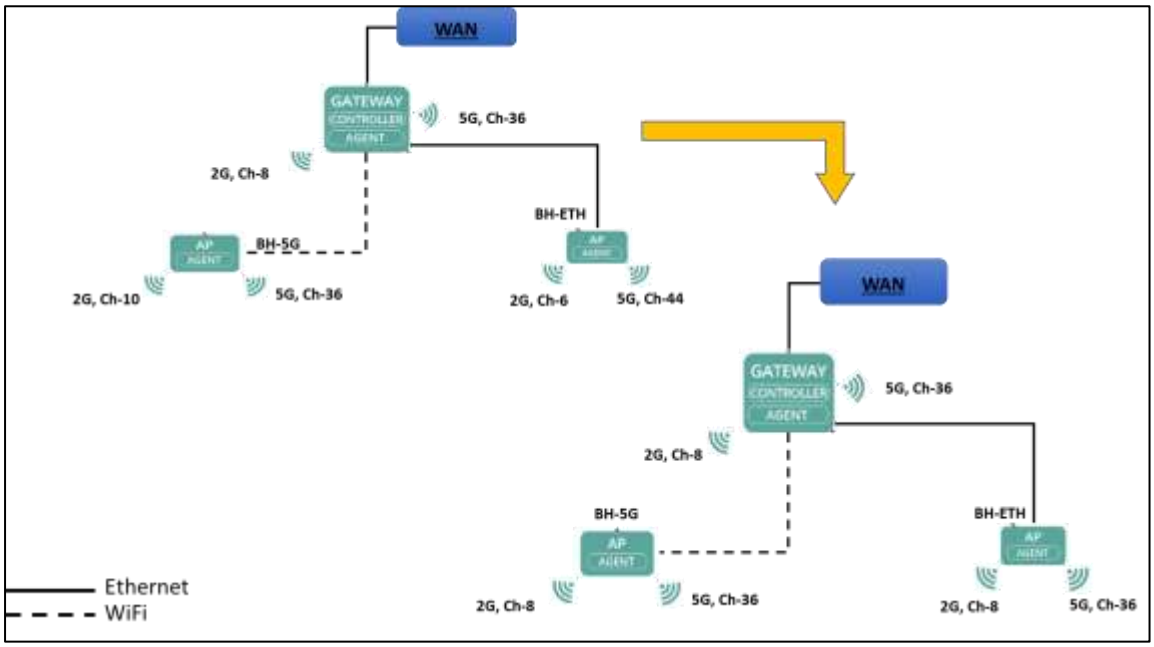


Figure 7- 5. Channel Planning run at bootup example

7.2.7.2 Channel Planning run due to User Prefer Channel command

If user decided to change the channel of all the devices including Controller. User can do this by using WEBGUI user config channel, and as per User specified channel, controllers channel planning algo changes channel of all devices. In this example User give command to change channel to 44 for 5G Interface via WEB UI.

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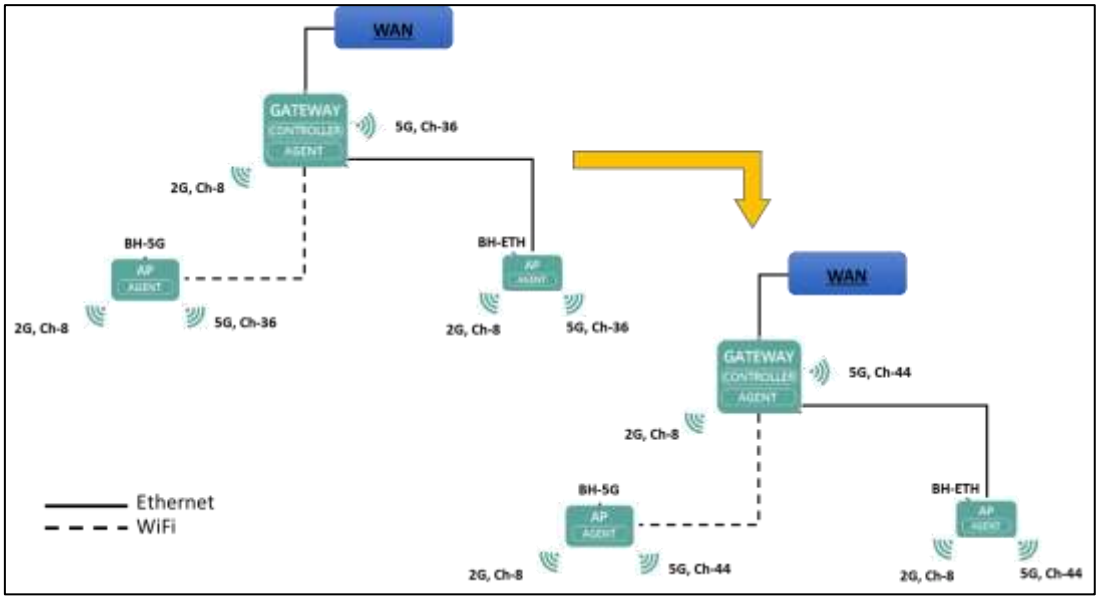


Figure 7- 6. User Preferred Channel command example

7.2.7.3 Channel Planning handling when Radar on Agent (Ethernet BH)

1. Radar found on ch-100, by agent with ETH BH.
2. Agent with BH as ETH on 5G Radio, detects the radar and mark ch-100 non operable locally.
3. Agent also change its 5G radio channel, to operable channel.
4. Later Agent provide info of radar to controller by marking ch-100 non-operable.
5. After receiving the non-operable info on ch-100, controller re-run channel planning and change channel on all device happens.

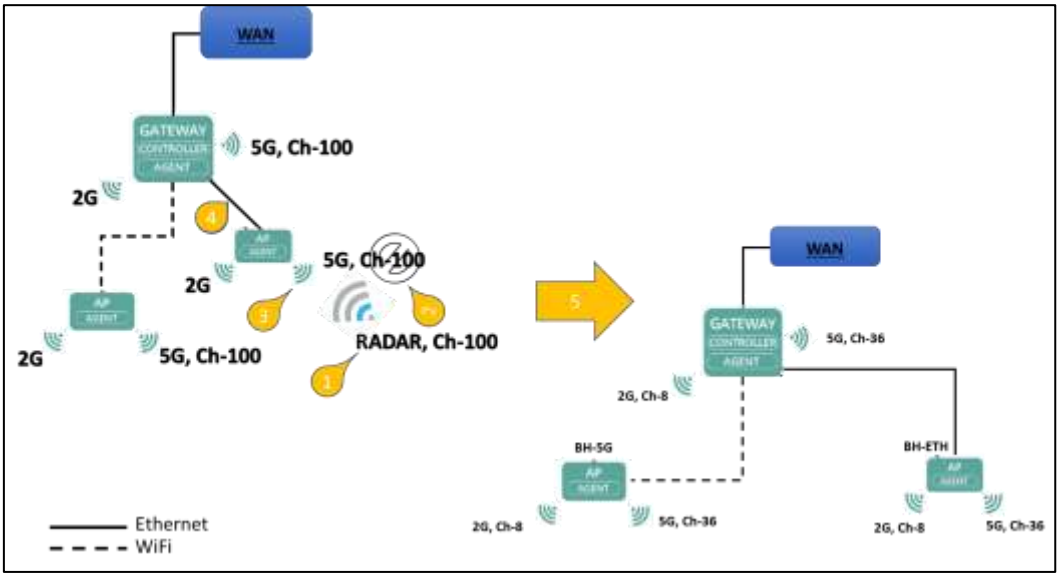


Figure 7- 7. Radar found on ethernet agent example

7.2.7.4 Channel Planning handling when Radar on Agent (WiFi BH)

1. Radar found on ch-100, with Agent on BH as WIFI 5G.
2. Agent with BH as wifi on 5G Radio, detects the radar and mark ch-100 non operable locally.

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3. Agent also change its 5G radio channel, to operable channel.
4. Due to step 3, BH connection breaks.
5. Due to step 4, a BH connection on 2G is establish.
6. Later after connection of agent, Agent provide info of radar to controller by marking ch-100 as non-operable.
7. After receiving the non-operable info on ch-100, controller re-run channel planning and change channel on all device happens.

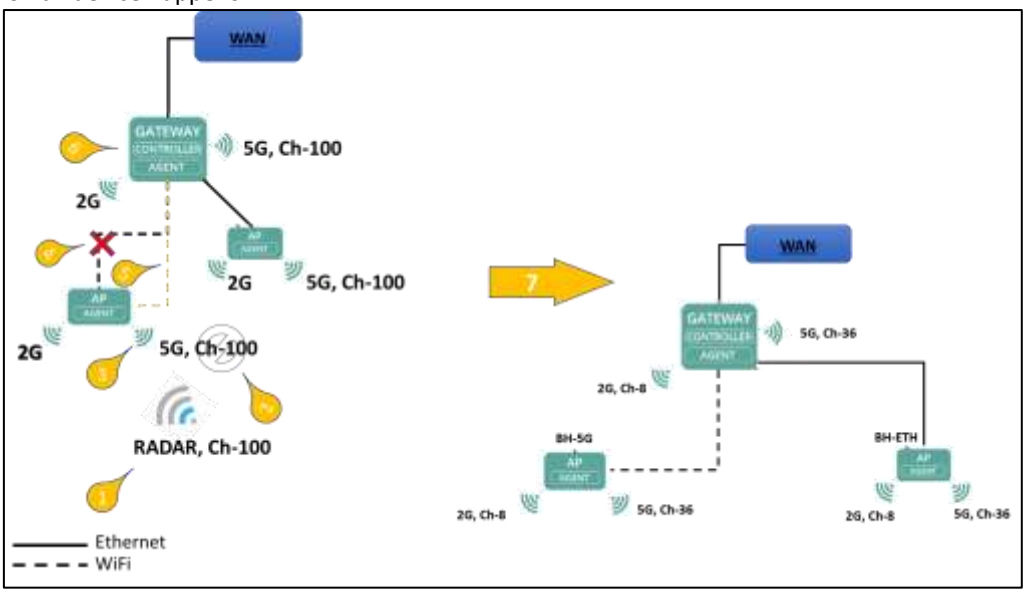


Figure 7- 8. Radar found on Agent with WiFi BH example

7.2.7.5 Channel Planning handling when Radar on Controller

1. Radar found on ch-100 by Controller on BH as WIFI 5G.
2. Then it will select a new random channel for 5G Hz from the valid channels in that band which can be DFS or non-DFS.
3. All agents switch to controller's new channel vois Controller beacon's Channel switch announcement.
4. If new channel chosen by Controller is a DFS channel , all devices will do CAC , so temporarily 2.4G BH may be formed by agent to controller
5. Later channel planning algo runs on controller and chooses a new best channel for the network.

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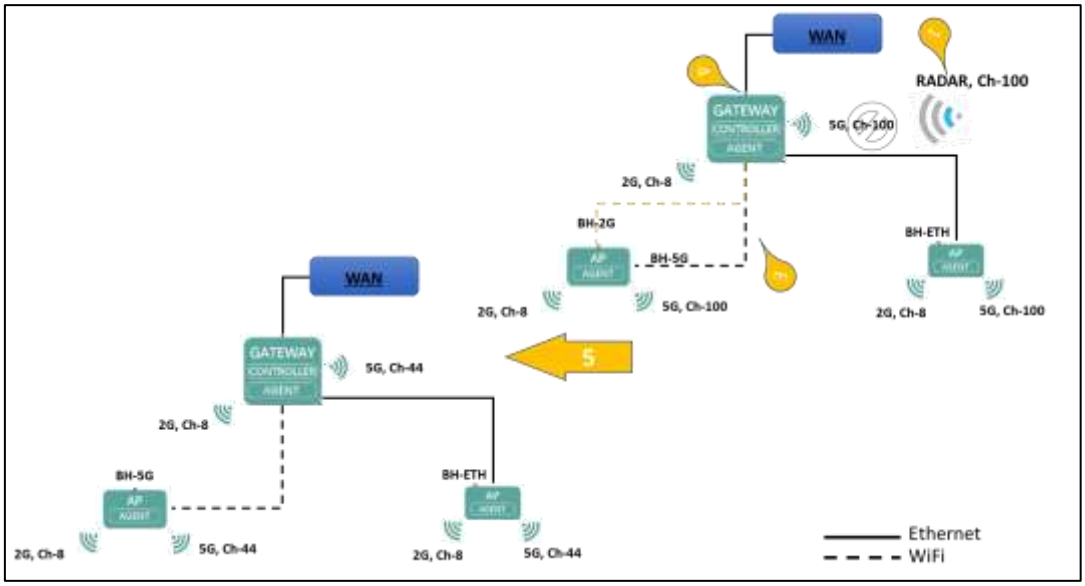


Figure 7- 9. Radar found on Controller example

7.2.8 Divergent channel planning

This MAP feature is used for forcefully switching a particular MAP device's channel by the user after giving channel and ALMAC for that device as input. The aim of this feature is to do a forceful channel switch, of any one device to a particular channel, on issuing the command manually.

Optionally, user can trigger divergent channel planning by using "forceChSwitch" command to configure channel for individual radio of an agent. The command will not be processed for a radio on which an ApCli interface exist in connected state. If the user triggers divergent channel planning API, convergent channel planning will be disabled unless it is explicitly enabled using enable_ch_planning API.Details and usage of divergent channel planning API are present in subsequent sections.

To use this feature in MAP R2, user will need an additional flag (flag name **DivergentChPlanning**) to be set before hand in mapd_user.cfg. By default this feature is disabled. The force channel switch mapd_cli command will be acceptable by daemon in MAP R2 case only if this flag is set. Similar to the MAP version R1 implementation this feature will mean that the convergent channel planning, based on channel conditions will not be able to run on all MAP devices. This is because if this feature flag is enabled then it is assumed that the user wants to configure each device's channel separately via web UI or command

To use this feature in MAP ver. R1 when MAP R2 code is compiled ON, user will need an additional flag to be set before hand in mapd_user.cfg. By default this feature is disabled, (flag name **DivergentChPlanning**).

When the mesh network is made up of Multi Link devices (MLDs) then this feature is not supported. This is because when MLD link is present between two Mesh devices then their channels should match on all links for data to be able to transmit.

7.2.8.1 Divergent Channel planning example

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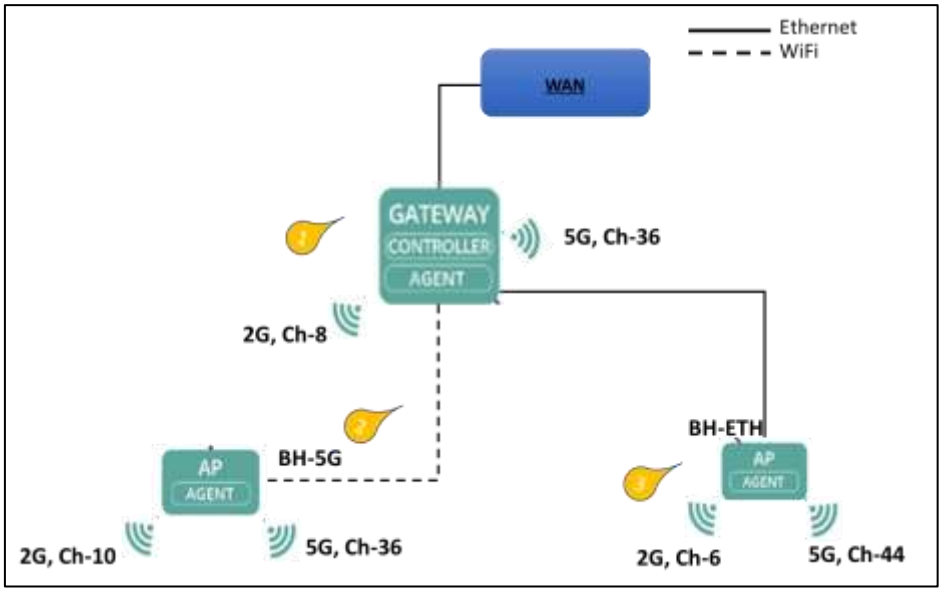


Figure 7- 10. Divergent Channel planning example

1. In Controller when **DivergentChPlanning=1,** Divergent channel planning is started.
2. Agent with BH as wifi, Radio on which the BH present has to be on the same channel as controller other radio can be on different channel then controller radio. In this example agent's 5G BH and channel is same as controller which is 36. But on 2G radio channel is 10 which is different then the controller 2G
3. As this agent is having BH as ETH both radio can be on different channels then Controller.

7.2.8.2 RADAR handling for Divergent channel planning

RADAR handling for agents in case of divergent channel planning remains same as in case of convergent channel planning. In case of Controller, once the RADAR is detected, channel planning will not be triggered. Instead, channel preference report will be indicated to the user application through an event. Detail format for the event is present in Appendix section 10.15.1 regarding user application event handling.

7.2.9 Static Puncturing handling in channel planning

Static Puncturing feature allows wireless device to use allotted BW efficiently, by puncturing interfered channel. In a mesh network where all the mesh devices are multi-link or non-multi-link devices, static puncturing helps in channel selection using static puncturing.



Figure 7- 11. Static puncturing of 20MHz channel when iner BSS traffix is detected

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Disabled Subchannel Bitmap provided punctured channel information:

The Disabled Subchannel Bitmap subfield is a 16-bit bitmap where the lowest numbered bit corresponds to the 20 MHz subchannel that lies within the BSS bandwidth and that has the lowest frequency of the set of all 20 MHz subchannels within the BSS bandwidth. Each successive bit in the bitmap corresponds to the next higher frequency 20 MHz subchannel. A bit in the bitmap is set to 1 to indicate that the corresponding 20 MHz subchannel is punctured and is set to 0 to indicate that the corresponding 20 MHz subchannel is not punctured.

	Control	CCFS0	CCFS1	Disabled Subchannel Bitmap
Octets:	1	1	1	0 or 2

Figure 7- 12. EHT Operation Information field format

- Each Agent or Controller device inform punctured channels information to Controller.
- Channel Planning Using Punctured Channels Information:
  i. Channel/BW selection algorithm need to consider punctured channel in channel planning algorithm i.e channels which are punctured should be avoided or needs to be considered based on below rules.
  ii. BW/Channel selection in channel planning:
    1. Selected BW group avoid having punctured channels i.e try to pick channel group which has less punctured channels, but score should be given high priority.
  iii. Punctured channel should not be picked as primary channel
    1. This will help if in future selected channel has interference then it can be punctured by the device.
    2. This will help in better utilization of channels.

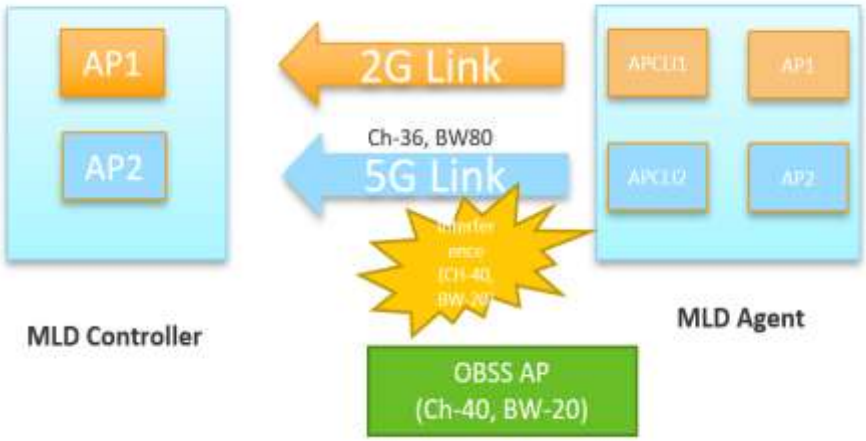


Figure 7- 13. Agent punctured Channel due to Interference

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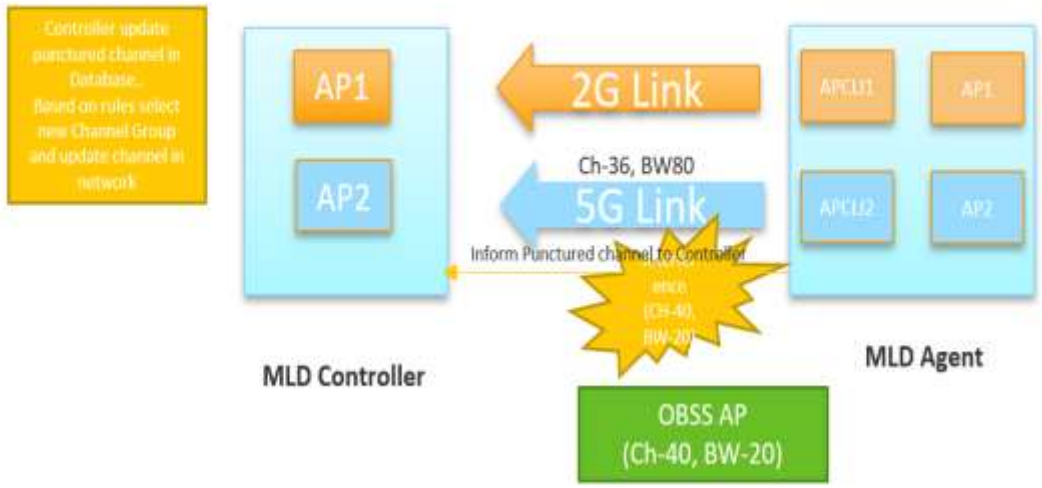


Figure 7- 14. Inform Punctured Channel Information to Controller

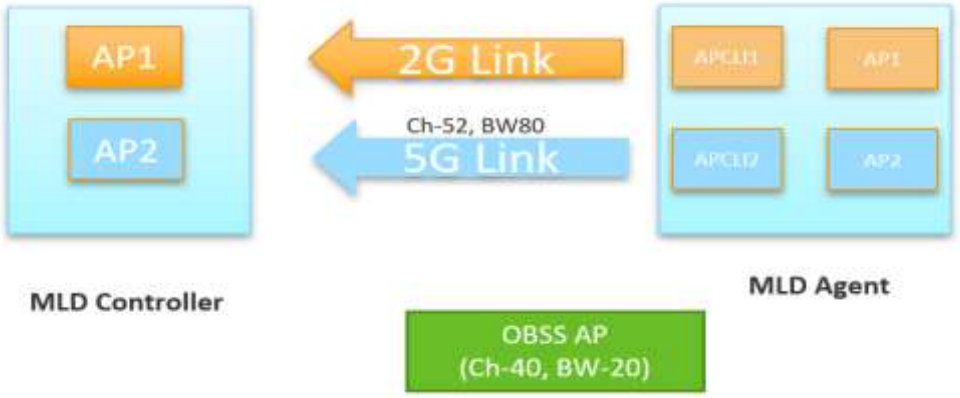


Figure 7- 15. Controller move to new channel based on static punctured when channel planning run

7.2.10 Feature Management Interfaces

7.2.10.1 Configuration file interface

On the board run time:
Below parameters are configurable through mapd_cfg to be used for OpenWRT:
1. In the file "/etc/map/mapd_default", check for values :

ChPlanningEnable=1 -> this is R1 channel planning
ChPlanningEnableR2= -> this is for R2 channel planning
By default ChPlanningEnableR2="empty". No need to do any setting as map will set it automatically based on map_version is R2 as per 1905d.cfg file. This parameter is only valid for the controller as entire ch planning algo runs only on controller.

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In order to give definition to a "**MAP network is idle**" we have two configurable parameters used by the application. These can be modified by the users before app starts. This has been picked up from MAP R1 channel planning itself. **If network activity is below a threshold level (configurable) for a predefined duration (configurable) then the MAP network is said to be idle.**

ChPlanningIdleByteCount
Configure number of Tx + Rx bytes per second below which system is considered idle

ChPlanningIdleTime
Configure consecutive number of seconds for which, if TX + RX byte count is below ChPlanningIdleByteCount, channel planning will be retriggered.

ChPlanningUserPreferredChannel5G
Configure user preferred channel for 5G radio

ChPlanningUserPreferredChannel2G
Configure user preferred channel for 2G radio

ChPlanningUserPreferredChannel6G
Configure user preferred channel for 6G radio

ChPlanningInitTimeout=120
This parameter which comes from MAP R1 channel planning is still functional in MAP R2 also. Its value determines, after how much time in secs after coming into the network will the channel planning algo run, aligning all devices to same channel on all bands.

ChPlanningEnableR2withBW=
By default **C**hPlanningEnableR2withBW ="empty". No need to do any setting as map will set it automatically based on map_version is R2 as per 1905d.cfg file. This parameter is only valid for the controller as entire ch planning algo runs only on controller. This parameter is enable by default internally if dev version is MAP R2, and comes in use only when "**ChPlanningEnableR2"** is enable.

DivergentChPlanning=0
This flag has been added for a special MAP R1 feature for setting the channel on a particular MAP device. By default this feature is disabled as convergent channel planning is enabled.

SetPSCChannel_6G=0
This flag is applicable only when 6G band is present. If user wants channel planning to choose only valid PSC channels from 6G band channel list then this flag is needed to be set. By default this flag is disabled , so channel planning will consider all 6G band channels to be potential primary channels, then out of all those channels, best channel will get selected based on channel statistics and score.

3. In the file **"/etc/mapd_strng.conf"**, there are the following parameters defined, the values of these can be modified as per the testers convenience. These parameters are only valid for the controller as entire ch planning algo runs on controller.

The default values are as given below:

ChPlanningChUtilThresh_24G=70
The value of Channel utilization threshold in percentage terms. If AP metric response from self or from agent has ch utilization value greater than this percentage then channel planning R2 algo can go for channel monitoring.

ChPlanningChUtilThresh_5GL=80
Same as above but for 5G interface

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ChPlanningChUtilThresh_6G=80
Same as above but for 6G interface

ChPlanningEDCCAThresh_24G=200
The value of EDCCA threshold in terms of normalized to 255 max. If AP metric response from self or from agent has EDCCA value greater than this value then channel planning R2 algo can go for channel monitoring.

ChPlanningEDCCAThresh_5GL=200
Same as above but for 5G interface

ChPlanningEDCCAThresh_6G=200
Same as above but for 6G interface

ChPlanningOBSSThresh_24G=200
The value of Other BSS load threshold in terms of normalized to 255 max. If AP metric response from self or from agent has OBSS load value greater than this value then channel planning R2 algo can go for channel monitoring.

ChPlanningOBSSThresh_5GL=200
Same as above but for 5G interface

ChPlanningOBSSThresh_6G=200
Same as above but for 6G interface

ChPlanningR2MonitorTimeoutSecs=300
This is time period in secs for which duration channel planning will monitor the current channel after detecting that a threshold has been crossed. At the end of this interval ch planning algo will compare the averaged data collected during this time with the thresholds and decide if there is need for going for channel planning scan or not.

ChPlanningR2MonitorProhibitSecs=900
This is the time period in seconds for which the channel planning algo (including ch planning monitoring) will be prohibited to run after last run.

ChPlanningR2MetricReportingInterval=10
This is the time period in secs, it is the frequency with which the controller wants agents to send the AP metric report during the monitoring period. This will help controller to gather sufficient data for taking decision whether to go for scan or not at the end of monitoring interval. This interval is sent to agents by means of ap metric policy. Controller also uses this interval for fetching its own metric values from WAPP. At the end of "**ChPlanningR2MonitorTimeoutSecs",** this value is restored back to the original metric reporting interval as its use is valid only during monitoring stage.

ChPlanningR2MinScoreMargin=10
This is percentage value of score difference. After completing channel planning scan the new best channel's score will be compared to the existing channel's score. If new channel score is better than current channel score by this much percentage then only the band will switch to the new channel.

7.2.10.2 Command Line Interface

1. Command to trigger channel scan for specific radio on one device. Input is the ALMAC of the device and band number based on below Band index:
   BAND_2G is 1
   BAND_5G is 2

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BAND_6G is 4mapd_cli /tmp/mapd_ctrl ch_scan_trigger_band <ALMAC> <BAND IDx>
Eg:
mapd_cli /tmp/mapd_ctrl ch_scan_trigger_band 02:0c:43:59:aa:61 4

2. Command to trigger channel planning forcefully after bypassing the monitor stage on a particular radio of all devices in the network. Index 1 is for 2.4G and Index 2 is for 5G and Index 4 is for 6G.

mapd_cli /tmp/mapd_ctrl ch_plan_R2 1
mapd_cli /tmp/mapd_ctrl ch_plan_R2 2
mapd_cli /tmp/mapd_ctrl ch_plan_R2 4

3. Command to get channel planning R2 information for debug purpose. Index 0 is for fetching only the states + score table. Index 1 is for getting scan results also along with basic R2 state info. Index 2 is for getting the channel preference db for all dev in network instead of the scan results.

mapd_cli /tmp/mapd_ctrl set get_ch_plan_info 0
mapd_cli /tmp/mapd_ctrl set get_ch_plan_info 1
mapd_cli /tmp/mapd_ctrl set get_ch_plan_info 2

4. Command to get channel planning R1and R2information for debug purpose. Each bit when set enables display for various ch planning arguments.

mapd_cli /tmp/mapd_ctrl show_ch_info 111111

5. Command to dump scan results info, especially for Web UI's graph display purpose. After giving this command results can be viewed in /tmp/dump_ch_scan.txt

mapd_cli /tmp/mapd_ctrl ch_scan_res_dump

6. Command to dump ch planning score info, especially for Web UI's graph display purpose. After giving this command results can be viewed in /tmp/dump_ch_score.txt

mapd_cli /tmp/mapd_ctrl ch_plan_score_dump

7. To enable/disable channel planning dynamically Enable_ch_planning

mapd_cli /tmp/mapd_ctrl enable_ch_planning 1
- Enable

mapd_cli /tmp/mapd_ctrl enable_ch_planning 0
- Disable

8. User_preferred_channel: to set user preferred channel

mapd_cli /tmp/mapd_ctrl set user_preferred_channel <channel>

Fro 6G band since channel number coincides with the channel number of 2G and 5G band , so to remove ambiguity we have extended the existing user preferred channel command to include band index also. Index 1 is for 2.4G and Index 2 is for 5G and Index 4 is for 6G.

mapd_cli /tmp/mapd_ctrl set user_preferred_channel <channel> <band_index>

9. Channel utilization threshold can be dynamically modified by below CLI command:

mapd_cli /tmp/mapd_ctrl set_ch_util_threshold <2G> <5G><5GH><6G>

Eg:

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mapd_cli /tmp/mapd_ctrl set_ch_util_threshold 80 100 80 80

7.2.10.3 WEB UI Interfaces

1. Channel scan:

If user wants to only do scan on a particular map R2 device (all radios), then this option can be used.

- Need to "Trigger All Channel Scan" from controller UI by choosing device
- After scan completes, click on "Display Channel Scan Result"
- Please note that the channel condition tab is not having valid results if you have just triggered scan via web UI. Its use and validity will be seen in Demo3.
- It is possible for the channels with neighbor number zero to still have some utilization. This is because these channels may be bonded to some other channels for operation in 40MHz/80MHz/160MHz BW. Hence there may be some traffic on these channels too.



Figure 7- 16. Channel statistics WEB UI button snapshot

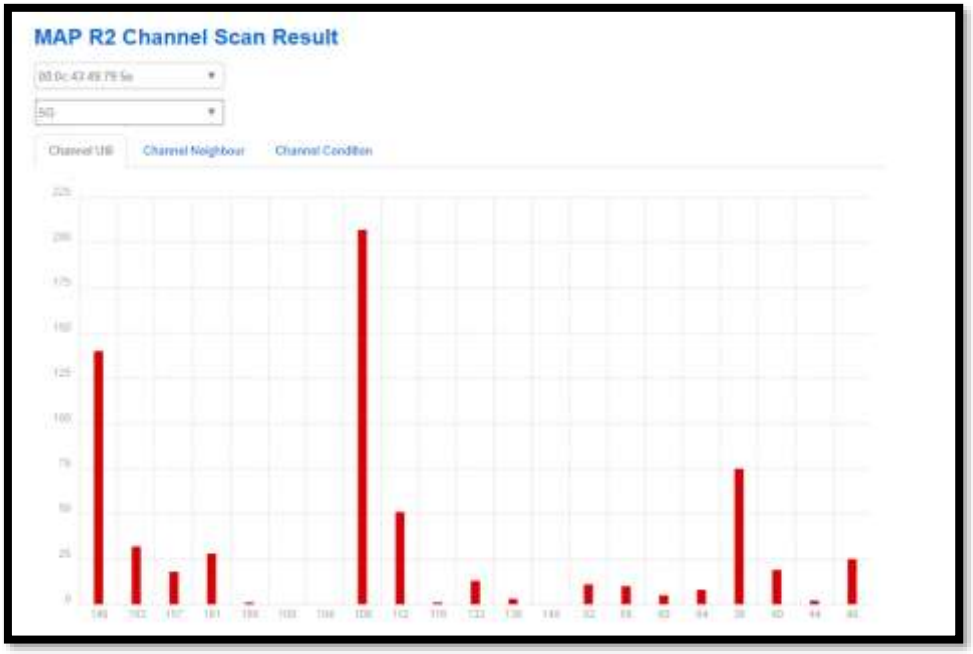


Figure 7- 17. Channel statistics example image 1

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- The x-axis of the branch shows the channel number and y axis shows the corresponding utilization on that channel as found by the device that did the all channel scanning.
- Channel utilization is an indicator of how much the channel is utilized by various transmissions occurring on it or any one of its bonded channel. It is normalized by factor of 255, hence the maximum value of channel utilization is around 255 as defined by the specification.

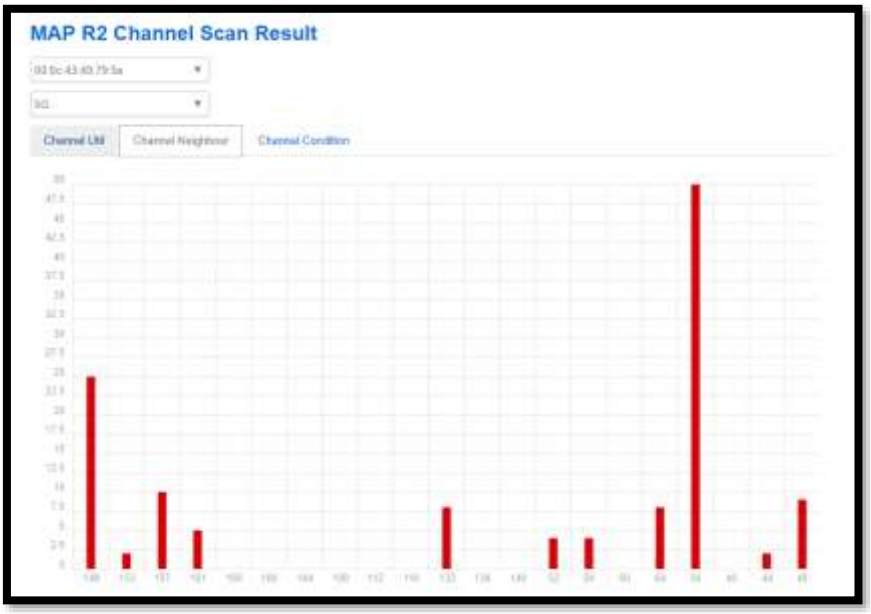


Figure 7- 18. Channel statistics example image 2



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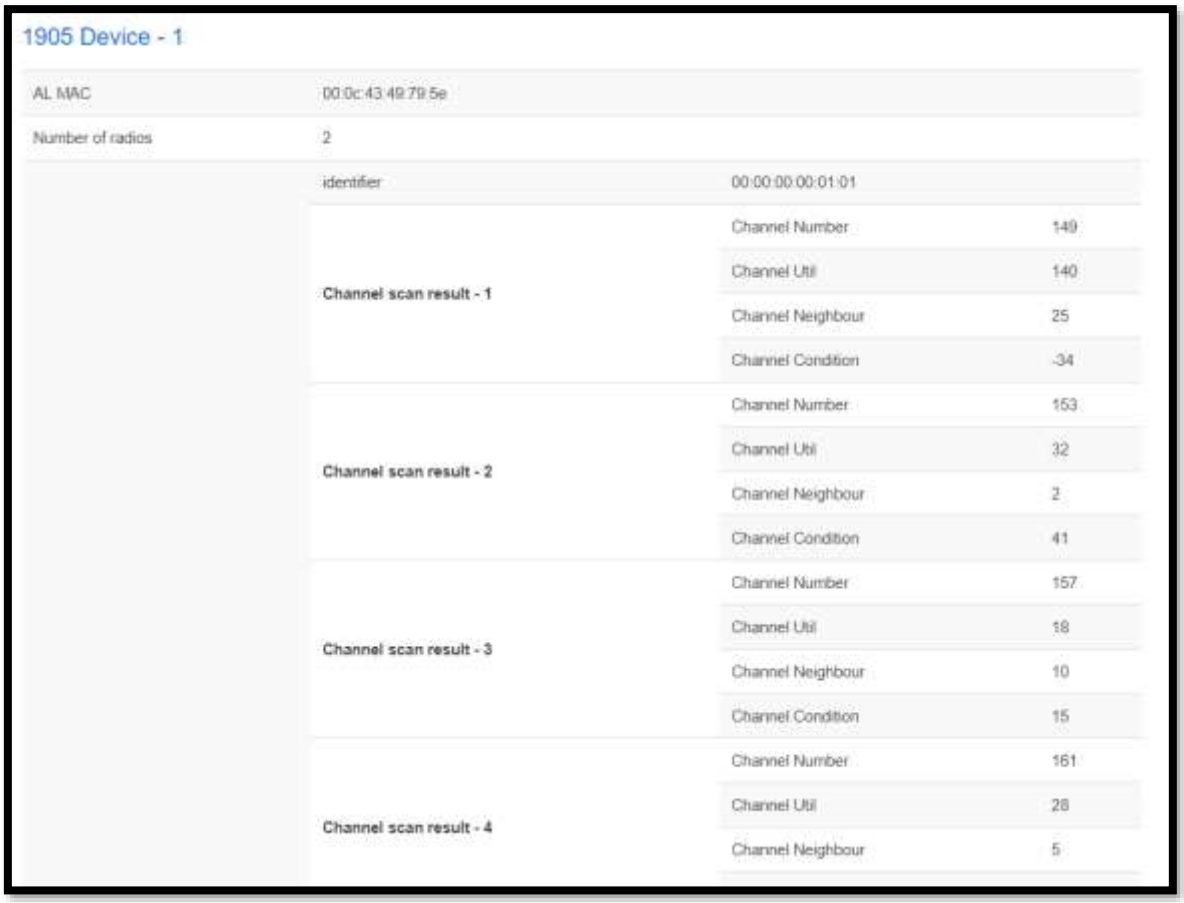


Figure 7- 19. Channel statistics example image 3

2. Channel Planning:

- At boot-up all the devices already attain a best channel. This happens after around 2 mins of complete network formation. 2 mins is considered as the idle time for the network to settle.
    - You can view the current operating channel values as below:



Figure 7- 20. Display Run time topology button on WEB UI

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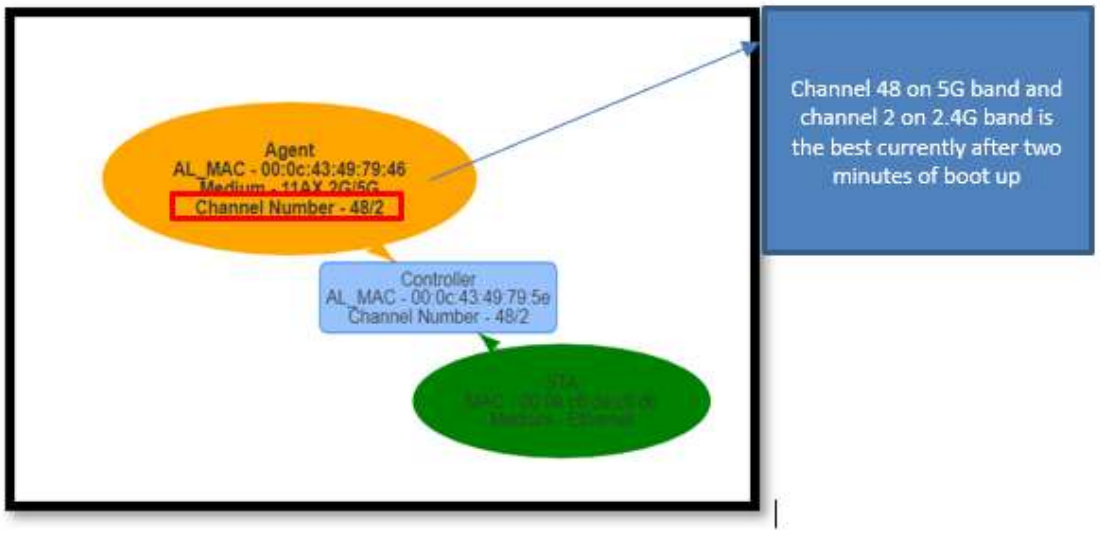


Figure 7- 21. Current operating channel value on WEB UI topology display example

- User triggered channel planning can be run by clicking on the following button on controller's advanced options Easymesh page.



Figure 7- 22. User triggered channel planning WEB UI button image 1



Figure 7- 23. User triggered channel planning image 2

- After some time you can see latest channel scores by clicking on:

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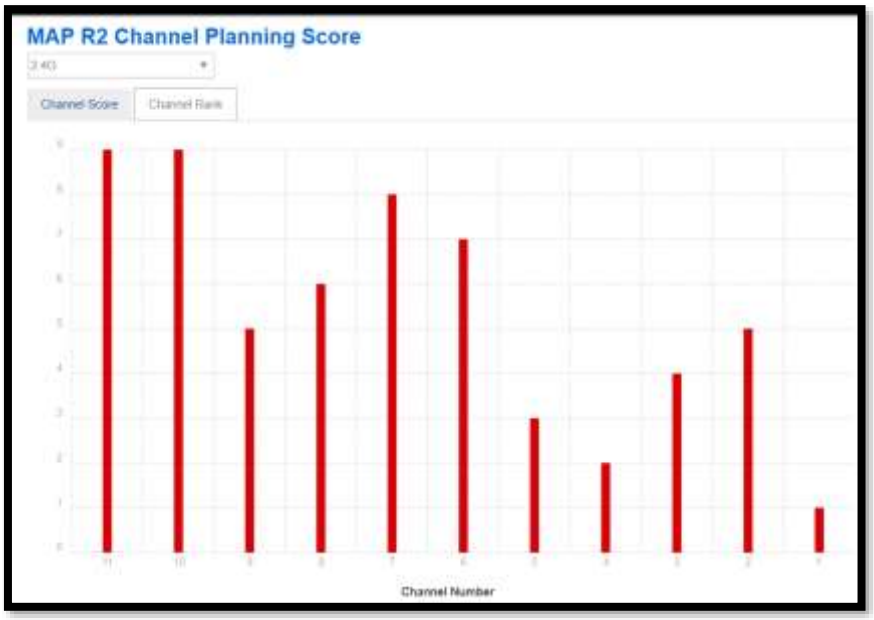


Figure 7- 24. User triggered channel planning image 3

- The higher the rank the better the channel. It is possible that two or more channels can have same rank/score. In this case any one of them can be chosen.
- Each channel's score and rank is calculated internally by the channel planning algorithm which makes use of the channel conditions found by scan results.
- NOTE: Please note that this display is valid for individual channel rank, provided the "ChPlanningEnableR2withBW" feature is turned off by the mapd_default file. Otherwise the channel to which you switch to may not match with the best channel with highest rank. By default the ChPlanningEnableR2withBW feature is enabled in mapd. So for demo purpose it is advisable to disable this advanced feature.

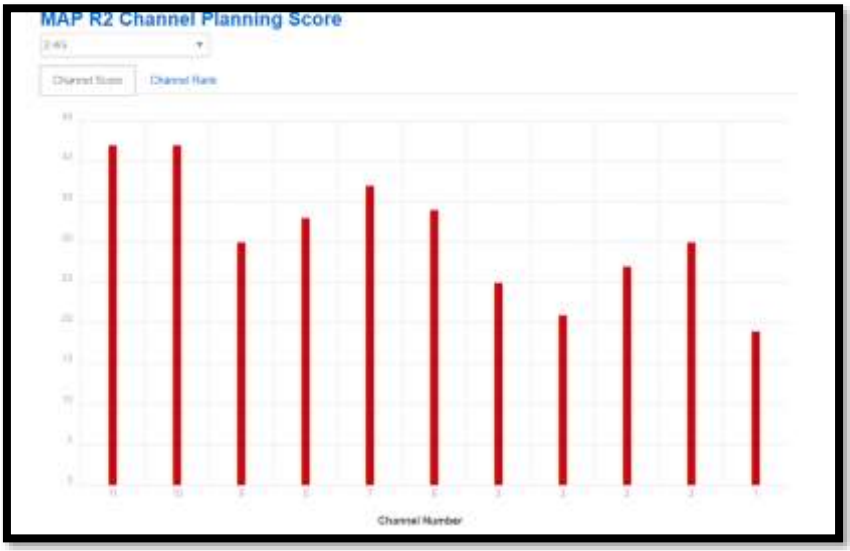


Figure 7- 25. User triggered channel planning image 4

- Also the devices in topology have got a new best channel as seen below:

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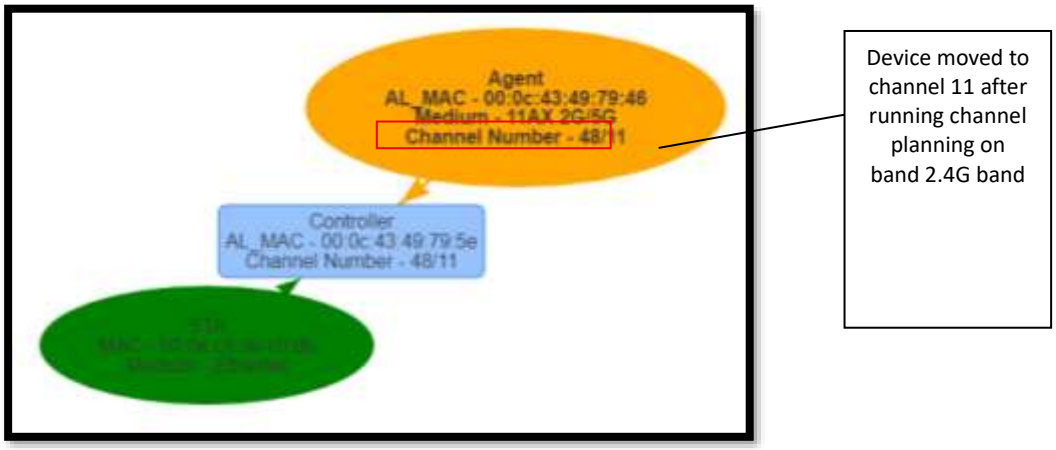


Figure 7- 26. User triggered channel planning image 5

7.2.10.4 UCI Interfaces

All Channel planning feature related parameters can be configured by UCI file system also in UCI supported devices:

```
root@OpenWrt:/# uci show mapd | grep ChPlan
mapd.mapd_cfg.ChPlanningIdleByteCount=' '
mapd.mapd_cfg.ChPlanningIdleTime=' '
mapd.mapd_cfg.ChPlanningUserPreferredChannel5G=' '
mapd.mapd_cfg.ChPlanningUserPreferredChannel5GH=' '
mapd.mapd_cfg.ChPlanningUserPreferredChannel2G=' '
mapd.mapd_cfg.ChPlanningUserPreferredChannel6G=' '
mapd.mapd_cfg.ChPlanningInitTimeout='120'
mapd.mapd_cfg.ChPlanningEnable='1'
mapd.mapd_cfg.ChPlanningEnableR2=' '
mapd.mapd_cfg.DivergentChPlanning='0'
mapd.mapd_cfg.ChPlanningScanValidTime='14400'
mapd.mapd_strng.ChPlanningChUtilThresh_24G='70'
mapd.mapd_strng.ChPlanningChUtilThresh_5GL='80'
mapd.mapd_strng.ChPlanningChUtilThresh_6G='80'
mapd.mapd_strng.ChPlanningEDCCAThresh_24G='200'
mapd.mapd_strng.ChPlanningEDCCAThresh_5GL='200'
mapd.mapd_strng.ChPlanningEDCCAThresh_6G='200'
mapd.mapd_strng.ChPlanningOBSSThresh_24G='200'
mapd.mapd_strng.ChPlanningOBSSThresh_5GL='200'
mapd.mapd_strng.ChPlanningOBSSThresh_6G='200'
mapd.mapd_strng.ChPlanningR2MonitorTimeoutSecs='300'
mapd.mapd_strng.ChPlanningR2MonitorProhibitSecs='900'
mapd.mapd_strng.ChPlanningR2MetricReportingInterval='10'
mapd.mapd_strng.ChPlanningR2MinScoreMargin='10'
```

Figure 7- 27. Channel planning parameters in UCI file system

Eg. Set User preferred channel on 2G:

- uci set mapd.mapd_cfg.ChPlanningUserPreferredChannel2G=' 6'
- uci commit mapd
- re-run Easymesh_openwrt.sh

Eg. Set Init timeout in channel plan:

- uci set mapd.mapd_cfg.ChPlanningIdleTime=' 120'
- uci commit mapd

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- re-run Easymesh_openwrt.sh

7.3 User Configurable Skip Channel List

The User Configurable Skip Channel List is a supporting feature of Channel planning. The user Configurable Skip Channel List configuration allows every MAP device to provide its own set of channel list to the controller which should be skipped, while selecting channel in controller. This is achieved by giving low score to channels in skiplist. This is a proprietary feature as only MTK devices support it.
This feature allows each MAP device to provide a set of channel skip list.
Each Device uses a Vendor message to send skip list to the controller, when any topology change happens.
Then controller will take the super set of each list and its own list, while deciding the score of each channel.

If a user preferred channel is given then channel skiplist will be avoided for that channel. Channel planning give first preference to user defined channel.

For 5G channel skiplist, MAX channel BW of MAP network will take care by controller.
Example: if 40 is present in channel skiplist and max channel BW is 40MHz. Then 36,40 both will not be allowed in channel selection.

At boot time of controller skiplist will be avoided, but later when channel planning runs it will take care of skip channel list.

If all channels are mentioned in the skip channel list, then boot time channel will be selected as the best channel.

If radar is found then skiplist will be considered but if all remaining channels are in skiplist then channel selected can be random. Later it when channel planning runs it will consider the channel skiplist again.

The flow chart of the channel skiplist feature:

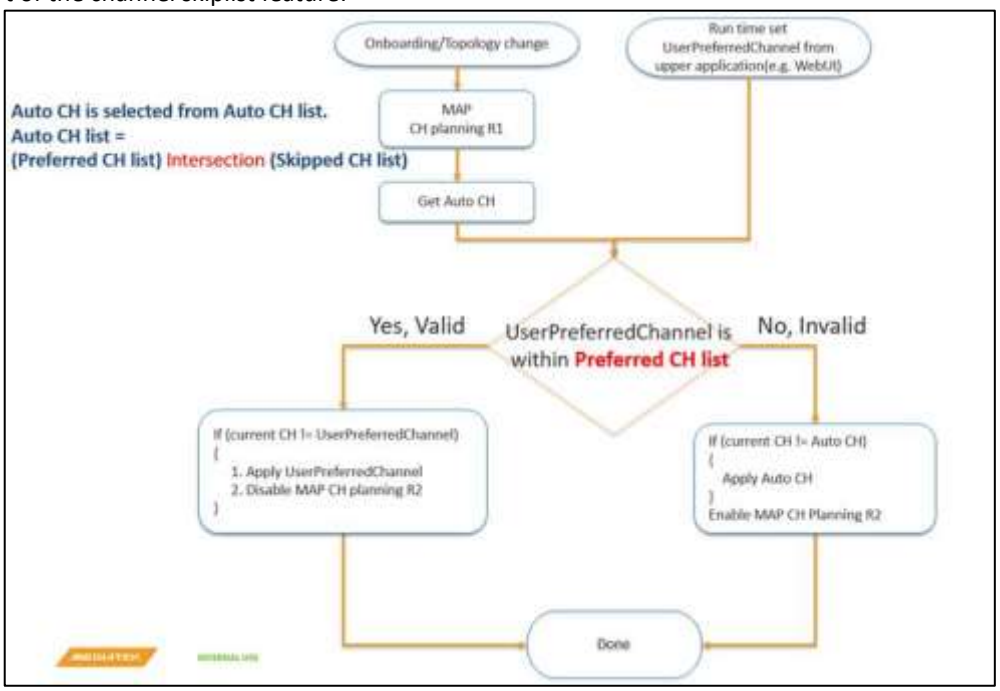


Figure 7- 28. Channel Skip List flow chart

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7.3.1 Feature Management Interfaces

7.3.1.1 Configuration file interface

Using .dat files Autochannelskiplist needs to configure. And reboot MAP network to allow it and for 6G radio add 6G channels in AutoChannelSkipList6G of the 6G interface .dat file.
Example:

AutoChannelSkipList=1;2;3

For 6G radio:
AutoChannelSkipList6G=1;37

7.3.1.2 Command Line Interface

Controller only commands.
1) To display complete skiplist:

mapd_cli /tmp/mapd_ctrl get_skip_list skip_list

2) To display complete preferred list:

mapd_cli /tmp/mapd_ctrl get_skip_list prefered_list

7.3.1.3 WEBUI Interface

To apply **user preferred channel** new drop down is introduced which contains a valid set of user preferred channels only. It is under MTK-> EasyMesh->Advanced



Figure 7- 29. WEB UI button to configure User Preferred channel using drop down

7.3.1.4 User configurable skip channel list Example

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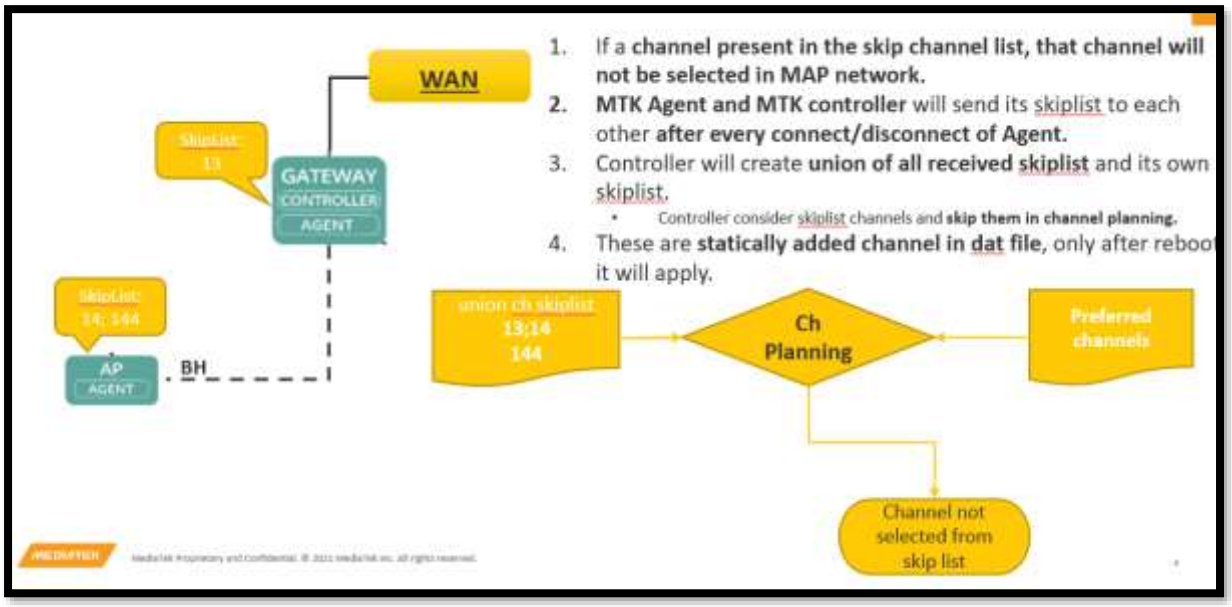


Figure 7- 30. User configurable skip channel list Example

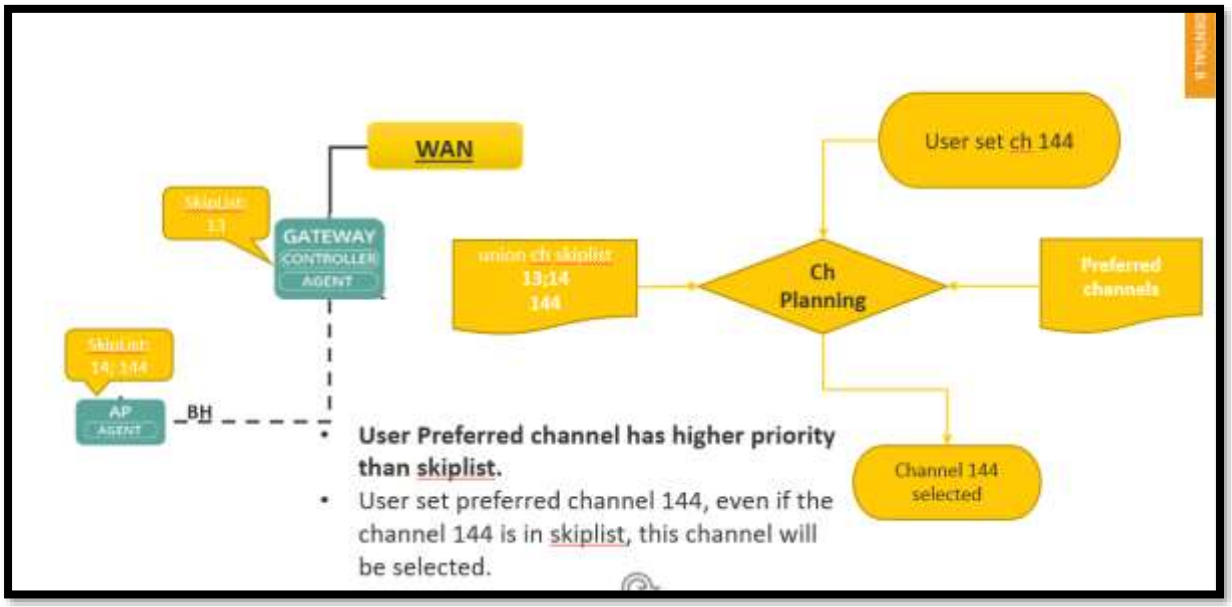


Figure 7- 31. User Configurable Skip channel list, with user preferred channel

7.4 DFS CAC

MAP R2 specification offers a mechanism to let the controller send channel availability check (CAC) request on a specific DFS channel to any agent and get the response. As specified in MAP R2 Specification there are various methods by which a device can perform CAC.
These methods are:

- Continuous CAC → Perform CAC on main radio
- Continuous with dedicated radio → Perform CAC on dedicated radio.
- MIMO dimension reduced(Not supported on MTK devices)
- Time sliced CAC (Not supported on MTK devices)

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7.4.1 Switching Channel to DFS

MAP R2 Turnkey uses Coordinated CAC while switching channel to a DFS channel. Unlike MAP R1, MAP R2 will first decide the device which would perform CAC and with what method using the Agent CAC Capabilities and max BW supported by device. The device to perform CAC is chosen in a manner so that the existing network is disturbed in the least possible manner. During CAC a device will not be able to Tx on the CAC channel for at least 65 secs and 605 secs for weather channels. If it is an agent that has to perform CAC on its main radio (continuous CAC), then this will impact its 5G BH connectivity.
Controller will decide which device to perform CAC in the below manner:

- Find max BW supported devices in the network.
- If Controller is MAP R2 and supports max BW in the networks, and DEDICATED radio is supported then
    - Controller will do CAC on dedicated radio.
- If Controller is MAP R2 and NO - DEDICATED radio is supported, then
    - Farthest R2 Agent of MAX BW supported of network with DEDICATED radio will do CAC.
    - Else, Farthest R2 Agent of MAX BW supported of network with REDUCED MIMO radio will do CAC.
    - Else, Farthest R2 Agent of MAX BW supported of network with CONTINUOUS radio will CAC.
    - Else, Controller R2 which supports MAX BW supported of network will directly switch to DFS channel and do CAC.
- If Controller is MAP R2 and NO- DEDICATED radio supported and all other agents are MAP R1 supported then
    - Controller will switch directly and do CAC on main radio.
- If Controller is MAP R1 and DEDICATED radio supported then
    - Controller will do CAC on dedicated radio
- If Controller is MAP R1 and NO - DEDICATED radio supported
    - Controller will switch directly to the DFS channel and do CAC on main radio.

If Controller decides other device (Agent) in network to perform CAC, then Controller will send CAC Request to that Agent. After this Agent will perform CAC either on dedicated radio or main radio according to its capability. On Successful CAC Completion, Agent will respond CAC Completion report to controller. Controller will then send the Channel Selection request to the agents starting from the last node of the network and corresponding to all the inner nodes, so that no disconnection occurs while switching channel.

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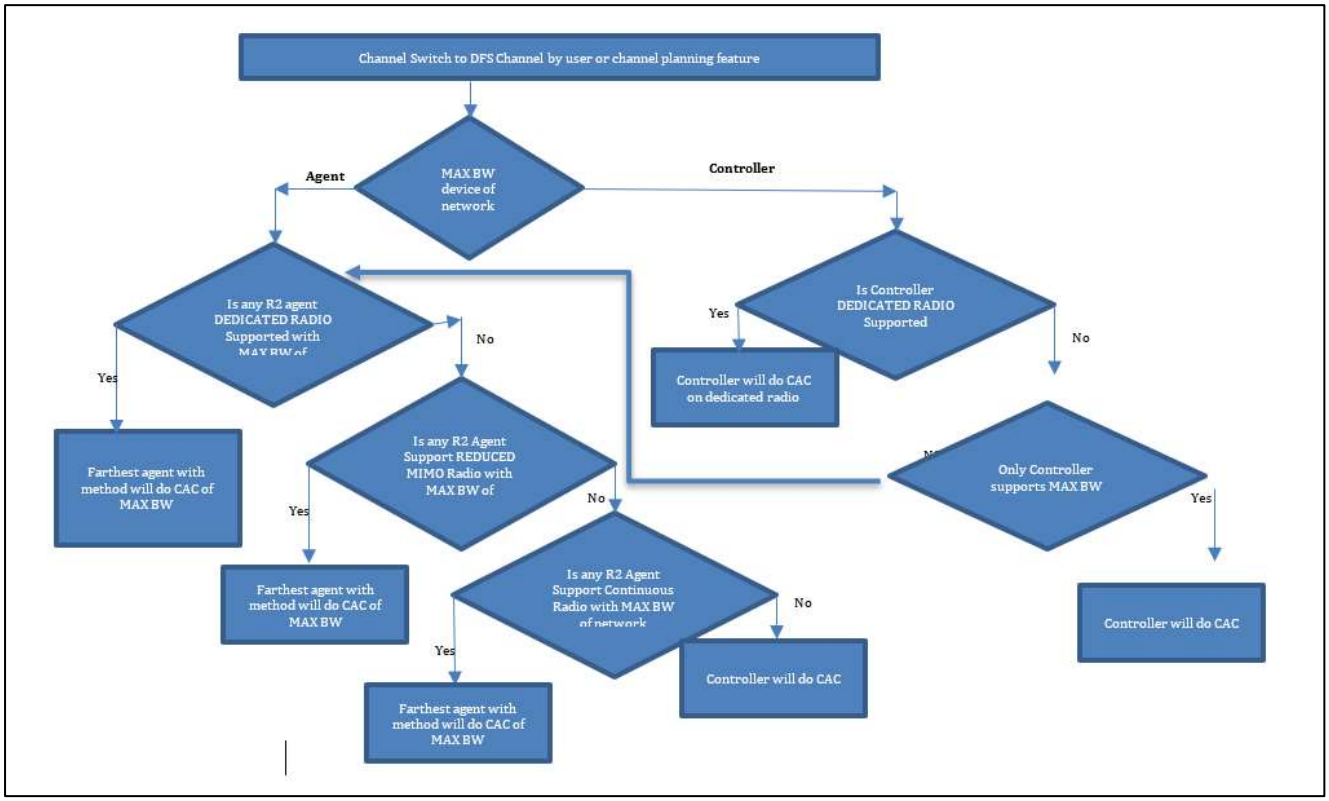


Figure 7- 32. Selecting CAC device Flow Chart

7.4.1.1 Which device will perform CAC algorithm Examples

CASE 1: Controller is MAP R2 and supports Dedicated Radio



Figure 7- 33. Device selection for CAC example case 1

CASE 2: Controller is MAP R2 and no Dedicated Radio. MAP R2 Agents with different capabilities with different positions



Figure 7- 34. Device selection for CAC example case 2

CASE 3: Controller is MAP R2 and no Dedicated Radio. 2 MAP R2 Agents with Dedicated Radio with different positions



Figure 7- 35. Device selection for CAC example case 3

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CASE 4: Controller is MAP R2 and no Dedicated Radio and all MAP R1 agents



Figure 7- 36. Device selection for CAC example case 4

CASE 5: Controller is MAP R1



Figure 7- 37. Device selection for CAC example case 5

CASE 6: Controller is MAP R2 and supports Dedicated Radio. BW of A1 and A3 is higher than controller. Agent 3 has continuous radio, but Agent 1 is dedicated Radio.



Figure 7- 38. Device selection for CAC example case 6

CASE 7: Controller is MAP R2 and supports Dedicated Radio. BW agent 3 is highest. Agent 3 do CAC.



Figure 7- 39. Device selection for CAC example case 7

7.4.2 CAC Enhancements

7.4.2.1 CAC by BW

- Driver and device are capable of doing CAC by bandwidth.
- For example: if Driver has done CAC for (BW = 80) on channel 52.
- Driver has taken care of checking RADAR on channel group of {52,56,60,64}
- So next time if MAPD wanted to change channel to 60, MAPD can use the old radar report so no need to do CAC again.

7.4.2.2 CAC at boot time

- If controller/Agent boots up at DFS channel, by default CAC is completed by driver.
- In this case if MAPD boots up before CAC is completed, and receives the CAC completion report from driver.

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- CAC done at boot time will be considered in Channel planning, which means if channel planning finds the best channel from the BW group on which the
- CAC is completed. No need to do CAC again.
- But this results only valid if any of the device is already on that CAC performed channel, if no device is on CAC performed channel.
- Old/boot time CAC report will be disqualified.

7.4.2.3 CAC reset after HWRDD moved

- HWRDD is Hardware block which takes care of finding if RADAR is present on DFS channel BW group.
- If we change channel out of the HWRDD group (even in case of non DFS), this got reset (non-DFS case)/moved (DFS case).
- So in this case CAC has to be done again, if moving back to same DFS channel.
- Also when MAPD is moving to another channel group by BW, old CAC result will be clear.

7.4.3 RADAR Detection

Radar Handling will change in MAP R2 due to the introduction of Coordinated CAC feature and dedicated radio.

7.4.3.1 Agent Radar Detection

- Any agent which detects RADAR on 5G BH will form link on 2.4G BH. Also Agent will select a new random channel for 5G Hz from the valid channels in that band which can be DFS or non-DFS channel. Connected clients should automatically move with it.
- Agent will send new channel preference report to controller using 2.4G BH informing presence of RADAR on a channel.
- As Controller detects RADAR from channel preference report, it triggers MAP R2 channel planning again to find an operable channel from channel preference report.
- If that Channel is non-DFS then Controller will switch channel on agents through Channel selection request and bring back to 5GH after triggering Network Optimization.
- If that Channel is DFS then Controller will decide which device will do CAC and by what CAC method. If any agent is selected then it will send CAC Request to that agent. If controller supports Dedicated radio then it will do CAC (Refer 7.4.2)
- On Successful CAC Completion Agent will respond CAC Completion report to controller.
- Controller will then send the Channel Selection request to the agents starting from the last node of the network and corresponding to all the inner nodes, so that no disconnection occurs while switching channel.
- On CAC Completion Controller will also trigger network optimization to bring back the connection from 2.4G to 5G

7.4.3.2 Controller Radar Detection

- If Controller detects Radar, then it will select a new random channel for 5G Hz from the valid channels in that band which can be DFS or non-DFS. If new channel is non-DFS then Controller will make BH on that new channel.
- If this new channel is DFS and Dedicated radio is supported on controller then, controller will do CAC on dedicated radio, making connection for further agents on default channel.
- If this new channel is DFS and Controller does not support Dedicated radio then, controller have to do CAC on main radio and will make BH on 2.4G.
- Controller triggers MAP R2 channel planning after 10 sec to find an operable channel from channel preference report.
- If that Channel is non-DFS then Controller will switch channel on agents through Channel selection request and bring back to 5GH after triggering Network Optimization.

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- If that Channel is DFS then Controller will decide which device will do CAC and by what CAC method. If any agent is selected then it will send CAC Request to that agent. If controller supports Dedicated radio then it will do CAC.
- On Successful CAC Completion Agent will respond CAC Completion report to controller.
- Controller will then send the Channel switch request to the agents starting from the last node of the network and corresponding to all the inner nodes, so that no disconnection occurs while switching channel.

7.4.4 DFS CAC in case of Bandwidth 160MHz

For platforms like AX6000 which support BW 160MHz for 5G interface, MAP also supports BW 160. Due to this there are two channel groups possible when BW is kept as 160. For example see picture of channel groups based on BW as below (source: internet):

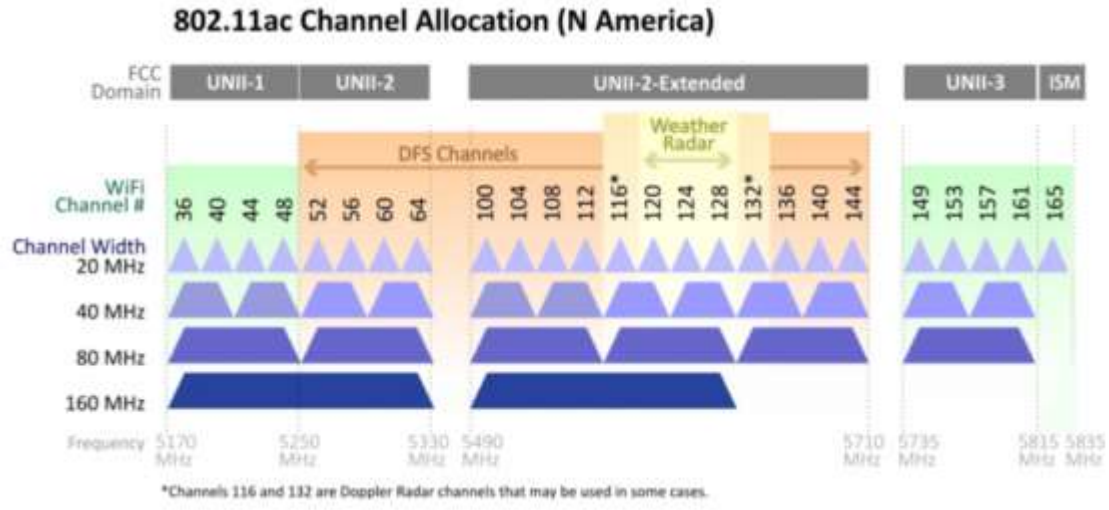


Figure 7- 40. Channel groups based on BW

Even if one channel in the entire channel group is DFS, then CAC needs to be done for the entire group always. So even if we want to switch to channel 36, still CAC will have to be done for the group 36 to 64(complete 160MHz) .

At bootup:
A device may bootup at a DFS or non-DFS channel as per setting inside the .dat file. But the BW setting in .dat file is important to note. If the channel group formed by that BW includes even one DFS channel , then the device will first do DFS CAC for that entire channel group. So you may experience some delay in connection allowed by the 5G Band because of this reason.
For example:

- If device bootups at channel 36 and 160MHz BW , and the first connection is made at 2.4G Backhaul, this may be due to case that 5G radio of controller is doing CAC at that time (even if primary channel is non-DFS). After network optimization run when CAC completes BH may shift on 5G BH based on network optimization decision. This is the expected behavior.

User preferred channel setting:
A user may set the primary channel of the device forcefully by mapd_cli commands or web UI. At this time the user can choose any channel (DFS or non-DFS). Attempt will be made to make device attain maximum bandwidth possible as per settings(channel preference). Hence upgrade or downgrade of Bandwidth is possible at this time. So DFS CAC will be done if needed in this case.
For example:

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- If initially device was operating on channel 149 (BW 80, since BW 160 is not possible on this channel) , then if channel is switched to channel 36 , and max possible BW of device is 160MHz, upgrade happens , then firstly CAC will be done on channel 36 to 64, then only channel switch will happen.

Radar case:
A radar signal can be detected at any DFS channel of the current operating Channel group. In this case all the DFS channels of that group will be marked as non-operable (NOP). When this happens, channel planning will make device shift to a new channel. New operating channel /channel group will be decided on the basis of channel groups rank as already explained in previous sections.
For example:

- If current primary channel is 64 say, and radar is detected on this, then all DFS channels in this group will be marked as NOP(52 to 64 will be marked as NOP), 36 to 48 can still be operable (in 20/40/80MHz) and 100 to 128 will be still operable (in 160MHz if weather channels are supported). If both groups have radar, then it will pick other non-DFS channels depending on group ranks, depending on country code and region (in lower BW). After NOP duration expires (after 30 mins), the entire group will become available again, but CAC will have to be done again.

7.4.5 Feature Management Interfaces

7.4.5.1 Configuration File Interface

Dedicated radio support can be enabled or disabled using .dat file as below:

To Enable

- **nvram_set rtdev DfsDedicatedZeroWait 1**
- **nvram_set rtdev DfsZeroWaitDefault 1**
-

To Disable

- **nvram_set rtdev DfsDedicatedZeroWait 0**
- **nvram_set rtdev DfsZeroWaitDefault 0**

For UCI supported SKUs following command will be use for DFS:

To Enable

- wireless.MT7990_1_2.dfs_dedicated_zero_wait='1'
- wireless.MT7990_1_2.dfs_zero_wait_default='1'

commit wireless

To Disable

- wireless.MT7990_1_2.dfs_dedicated_zero_wait='0'
- wireless.MT7990_1_2.dfs_zero_wait_default='0'
- uci commit wirelss

After setting above value reboot the board.

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8 Traffic Separation and Service Prioritization

8.1 Traffic Separation

8.1.1 Introduction

Traffic separation is mainly used to divide the whole MAP network into several sub networks by VLAN technique. The sub networks with different VLAN id cannot communicate with each other. However all the sub networks can access to the internet. There is one primary VLAN network and zero or more secondary VLAN networks. Controller would announce the traffic separation policy to all the agents by auto-configuration or renew procedure to partition VLAN networks. The policy contains a series of the SSID names and the corresponding VLAN ids together with one primary VLAN id. Agents will assign the VLAN id to all the fronthaul BSSs based on the relationship between SSID names and VLAN ids by policy. Ethernet LAN network belongs to primary network, which means local Ethernet network can only communicate with WIFI primary network.

Here is a picture to help you understand more of the traffic separation feature.

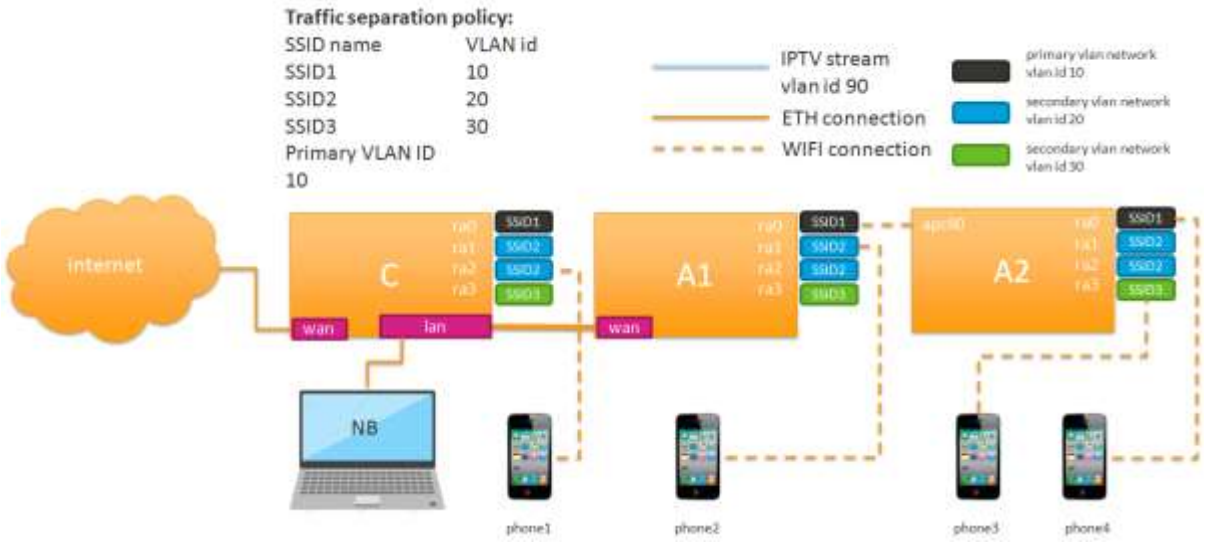


Figure 8- 1. Traffic Separation Image 1

Like the above network layout showing by picture, every MAP device starts up 4 WIFI interfaces and one of them belongs to the primary network (SSID1), two of them belong to the secondary network (SSID2) and one of them belong to the other secondary network (SSID3). The traffic flows are expected as below:

a. NB and phone4 belongs to the primary network. They can communicate with each other but cannot communicate with phone1 ~phone3.
b. Phone1 and phone2 belongs to the secondary network (vlan 20). They can communicate with each other but cannot communicate with phone3, phone4 and NB.
c. Phone3 belongs to the other secondary network (vlan 30). It cannot communicate with phone1, phone2, phone4 and NB.
d. All the STA devices, including phone1 ~phone4 and NB can access to internet.

8.1.2 Operating Mode

There are two types of operating mode, bridge mode and router mode. On bridge mode, WAN interface (eth1) is bonded to bridge interface (br-lan). All the packets from WAN interface will be transferred directly to other

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interface by bridge. In most of the case, this mode is set to agent. However in the case of controller connecting to GPON, this mode will also be set to controller. On router mode, WAN interface (eth1) isn't bonded to bridge interface (br-lan). All the packets from WAN interface need go through IP layer. This mode can only be set to controller. The packets with transparency vlan id will be dropped by protocol when they are coming to IP layer.

8.1.3 IPTV Service

Some of operators will provide IPTV (television based on IP service) service which also utilizes VLAN technique. The operator will assign one unique VLAN id to IPTV streaming and the streaming will flow from WAN network to IPTV connecting to MAP network. Normally the topology is like below:

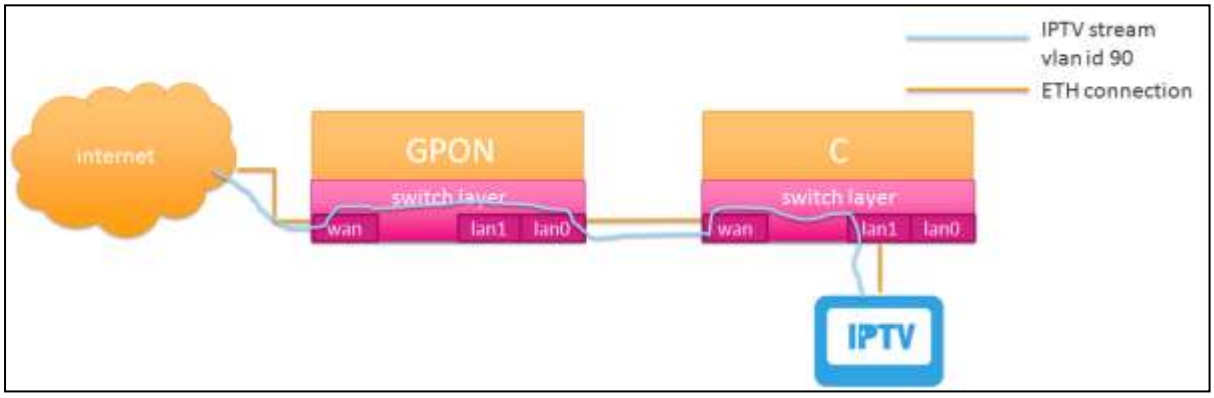


Figure 8- 2. Traffic Separation Image 2

The operator will set bunches of switch command to let IPTV stream transferred directly from controller WAN port to IPTV port lan1 to improve efficiency.

In order to let traffic separation feature work with IPTV feature, we provide two solutions. Customer can take any of them based on own network deployment.

1) All the backhaul connection among MAP networks and IPTV connection are Ethernet. In this scenario, IPTV stream can directly transferred through switch. As the IPTV stream will not go up to kernel, it will get high performance experience.

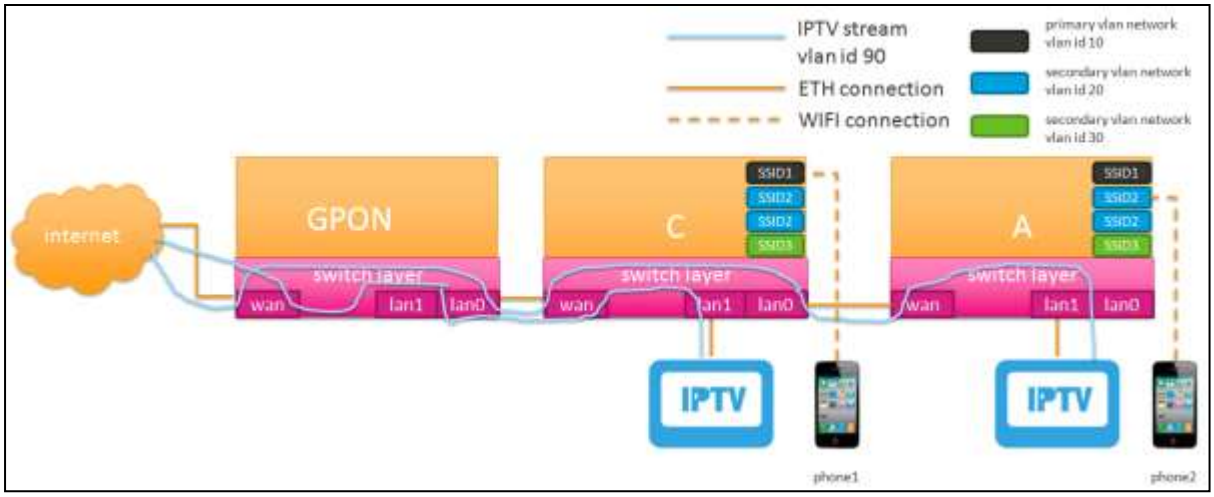


Figure 8- 3. Traffic Separation Image 3

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Some of additional switch commands needed be set on both controller and agent. We write a script to simplify the process. On the controller, “tp_ts_switch.sh 90 20,30 1-1 0” needed be executed. On the agent, “tp_ts_switch.sh 90 20,30 1-1 x” needed be executed. The explanation of the parameters is list as below:

a. The first parameter represents the IPTV VLAN id.
b. The second parameter represents the all the secondary traffic separation VLAN id.
c. The third parameter represents the switch port used to connect IPTV and if need peel the IPTV VLAN header. 1-1 means port1 used to connect IPTV and all the packets go out this port need maintain IPTV VLAN header. 1-0 means port1 used to connect IPTV and all the packets go out this port need remove IPTV VLAN header.
d. The fourth parameter represents the switch port used to connect to downstream MAP device and the downstream MAP device will provide IPTV service. x means no downstream MAP device or no IPTV service on downstream MAP device.

2) There is WIFI backhaul connection or WIFI IPTV connection.

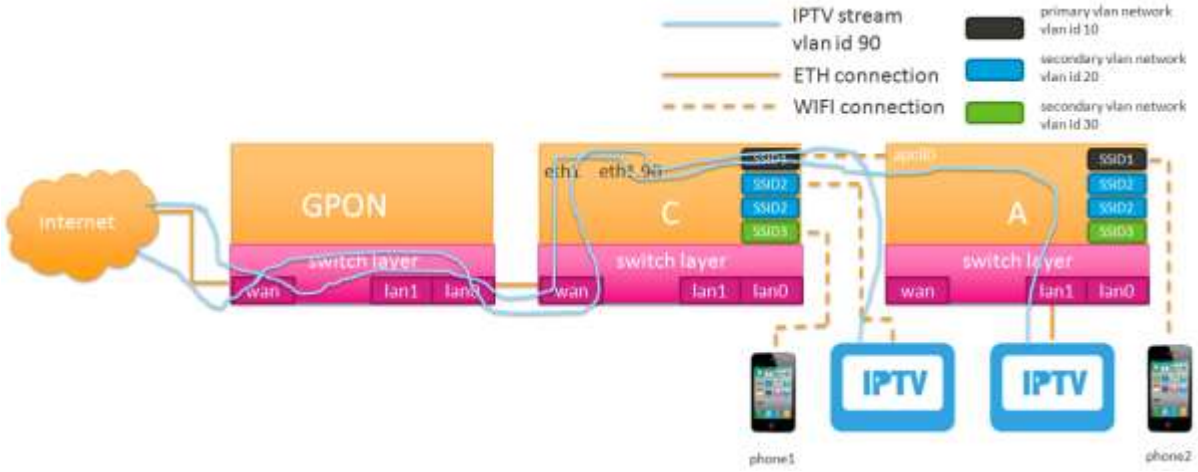


Figure 8- 4. Traffic Separation Image 4

In this scenario, IPTV stream will go up to kernel by WAN interface. It need create a VLAN interface based on WAN interface with the same IPTV VLAN id to receive the VLAN stream if the controller works on router mode. The commands are list as below:

```
vconfig add eth1 90
ifconfig eth1.90 hw ether xx:xx:xx:xx:xx:xx
ifconfig eth1.90 up
brctl addif br-lan eth1.90
```

Besides it need set “transparent_vids=90;” in 1905d.cfg on the controller and rerun EasyMesh_openwrt.sh to announce new IPTV policy. Only need set “transparent_vids=90;” in 1905d.cfg on the controller and rerun EasyMesh_openwrt.sh if controller works on bridge mode. Currently this method only supports IPTV with VLAN function, which means all the traffic in and out IPTV will carry VLAN header. This method is less efficient than method one but more extensible and simpler than previous one.

8.1.4 Topology and command combination

Below command is a reference for customer to correctly set up the traffic separation and IPTV service. Customer can check the script to get more details.

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1) Scenario 1. All the backhaul connection and IPTV connection are Ethernet

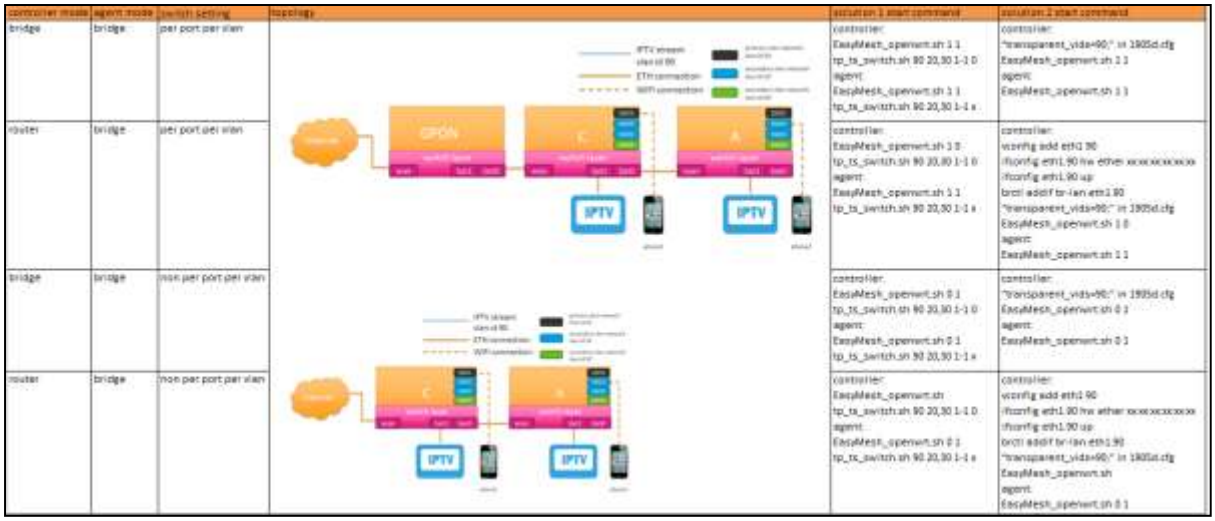


Figure 8- 5. Traffic Separation Image 5

2) Scenario 2. There is WIFI backhaul connection or WIFI IPTV connection.

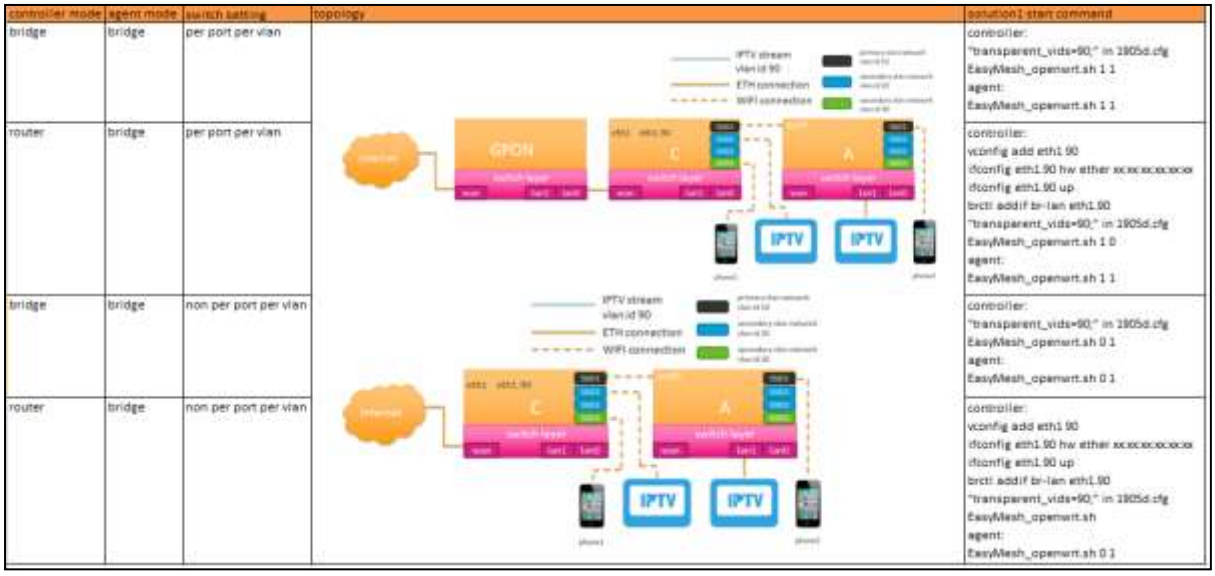


Figure 8- 6. Traffic Separation Image 6

8.1.5 Limitations

Traffic separation does not support the following two topology.

1) Traffic separation does not support the topology that all the MAP devices connect directly to GPON. Packets with VLAN header cannot be removed on agents when sending out by WAN port. It cannot get service from internet.

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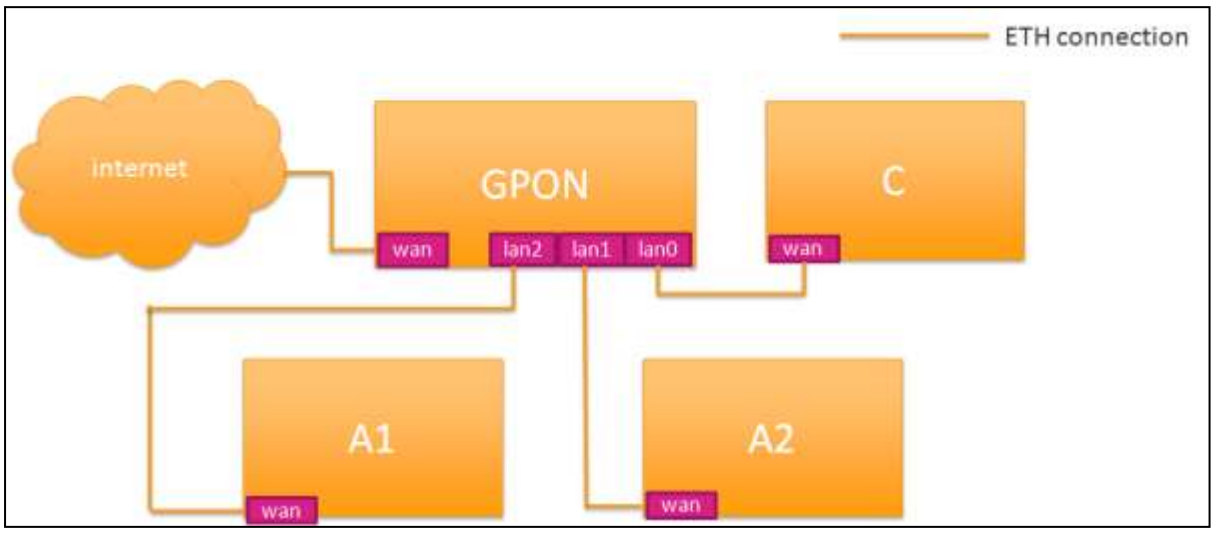


Figure 8- 7. Traffic Separation Limitation Image 1

2) Traffic separation does not support the topology that both the IPTV witch support VLAN function and the IPTV which do not support VLAN function connect to the same BSS.

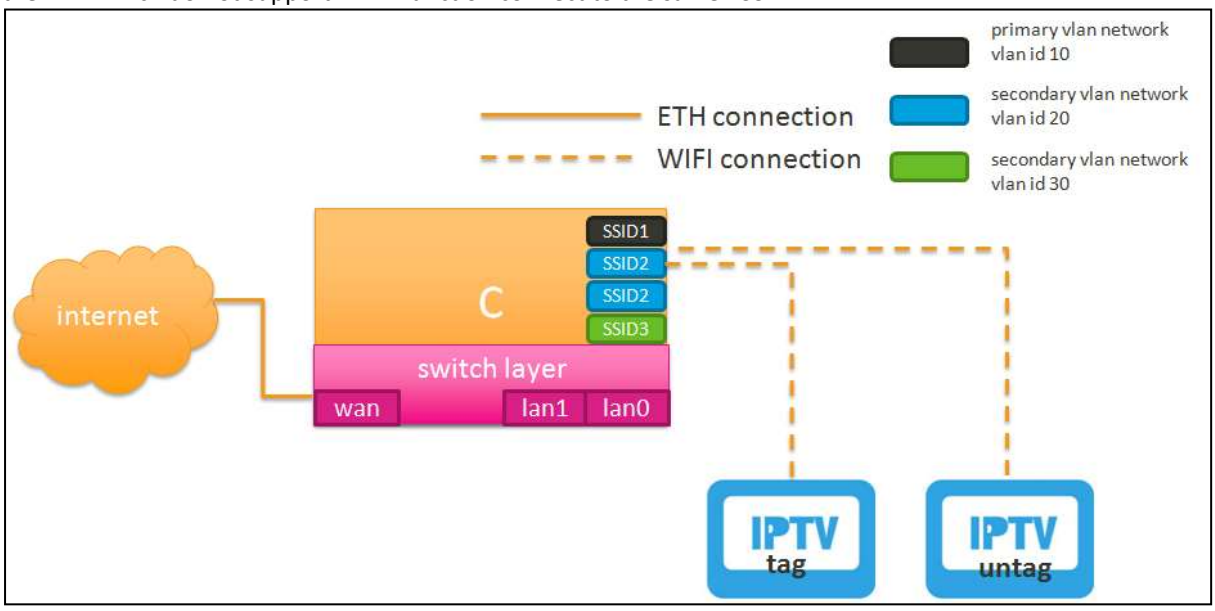


Figure 8- 8. Traffic Separation Limitation Image 2

3) Traffic separation does not support the topology that controller working as router mode connecting to the PON without MAP service. In this scenario, it will have DHCP issue.

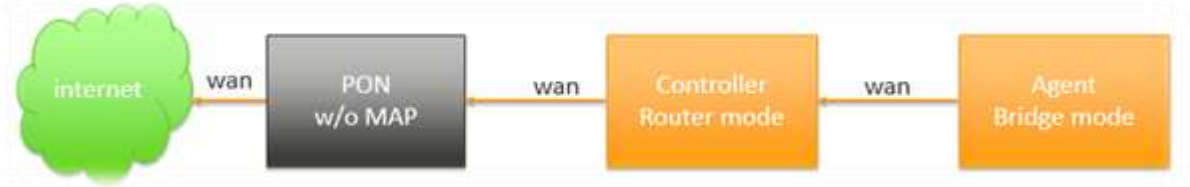


Figure 8- 9. Traffic Separation Limitation Image 3

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8.1.6 Feature Management Interfaces

8.1.6.1 Configuration Parameters

The last three parameters in wts_bss_info_config file are related to traffic separation feature.

```
1,FF:FF:FF:FF:FF:FF 8x AX3200-24G-1 0x0020 0x0008 maprocks1 1 1 hidden-N 10 N/A N/A
2,FF:FF:FF:FF:FF:FF 8x AX3200-24G-2 0x0020 0x0008 maprocks2 0 1 hidden-N 20 N/A N/A
3,FF:FF:FF:FF:FF:FF 8x Multi-AP-24G-3 0x0020 0x0008 maprocks3 0 1 hidden-N 4095 N/A N/A
4,FF:FF:FF:FF:FF:FF 8x Multi-AP-24G-4 0x0020 0x0008 maprocks4 0 1 hidden-N 4095 N/A N/A
5,FF:FF:FF:FF:FF:FF 12x Multi-AP-5HG-1 0x0020 0x0008 maprocks1 1 1 hidden-N 4095 N/A N/A
6,FF:FF:FF:FF:FF:FF 12x Multi-AP-5HG-2 0x0020 0x0008 maprocks2 0 1 hidden-N 4095 N/A N/A
7,FF:FF:FF:FF:FF:FF 12x Multi-AP-5HG-3 0x0020 0x0008 maprocks3 0 1 hidden-N 4095 N/A N/A
8,FF:FF:FF:FF:FF:FF 12x Multi-AP-5HG-4 0x0020 0x0008 maprocks4 0 1 hidden-N 4095 N/A N/A
9,FF:FF:FF:FF:FF:FF 11x AX3200-5LG-1 0x0020 0x0008 maprocks1 1 1 hidden-N 10 pvid 5
10,FF:FF:FF:FF:FF:FF 11x AX3200-5LG-2 0x0020 0x0008 maprocks2 0 1 hidden-N 20 N/A N/A
11,FF:FF:FF:FF:FF:FF 11x Multi-AP-5LG-3 0x0020 0x0008 maprocks3 0 1 hidden-N 4095 N/A N/A
12,FF:FF:FF:FF:FF:FF 11x Multi-AP-5LG-4 0x0020 0x0008 maprocks4 0 1 hidden-N 4095 N/A N/A
```

Figure 8- 10. Traffic Separation feature configuration via wts_bss_file example

User sets the primary vlan id and secondary vlan id on UI. UI will write the value in wts_bss_info_config file. The third parameter from the bottom is VLAN id. The second parameter from the bottom stands for primary VLAN id. “pvid” means the VLAN id in this line is primary VLAN id.
The last parameter means PCP, only valid for primary VLAN id. PCP is not used for traffic separation.
4095 is invalid VLAN id. It means user doesn’t set the vlan id to the SSID explicitly, so it will be taken as primary VLAN id implicitly. Then user clicks save-and-apply button to trigger renew procedure to update traffic separation policy to the whole MAP network.

If user need use IPTV service and its topology is shown like scenario two, it need set IPTV VLAN id in 1905d.cfg.

```
# Controller's ALID
map_controller_alid=00:0c:43:48:f6:51

# Has MAP agent on this device
map_agent=0

# This device is a MAP root
map_root=1

# Agent's ALID
map_agent_alid=00:0c:43:48:f6:51

# Default Backhault Type
bh_type=eth

#Config Band setting of each Radio
radio_Band=24G;5G;5G

#bridge interface
br_inf=br-lan

#lan interface
transparent_vids=90;
map_ver=2

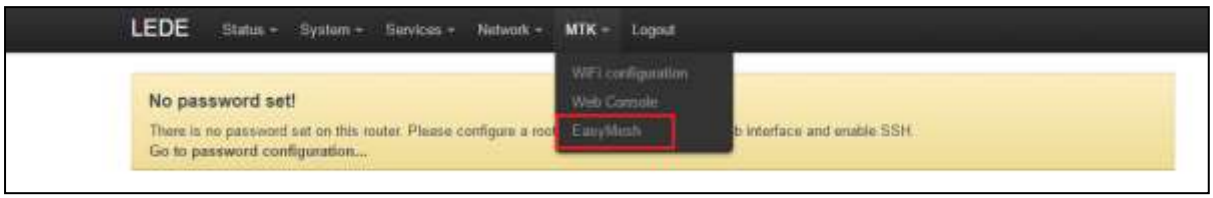
bss_config_priority=ra0;ra1;apcli0;rai0;rai1;apcli1

#ethernet device name used to read the switch table
#do not set if not understanding
#ethernet_dev_name=
lan=eth0
```

Figure 8- 11. Traffic Separation parameter in 1905d.cfg snapshot

8.1.6.2 WEB UI Interfaces

1) Click MTK→EasyMesh to enter in Easymesh configuration page



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Figure 8- 12. Traffic Separation WEB UI Image 1

2) Click Advanced→Configure BSS to enter in BSS configuration page

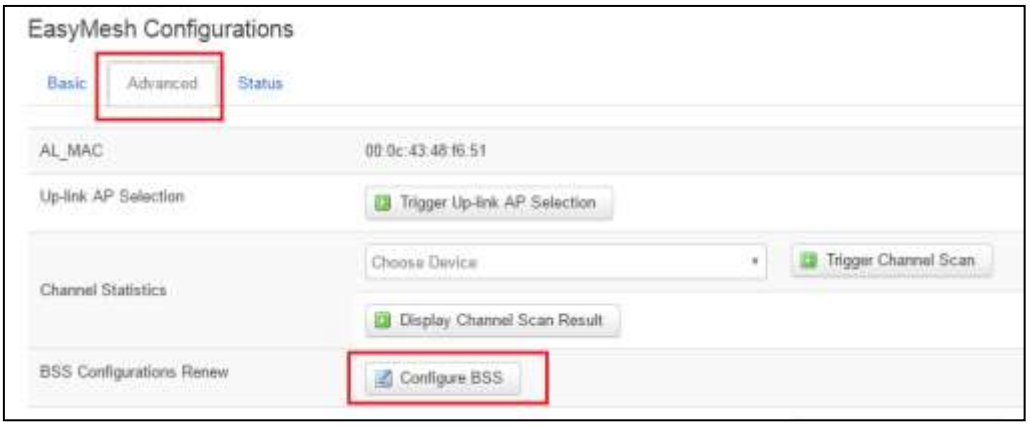


Figure 8- 13. Traffic Separation WEB UI Image 2

3) Click Edit BSS to enter in BSS information configuration page

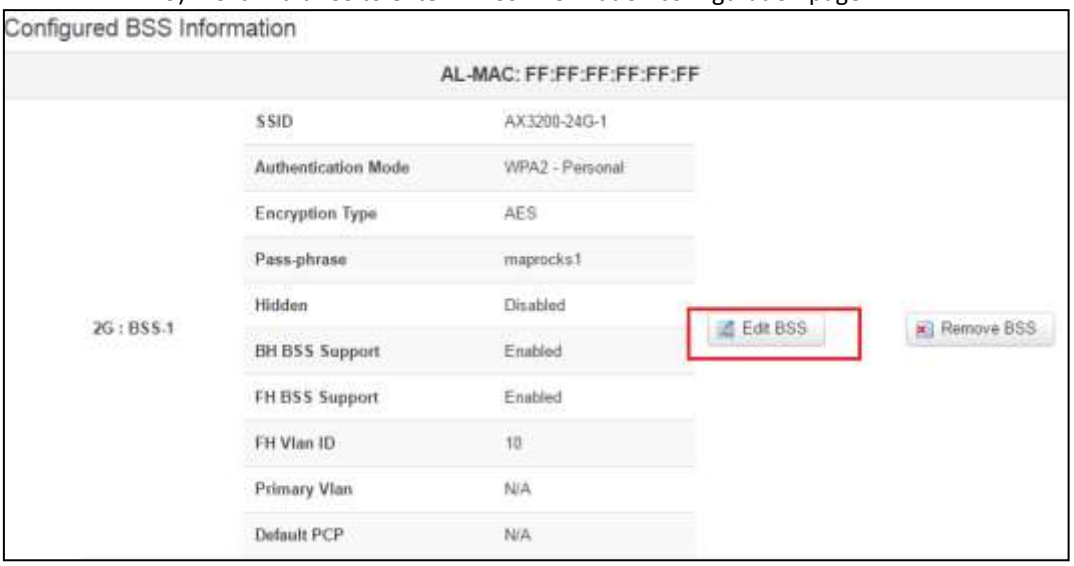


Figure 8- 14. Traffic Separation WEB UI Image 3

4) Fill in “FH VLAN ID”. If this VLAN id is primary VLAN id, you need tick off the “Primary Vlan” and fill in “Default PCP”. Keep in mind only one Primary VLAN id in the whole MAP network. If you choose another VLAN id as Primary VLAN id, the previous one you have chosen will be cancelled. Click “Save” button to save the setting and move to another BSS to configure VLAN. If you finish the entire configuration, click “Save and Apply” button to trigger renew procedure to update traffic separation policy to the whole MAP network.

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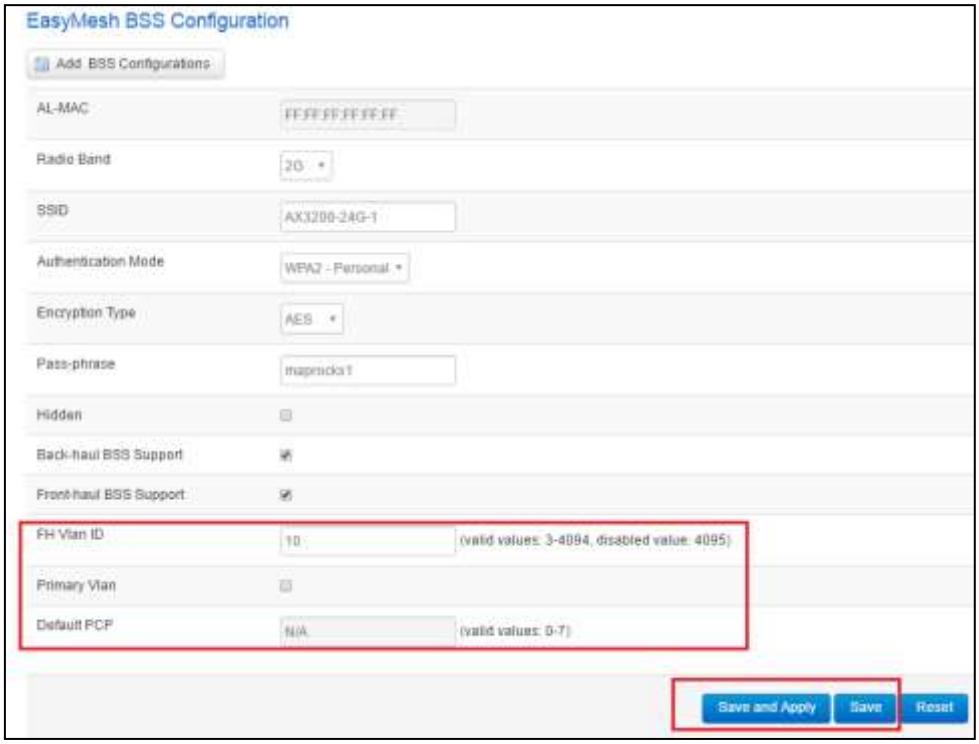


Figure 8- 15. Traffic Separation WEB UI Image 4

8.2 Service prioritization

8.2.1 Introduction

MAP R3 provides the service prioritization (SP) for data stream in MAP network. MAP dev will check the ingress packet, apply the priority of ingress packet according to the special rule by setting the vlan pcp (Priority Code Point) and IP dscp (Different Service Code Point), this will speed up the packet transmission through the MAP network.

8.2.2 Rule category

8.2.2.1 Static rule

User can configurate the transmission priority of one special WIFI station, or SSID, or protocol, or IP address related data stream. MAP network can speed up the related upstream and downstream packet which matching the special setting rule.
Take station 3 for example to introduce the data flow:

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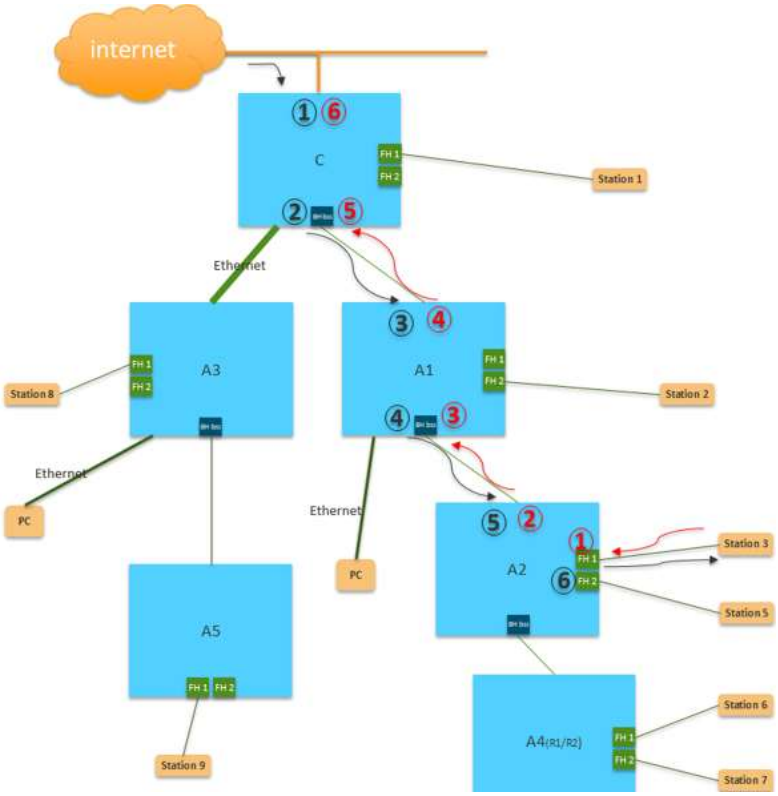


Figure 8- 16. Service Prioritization Data Flow with Static Rule

For upstream:

a. When A2 receive the upstream data from station 3, at point (1), A2 will get the priority according to static rule, then apply to the vlan pcp value and IP dscp value.

b. At point (2), A2 will speed up the data transmission about station 3 to A1 according to the vlan pcp value setting.

c. At point (3), A1 keeps the vlan header setting when receiving the upstream data, t point (4), A1 will speed up the data transmission about station 3 to C according to the vlan pcp value setting.

d. At point (5), C will remove the vlan header of packet, At point (6), station 3 upstream data leave the MAP network.

e. So at point (2)→(3), (4)→(5), upstream data can be speed up to transmission according to the sp rule setting.

For downstream:

a. When downstream data from internet to MAP network, at point (1), controller will get the priority according to static rule, then apply to the vlan pcp and ip dscp.

b. At point (2), C will speed up the data transmission about station 3 to A1 according to the vlan pcp value setting。

c. At point (3), A1 keeps the vlan header setting when receiving the downstream data. At point (4), A1 will speed up the data transmission about station 3 to C according to the vlan pcp value setting.

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d. At point ⑤ C will remove the vlan header of packet. At point ⑥, downstream data leave the MAP network to station 3.

e. So at point ②→③、④→⑤ downstream data can be speed up to transmission according to the sp rule.

8.2.2.2 Dynamic rule

When enable the dynamic rule feature, MAP dev can study the upstream data priority of the wifi sta withing the QOS capability, set the priority to vlan pcp and IP dscp. Then add the tcp quadruple of the sta pkt and priority to dynamic rule, when MAP network receives the internet data to the special wifi sta, the vlan pcp can also modified by the dynamic rule. This also can speed up the related upstream and downstream packet.
Take station 3 for example to introduce the data flow:

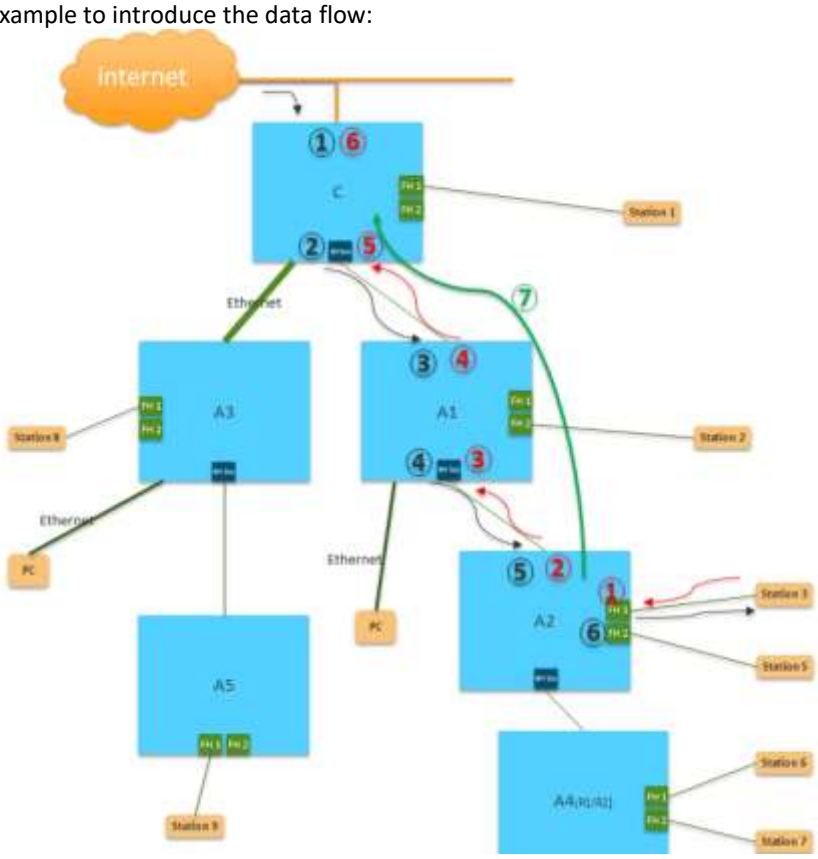


Figure 8- 17. Service Prioritization Data Flow with Dynamic Rule

For upstream:

a. When A2 receive the upstream data from station 3, at point ①, A2 will study the upstream data priority of the wifi sta, apply to the vlan pcp value and IP dscp value. Make the dynamic rule according to the tcp quadruple of the sta pkt and priority, also notify controller about the new learning dynamic rule.

b. At point ②, A2 will speed up the data transmission about station 3 to A1 according to the vlan pcp value setting.

c. At point ③, A1 keeps the vlan header setting when receiving the upstream data, t point ④, A1 will speed up the data transmission about station 3 to C according to the vlan pcp value setting.

d. At point ⑤, C removes the vlan header of packet. At point ⑥, station 3 upstream data leave the MAP network.

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e. So at point ② → ③, ④ → ⑤, upstream data can be speed up to transmission according to the sp rule setting.

For downstream:

a. When downstream data from internet to MAP network, at point ①, controller will get the priority according to new dynamic rule, then apply to the vlan pcp and ip dscp.

b. At point ②, C will speed up the data transmission about station 3 to A1 according to the vlan pcp value setting。

c. At point ③, A1 keeps the vlan header setting when receiving the downstream data, at point ④, A1 will speed up the data transmission about station 3 to C according to the vlan pcp value setting.

d. At point ⑤, C removes the vlan header of packet. At point ⑥, downstream data leave the MAP network to station 3.

e. So at point ② → ③, ④ → ⑤ downstream data can be speed up to transmission according to the sp rule.

8.2.2.3 Dscp to pcp mapping table

User can configuration the priority for special dscp to pcp mapping, when ingress packet comes to MAP network, MAP dev will parse the dscp value from IP header, get the special pcp value according the user mapping table by dscp value, then setting to the vlan pcp. This also can speed up the related packet.

8.2.3 SP configuration file

Sp configuration file is named as sp_rule.cfg under /etc/map folder.
Rule format in sp config file is as below:
Rule_index.Rule_name:[AL_MAC] +[SSID]+[Client's mac address]+[Protocol]+[destination IP address]+[source IP address]+[destination port]+[source port]=output value(0~7)

The elements in square brackets are described below:

Item	Value	Description
AL_MAC	00:01:02:03:04:05	MAP dev AL MAC
SSID	Multi_AP_5G_1	The SSID of one Front Haul BSS in MAP network
Client's mac address	00:0C:43:12:34:56	Wifi station mac
Protocol	TCP	Protocol type, such as TCP/UDP/ICMP…
destination IP address	8.8.8.8	Destination IP addr from uplink stream side
source IP address	192.168.1.200	Source IP addr from uplink stream side
destination port	8080	Destination port from uplink stream side
source port	123	Source port from uplink stream side

Example:

```
1.http_service:[00:01:02:03:04:05]+[]+[]+[]+[]+[]+[]+[]=7
2.FH1_service:[]+[FH1]+[]+[]+[]+[]+[]+[]=6
3.game_session:[]+[]+[]+[TCP]+[10.18.12.123]+[192.168.1.200]+[8080]+[8080]=7
DSCP_PCP_MAPPING=0.1.2.3.4.0.6.7.  0.1.2.3.4.0.6.7.  0.1.2.3.4.0.6.7.  0.1.2.3.4.0.6.7.  0.1.2.3.4.0.6.7.  0.1.2.3.4.0.6.7.
0.1.2.3.4.0.6.7. 0.1.2.3.4.0.6.7
```


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8.2.4 Feature Management Interfaces

8.2.4.1 WEB UI Interfaces

Click MTK→EasyMesh to enter in Easymesh configuration page on controller UI, then switch to MAP QOS page

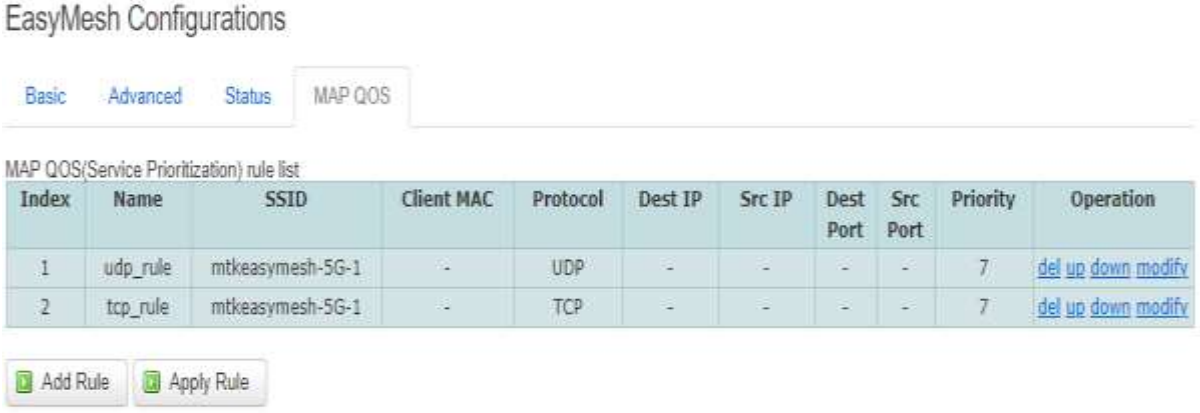


Figure 8- 18. Service Prioritization UI Setting

Add or modify the rule by any parameters list above, then push button “Apply Rule”, SP rule sync will be triggered between Controller and Agent as a precondition.

8.2.4.2 8.10.4.2 Command Line Interfaces

mapd_cli provides several commands to set and modfiy the sp rule, details as below:

Table 8- 2. Service Prioritization Setting Cmd

Item	CMD
1	Mapd_cli /tmp/mapd_ctrl sp_add_rule 10.game_session:[]+[]+[07:06:FE:Ac:34:ac]+[TCP]+[]+[192.168.1.200]+[]+[8080]=7
2	Mapd_cli /tmp/mapd_ctrl sp_set_rule 10.game_session:[]+[]+[]+[TCP]+[]+[192.168.1.200]+[]+[8080]=7
3	Mapd_cli /tmp/mapd_ctrl sp_move_rule_forward 10
4	Mapd_cli /tmp/mapd_ctrl sp_move_rule_backward 10
5	Mapd_cli /tmp/mapd_ctrl sp_rm_rule 10
6	Mapd_cli /tmp/mapd_ctrl sp_reorder_rule 3 1
7	Mapd_cli /tmp/mapd_ctrl sp_list_rule
8	Mapd_cli /tmp/mapd_ctrl sp_get_rule 1
9	Mapd_cli /tmp/mapd_ctrl sp_enable_dynamic_rule true true
10	Mapd_cli /tmp/mapd_ctrl sp_dscp_pcp_table 07.07.07.05.05.05……05.05
11	Mapd_cli /tmp/mapd_ctrl sp_rm_dscp_pcp_table
12	Mapd_cli /tmp/mapd_ctrl sp_config_done

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9 Stand-Alone Band Steering (Easy Mesh Disabled)

9.1 Introduction

Band Steering feature can work even if EasyMesh (MAP) is disabled.
This can be used in a:-

1. Single AP deployment
2. Legacy repeater network, where MAP is disabled on all the Access Points.

The algorithm works in exactly the same way as Agent Initiated Band Steering in a MAP network, with the exception that the AP devices present in the network do not have to wait for a Steering Opportunity as in a MAP network.
Every AP would perform Band Steering independently.

9.2 Components

Stand Alone Band Steering uses the same components as EasyMesh except that it doesn't require the 1905Daemon and full-fledged MAP daemon(mapd). Following components are required: -

- **bs20**: The Band Steering daemon, which is mapd minus the 1905 interface and with Multi-AP steering and MAP features disabled at run-time.
- **wapp**
- **bs_cli**
- **Driver**

9.3 Feature details

There are three primary types of steering that are supported: -

- Pre-association steering
- Idle post-association steering
- Active post-association steering.
  - Active Non-Overloaded Steering
  - Active Overloaded Steering

9.3.1 Pre-association steering

-
- Pre-association steering aims to connect dual band stations to a channel that is not overloaded. A channel is considered overloaded when its average medium exceeds pre-defined *CUOverloadThreshold*_XX (e.g. 70%) or the connected clients number exceed *MaxClientOverloaded*. While this condition holds, if the AP sees a dual band STA sending probe requests and the probe requests on the non-overloaded channel exceed a configurable threshold *MinRSSIOverloaded* (e.g. 20 dB), a blacklist is installed and auth requests are rejected on the overloaded channel to encourage the STA to associate on the non-overloaded channel.
- If *PHYBasedSelection* option is set, the AP will also install blacklist statically on 5G channels to those dual band non-HT STAs.
- If that STAs try to connect to blacklisted BSS(es) more than *AuthRejMax* times, the existing pre-association should be aborted and allow the connection.

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- **Load balancing in case of triband solution –** Pre-assoc overload steering is performed in case of triband only if at least two bands are not overloaded. If two or more bands are overloaded, pre-assoc overload steering will not be performed.

Table 9- 1. Pre-Assoc Steering action

Channel Conditions	Pre-Association Steering Action	
	PHYBasedSelection=ON	**PHYBasedSelection=OFF**
All channels Overloaded	Disallow legacy clients on 5GHz	Allow all clients on all channels
All channels Non-Overloaded	Disallow legacy clients on 5GHz	Allow all clients on all channels
Some channel(s) Overloaded and some are Non-Overloaded	Disallow legacy clients on 5GHz. Disallow all HT/VHT/HE clients on Overloaded channels with recent and good enough RSSI	Disallow clients on Overloaded channels with recent and good enough RSSI

9.3.2 Idle post-association steering

- An idle client is defined as a client for which the sum of Avg Tx Bytes/sec and Avg Rx Bytes/sec is less that *Activitythreshold*
- An idle client can be steered by either of the two methods:-
  - Forced Steering: Disconnect + Deauth
  - BTM Steering: IEEE80211v BSS Transition Management
- IDLE Clients with recent enough RSSI on 5G band, whose estimated non-serving RSSI(2.4Ghz) has fallen below *RSSISteeringEdge_DG* are forced steered/ BTM steered to 2.4G band if the 2.4G channel is non-overloaded. If the RSSI is quite old, the RSSI is refreshed by triggering a series of NULL frames to the IDLE client. If IDLE client is known to fail BTM steering, forced steering is attempted.
- IDLE Clients with recent enough RSSI on 2.4Ghz band, whose non-serving RSSI(RSSI on 5G) has increased above *RSSISteeringEdge_UG* are forced steered/ BTM steered to 5G band if the 5G channel is non-overloaded. If the RSSI is quite old, the RSSI is refreshed by triggering a series of NULL frames to the IDLE client. If IDLE client is known to fail BTM steering, forced steering is attempted.

Table 9- 2. Post-Assoc Steering action

Client condition	• 2.4G Channel Condition	• 5GL Channel Condition	• 5GH channel Condition	• PHYBasedSelection	• Steering Action
Client Connected on 2.4G with Estimated 5G RSSI > *RSSISteeringEdge_UG*	Don't care	NOL	NOL	ON	If the client is VHT steer it to 5GH otherwise steer to 5GL
	Don't' care	NOL	NOL	OFF	Steer to least utilized 5G Channel
	Don't care	NOL	OL	Don't care	Steer to 5GL
	Don't Care	OL	OL	Don't' care	Do not Steer
	Don't Care	OL	NOL	Don't' care	Steer to 5GH channel

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Client Connected on 2.4G with Estimated 5G RSSI < *RSSISteeringEdge_UG*	Don't' care	Don't' care	Don't' care	Don't' care	Do not Steer
Client Connected on 5GL/5GH, with Estimated 2.4G RSSI < *RSSISteeringEdge_DG*	OL	Don't' care	Don't' care	Don't' care	Do not steer
	NOL	Don't' care	Don't' care	Don't' care	Steer to 2.4 channel
Client Connected on 5GL with Estimated 2.4G RSSI > *RSSISteeringEdge_DG*	Don't' care	Don't' care	OL	Don't Care	Do not steer
	Don't' care	Don't' care	NOL	OFF	Do not steer
	Don't Care	Don't Care	NOL	ON	If the client is VHT, steer it to 5GH, otherwise do nothing
Client Connected on 5GH with Estimated 2.4G RSSI > *RSSISteeringEdge_DG*	Don't Care	OL	Don't Care	Don't Care	Do not steer
	Don't Care	NOL	Don't Care	OFF	Do not steer
	Don't Care	NOL	Don't Care	ON	If the client is Non-VHT, steer it to 5GL, otherwise do nothing

9.3.3 Active post-association steering.

- An ACTIVE client is defined as a client for which the sum of Avg Tx Bytes/sec and Avg Rx Bytes/sec is greater than *ActivityThreshold*.
- Only ACTIVE clients that are both 802.11K and 802.11V capable are eligible for steering
- Only BTM steering is used for ACTIVE clients to cause minimum service disruption.
- There are two types of steering that are defined for Active Clients:-
    - Active Non-Overloaded Steering
    - Active Overloaded Steering[Offloading]

9.3.3.1 Active Non-Overloaded Steering

- Active clients whose serving RSSI and DL MCS have increased above specific thresholds, are BTM steered to 5G band provided that the 5G channel is not overloaded and the steering of Active client doesn't make the 5G channel as overloaded. 11k helps to predict the impact of moving the client to the target channel. If the Active client is known to fail BTM steering, then no steering is attempted.
- Active clients whose serving RSSI and DL MCS have increased above specific thresholds, are BTM steered to 5G band provided that the 5G channel is not overloaded and the steering of Active client doesn't make the 5G channel as overloaded. 11k helps to predict the impact of moving the client to the target channel. If the Active client is known to fail BTM steering, then no steering is attempted

9.3.3.2 Active Overloaded Steering [Offloading]

Active clients connected to an Overloaded channel are BTM steered to a non-overloaded channel provided that the steering of Active client doesn't make the target channel as overloaded. 11k helps to predict the impact of moving the client to the target channel. If the Active client is known to fail BTM steering, then no steering is attempted

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9.4 Limitations

- No Blacklisting coordination between the APs, if the RootAP and Repeaters share the same SSID.
- As per design, Pre-association steering cannot happen till it it's deduced that a client is Dual-Band. Therefore, if the STA is using randomized address to scan, dual band capability of the client can only be deduced after it connects to the AP.
- Pre-association Steering based on RSSI and channel load conditions (Non PHYbasedSelection pre-steering) cannot happen, if the STA doesn't send a Probe Request, before Sending 802.11 Authentication request. Probe request is required for the STA's recent RSSI.
- Pre-association steering can sometimes fail with stations that use random mac address to scan, and the time gap between their join probe-request and authentication request is too short for our steering design to take a decision for blacklisting and rejecting authentication. One such case has been observed with Huawei mate20 STA in which the Auth request is sent within 4 ms of the Probe request. In such a scenario, some-times the STA is rejected Association, and it never re-tries association.
- 802.11r based fast transition not supported across bands. If FT is enabled on the DBDC AP, FT capable station, when band steered, would fall back on non-FT Association.

9.5 Feature Management Interfaces

9.5.1 Enable Bandsteering 2.0

- Configuration File Interface:

1) Set MapMode=2 in the driver dat file.
2) Set DeviceRole=1 in /etc/map/mapd_user.cfg
3) Set CentralizedSteering=0 in /etc/map/mapd_user.cfg

9.5.2 Configuration Parameters

The steering configuration parameters can be over-ridden by specifying them in /etc/mapd_strng.conf. The parameters and their default values are as below:-

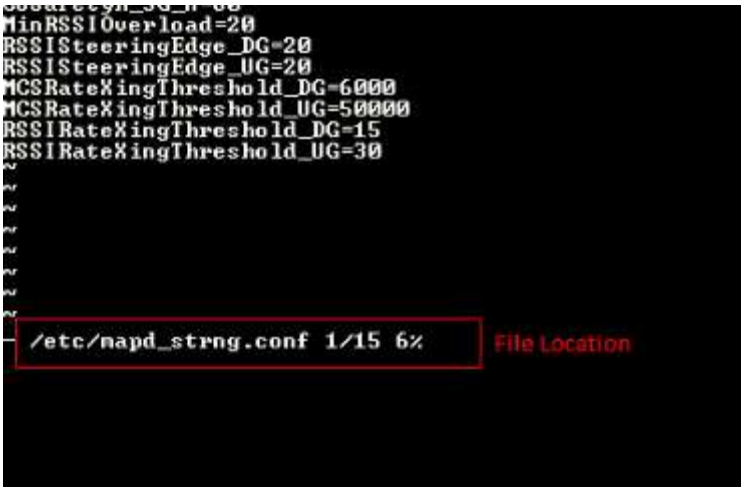


Figure 9- 1. Mapd_strng.conf file example

For UCI supported SKUs following value will be updated in /etc/config/mapd files as shown below

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```
config mapd_strng 'mapd_strng'
        option CUOverloadTh_5G_H '80'
        option CUOverloadTh_6G '70'
        option MetricPolicyChUtilThres_24G '70'
        option MetricPolicyChUtilThres_5GL '80'
        option MetricPolicyChUtilThres_5GH '80'
        option MetricPolicyChUtilThres_6G '80'
        option ChPlanningChUtilThresh_24G '70'
        option ChPlanningChUtilThresh_5GL '80'
        option ChPlanningChUtilThresh_6G '80'
        option ChPlanningEDCCAThresh_24G '200'
        option ChPlanningEDCCAThresh_5GL '200'
        option ChPlanningEDCCAThresh_6G '200'
        option ChPlanningOBSSThresh_24G '200'
        option ChPlanningOBSSThresh_5GL '200'
        option ChPlanningOBSSThresh_6G '200'
        option ChPlanningR2MonitorTimeoutSecs '300'
        option ChPlanningR2MonitorProhibitSecs '900'
        option ChPlanningR2MetricReportingInterval '10'
        option ChPlanningR2MinScoreMargin '10'
        option MetricRepIntv '10'
        option MetricPolicyRcpi_24G '100'
        option MetricPolicyRcpi_5GL '100'
        option MetricPolicyRcpi_5GH '100'
        option MetricPolicyRcpi_6G '100'
        option LowRSSIAPSteerEdge_RE '40'
        option CUOverloadTh_2G '70'
        option CUOverloadTh_5G_L '80'
```

Table 9- 3. Band Steering Configuration parameters

Parameter	Description	Default Value
ActivityThreshold	The Avg sum of Tx Bytes and Rx Bytes per second above which the client is considered as ACTIVE.	3000 Bps
RSSISteeringEdge_UG	The point at which the measured or estimated RSSI on 5 GHz dictates a node associated on 2.4G should be steered to 5 GHz	20db
RSSISteeringEdge_DG	The point at which the measured or estimated RSSI on 2.4 GHz dictates a node associated on 5 GHz should be steered to 2.4 GHz	20db
MinRSSIOverloaded	Uplink RSSI (in dB) above which pre-association steering and post-association offloading is allowed.	20db
PHYBasedSelection	In pre-association, put non HT client to 2.4G channel. In post-association, to prioritize VHT capable clients to high power 5G channel, and put non-VHT clients to other channels.	0
MaxClientOverloaded	The maximum connected client number to indicate potential overloaded.	100
ForcedRssiUpdate	Force RSSI Update enable/disable	0(disabled)

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RSSIAgeLim	Number of seconds to check if last record RSSI is fresh enough.	5s
RSSIMeasureSamples	Number of NULL frames to be sent for client uplink RSSI measurement.[Only applicable when [ForcedRssiUpdate=1]	5
MCSCrossingThreshold_UG	The rate (in Kbps) above which an active client is a potential candidate for upgrade to 5 GHz.	50000
RSSICrossingThreshold_UG	The value (in dB) the uplink RSSI must be above to be considered for active steering to 5 GHz. This threshold is in conjunction with the MCSCrossingThreshold_UG.	30db
MCSCrossingThreshold_DG	The rate (in Kbps) at which a rate crossing event should be generated for a potential active client downgrade to 2.4 GHz. The default basically disable downgrade steering.	6000
RSSICrossingThreshold_DG	The value (in dB) the uplink RSSI must be below to be considered for active steering to 2.4 GHz. This threshold is an additional trigger (an OR condition) for downgrade steering.	15db
CUOverloadThreshold_2G	Medium utilization threshold (in percentage %) for an overload condition on 2.4 GHz	80
CUOverloadThreshold_5G_L	Medium utilization threshold (in percentage %) for an overload condition on the first 5G radio	70
CUOverloadThreshold_5G_H	Medium utilization threshold (in percentage %) for an overload condition on the second 5G radio	70

9.5.3 Command Line Interface

1. How to check Channel Utilization on each Radio and BSS info.

mapd_cli /tmp/mapd_ctrl mib

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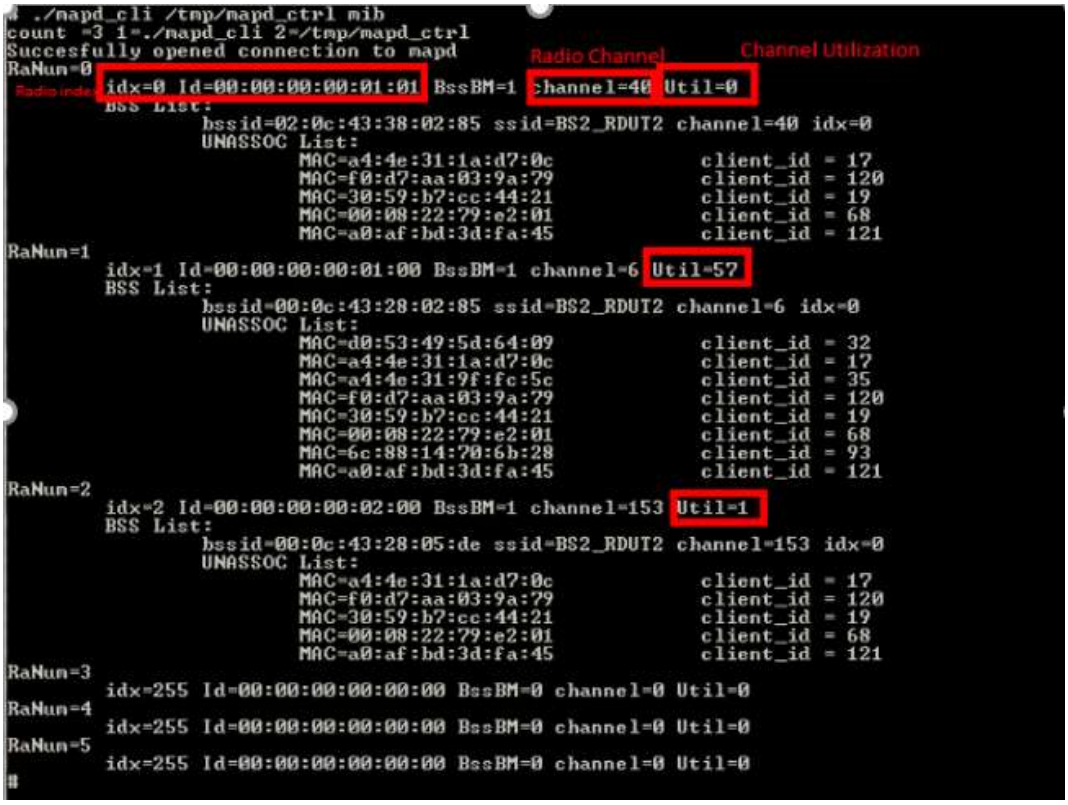


Figure 9- 2. Mapd_cli mib command console logs

RaNum	Radio number
Idx	Radio Index
Id	Radio Identifier
BssBM	BSS bitmap
channel	Channel Number
Util	Channel utilization
Bss List	List of all MBSS for this radio

2. How to check the clients that are monitored by the mapd for steering

*mapd_cli/tmp/mapd_ctrl mib sta_see*n_list

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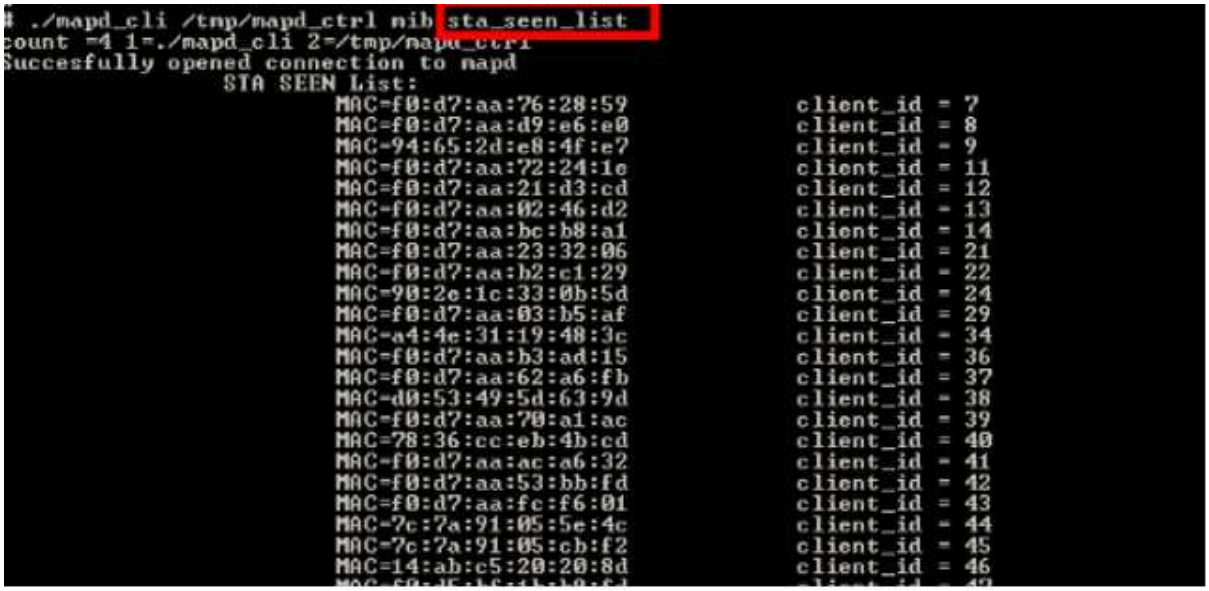


Figure 9- 3. Mapd_cli mib sta_seen_list command console logs



Figure 9- 4. Mapd_cli mib sta <id> command console logs

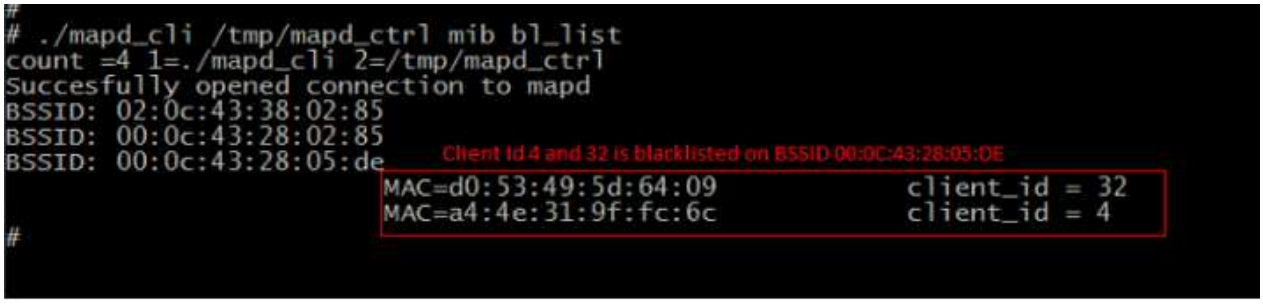


Figure 9- 5. Mapd_cli mib bl_list command console logs

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9.6 Examples

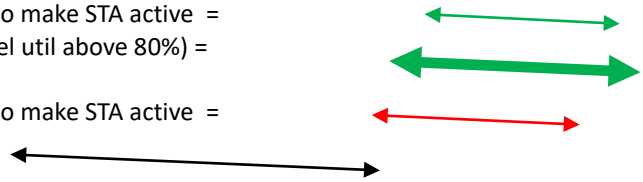
1: agent_post_assoc_ol_active_dg:

Steps:
a. Configure MAP agent and controller in HE mode wpa2_psk_aes and keep centralized steering enabled, phy based steering and nol_multiap steering disabled on both.
b. Onboard the agent with the controller.
c. Connect STA1(which supports 5G only) and 11KV dual band STA2 both to 5G interface of the agent.
d. Run traffic with STA2, just to make it active and make 5G OL with the help of heavy traffic with STA1 and keep 2G NOL.

Expected result:
Check the dual band STA2 get steered to 2G interface of the agent.

Legends used in below diagrams:
5G connection with just nominal traffic to make STA active =
5G connection with heavy traffic (channel util above 80%) =

2G connection with just nominal traffic to make STA active =
MAP onboarding =



Initial setup for the test:

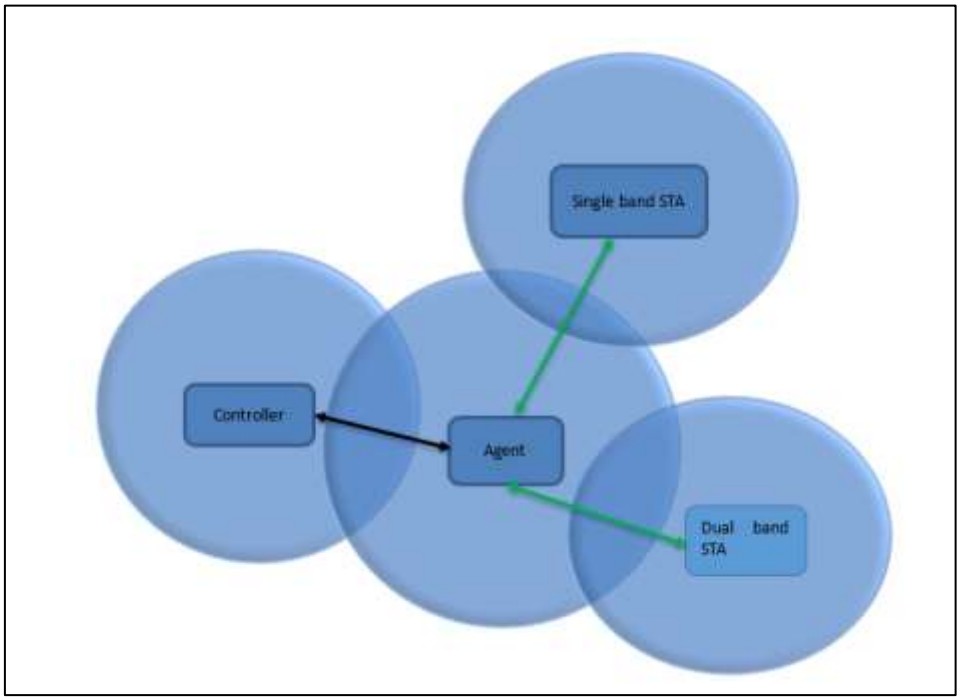


Figure 9- 6. Band Steering example Image 1

Just before band steering supposed to happen:

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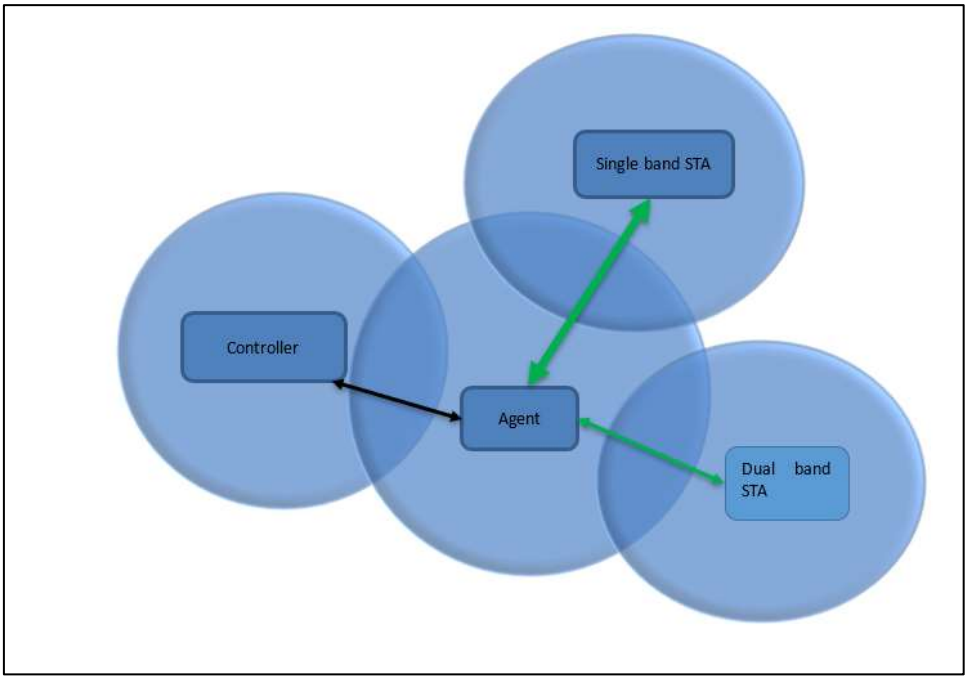


Figure 9- 7. Band Steering example Image 2

After Band steering is performed (the STA got steered to 2G interface of the Agent):

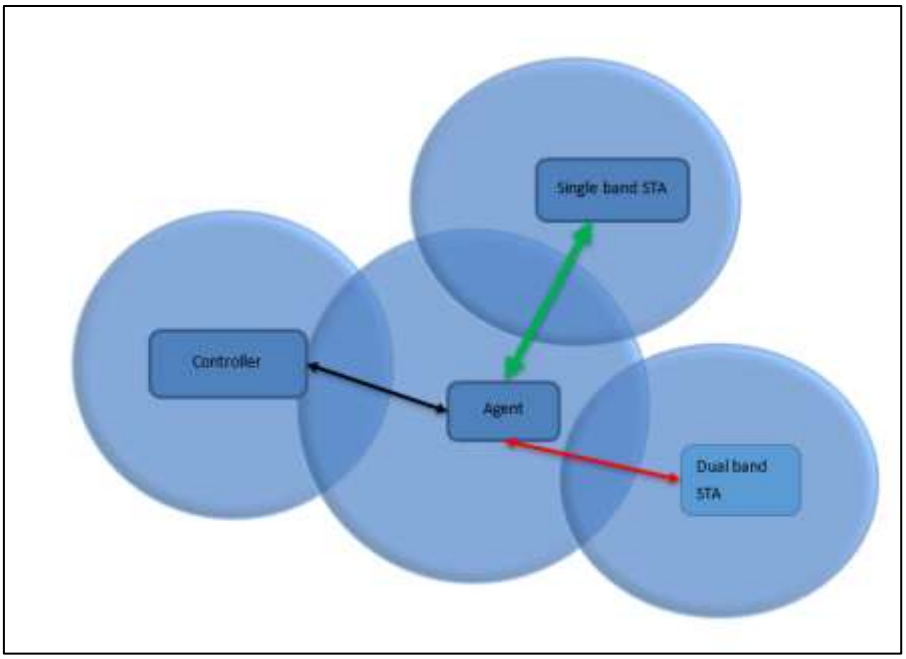


Figure 9- 8. Band Steering example Image 3

2: agent_post_assoc_rssi_active_ug:
Steps:
1. Configure MAP agent and controller in HE mode wpa2_psk_aes and keep centralized steering enabled, phy based steering and nol_multiap steering disabled on both.
2. Onboard the agent with the controller.
3. Blacklist the 11KV dual band STA for some certain time on 5G interface of the agent.

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4. Connect the STA to 2G interface of the agent.
5. Run traffic with STA, just to make it active
6. Wait for the blacklist to be over.
7. Now make RSSI good in 5G.

Expected result:
Check the dual band STA get steered to 5G interface of the agent.

Legends used in below diagrams:
5G connection with just nominal traffic to make STA active =
2G connection with just nominal traffic to make STA active =

MAP onboarding =

When the STA is out of Agent's 5G range and within Agent's 2G range:

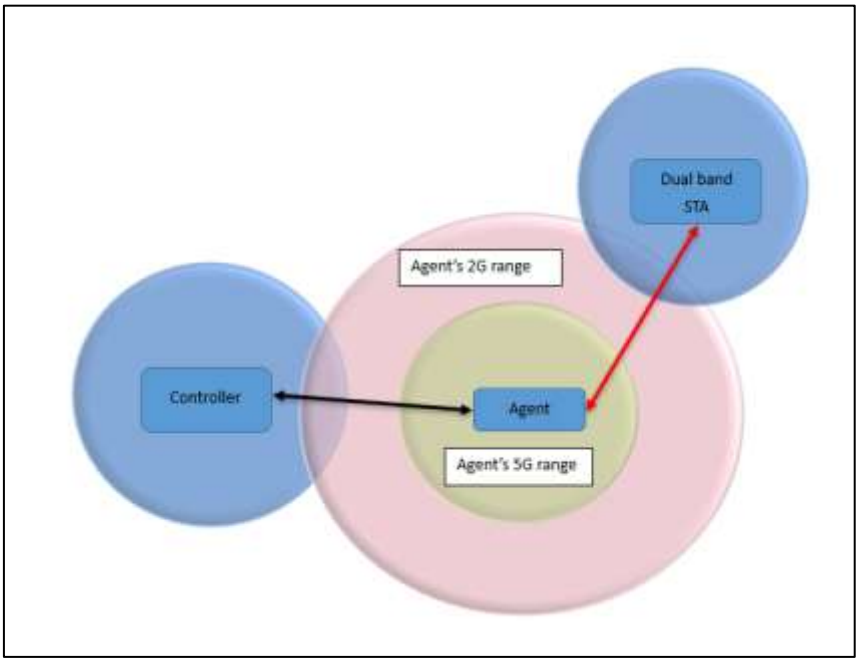


Figure 9- 9. Band Steering example Image 4

When the STA is within Agent's 5G range (the STA got steered to 5G interface of the Agent):

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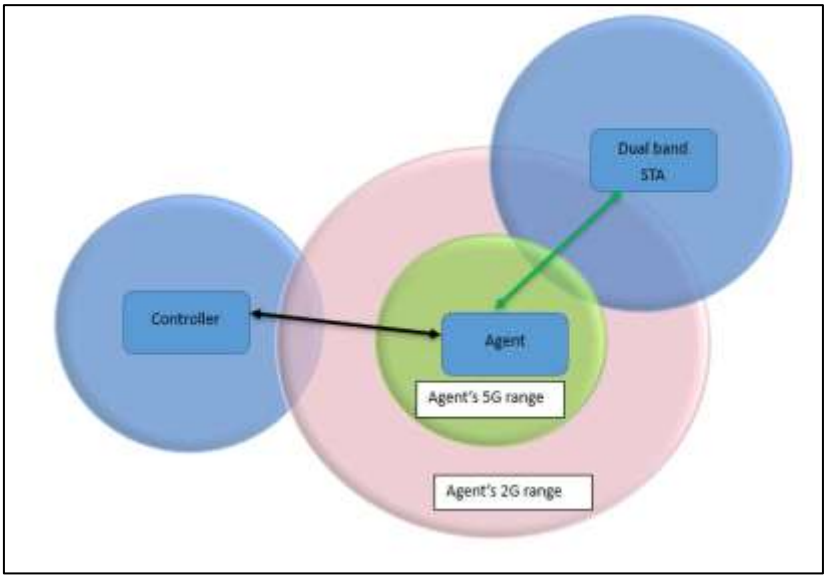


Figure 9- 10. Band Steering example Image 5

9.7 Known IOT Issues

Table 9- 4. Known IOT issue for steering

Issue	Impact	Seen with	Root Cause	Resolution
802.11k failure with some phones when PMF is enabled	MAP Steering; Post Association Active Steering; Offloading	• Iphone 6s: MN8X2HN/A; IOS Version 12.1.4 • Samsung Galaxy S8: SM-G950FD, Android 9.0(Feb 15 2019) • Samsung Galaxy S9	Problem with third party STA wherein they do not encrypt the RRM response frames •	Disable PMF on the APUT
BTM Failure: Some phones reject BTM requests most of the times	BTM Steering(and hence Active Steering) not possible	• Samsung Galaxy S8: SM-G950FD, Android 9.0(Feb 15 2019) • Samsung Galaxy S9. • Samsung Galaxy S10. •	Unknown	None; If the Phone is rejecting BTM request, it will not be active steered
Some STAs advertise their capability to support DFS channel (through 11k), but never honor the BTM request to move to DFS Channel BSS.	BTM Steering(and hence Active Steering) not possible to the DFS channel	• Nokia 5.1: TA-1102; Android 9.0	Problem with third party STA	None; If the Phone is rejecting BTM request, it will not be active steered

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BTM Failure: Some stations respond to BTM requests with BTM response success but do not steer		• Intel AX 200		None; If the stations is not steering after BTM Response, it will not be active steered
No BTM response for some STA	Impact on BTM steering	• Realme XT when MBO enabled • Oneplus when PMF enabled	Problem with third party STA	None; If the stations is not responding steering cannot be done

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10 3rd party controller co-exists with MTK agent

10.1 Introduction

The MTK agent and a third-party controller application can operate on the same device. The third-party controller uses the Multi-AP algorithm/1905 message to control all Multi-AP agents, both local and remote ones.

- The controller will utilize the MAP algorithm and send a 1905 message to both local (Agent1) and remote agents.
- Agent1 received a 1905 message from the bridge for controlling local devices.

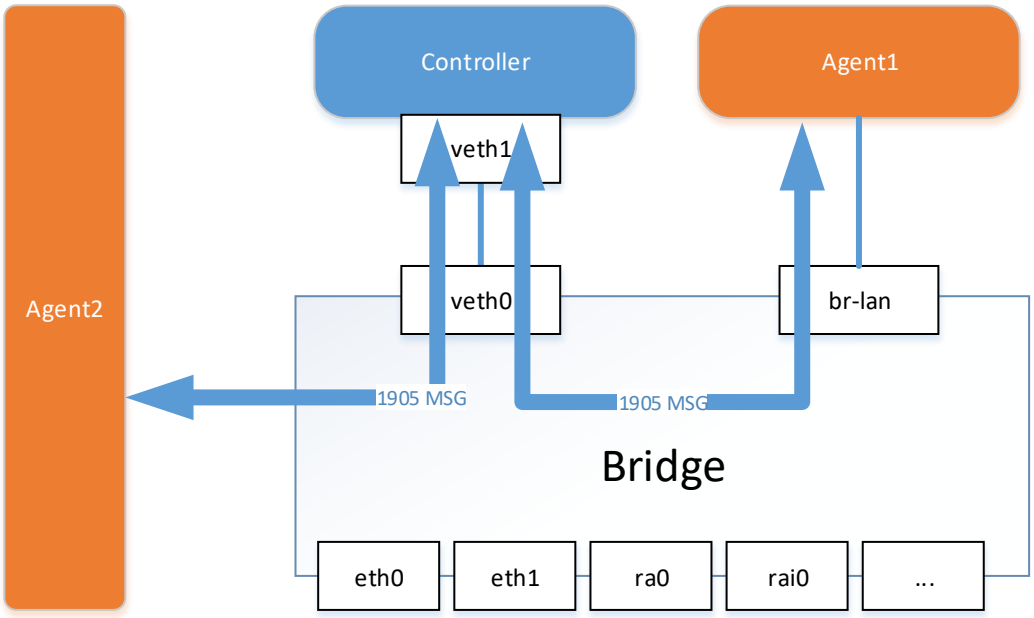


Figure 10- 1. 3rd party controller architecture overview

10.2 Setup Environment

10.2.1 Add virtual interface for controller

Please enter the set command in the console after the controller device has finished booting up:

1. *ip link add veth0 type veth peer name veth1*
2. *ip link set dev veth0 up*
3. *ip link set dev veth1 up*
4. *ifconfig veth1 xx.xx.xx.xx*// set IP address for veth1, for example: 192.168.1.100
5. *brctl addif br-lan veth0*

The controller application must create a socket connection to the veth1, which will be utilized for sending and receiving 1905 messages to both local and remote agents.

10.2.2 Add veth0 to the interface list of agent1

When Agent1 receives a 1905 message, it is required to identify from which interface it was received and record that information. To initiate an interface list during launch, Agent1 reads the configuration file located at /etc/map/1905d.cfg. The user must add the following two lines that specify the new veth interface:

#veth0
veth=veth0

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10.3 Onboarding and Auto Configuration

10.3.1 Agent1 onboarding and auto configuration

1. Execute the EashMesh_openwrt.sh script to enable Easy Mesh on Agent1.
2. Launch 3rd controller application.
3. Please initiate the onboarding process for Agent1 using the following command:
   wappctrl ra0 map set_bh_type eth

Upon completion of Step 3, Agent1 onboarding will be finalized, triggering an automatic configuration process. Both Backhaul and Fronthaul BSS will be configured accordingly.

10.3.2 Agent2 and other enrollee onboarding and auto configuration

10.3.2.1 Wireless onboarding

1. Trigger WPS on Agent1 with command:
   wappctrl ra0 wps_pbc
2. Trigger WPS on Agent2 with command:
   wappctrl apcli0 wps_pbc

The onboarding process for Agent2 will be successful, triggering the auto configuration.

10.3.2.2 Ethernet onboarding

Connecting Agent2 and the controller using Ethernet will trigger an automatic configuration process.

10.4 Get Topology

Once Agent1 has successfully completed onboarding, the topology will be displayed on webUI:
Controller ←→ Agent1
After Agent2 onboarding success, the topology would be *Controller ←→Agent1 ←→Agent2*.

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11 Appendix

11.1 Configuration difference between LSDK and OpenWRT

MAP LSDK does not support persistent file system to store configurable parameters. Hence all configurations are first stored onto the non-volatile memory space using the supported APIs. These configurations are then read back using supported APIs from NVRAM and written into a file after reboot.

However OpenWRT SDK supports persistent file system. It is not required to write the configurations into the NVRAM first. User can directly write into the configuration files to change the configuration and it will be retained across reboot.

Majority of configuration examples present in this document are written keeping LSDK in mind. When extended to OpenWRT SDK, the commands should be changed accordingly. The appropriate file where the configuration parameter should be present against the configuration parameter.

An example of how a parameter is to be configured on two SDKs is below:

for LSDK

- nvram_set 2860 DeviceRole 2

for OpenWRT

- wificonf -f <config file path> set DeviceRole 2

11.2 How to enable MAP

Please enable the macro MAP_SUPPORT to enable MAP function in driver Kconfig file.

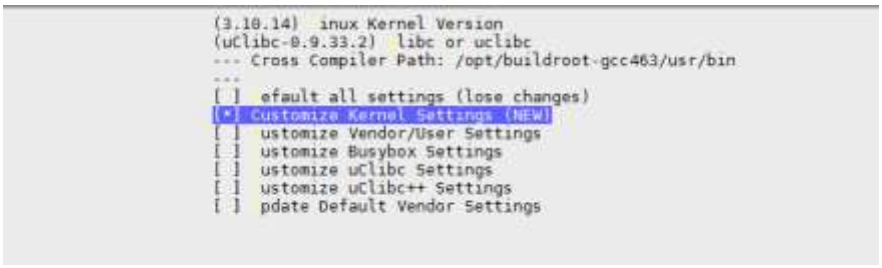


Figure 11- 1. How to enable MAP Image 1



Figure 11- 2. How to enable MAP Image 2

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11.3 How to enable MAP R2

Mediatek MAP R2 Turnkey solution utilizes all the MAP R1 turnkey features and further enhances them by making use of the newly define frames in MAP Specification version R2. Apart from this there are a few newly added features also. All these features are explained in details in respective chapters above.

1. Data Elements
2. Channel Planning
3. DFS CAC
4. WPA3 support with MAP
5. Traffic separation

This section describes briefly the changes that have been done for compile time options in various SDKs and run time option to enable MAP R2.

11.3.1.1 Compile time flag

MAP R2 feature compilation flags needs to be enabled for driver and applications for different SDKs in the below manner:

11.3.1.2 AX1800

1. Autobuild option :
   - Path : **autobuild/mt7621-mt7915-AP-AX1800/.config** and **autobuild/mt7621-mt7915-AP-AX1800/config-4.4**
   - Driver Compile flag in autobuild config-4.4: **CONFIG_MAP_R2_VER_SUPPORT**
   - Application Compile flag in autobuild .config: **CONFIG_MAP_R2_SUPPORT**
2. make menuconfig/ make kernel_menuconfig option
   - To enabled MAP R2 on driver use command "**make kernel_menuconfig"**
     - Enable "**Multi-AP R2 version support**" under MAP Support.
   - To enable MAP R2 on applications use command "**make menuconfig"**
     - Enable **"MAPR2"** under mapd daemon check

11.3.1.3 AX3600

1. Autobuild option
   - Path : **autobuild/mt7622-mt7915-AP-AX3600/.config**
   - Driver Compile flag in autobuild .config: **CONFIG_MTK_MAP_R2_VER_SUPPORT**
   - Application Compile flag in autobuild .config: **CONFIG_MAP_R2_SUPPORT**
2. make menuconfig option
   - To enabled MAP R2 on driver use command "**makmenuconfig"**
   - Enable **"Multi-AP R2 version support"** under MAP Support.
   - To enable MAP R2 on applications use command "**make menuconfig"**
   - Enable **"MAPR2"** under mapd daemon check

11.3.1.4 AX3200

1. Autobuild option
   - Path : **autobuild/mt7622-mt7915-AP-AX3200/.config**
   - 7622 sku Driver Compile flag in autobuild .config: **CONFIG_MTK_APSOC_WIFI_MAP_R2_VER_SUPPORT**
   - 7915 sku Driver Compile flag in autobuild .config: **CONFIG_MTK_MAP_R2_VER_SUPPORT**
   - Application Compile flag in autobuild .config: **CONFIG_MAP_R2_SUPPORT**

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2. make menuconfig option

- To enabled MAP R2 on driver use command "**make menuconfig"**
- Enable **"Multi-AP R2 version support"** under MAP Support.
- To enable MAP R2 on applications use command "**make menuconfig"**
- Enable **"MAPR2"** under mapd daemon check

For Driver user will find such type of option in menuconfig

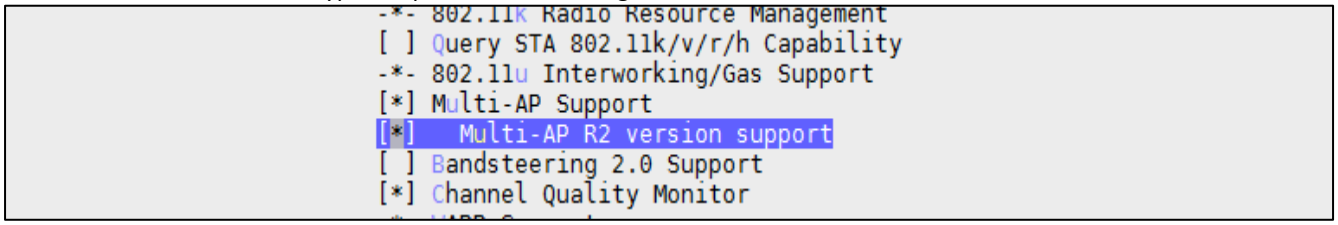


Figure 11- 3. How to enable MAP R2 Image 1

For applications use will find such option in menuconfig

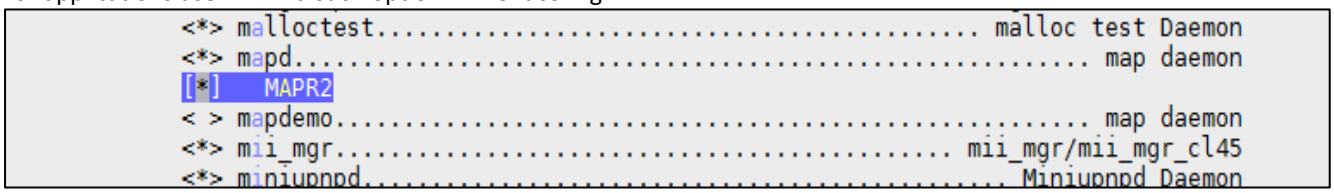


Figure 11- 4. How to enable MAP R2 Image 2

11.3.1.5 AX7800

1. Enable TurnKey by dat files(manually).
Need to add MapMode=1 in all dat files present in all dat files of ax7800.

```
root@OpenWrt:~# cat /etc/wireless/mediatek/mt7986-ax7800.dbdc.b0.dat | grep MapM
MapMode=1
root@OpenWrt:~#
root@OpenWrt:~#
root@OpenWrt:~#
root@OpenWrt:~# cat /etc/wireless/mediatek/mt7986-ax7800.dbdc.b1.dat | grep MapM
MapMode=1
root@OpenWrt:~# cat /etc/wireless/mediatek/mt7916-ax7800.dbdc.b1.dat | grep MapM
MapMode=1
root@OpenWrt:~# cat /etc/wireless/mediatek/mt7916-ax7800.dbdc.b0.dat | grep MapM
MapMode=1
root@OpenWrt:~#
```

Figure 11- 5. How to enable Turnkey via MapMode in ax7800

2. BssidNum=1 in cat /etc/wireless/mediatek/mt7986-ax7800.dbdc.b0.dat, should not be increased Hardware limitation.



Figure 11- 6. Don't increase BSSID for ra0 radio in ax7800.

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11.3.2 Run Time option

On the board run time:

- In the file "/etc/map/1905d.cfg" , make sure for R2
  - **map_ver=R2**
- For R1 this value must be
  - **map_ver=R1**

- Need to do this setting at both controller and agent to identify the map version of the device. Run Easymesh after adding the setting.
- Note: If user wants to change the map version of already onboarded device, then 1905d.cfg needs to be changed accordingly and Easymesh script needs to be run on all the devices in the network.

11.3.3 Dat file settings on each board

After loading the image on the board first time, you need to ensure that the MAP Mode is set as turnkey. This can be enabled via the Web UI EasyMesh enable page or by manually giving the following nvram set commands.

- nvram_set rtdev MapMode 1
- nvram_set 2860 MapMode 1

On 5G BH interface you might want to enable dedicated radio support to do zero wait DFS, this can be done by setting: (see section 7.4.1 for more details)

- nvram_set rtdev DfsDedicatedZeroWait 1
- nvram_set rtdev DfsZeroWaitDefault 1

On AX3200 and AX3600 platforms MBO is not enabled by default, but MBO is a mandatory feature for MAP R2, so it can be enabled as follows for each BSS:

- nvram set rtdev MboSupport 1;1;1;1
- nvram_set 2860 MboSupport 1;1;1;1

- After these settings are done, each device must be rebooted once.

11.4 How to enable MAP R3

This section describes briefly the changes that have been done for compile time options in various SDKs and run time option to enable MAP R3.

11.4.1.1 Compile time flag

MAP R3 feature compilation flags needs to be enabled for driver and applications for different SDKs in the below manner:

11.4.1.1.1 Panther AX6000

a. Auto build option: Multiple autobuild options available for Panther, user need to refer correct panther autobuild folder for configuration.
   ii. Below show example of one of the Panther autobuild config:
      - Path : **autobuild/mt7986-AX6000/.config**
      - Driver Compile flag in autobuild .config: **CONFIG_MAP_R3_VER_SUPPORT**
      - Application Compile flag in autobuild .config: **CONFIG_MAP_R3_SUPPORT**

b. make menuconfig/ make kernel_menuconfig option
   - To enabled MAP R3 on driver use command "**make kernel_menuconfig"**
     - Enable "**Multi-AP R3 version support**" under MAP Support.
   - To enable MAP R3 on applications use command "**make menuconfig"**

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- Enable **“MAPR3”** under mapd daemon check

For Driver user will find such type of option in menuconfig

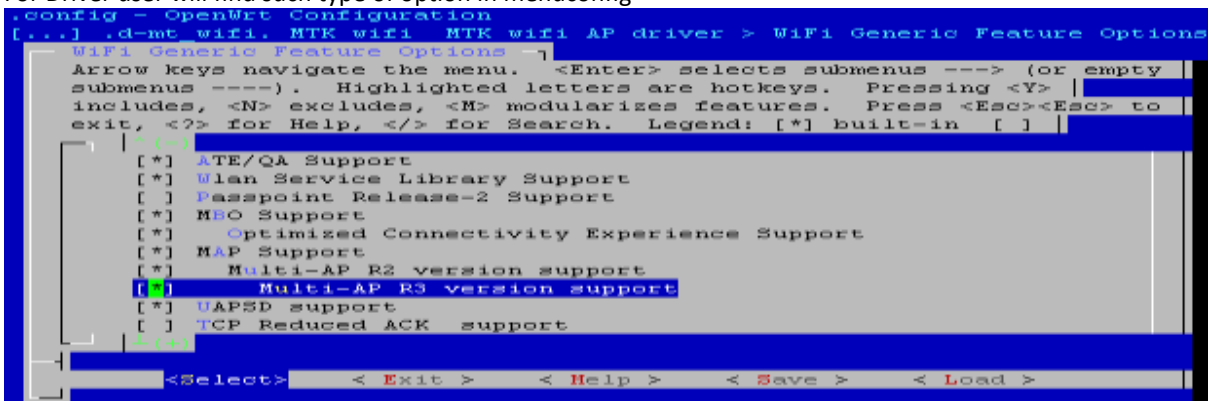


Figure 11- 7. How to enable MAP R3 Image 1

For applications use will find such option in menuconfig

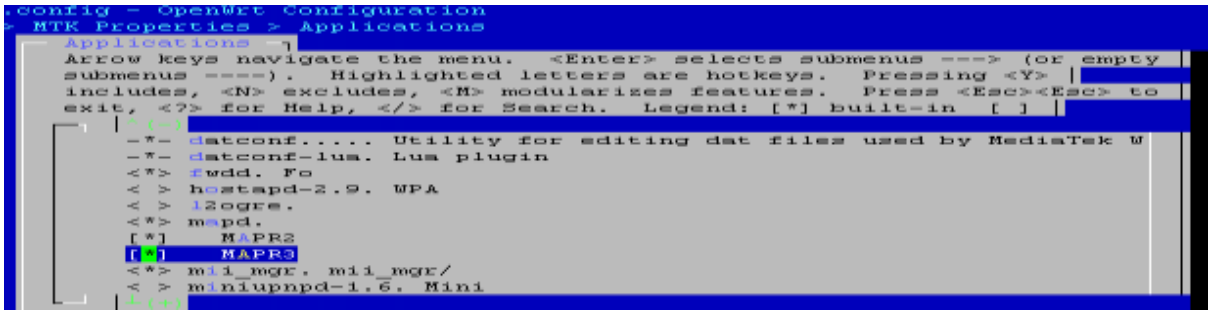


Figure 11- 8. How to enable MAP R3 Image 2

11.4.1.1.2 Compile Time Options

Below options are enabled using autobuild settings in mediatek provided image.

Mapd:
CONFIG_MAP_R3_SUPPORT=y in mapd_config
1905daemon:
MAP_R3_SUPPORT=y in config.mk
Wappd:
CONFIG_DPP_SUPPORT=y in wappd_config
Driver: In driver makefile, following option should be enabled:

```
ifeq ($(CONFIG_MAP_R3_VER_SUPPORT),y)
  EXTRA_CFLAGS += -DDPP_SUPPORT
  EXTRA_CFLAGS += -DDPP_R2_SUPPORT
  EXTRA_CFLAGS += -DMAP_R3_WF6
  EXTRA_CFLAGS += -DMAP_R3
endif
```

In driver kconfig, following should be set:

```
config MAP_R3_VER_SUPPORT
    bool "Multi-AP R3 version support"
    depends on MT_AP_SUPPORT
    depends on MAP_SUPPORT
    depends on MAP_R2_VER_SUPPORT
    default n
```


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11.4.2 Run Time option

On the board run time:

- In the file "/etc/map/1905d.cfg" , make sure for R3
  - **map_ver=R3**
- For R2 this value must be
  - **map_ver=R2**

- Need to do this setting at both controller and agent to identify the map version of the device. Run Easymesh after adding the setting.
-
- Note: If user wants to change the map version of already onboarded device, then 1905d.cfg needs to be changed accordingly and Easymesh script needs to be run on all the devices in the network.

11.5 How to enable MAP R4

This section describes briefly the changes that have been done for compile time options in various SDKs and run time option to enable MAP R4. It is mandatory for MAP R1, R2 and R3 to be supported for a MAP R4 SKU , so all the existing lags relevant for MAP R1, R2 and R3 will need to be enabled beforehand itself.

11.5.1.1 Compile time flag

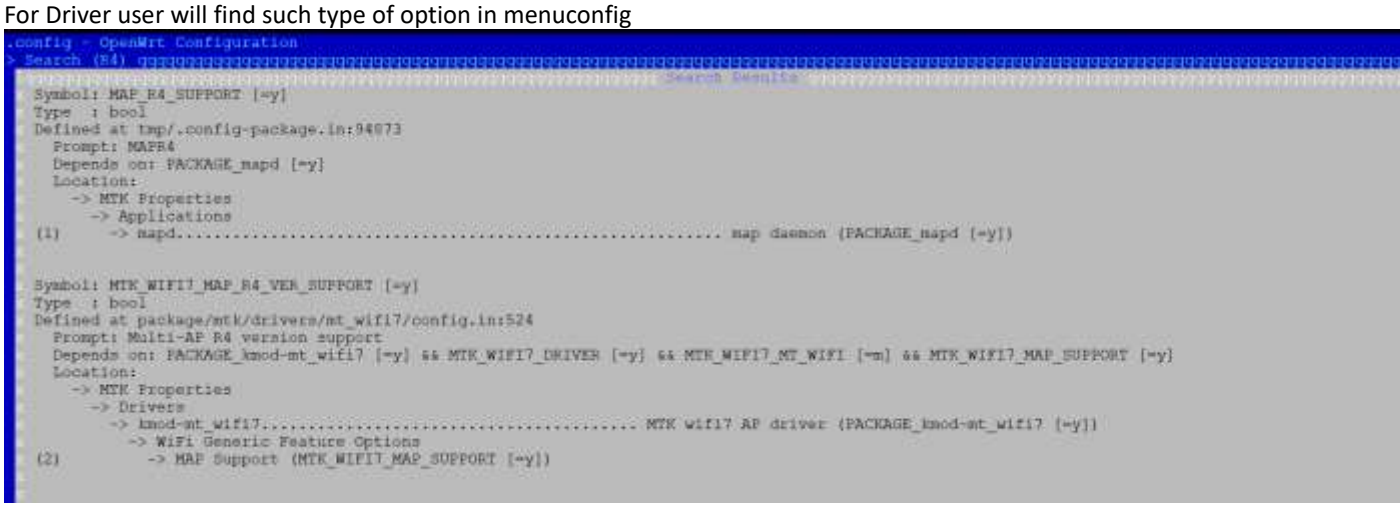
MAP R4 feature compilation flags needs to be enabled for driver and applications for different SDKs in the below manner:

11.5.1.1.1 Eagle

a. Auto build option: Multiple autobuild options available for Panther, user need to refer correct panther autobuild folder for configuration.
   i. Below show example of one of the Eagle autobuild config:
      - Path : **autobuild/mt7988-mt7990-BE19000/.config**
      - Driver Compile flag in autobuild .config: **CONFIG_MAP_R4_VER_SUPPORT**
      - Application Compile flag in autobuild .config: **CONFIG_MAP_R4_SUPPORT**

b. make menuconfig/ make kernel_menuconfig option
   - To enabled MAP R4 on driver use command "**make kernel_menuconfig"**
     - Enable "**Multi-AP R4 version support**" under MAP Support.
   - To enable MAP R4 on applications use command "**make menuconfig"**
     - Enable **"MAPR4"** under mapd daemon check

For Driver user will find such type of option in menuconfig



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Figure 11- 9. How to enable MAP R4 Image 1

For applications user will find below option in menuconfig

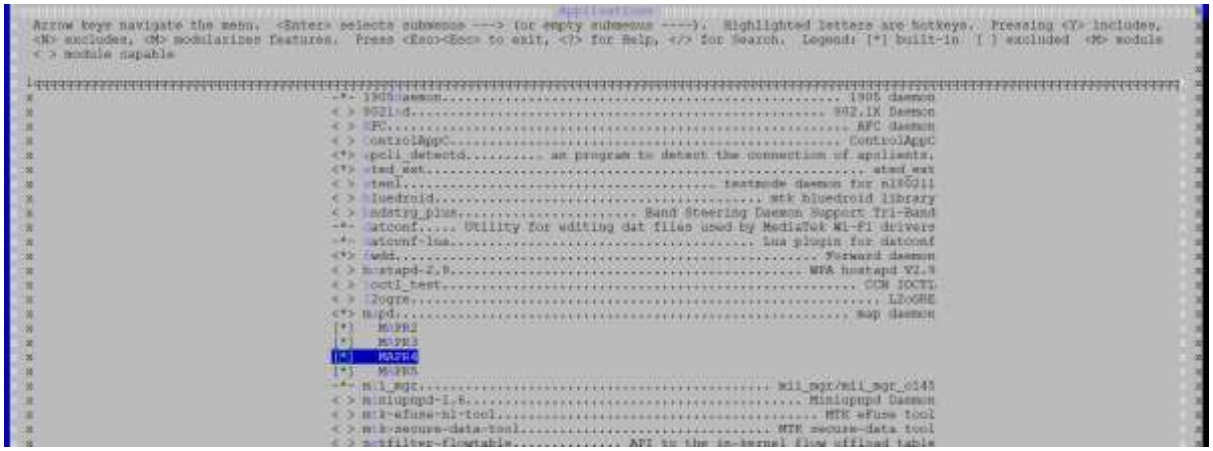


Figure 11- 10. How to enable MAP R4 Image 2

11.5.1.1.2 Compile Time Options

Below options are enabled using autobuild settings in mediatek provided image.

Mapd:

```
CONFIG_MAP_R4_SUPPORT=y
CONFIG_MTK_WIFI7_MAP_HOSTAPD_SUPPORT= y

ifeq ($(CONFIG_MTK_WIFI7_MAP_HOSTAPD_SUPPORT), y)
    CFLAGS += -DMTK_HOSTAPD_SUPPORT
    CFLAGS += -DMAP_EHT_SUPPORT
    CFLAGS += -DMTK_MLO_MAP_SUPPORT
Endif
ifeq ($(CONFIG_MAP_R4_SUPPORT),y)
CFLAGS += -DMAP_R4
endif
```

1905daemon:

```
CONFIG_MTK_WIFI7_MAP_HOSTAPD_SUPPORT= y
ifeq ($(CONFIG_MTK_WIFI7_MAP_HOSTAPD_SUPPORT), y)
    CFLAGS += -DMTK_MLO_MAP_SUPPORT
endif
```

Wappd:

```
CONFIG_MTK_WIFI7_MAP_HOSTAPD_SUPPORT=y
ifeq ($(CONFIG_MTK_WIFI7_MAP_HOSTAPD_SUPPORT), y)
    CFLAGS += -DMTK_HOSTAPD_SUPPORT
    CFLAGS += -DMAP_EHT_SUPPORT
    CFLAGS += -DMTK_TK_HOSTAPD_SUPPORT
    CFLAGS += -DMTK_MLO_MAP_SUPPORT
endif
```

11.5.2 Run Time option

On the board run time:

- In the file "/etc/map/1905d.cfg", **make sure for R4 also the configuration must remain as R3.**

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- **map_ver=R3**

- Need to do this setting at both controller and agent to identify the map version of the device. Run Easymesh after adding the setting.

Note: If user wants to change the map version of already onboarded device, then 1905d.cfg needs to be changed accordingly and Easymesh script needs to be run on all the devices in the network.

11.6 How to use WAPP daemon

Start command:
wapp -d1 -v2 –c<INF1_NAME> –c<INF2_NAME>

INF1_NAME and INF2_NAME stands for two BSS interfaces on which 802.11R is supported.
-c option is not required if 802.11R is not enabled on any interface.

There are two processes created with different PID but same COMMAND name.
WAPP supports multiple configuration files for different Wi-Fi interface (ex. ra0/ra1/ra2/rai0/rai1). Configuration files are used to initialize WAPP functions and default settings. It can be added in the file wapp_ap.conf. Currently, by default, only ra0 (wapp_ap_ra0.conf) is used to load all WAPP settings.

11.7 How to use 1905 daemon

This file is put in /etc/map/ and is shown as figure below:

```
# Controller's ALID
map_controller_alid=00:0c:43:88:88:88

# Agent's ALID
map_agent_alid=00:0c:43:22:22:22

#bridge interface
br=br-lan

#lan interface
lan=eth0
```

Figure 11- 11. 1905daemon uses 1905d.cfg file

If 1905daemon starts up as controller, then map_controller_alid will be needed. If 1905daemon starts up as agent, then map_agent_alid will be needed. User can add –d to add more log info in below command.
The cmd to start up 1905 daemon is:

Controller:
p1905_managerd -r0&
Agent:
p1905_managerd -r1 &

11.8 How to use MAP Daemon

MAP Daemon has to be started after wapp and 1905 daemon has been started.
To start MAP Daemon, following command needs to be given:

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mapd -I "/etc/map/mapd_cfg" -O "/etc/mapd_strng.conf" > /dev/console&

- /tmp/mapd_ctrl is the address for creating a socket for mapd_cli (control interface.)
- */etc/map/mapd_cfg* is path for MAP configuration file.**
- */etc/mapd_strng.conf* is path for AP steering configuration file.**
- User can add –d command line argument to enable more log info.

**Snapshot of the contents of these files are present below. Description of each parameter/flag present in the configuration file will be introduced in the respective feature's chapter.

11.9 MAP device Bring-up

11.9.1 EasyMesh (LSDK)

Steps to add binaries in LSDK

1. Precondition: Build the SDK once with the driver and make sure romfs folder is created.
    a. cd MTK_APSoC_SDK/source
2. Copy the following binaries in romfs/sbin
    a. mapd
    b. p1905_managerd
    c. wapp
    d. wappctrl
    e. mapd_cli
    f. EasyMesh.sh
3. Copy the following files in romfs/etc_ro
    g. wts_bss_info_config
    h. mapfilter.ko
4. Copy the following files in romfs/lib/
    i. libmapd_interface_client.so
5. Build the image again.

11.9.1.1 Steps to bring up MAP device (LSDK)

1. Pre-Condition: Build the image with MAP enabled driver and flash it to the board.
2. Ensure that following files are present in the image:
    a. mapd : MAP daemon
        i. /sbin/mapd
    b. p1905_managerd : 1905 Daemon
        i. /sbin/p1905_managerd
    c. wapp : WAPP application
        i. /sbin/wapp
    d. wappctrl : WAPP control interface
        i. /sbin/wappctrl
    e. mapd_cli : MAP Daemon control interface
        i. /sbin/mapd_cli
    f. libmapd_interface_client.so : library for interaction with mapd
        i. /lib/ libmapd_interface_client.so
    g. wts_bss_info_config : only for controller functionality
        i. Should be present at /etc_ro/ wts_bss_info_config
    h. mapfilter.ko
        i. /etc_ro/mapfilter.ko
3. Ensure that all the interfaces are part of the bridge.
4. Initial settings :
    a. nvram_set 2860 MapEnable 1

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i. This is used to enable the MAP feature.(command should be repeated on other nvram zones for other bands)
ii. The parameter is required in both driver .dat file and MAP configuration file. Hence nvram_set command is required in OpenWRT also.

b. nvram_set 2860 MAP_Turnkey 1
i. This is used to enable/disable MAP turnkey features such as Automatic BH prioritization, Best Uplink AP Selection on first band.(command should be repeated on other nvram zones for other bands)

c. nvram_set 2860 APSteerRssiTh -54
i. Set the default value of RSSI threshold

d. nvram set 2860 ChPlanningEnable 1:
i. To enable channel planning feature.

e. nvram_set 2860 radio_band "band_list"
i. To specify the bss list for configuring the BSS of devices in accordance from wts_bss_info_config file
ii. Format for band_list: "24G;5G;5G;"
iii. For Dual band: "24G;5G;"
iv. For Triband: "24G;5GL;5GH;"
v. By default value: "24G;5G;5G;"

5. To start controller functionality
a. nvram_set 2860 DeviceRole 1
b. nvram_set 2860 APSteerRssiTh -54
c. nvram_set 2860 SteerEnable 1
d. nvram_set 2860 CUOverloadTh_2G 80
e. nvram_set 2860 CUOverloadTh_5G_L 70
f. nvram_set 2860 CUOverloadTh_5G_H 70
i. This is the default value.
ii. Need to trigger reboot after giving nvram_set commands

g. Update /etc/map/wts_bss_info_config
i. If specific configuration needs to be given to an Agent, then add entries for that ALID
ii. For all agents to which common configuration need to be given, add entries with wildcard ALID (FF:FF:FF:FF:FF:FF)

h. Run EasyMesh.sh

6. To start agent functionality
a. nvram_set 2860 DeviceRole 2
i. Need to trigger reboot after this.

b. nvram_set 2860 SteerEnable 1
c. nvram_set 2860 CUOverloadTh_2G 80
d. nvram_set 2860 CUOverloadTh_5G_L 70
e. nvram_set 2860 CUOverloadTh_5G_H 70
f. Run EasyMesh.sh

11.9.2 EasyMesh (OpenWRT)

A sample script is provided to modify/prepare configuration files for all SKU and trigger all related daemons. Major responsibility covered by EasyMesh_openwrt.sh (or EasyMesh_openwrt_5_x.sh for SDK version 5.x and above) script are as below:

- Prepares list of WiFi interfaces available. List prepared is used to configure “bss_config priority” parameter in mapd_cfg and 1905d.cfg files.
- Initialize platform specific variables for mapd_cfg and 1905d.cfg files
- Prepares mapd_cfg file from mapd_user.cfg and mapd_default.cfg

a. Mapd_default.cfg is used to preserve all default parameter values.

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b. Mapd_user.cfg is used to preserve all parameters configured by the user. This file is used to override configuration parameters already present in mapd_default.cfg
c. If you are doing the settings by Web UI then the parameters inside mapd_user.cfg will be saved via web UI and used by MAPD.
d. But If you are doing device selection setting without web UI, then you need to add below lines in corresponding files to make the role of device as example :

i. To make a Controller :
   1. wificonf -f /etc/map/mapd_user.cfg set DeviceRole 1
   2. wificonf -f /etc/dpp_cfg.txt set allowed_role 2

ii. To make agent (setting present by default)
   1. wificonf -f /etc/map/mapd_user.cfg set DeviceRole 2
   2. wificonf -f /etc/dpp_cfg.txt set allowed_role 1

11.9.2.1 Steps to bring up MAP device (OpenWrt)

1. Pre-Condition: Build the image with MAP enabled driver and flash it to the board.
2. Ensure that following files are present in the image :
   a. mapd : MAP daemon
      i. /sbin/mapd
   b. p1905_managerd : 1905 Daemon
      i. /sbin/p1905_managerd
   c. wapp : WAPP application
      i. /sbin/wapp
   d. wappctrl : WAPP control interface
      i. /sbin/wappctrl
   e. mapd_cli : MAP Daemon control interface
      i. /sbin/mapd_cli
   f. libmapd_interface_client.so : library for interaction with mapd
      i. /lib/ libmapd_interface_client.so
   g. wts_bss_info_config : only for controller functionality
      i. Should be present at /etc/map/wts_bss_info_config
   h. Mapd_user.cfg : configuration file to preserve user specific configurations
      i. /etc/map/mapd_user.cfg
   i. Mapd_default.cfg : configuration file to preserve default configurations
      i. /etc/map/mapd_default.cfg
   j. 1905d.cfg : configuration file for 1905 daemon parameters
      i. /etc/map/1905d.cfg
3. Ensure that all the interfaces are part of the bridge.
4. Need to set appropriate ChannelGrp for triband system
5. Set MAP Mode =1 (for turnkey) in .dat file of all the radios and reboot the board.
   a. For ax7800 refer section 10.3.1.5.
6. Run easyMesh_openwrt.sh (or runs automatically) via startup scripts on latest versions.
7. Check via “ps” command that all daemons are up and running.
8. Trigger onboarding procedure on controller and agent

11.10 Steps to bring up Stand Alone Band Steering

To enable Stand Alone Band Steering (with EasyMesh disabled): -

1. Copy the following binaries in romfs/sbin
   a. bs20
   b. wapp

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c. wappctrl
d. mapd_cli
e. EasyMesh.sh

2. Copy the following files in romfs/lib/
   a. libmapd_interface_client.so
3. Build the image again.

- Pre-Condition: Build the image with MAP enabled driver and flash it to the board.
- Ensure that following files are present in the image :
  - bs20 : Band Steering Daemon
    1. /sbin/bs20
  - wapp : WAPP application
    1. /sbin/wapp
  - mapd_cli : MAP Daemon control interface
    1. /sbin/mapd_cli
  - libmapd_interface_client.so : library for interaction with bs20
    1. /lib/ libmapd_interface_client.so
- Initial settings :
  - **OpenWRT Platform**
    - wificonf -f /etc/map/mapd_user.cfg set DeviceRole 1
    - wificonf -f /etc/map/mapd_user.cfg set APSteerRssiTh -100
    - **nvram_set 2860 MapMode 2**
    - **nvram_set rtdev MapMode 2**
    - **nvram_set wifi3 MapMode 2(for Triband system)**
    - reboot
  - **LSDK(MT7621 + MT7615D)**
    - nvram_set 2860 MapEnable 0
    - nvram_set 2860 MAP_Turnkey 0
    - nvram_set 2860 BSEnable 1
    - nvram_set 2860 DeviceRole 1
    - nvram_set 2860 APSteerRssiTh -100
    - nvram_set 2860 MAP_Ext “0;0;0;0;0;0;0;0”
    - reboot

11.11 Sample data of wts_bss_info_config file

Sample data in this file could look like:

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```
#ucc_bss_info
1,00:0C:43:22:22:22 11x Multi-AP-5LG-1 0x0020 0x0008 maprocks1 1 0 hidden-N
2,00:0C:43:22:22:22 11x Multi-AP-5LG-2 0x0020 0x0008 maprocks2 0 1 hidden-Y
3,00:0C:43:22:22:22 11x Multi-AP-5LG-3 0x0020 0x0008 maprocks3 0 1 hidden-N
4,00:0C:43:22:22:22 11x Multi-AP-5LG-4 0x0020 0x0008 maprocks4 0 1 hidden-N
5,00:0C:43:22:22:22 8x Multi-AP-24G-1 0x0020 0x0008 maprocks1 1 0 hidden-N
6,00:0C:43:22:22:22 8x Multi-AP-24G-2 0x0020 0x0008 maprocks2 0 1 hidden-Y
7,00:0C:43:22:22:22 8x Multi-AP-24G-3 0x0020 0x0008 maprocks3 0 1 hidden-N
8,00:0C:43:22:22:22 8x Multi-AP-24G-4 0x0020 0x0008 maprocks4 0 1 hidden-N
9,00:0C:43:22:22:22 12x Multi-AP-5HG-1 0x0020 0x0008 maprocks1 1 0 hidden-N
10,00:0C:43:22:22:22 12x Multi-AP-5HG-2 0x0020 0x0008 maprocks2 0 1 hidden-Y
11,00:0C:43:22:22:22 12x Multi-AP-5HG-3 0x0020 0x0008 maprocks3 0 1 hidden-N
12,00:0C:43:22:22:22 12x Multi-AP-5HG-4 0x0020 0x0008 maprocks4 0 1 hidden-N
13,ff:ff:ff:ff:ff:ff 11x Multi-AP-5LG-1 0x0020 0x0008 maprocks1 1 0 hidden-N
14,ff:ff:ff:ff:ff:ff 11x Multi-AP-5LG-2 0x0020 0x0008 maprocks2 0 1 hidden-Y
15,ff:ff:ff:ff:ff:ff 11x Multi-AP-5LG-3 0x0020 0x0008 maprocks3 0 1 hidden-N
16,ff:ff:ff:ff:ff:ff 11x Multi-AP-5LG-4 0x0020 0x0008 maprocks4 0 1 hidden-N
17,ff:ff:ff:ff:ff:ff 8x Multi-AP-24G-1 0x0020 0x0008 maprocks1 1 0 hidden-N
18,ff:ff:ff:ff:ff:ff 8x Multi-AP-24G-2 0x0020 0x0008 maprocks2 0 1 hidden-Y
19,ff:ff:ff:ff:ff:ff 8x Multi-AP-24G-3 0x0020 0x0008 maprocks3 0 1 hidden-N
20,ff:ff:ff:ff:ff:ff 8x Multi-AP-24G-4 0x0020 0x0008 maprocks4 0 1 hidden-N
21,ff:ff:ff:ff:ff:ff 12x Multi-AP-5HG-1 0x0020 0x0008 maprocks1 1 0 hidden-N
22,ff:ff:ff:ff:ff:ff 12x Multi-AP-5HG-2 0x0020 0x0008 maprocks2 0 1 hidden-Y
23,ff:ff:ff:ff:ff:ff 12x Multi-AP-5HG-3 0x0020 0x0008 maprocks3 0 1 hidden-N
24,ff:ff:ff:ff:ff:ff 12x Multi-AP-5HG-4 0x0020 0x0008 maprocks4 0 1 hidden-N
```

Figure 11- 12. wts_bss_info_config file example 1

Or in MAP R3/R4 device with 6G radio support it may look like :

```
#ucc_bss_info
1,ff:ff:ff:ff:ff:ff 11x Multi-AP-1 0x00E0 0x0008 maprocks1 1 0 hidden-N 4095 pvid 5
2,ff:ff:ff:ff:ff:ff 11x Multi-AP-5LG-2 0x0060 0x0008 maprocks2 0 1 hidden-N 4095 N/A N/A
3,ff:ff:ff:ff:ff:ff 11x Multi-AP-5LG-3 0x0060 0x0008 maprocks3 0 1 hidden-N 4095 N/A N/A
4,ff:ff:ff:ff:ff:ff 11x Multi-AP-5LG-4 0x0060 0x0008 maprocks4 0 1 hidden-N 4095 N/A N/A
5,ff:ff:ff:ff:ff:ff 8x Multi-AP-1 0x00E0 0x0008 maprocks1 1 0 hidden-N 4095 N/A N/A
6,ff:ff:ff:ff:ff:ff 8x Multi-AP-24G-2 0x0060 0x0008 maprocks2 0 1 hidden-N 4095 N/A N/A
7,ff:ff:ff:ff:ff:ff 8x Multi-AP-24G-3 0x0060 0x0008 maprocks3 0 1 hidden-N 4095 N/A N/A
8,ff:ff:ff:ff:ff:ff 8x Multi-AP-24G-4 0x0060 0x0008 maprocks4 0 1 hidden-N 4095 N/A N/A
9,ff:ff:ff:ff:ff:ff 12x Multi-AP-1 0x00E0 0x0008 maprocks1 1 0 hidden-N 4095 N/A N/A
10,ff:ff:ff:ff:ff:ff 12x Multi-AP-5HG-2 0x0060 0x0008 maprocks2 0 1 hidden-N 4095 N/A N/A
11,ff:ff:ff:ff:ff:ff 12x Multi-AP-5HG-3 0x0060 0x0008 maprocks3 0 1 hidden-N 4095 N/A N/A
12,ff:ff:ff:ff:ff:ff 12x Multi-AP-5HG-4 0x0060 0x0008 maprocks4 0 1 hidden-N 4095 N/A N/A
13,ff:ff:ff:ff:ff:ff 13x Multi-AP-1 0x00C0 0x0008 maprocks1 1 0 hidden-N 4095 N/A N/A
14,ff:ff:ff:ff:ff:ff 13x Multi-AP-6G-2 0x0040 0x0008 maprocks2 0 1 hidden-N 4095 N/A N/A
15,ff:ff:ff:ff:ff:ff 13x Multi-AP-6G-3 0x0040 0x0008 maprocks3 0 1 hidden-N 4095 N/A N/A
16,ff:ff:ff:ff:ff:ff 13x Multi-AP-6G-4 0x0040 0x0008 maprocks4 0 1 hidden-N 4095 N/A N/A
```

Figure 11- 13. wts_bss_info_config file example 2

Note : If user needs to configure more than 4 MBSS then user needs to add corresponding entries in “wts_bss_info_config” file. Below example is for 5 MBSS in which entries has been added with SSID Multi-AP-5LG-5, Multi-AP-24G-5 and Multi-AP-5HG-5 respectively.

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```
root@LEDE:/# cat /etc/map/wts_bss_info_config
#ucc_bss_info
1,ff:ff:ff:ff:ff:ff 11x Multi-AP-5LG-1 0x0020 0x0008 maprocks1 1 1 hidden-N 4095 pvid 5
2,ff:ff:ff:ff:ff:ff 11x Multi-AP-5LG-2 0x0020 0x0008 maprocks2 0 1 hidden-N 4095 N/A N/A
3,ff:ff:ff:ff:ff:ff 11x Multi-AP-5LG-3 0x0020 0x0008 maprocks3 0 1 hidden-N 4095 N/A N/A
4,ff:ff:ff:ff:ff:ff 11x Multi-AP-5LG-4 0x0020 0x0008 maprocks4 0 1 hidden-N 4095 N/A N/A
5,ff:ff:ff:ff:ff:ff 11x Multi-AP-5LG-5 0x0020 0x0008 maprocks5 0 1 hidden-N 4095 N/A N/A
6,ff:ff:ff:ff:ff:ff 8x Multi-AP-24G-1 0x0020 0x0008 maprocks1 1 1 hidden-N 4095 N/A N/A
7,ff:ff:ff:ff:ff:ff 8x Multi-AP-24G-2 0x0020 0x0008 maprocks2 0 1 hidden-N 4095 N/A N/A
8,ff:ff:ff:ff:ff:ff 8x Multi-AP-24G-3 0x0020 0x0008 maprocks3 0 1 hidden-N 4095 N/A N/A
9,ff:ff:ff:ff:ff:ff 8x Multi-AP-24G-4 0x0020 0x0008 maprocks4 0 1 hidden-N 4095 N/A N/A
10,ff:ff:ff:ff:ff:ff 8x Multi-AP-24G-5 0x0020 0x0008 maprocks5 0 1 hidden-N 4095 N/A N/A
11,ff:ff:ff:ff:ff:ff 12x Multi-AP-5HG-1 0x0020 0x0008 maprocks1 1 1 hidden-N 4095 N/A N/A
12,ff:ff:ff:ff:ff:ff 12x Multi-AP-5HG-2 0x0020 0x0008 maprocks2 0 1 hidden-N 4095 N/A N/A
13,ff:ff:ff:ff:ff:ff 12x Multi-AP-5HG-3 0x0020 0x0008 maprocks3 0 1 hidden-N 4095 N/A N/A
14,ff:ff:ff:ff:ff:ff 12x Multi-AP-5HG-4 0x0020 0x0008 maprocks4 0 1 hidden-N 4095 N/A N/A
15,ff:ff:ff:ff:ff:ff 12x Multi-AP-5HG-5 0x0020 0x0008 maprocks5 0 1 hidden-N 4095 N/A N/A
root@LEDE:/#
```

Figure 11- 14. wts_bss_info_config file example 3

Important note about MLO supported devices

For Multi link operation (MLO) supported SKU's the "wts_bss_configuration" file should follow this note.
The security configurations like SSID, Authmode, passphrase, encryption type should be same for the interfaces associated with one MLD groups.

Example:
If the MLD groups are formed as below setting:

ra0, rai0, rax1 have config as "Multi-AP-1", "0x00E0 / 0x00C0 ", "maprocks1"
ra1, rai1, rax1 have config as "Multi-AP-2", "0x0060 / 0x0040 ", "maprocks2"
ra2, rai2, rax2 have config as "Multi-AP-3", "0x0060 / 0x0040 ", "maprocks3"
ra3, rai3, rax3 have config as "Multi-AP-4", "0x0060 / 0x0040 ", "maprocks4"

- ➔ Here since on 6G radio the minimum Authmode support is SAE only so we need to keep the 2G and 5G band I WPA2+WPS3 transition mixed mode support.
- ➔ SSID is same for each BSS of the MLD group
- ➔ Password is same for each BSS of the MLD group
- ➔ Similarly other configurations must also match.

```
root@OpenWrt:/# cat /etc/map/wts_bss_info_config
#ucc_bss_info
1,ff:ff:ff:ff:ff:ff 11x Multi-AP-1 0x00E0 0x0008 maprocks1 1 0 hidden-N 4095 pvid 5
2,ff:ff:ff:ff:ff:ff 11x Multi-AP-2 0x0060 0x0008 maprocks2 0 1 hidden-N 4095 N/A N/A
3,ff:ff:ff:ff:ff:ff 11x Multi-AP-3 0x0060 0x0008 maprocks3 0 1 hidden-N 4095 N/A N/A
4,ff:ff:ff:ff:ff:ff 11x Multi-AP-4 0x0060 0x0008 maprocks4 0 1 hidden-N 4095 N/A N/A
5,ff:ff:ff:ff:ff:ff 8x Multi-AP-1 0x00E0 0x0008 maprocks1 1 0 hidden-N 4095 N/A N/A
6,ff:ff:ff:ff:ff:ff 8x Multi-AP-2 0x0060 0x0008 maprocks2 0 1 hidden-N 4095 N/A N/A
7,ff:ff:ff:ff:ff:ff 8x Multi-AP-3 0x0060 0x0008 maprocks3 0 1 hidden-N 4095 N/A N/A
8,ff:ff:ff:ff:ff:ff 8x Multi-AP-4 0x0060 0x0008 maprocks4 0 1 hidden-N 4095 N/A N/A
9,ff:ff:ff:ff:ff:ff 12x Multi-AP-1 0x00E0 0x0008 maprocks1 1 0 hidden-N 4095 N/A N/A
10,ff:ff:ff:ff:ff:ff 12x Multi-AP-5GH-2 0x0060 0x0008 maprocks2 0 1 hidden-N 4095 N/A N/A
11,ff:ff:ff:ff:ff:ff 12x Multi-AP-5GH-3 0x0060 0x0008 maprocks3 0 1 hidden-N 4095 N/A N/A
12,ff:ff:ff:ff:ff:ff 12x Multi-AP-5GH-4 0x0060 0x0008 maprocks4 0 1 hidden-N 4095 N/A N/A
13,ff:ff:ff:ff:ff:ff 13x Multi-AP-1 0x00C0 0x0008 maprocks1 1 0 hidden-N 4095 N/A N/A
14,ff:ff:ff:ff:ff:ff 13x Multi-AP-2 0x0040 0x0008 maprocks2 0 1 hidden-N 4095 N/A N/A
15,ff:ff:ff:ff:ff:ff 13x Multi-AP-3 0x0040 0x0008 maprocks3 0 1 hidden-N 4095 N/A N/A
16,ff:ff:ff:ff:ff:ff 13x Multi-AP-4 0x0040 0x0008 maprocks4 0 1 hidden-N 4095 N/A N/A
root@OpenWrt:/#
```

Figure 11- 15. wts_bss_info_config file example 3

11.12 DPP APIs

Below are some APIs to be used for the DPP onboarding. Some APIs are to be used on controller, some on agents and some can be used on both (mentioned below in detail). Interface used in all commands is ra0. No other interface is to be mentioned after wappctrl.

- **dpp_bootstrap_gen:**

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Description: To Generate URI (DPP QR Code) on Agent. The only mandatory parameter for this command is type. All other parameters are optional, if user want to add in URI.

Device: Agent

Command: wappctrl ra0 dpp dpp_bootstrap_gen [Type] [Chan-list] [MAC Address] [Info] [Private Key]

Parameters:

- type: Type of bootstrapping. Currently only qrcode is supported
- chan-list: Channel to be added in the URI. Use the format opclass/channel number.
  Usage: chan=81/6,115/36
- MAC Address: Mac address of the device.
  Usage: mac=00:0C:43:62:44:BB
- INFO: Additional information about device(Like Serial number)
  Usage: info=SN=4774LH2b4044
- Private key: Private key of the generated URI if user want to use same URI. By default this key will be stored in DPP CFG file.
  Usage:
  key=30770201010420deac3fa1e062f67f49f4e4598a2f4ba39ece390654ef6421ab616592406d60dfa00a06082a8648ce3d030107a14403420004a98b1ee10a7d161155b557ceed6eaebcb6da99f060cf128e8c9b645bc561c1d5d52afb31d40943fe82ffeab82bc3f41e595c961e2bf59512a91c82bd3b2ee699

Example: wappctrl ra0 dpp dpp_bootstrap_gen type=qrcode



Figure 11- 16. Console log output example of command dpp dpp_bootstrap_gen

- **dpp_bootstrap_get_uri :**

Description: To retrieve the DPP URI with index. Index 1 would be own URI (Agent/Controller) and on controller user can get agent's URI with index greater than 1.

Device: Both controller and Agent

Command: wappctrl ra0 dpp dpp_bootstrap_get_uri "URI_INDEX"

Parameter:

- URI_INDEX : index corresponding to URI on either controller or agent

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Example: wappctrl ra0 dpp dpp_bootstrap_get_uri 1



Figure 11- 17. Console log output example of command dpp dpp_bootstrap_get_uri

- **dpp_bootstrap_remove :**

  Description: To remove the own URI on agent or Agent's URI on the controller. Index 1 would be own URI (Agent) and on controller user can remove agent's URI with index greater than 1.

  Device: Both Controller and Agent

  Command: wappctrl ra0 dpp dpp_bootstrap_remove "URI_INDEX"

  Parameter:
  - URI_INDEX : index corresponding to URI to be removed on controller

  Example: wappctrl ra0 dpp dpp_bootstrap_remove 1



Figure 11- 18. Console log output example of command dpp_bootstrap_remove

- **dpp_qr_code :**

  Description: To provide the DPP URI of agents to the controller. Multiple URIs can be provided to the controller for multiple Agents onboarding.

  Device: Controller

  Command: wappctrl ra0 dpp dpp_qr_code "DPP URI"

  Parameter:
  - DPP URI: Agent's generated URI to be onboarded. Multiple Agent's URI can be added.

  Example: wappctrl ra0 dpp dpp_qr_code "DPP:V:2;K:MDkwEwYHKoZIzj0CAQYIKoZIzj0DAQcDIgADezecPyVDIgxxRxGpNU7x9cHFBwjVFefryiAlBi6Ix+E=;;"

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Figure 11- 19. Console log output example of command dpp_qr_code

- **dpp_cce_indication :**

  Description: To Enable/Disable the CCE IE indication to R3 onboarded agents. This command is to be given on the controller. Almac field is optional for this command, If no almac is mentioned the CCE indication will be broadcast.

  Device: Controller

  Command: wappctrl ra0 dpp dpp_cce_indication [flag] [almac]

  Parameter:
  - Flag: To Enable/Disable CCE-IE indication corresponding to value 1/0.
    flag 0: To disable CCE-IE
    flag 1: To enable CCE-IE

  - Almac: almac of the device to send the CCE indication to. If no almac is specified then this message will be broadcasted.

  Example:
  - To enable the CCE IE for the almac 00:0c:43:49:a1:5c.
    wappctrl ra0 dpp dpp_cce_indication flag=1 almac= 00:0c:43:49:a1:5c

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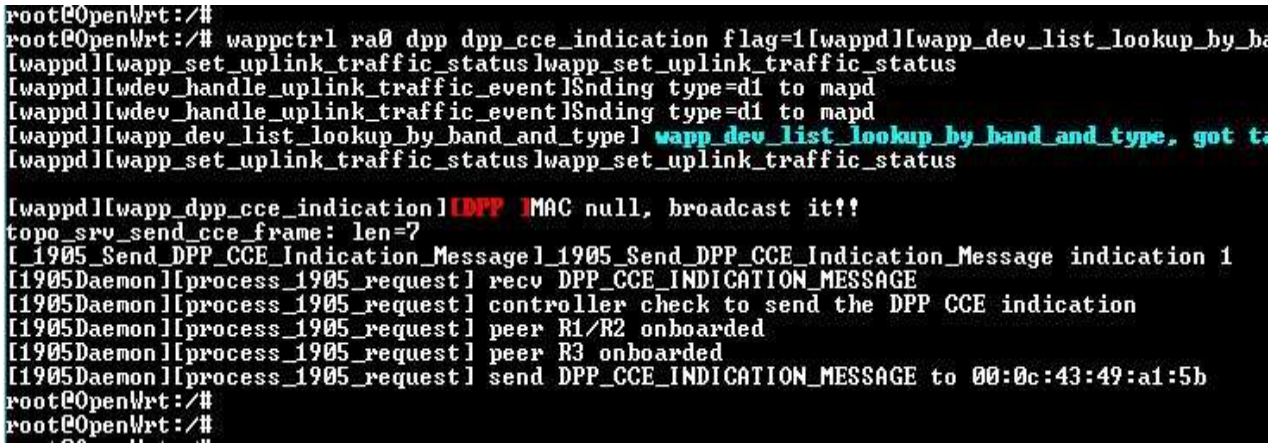


Figure 11- 20. Console log output example of command dpp_cce_indication

- **dpp_onboard_type :**

Description: To select the R3 device onboarding type. If user want to change the onboarding type of R3 device to non-DPP onboarding i.e. Legacy WPS onboarding then this command will be used. By default for R3 device, the DPP using WPS onboarding will be done if onboarding type is not set.
This command is only needed of the onboarding is triggered using WPS.

Device: Both Agent and Controller

Command: wappctrl ra0 dpp dpp_onboard_type [type]

Parameter:

- type: To select the onboarding type of the agent before triggering WPS onboarding.
  type 1: Legacy WPS
  type 0: Default DPP

Example:

- To set the onboarding type to legacy WPS for R3 device.
  wappctrl ra0 dpp dpp_onboard_type type=1



Figure 11- 21. Console log output example of command dpp_onboard_type

- **dpp_start :**

Description: To Trigger R3 DPP onboarding on the agent.

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Device: Agent

Command: wappctrl ra0 dpp dpp_start

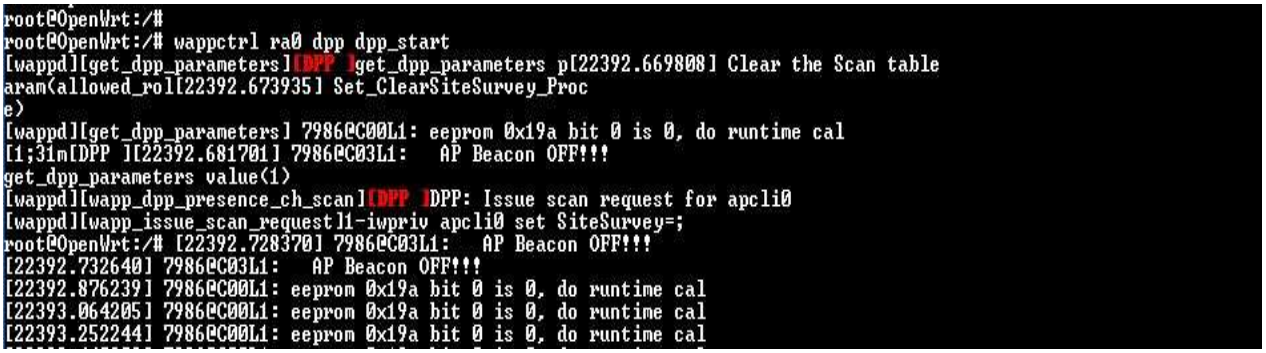


Figure 11- 22. Console log output example of command dpp dpp_start

- **dpp_chirp_stop :**

  Description: To stop the ongoing presence announcement on the Agent.

  Device: Agent

  Command: wappctrl ra0 dpp dpp_chirp_stop



Figure 11- 23. Console log output example of command dpp_chirp_stop

- **dpp_chirp_ch_en :**

  Description: To add/remove the channel in the channel list for the presence announcement.

  Device: Agent

  Command: wappctrl ra0 dpp dpp_chirp_ch_en [flag] [chlist]

  Parameter:
  - Flag: To Enable/Disable CCE-IE indication corresponding to value 1/0.
    flag 0: for removing channels from list
    flag 1: for adding channels in list
  - Chlist: Channel list to added/removed.
    Example: 40:44

  Example:
  - To add channel number 40 and 44 in the presence announcement channels.
    wappctrl ra0 dpp dpp_chirp_ch_en en=1 chlist=40:44

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Figure 11- 24. Console log output example of command dpp_chirp_ch_en en

- **dpp_chirp_notif :**

  Description: To clear the DPP onboarding state machine of Enrollee Multi-AP Agents from the memory of the existing Multi-AP Proxy Agents, or dropping/canceling any ongoing DPP sessions. Peer mac Is the mac address of the enrollee and almac is the optional field

  Device: Controller

  Command: wappctrl ra0 dpp dpp dpp_chirp_notif [hash_validity][flag][peer_mac][almac]

  Parameter:
  - Hash validity: set to zero to drop the ongoing DPP connection with mentioned Peer mac address.
  - Flag: To indicate the specific almac or broadcast almac to which chirp notification to be sent.
    flag 0: broadcast the message
    flag 1: send to specified almac
  - Peer mac: The mac address of the enrollee for which the connection needs to be dropped.
  - Almac: If the flag value in the second parameter is set then specify specific almac address of the proxy agent on which indication needs to be sent.

  Example:
  - wappctrl ra0 dpp dpp_chirp_notif 1 0 peer_mac
  - wappctrl ra0 dpp dpp_chirp_notif 1 1 peer_mac almac



Figure 11- 25. Console log output example of command dpp dpp_chirp_notif

- **dpp_reset_dpp_config_file:**

  Description: To clear the parameters like qr_count , akm , dpp_private_key , 1905valid , bhcredlen , fhcredlen , connector , netaccesskey from dpp_cfg.txt file

  Device : Controller and Agent

  Command: wappctrl ra0 dpp dpp_reset_dpp_config_file

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Figure 11- 26. Console log output example of command dpp_reset_dpp_config

11.13 List of Configuration file parameters

The parameters should be present in configuration file before MAP Turnkey applications are started. Following table list down the parameters along with the Description and Default Values. User can modify the parameters as per the requirement. However, the change will be reflected only after restarting the MAP Turnkey applications.

Table 11- 1. Configuration file parameters description

Parameter	Description	Default Value
DeviceRole	Decide device role as Controller/Agent/Auto	Agent
APSteerRssiTh	Minimum RSSI of a station below which APSteering logic will be triggered at controller	-54db
ChPlanningEnable	Enable/Disable convergent channel planning	Enabled
ChPlanningIdleByteCount	Configure number of Tx + Rx bytes per second below which system is considered idle	65536
ChPlanningIdleTime	Consecutive number of seconds for which, if TX + RX byte count is below ChPlanningIdleByteCount, channel planning will be retriggered.	1800
ChPlanningUserPreferredChannel5G	User preferred channel for 5G radio	Controller's current channel
ChPlanningUserPreferredChannel2G	User preferred channel for 5G radio	Controller's current channel
BhPriority2G	Bh connection priority for 2G radio	Minimum
BhPriority5GL	Bh connection priority for 5GL radio	Maximum
BhPriority5GH	Bh connection priority for 5GL radio	Maximum
BhPriority6G	Bh connection priority for 6G radio	Maximum
ThirdPartyConnection	When controller wants to connect to a third party AP via Wifi to gain controller connectivity then need to sent this flag at all network devices	0

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MAP_QuickChChange	This feature must remain enable so that set channel in driver happens via MAPChannel command	1
DualBH	Internal flag, feature not supported , must not be changed	0
MaxAllowedScan	This flag is not needed anymore. This was a parameter only used in mapd for dual backhaul, not needed now, since dual backhaul feature is disabled	blank
BHSteerTimeout	This flag is used in mapd to configure the timeout for BH steer command , after this timeout BH steer failure response to controller.	120secs
role_detection_external	Used in wappd in case of auto role, when a controller is moving back to become an agent, it should be able to use its existing configuration to connect to other bss. So controller must be able to save the profile on itself also. This feature only means that "write_backhaul_configs_all" should be called irrespective of device is a controller or agent.	Default value is 0. May be set if user wants below functionality
NonMAPAPEnable	Internal flag , not to be changed	1
SteerEnable	Enable/Disable AP Steering	Enabled
CUOverloadTh_2G	Percentage channel utilization above which the 2.4G radio is considered as overloaded	70%
CUOverloadTh_5G_L	Percentage channel utilization above which the 5GL radio is considered as overloaded	80%
CUOverloadTh_5G_H	Percentage channel utilization above which the 5GH radio is considered as overloaded	80%
ChPlanningInitTimeout	Duration after which channel planning will be triggered on a new device in a network	120 seconds
ChPlanningScanValidTime	Time after which channel planning R2 scan and score calculation will be rerun automatically after bootup channel planning is completed.	Default value is 14400secs (4 hours)

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DivergentChPlanning	Refer to AN section 7.2.8 for details	0
DhcpCtl	Enable/Disable DHCP control feature	1
NetworkOptimizationEnabled	Enable/Disable Network Optimization Feature	Enabled
NtwrkOptConnectWaitTime	The time (seconds) for which controller will wait after a new connection to trigger network optimization.	45 seconds
NtwrkOptDisconnectWaitTime	The time (seconds) for which controller will wait after a disconneciton to trigger network optimization	45 seconds
AutoBHSwitching	Enable/Disable Ethernet to WiFi switching automatically	Enabled
BandSwitchTime	Duration after which a connection will be attempted on a higher priority BH if connection is formed on low priority BH. Value 0 means connection to higher priority BH will not be attempted	120 Seconds
NtwrkOptPostCACTriggerTime	Duration after which network opt will be triggered once CAC gets completed on controller/agent	25 Seconds
NetworkOptPrefer5Gover2G	Enable/disable a fail safe method to trigger network optimization if BH stays on 2G for more than certain duration(default: 2 min)	Disabled
NetworkOptPrefer5Gover2GRetryCnt	Count for number of times network optimization will be triggered if BH stays on 2G	0
MaxStaAllowed	By default not present, Can be set to limit max number of STA associated to a device from 1 to 64	64
NtwrkOptDataCollectionTime	The maximum time for which controller will wait to get scan response for scan request in network optimization	300 seconds
NetOptUserSetPriority	Set “NetOptUserSetPriority” as 1 to consider BH priority on Network Optimzation.	0
DEStaConEventPath	Related to DE Standalone support, not needed in MAP Turnkey.	Disabled by default
CentralizedSteering	Internal flag , must not be changed	1

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SetPSCChannel_6G	This flag is applicable only when 6G band is present. If user wants channel planning to choose only valid PSC channels from 6G band channel list then this flag is needed to be set.	0
ChPlanningEnableR2withBW	Internal feature flag for considering available BW with Channel combination during channel planning best channel selection . Avoid changing default value of this flag.	By default ChPlanningEnableR2withBW ="empty". No need to do any setting as map will set it automatically based on map_version is R2 as per 1905d.cfg file. This parameter is only valid for the controller as entire ch planning algo runs only on controller. This parameter is enable by default internally if dev version is MAP R2, and comes in use only when "ChPlanningEnableR2" is enable
ScanThreshold2g	Please refer section 4.3.2.2.1	-88
ScanThreshold5g	Please refer section 4.3.2.2.1	-88
ScanThreshold6g	Please refer section 4.3.2.2.1	-88
NtwrkOptMldBhPriority	Please refer section 4.4.4, poin:6	1

11.14 List of MAPD_CLI Commands

Following table list down the APIs along with the Description and Default Values. User can use the commands to configure the behavior or query the status at run time.

Table 11- 2. MAPD_CLI command list

Command	Explanation	Usage
Dump topology	Show the topology database info	mapd_cli /tmp/mapd_ctrl dump_topology
		mapd_cli /tmp/mapd_ctrl dump_topology_v1
Show device stats	Show device info such as :	
	1. Client database	mapd_cli /tmp/mapd_ctrl mib
	2. Radios supported by device (radio index, id, channel)	
	3. BSS list (BSSID, SSID, channel)	

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Show clients being monitored for steering	Show the clients that are being monitored for steering	mapd_cli /tmp/mapd_ctrl mib sta_seen_list
Show blacklisted clients	Show the clients that are blacklisted for steering or other purpose	· mapd_cli /tmp/mapd_ctrl mib bl_list
Client statistics	Show client statistics such as bands supported, phy rate etc.	· mapd_cli /tmp/mapd_ctrl mib sta <client_id>
Enable channel planning	Dynamically enable/disable channel planning.	· mapd_cli /tmp/mapd_ctrl enable ch_planning <1/0>
Forcefully run channel planning	to trigger channel planning through command line interface	mapd_cli /tmp/mapd_ctrl ch_plan_R2 <band> 1 for 2.4G , 2 for 5G, 4 for 6G
Channel plan dump info	1.Trigger R2 scan 2. Dump scan results info	mapd_cli /tmp/mapd_ctrl ch_scan_trigger 00:0c:43:22:22:0c mapd_cli/tmp/mapd_ctrl ch_scan_res_dump
	3. Dump ch planning score info	mapd_cli/tmp/mapd_ctrl ch_plan_score_dump
Set Blacklist for a client	Mode 0 : Set Blacklist on all BSS of all devices in the network Mode 1: Set Blacklist on all BSS of 1 device in the network Mode 2: Set Blacklist on 1 BSS of 1 device in the network	mapd_cli /tmp/mapd_ctrl block_acl <Mode number> <Block/Unblock> <duration> <rest of parameters> Details : **Mode 0 :** mapd_cli /tmp/mapd_ctrl block_acl 0 0<duration> <client_addr> **Mode 1 :** mapd_cli /tmp/mapd_ctrl block_acl 1 0<duration> <cli_addr> <al_mac> **Mode 2:** mapd_cli /tmp/mapd_ctrl block_acl 2 0<duration> <cli_addr> <bssid>

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		• Rest of parameters means this parameters can vary depending on the different modes • For <block/unclock> ○ 0 means block ○ 1 means unblock ○ When we give command 1 instead of 0 to unblock then, <duration> is not needed Eg: If we want to unblock then give 1 instead of 0, in this case duration needs to be skipped Eg : **mapd_cli /tmp/mapd_ctrl block_acl 0 1 06:0c:43:46:35:40 06:0c:43:46:35:10**
For MLD we can block complete device by using MLD MAC address mapd_cli /tmp/mapd_ctrl block_acl <Mode 0> <block/unblock> <duration> <cli_addr> <mld_mac> **Details :** • Tested for Mode 0 with MLD groups • For <block/unclock> ○ 5 means MLD block ○ 6 means MLD unblock ○ When we give command 6 instead of 5 to unblock then, <duration> is not needed **Example :** **Block :** mapd_cli /tmp/mapd_ctrl block_acl 0 5 100 06:0c:43:46:35:40 06:0c:43:46:35:10 **Unblock:** mapd_cli /tmp/mapd_ctrl block_acl 0 6 06:0c:43:46:35:40 06:0c:43:46:35:10		
BackhaulSTA info	Get backhaulSTA info all devices in the network. It will show the bssid and medium type	mapd_cli /tmp/mapd_ctrl bh_info

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STA Medium info	Show the Connected clients and the band on which all the connected STA are present	mapd_cli /tmp/mapd_ctrl sta_med_info
Send renew	Handle the renew operation at local and send renew message to remote agents.	mapd_cli /tmp/mapd_ctrl renew
Onboarding and auto-config status	Get the onboarding and auto-configuration status of the own device. It will print two integer values as defined by below structure:	
	#define STATUS_BHSTA_UNCONFIGURED 0	mapd_cli /tmp/mapd_ctrl conn_status
	#define STATUS_BHSTA_BH_READY 1	
	#define STATUS_BHSTA_CONFIGURED 2	
	#define STATUS_BHSTA_CONFIG_PENDING 3	
	#define STATUS_BHSTA_WPS_TRIGGERED 4	
	#define STATUS_BHSTA_WPS_FAILED 5	
	#define STATUS_FHBSS_UNCONFIGURED 0	
	#define STATUS_FHBSS_WPS_TRIGGERED 1	
	#define STATUS_FHBSS_CONFIGURED 2	
	#define STATUS_FHBSS_WPS_FAILED 3	

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	#define STATUS_FHBSS_WPS_SUCCESSFULL 4 typedef struct GNU_PACKED _wapp_device_status { u32 status_fhbss; u32 status_bhsta; }wapp_device_status;	
Change debug level	Change debug level for Map Daemon	mapd_cli /tmp/mapd_ctrl set log_level <value>
Get Version Info	To get version info of the release/image	mapd_cli /tmp/mapd_ctrl version_info
Set the backhaul band priority dynamically	Set the backhaul STA band priority dynamically	mapd_cli /tmp/mapd_ctrl bh_priority <2G_priority> <5GL_priority> <5GH_priority >
ForceChannelSwitch	Set channel for an agent/controller This command is not valid if MAP network has MLD devices.	mapd_cli /tmp/mapd_ctrl forceChSwitch <ALMAC> <24G_channel> <5GL_channel> <5GH_channel>
Send higher layer data over 1905	Tx higher layer data over 1905 abstraction layer	mapd_cli /tmp/mapd_ctrl tx_higher_layer_data <ALMAC> <Protocol> <Payload_len> <Payload>
Receive higher layer data over 1905	Rx higher layer data over 1905 abstraction layer. Application will register a socket interface to receive data.	mapd_cli /tmp/mapd_ctrl rx_higher_data_reg_inf <unix_socket_path>

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Black list timeout	Black listing APs for this particular time period in case APs are not allowing connections (For e.g. RSN IE mismatch, MIC error) so that Cli is not stuck making connections on the same AP	mapd_cli /tmp/mapd_ctrl bl_timeout <secs>
Controller's Metric Policy parameters update	radio band 0/1/2	mapd_cli /tmp/mapd_ctrl metric_policy_set <radio band > <parameter ID> <value>
	BAND_2G 0x01	
	BAND_5GL 0x02	
	BAND_5GH 0x03	
	parameter ID 0 to 4	
	PARAM_ID_RCPI_THRESH 0	
	PARAM_ID_HYS 1	
	PARAM_ID_METRIC_INCLUSION 2	
	PARAM_ID_TRAFFIC_INCLUSION 3	
	PARAM_ID_CH_UTIL_THRESH 4	
User preferred skiplist and preferred list print cmd	To print user skip channel list and preferred channel list	To display complete skiplist:
		mapd_cli /tmp/mapd_ctrl get_skip_list skip_list
		To display complete preferred list:
		mapd_cli /tmp/mapd_ctrl get_skip_list prefered_list
Restrict BSS for steering	User can target BSS for steer by putting other BSS in restrict list	To Add the bssid in the restrict bssid list:
		mapd_cli /tmp/mapd_ctrl steer_restrict_bss ADD <BSSID>
		Example:
		mapd_cli /tmp/mapd_ctrl steer_restrict_bss ADD 11:22:33:44:55:66
		To Remove the bssid from the restrict bssid list:

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		mapd_cli /tmp/mapd_ctrl steer_restrict_bss REMOVE <BSSID>
		Example:
		mapd_cli /tmp/mapd_ctrl steer_restrict_bss REMOVE 11:22:33:44:55:66
Change the RSSI Threshold	User can change the RSSI threshold for Scan results run time	mapd_cli /tmp/mapd_ctrl scan_thresh_2g <RSSI>
		mapd_cli /tmp/mapd_ctrl scan_thresh_5g <RSSI>
		Example:
		mapd_cli/tmp/mapd_ctrl scan_thresh_2g -75
		mapd_cli /tmp/mapd_ctrl scan_thresh_5g -75
ACL CTRL	Set acl ctrl on particular device The <type> and <acl_cmd> has following values:	STA add/delete to ACL list:
		mapd_cli /tmp/mapd_ctrl acl_ctrl <type> <acl_cmd> <sta_mac> <bssid/almac>
	<type> Description	Configure ACL list:
	0 acl_cmd for all dev in n/w	mapd_cli /tmp/mapd_ctrl acl_ctrl <type> <acl_cmd> <bssid/almac>
	1 acl_cmd for dev	
	2 acl_cmd for bss	Example:
		▪ To add/del sta(06:0c:43:28:a1:63) on all bss of dev
	<acl_cmd> Description	mapd_cli /tmp/mapd_ctrl acl_ctrl 0 0 06:0c:43:28:a1:63
	0 Add entry to list	mapd_cli /tmp/mapd_ctrl acl_ctrl 0 1 06:0c:43:28:a1:63
	1 Del entry from list	
	2 Flush list	
	3 Set ACL policy to 0	
	4 Set ACL policy to 1 (Whitelist)	
	5 Set ACL policy to 2 (Blacklist)	
	6 Show ACL list in mapd	
Mandate client steer	trigger mandate steering	Example:
		mapd_cli /tmp/mapd_ctrl steermand 44:74:6c:b0:00:7d 00:0C:43:0C:0C:55
Dump Data Element	User can get dump the data element	Example:
		mapd_cli /tmp/mapd_ctrl get_de_stats 00:0c:43:48:f4:f0
BH steer command	To change the current BH of an agent	mapd_cli /tmp/mapd_ctrl steerbh < APCLI MAC Address of current connected BH> <target BSSID of other AP BH, may or may not be of same band as previous BH>

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Assoc sta list display with client id	To display the associated client list with client ID	mapd_cli /tmp/mapd_ctrl mib assoc_sta_list
STA Metric	To get the STA metric for all the STAs	mapd_cli /tmp/mapd_ctrl show_sta_metric
Bh Link Metric	To get the backhaul link metric for all wireless Agents	mapd_cli /tmp/mapd_ctrl show_bh_link_metric

11.15 Things to remember

2. MAP controller link with Non-MAP gateway:
   a. Controller can only be connected to a NonMAP gateway through an Ethernet link
   b. Non-MAP wifi link between Controller and Non-MAP gateway is not supported
   c. No Agent can connect directly to Non-MAP gateway. Only controller can connect directly to non-MAP gateway.

11.16 Sample Application for receiving events from MAP Daemon

A Sample application by the name "mapd_user_iface" is available which will continuously monitor user events coming from MAP Daemon and print the Event type. The end user is free to update the action to be taken when receiving the event, by modifying the sample application's code.
To start this application use the below command:

./mapd_user_iface /tmp/mapd_ctrl

After giving this command the "mapd_user_iface" will continuously monitor all the supported user events. The application parses the events in the below structure:
User event structure:

```
struct mapd_user_event
{
        int event_id;
        char event_body[0];
};
```


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11.16.1 List of Currently Supported User Events

Table 11- 3. Sample Application user events list

Event ID	Description	Structure format
WIRELESS_CLIENT_JOIN_NOTIF 1	Whenever wireless client joins	struct mapd_user_iface_wireless_client_event{ unsigned char sta_mac[ETH_ALEN]; unsigned char bssid[ETH_ALEN]; char ssid[MAX_SSID_LEN + 1];};
WIRELESS_CLIENT_LEAVE_NOTIF 2	Whenever wireless client leaves	struct mapd_user_iface_wireless_client_event{ unsigned char sta_mac[ETH_ALEN]; unsigned char bssid[ETH_ALEN]; char ssid[MAX_SSID_LEN + 1];};
ETHERNET_CLIENT_JOIN_NOTIF 3	Whenever ethernet client joins	struct mapd_user_iface_eth_client_event{ unsigned char sta_mac[ETH_ALEN]; unsigned char almac[ETH_ALEN];};
ETHERNET_CLIENT_LEAVE_NOTIF4	Whenever ethernet client leaves	struct mapd_user_iface_eth_client_event{ unsigned char sta_mac[ETH_ALEN]; unsigned char almac[ETH_ALEN];};
CH_PREF_NOTIF 5	Whenever channel preference report is changed	struct mapd_user_iface_ch_pref_event{ unsigned char radio_id[ETH_ALEN]; unsigned char almac[ETH_ALEN]; unsigned char op_class; unsigned char ch_num; unsigned char ch_list[MAX_CH_NUM]; unsigned char perference; unsigned char reason;};

HIGHER_LAYER_PAYLOAD 6	When user wants to send higher layer data	struct mapd_user_higher_layer_data_event
		{
		unsigned int buf_len;
		unsigned char buf[4096];
		};

ONBOARDING_STATUS_NOTIF 7	Whenever Auto Config Start and Auto Config Done	struct mapd_user_onboarding_event
		{
		unsigned int bh_type; //Eth-0, WiFi-1
		unsigned int onboarding_start_stop; //Start – 0, Done -1
		};

TRIGGER_MAP_STEERING_NOTIF 8	Whenever Steer notification received	struct mapd_user_iface_trigger_map_steering_event { unsigned char band; unsigned char sta_mac[ETH_ALEN];

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		char ssid[MAX_SSID_LEN + 1]; unsigned char cap_11v; char ul_rssi; };

GET_BEACON_REPORT_NOTIF 9	Whenever Beacon report recieved	struct mapd_user_iface_get_beacon_report_event
		{
		unsigned char band;
		unsigned char sta_mac[ETH_ALEN];
		char ssid[MAX_SSID_LEN + 1];
		unsigned char bssid[ETH_ALEN];
		char dl_rssi;
		unsigned char target_bssid[ETH_ALEN];
		char target_dl_rssi;
		};

GET_BTM_RESPONSE_NOTIF 10	Whenever BTM response received	struct mapd_user_iface_get_btm_response_event
		{
		unsigned char band;
		unsigned char sta_mac[ETH_ALEN];
		char ssid[MAX_SSID_LEN + 1];
		unsigned char status_code;
		};

SEC_CONNECTION_FAILURE 11	Whenever user connection fails	struct mapd_user_conn_fail_event
		{
		unsigned char reason[64];
		};

USER_FAIL_EVENTS 12	Whenever user events fail	struct mapd_user_fail_event
		{
		unsigned char reason_id;
		unsigned char reason[64];
		};

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11.17 Sample mapd_default.txt file

```
root@OpenWrt:/# cat /etc/map/mapd_default.cfg
mode=1
lan_interface=eth0
wan_interface=eth1
DeviceRole=2
APSteerRssiTh=-54
BhPriority2G=1
BhPriority5GL=3
BhPriority5GH=3
ChPlanningIdleByteCount=
ChPlanningIdleTime=
ChPlanningUserPreferredChannel5G=
ChPlanningUserPreferredChannel5GH=
ChPlanningUserPreferredChannel2G=
ChPlanningInitTimeout=120
NtwrkOptBootupWaitTime=45
NtwrkOptConnectWaitTime=45
NtwrkOptDisconnectWaitTime=45
NtwrkOptPeriodicity=3600
NetworkOptimizationScoreMargin=100
BandSwitchTime=
ScanThreshold2g=-75
ScanThreshold5g=-75
LowRSSIAPSteerEdge_RE=40
CUOverloadTh_2G=70
CUOverloadTh_5G_L=80
CUOverloadTh_5G_H=80
BhProfile0Valid=
BhProfile0Ssid=
BhProfile0AuthMode=
BhProfile0EncrypType=
BhProfile0WpaPsk=
BhProfile0RaID=
BhProfile1Ssid=
BhProfile1AuthMode=
BhProfile1EncrypType=
BhProfile1WpaPsk=
BhProfile1Valid=
BhProfile1RaID=
BhProfile2Ssid=
BhProfile2AuthMode=
BhProfile2EncrypType=
BhProfile2WpaPsk=
BhProfile2Valid=
BhProfile2RaID=
ChPlanningEnable=1
ChPlanningEnableR2=
SteerEnable=1
NetworkOptimizationEnabled=1
AutoBHSwitching=1
DhcpCtl=1
ThirdPartyConnection=0
```

Figure 11- 27. mapd_default.txt file sample

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11.18 Sample mapd_strng.txt file

```
root@OpenWrt:/# cat /etc/mapd_strng.conf
LowRSSIAPSteerEdge_RE=40
CUOverloadTh_2G=70
CUOverloadTh_5G_L=80
CUOverloadTh_5G_H=80
MetricPolicyChUtilThres_24G=70
MetricPolicyChUtilThres_5GL=80
MetricPolicyChUtilThres_5GH=80
ChPlanningChUtilThresh_24G=70
ChPlanningChUtilThresh_5GL=80
ChPlanningEDCCAThresh_24G=200
ChPlanningEDCCAThresh_5GL=200
ChPlanningOBSSThresh_24G=200
ChPlanningOBSSThresh_5GL=200
ChPlanningR2MonitorTimeoutSecs=300
ChPlanningR2MonitorProhibitSecs=900
ChPlanningR2MetricReportingInterval=10
ChPlanningR2MinScoreMargin=10
MetricRepIntv=10
MetricPolicyRcpi_24G=100
MetricPolicyRcpi_5GL=100
MetricPolicyRcpi_5GH=100
```

Figure 11- 28. Mapd_strng.conf file sample

11.19 Sample 1905d.cfg file

```
root@OpenWrt:/# cat /etc/map/1905d.cfg
# Controller's ALID
map_controller_alid=00:0c:43:49:c0:bb

# Has MAP agent on this device
map_agent=1

# This device is a MAP root
map_root=0

# Agent's ALID
map_agent_alid=00:0c:43:49:c0:bb

# Default Backhault Type
bh_type=eth

#Config Band setting of each Radio
radio_band=24G;5G;5G

#bridge interface
br_inf=br-lan

#lan interface

map_ver=R3

#fixed al_mac of al_inf(only wifi inf) mac
#al_inf=ra0

bss_config_priority=ra0;ra1;ra2;ra3;apcli0;rax0;rax1;rax2;rax3;apclix0;rai0;rai1;rai2;rai3;apclii0

#ethernet device name used to read the switch table
#do not set if not understanding
#ethernet_dev_name=

decrypt_fail_threshold=10
gtk_rekey_interval=3600

lan=lan1
lan=lan2
lan=lan3
lan=lan4
wan=eth1
root@OpenWrt:/#
```

Figure 11- 29. 1905d.cfg file sample

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11.20 MAP integration with UCI system

The abbreviation UCI stands for Unified Configuration Interface and is a system to centralize the configuration of OpenWrt services. Starting from the Eagle SKU , the Easymesh daemons will be using this UCI method to do all configuration.

The main modifications done to support MAP configuration via UCI is as below :

1. In new design we replace all easymesh file in UCI format so if any value written from WEB UI it should written in UCI file as shown below in /etc/config/ directory :
2. Only the configurable parameters will get updated in these files.
3. A new script /etc/uci2map.lua is created to convert the UCI files.

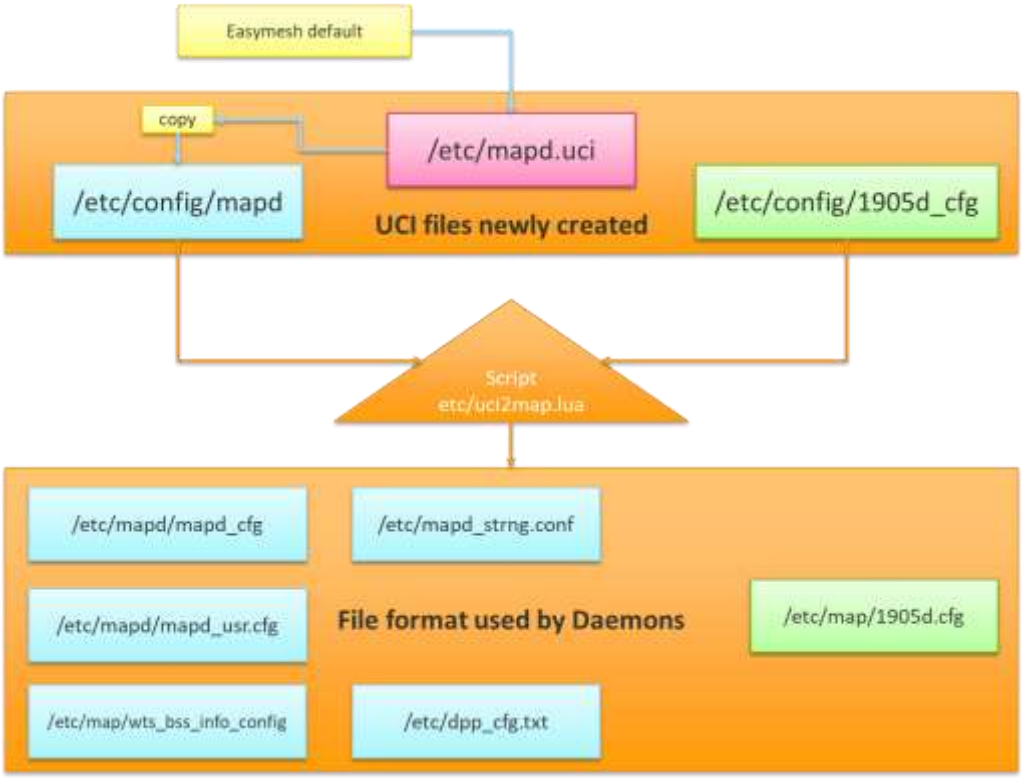


Figure 11- 30. UCI system for Easymesh

4. Internally from the wapp daemon all the wireless related configuration to be set after onboarding will now be written in etc/config/wireless instead of .dat files as earlier implementation. This is because latest driver uses etc/config/wireless configuration file instead of .dat files.

11.20.1 Sample mapd UCI file

This file will be used to configure map parameters for UCI supported SKU

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```
ot@OpenWrt:/# cat /etc/config/mapd

nfig mapd_cfg 'mapd_cfg'
        option lan_interface 'eth0'
        option wan_interface 'eth1'
        option APSteerRssiTh '-54'
        option BhPriority2G '3'
        option BhPriority5GL '1'
        option BhPriority5GH '1'
        option BhPriority6G '1'
        option ChPlanningIdleByteCount ' '
        option ChPlanningIdleTime ' '
        option ChPlanningUserPreferredChannel5G ' '
        option ChPlanningUserPreferredChannel5GH ' '
        option ChPlanningUserPreferredChannel2G ' '
        option ChPlanningUserPreferredChannel6G ' '
        option ChPlanningInitTimeout '120'
        option NtwrkOptBootupWaitTime '45'
        option NtwrkOptConnectWaitTime '45'
        option NtwrkOptDisconnectWaitTime '45'
        option NtwrkOptPeriodicity '3600'
        option NetworkOptimizationScoreMargin '100'
        option BandSwitchTime ' '
        option ScanThreshold2g '-88'
        option ScanThreshold5g '-88'
        option ScanThreshold6g '-88'
        option LowRSSIAPSteerEdge_RE '40'
        option BhProfile0Valid ' '
        option BhProfile0Ssid ' '
        option BhProfile0AuthMode ' '
        option BhProfile0EncrypType ' '
        option BhProfile0WpaPsk ' '
        option BhProfile0RaID ' '
        option BhProfile1Ssid ' '
        option BhProfile1AuthMode ' '
        option BhProfile1EncrypType ' '
        option BhProfile1WpaPsk ' '
        option BhProfile1Valid ' '
        option BhProfile1RaID ' '
        option BhProfile2Ssid ' '
        option BhProfile2AuthMode ' '
        option BhProfile2EncrypType ' '
        option BhProfile2WpaPsk ' '
        option BhProfile2Valid ' '
        option BhProfile2RaID ' '
        option ChPlanningEnable '1'
        option ChPlanningEnableR2 ' '
        option SteerEnable '1'
        option NetworkOptimizationEnabled '1'
```


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```
    option AutoBHSwitching '1'
    option DhcpCtl '1'
    option ThirdPartyConnection '0'
    option MAP_QuickChChange '1'
    option bss_config_priority 'ra0;ra1;apcli0;rai0;apcl
    option DualBH ' '
    option MetricRepIntv '60'
    option MaxAllowedScan ' '
    option BHSteerTimeout '120'
    option NtwrkOptPostCACTriggerTime '30'
    option role_detection_external '0'
    option NetworkOptPrefer5Gover2G '0'
    option NetworkOptPrefer5Gover2GRetryCnt '0'
    option NonMAPAPEnable '1'
    option CentralizedSteering '1'
    option DivergentChPlanning '0'
    option LastMapMode '1'
    option NtwrkOptDataCollectionTime '60'
    option ChPlanningScanValidTime '14400'
    option NetOptUserSetPriority '0'
    option DESerialNumber 'v3.0'
    option DESoftwareVersion 'v3.0'
    option DEExecutionEnv 'Linux-Openwrt'
    option DEChipsetVendor 'MediaTek'
    option SetPSCChannel_6G '0'
    option NtwrkOptChUtilCollectionTime6G '10'
    option NtwrkOptDataChUtilThresh6G '20'
    option NtwrkOptDataChUtilEnDisable6G '1'
    option MapMode '1'
    option CUOverloadTh_2G '70'
    option CUOverloadTh_5G_H '80'
    option CUOverloadTh_5G_L '80'
    option CUOverloadTh_6G '70'
    option DeviceRole '1'
    option index '1'
    option mode '0'
    option priority '1'
    option TriBand '0'
```

```
config mapd_strng 'mapd_strng'
        option CUOverloadTh_5G_H '80'
        option CUOverloadTh_6G '70'
        option MetricPolicyChUtilThres_24G '70'
        option MetricPolicyChUtilThres_5GL '80'
        option MetricPolicyChUtilThres_5GH '80'
        option MetricPolicyChUtilThres_6G '80'
        option ChPlanningChUtilThresh_24G '70'
        option ChPlanningChUtilThresh_5GL '80'
        option ChPlanningChUtilThresh_6G '80'
        option ChPlanningEDCCAThresh_24G '200'
        option ChPlanningEDCCAThresh_5GL '200'
        option ChPlanningEDCCAThresh_6G '200'
        option ChPlanningOBSSThresh_24G '200'
        option ChPlanningOBSSThresh_5GL '200'
        option ChPlanningOBSSThresh_6G '200'
        option ChPlanningR2MonitorTimeoutSecs '300'
        option ChPlanningR2MonitorProhibitSecs '900'
        option ChPlanningR2MetricReportingInterval '10'
        option ChPlanningR2MinScoreMargin '10'
        option MetricRepIntv '10'
        option MetricPolicyRcpi_24G '100'
        option MetricPolicyRcpi_5GL '100'
        option MetricPolicyRcpi_5GH '100'
        option MetricPolicyRcpi_6G '100'
        option LowRSSIAPSteerEdge_RE '40'
        option CUOverloadTh_2G '70'
        option CUOverloadTh_5G_L '80'

config mapd_user 'mapd_user'
        option APSteerRssiTh '-54'
        option MapMode '1'
        option DeviceRole '1'
        option mode '0'
        option TriBand '0'

config dpp_cfg 'dpp_cfg'
        option allowed_role '2'
        option presence_band_priority '5g'
```


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```
config iface '1'
        option mac 'ff:ff:ff:ff:ff:ff'
        option radio '11x'
        option ssid 'ddmulti-1'
        option authmode '0x00E0'
        option EncryptType '0x0008'
        option PSK 'maprocks1'
        option bhbss '1'
        option fhbss '0'
        option hidden 'N'
        option vlan '4095'
        option pvid '5'
        option pcp 'N/A'

config iface '2'
        option mac 'ff:ff:ff:ff:ff:ff'
        option radio '11x'
        option ssid 'ddmulti-5LG-2'
        option authmode '0x0060'
        option EncryptType '0x0008'
        option PSK 'maprocks2'
        option bhbss '0'
        option fhbss '1'
        option hidden 'N'
        option vlan '4095'
        option pvid 'N/A'
        option pcp 'N/A'

config iface '3'
        option mac 'ff:ff:ff:ff:ff:ff'
        option radio '11x'
        option ssid 'ddmulti-5LG-3'
        option authmode '0x0060'
        option EncryptType '0x0008'
        option PSK 'maprocks3'
        option bhbss '0'
        option fhbss '1'
        option hidden 'N'
        option vlan '4095'
        option pvid 'N/A'
        option pcp 'N/A'
```

Figure 11- 31. etc/config/mapd file example

Following steps are required to configure any map related parameter change:

- uci set mapd.<section name>.<parameter name>='<value>'
- uci commit mapd
- re-run Easymesh_openwrt script or reboot the board.

Eg. Need to set value of CUOverloadTh_6G in mapd_strng section

- uci set mapd.mapd_strng.CUOverloadTh_6G='70'
- uci commit mapd
- re-run Easymesh_openwrt script or reboot the board.

11.20.2 Sample 1905d_cfg UCI file

This file will be used to configure map version parameter for UCI supported SKUs

```
root@OpenWrt:/# cat /etc/config/1905d_cfg
config iface map
        option map_ver 'R3'
root@OpenWrt:/#
root@OpenWrt:/#
root@OpenWrt:/#
```

Figure 11- 32. etc/config/1905d_cfg example

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Eg. Set map ver R2

- uci set 1905d_cfg.map.map_ver='R2’
- uci commit 1905d_cfg

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